



UnityCloud Deployment Guide

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Affirmed Networks Inc.

A Microsoft Company

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Preface

This guide describes how to deploy Kubernetes clusters on various cloud platforms.

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide), installation team staff, and other qualified service personnel responsible for installing, provisioning, and monitoring the system.

The procedures in this guide require you to understand and follow the safety practices at your site, as well as those identified in this guide.

Release Information

See the UnityCloud™ Software Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. You can access Affirmed Networks Guides and Release Notes on the [Manuals](#) site. From here, you can also download a PDF version of an individual guide, or the full manual set for a product. For access and instructions, please contact Support.

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Deploy UnityCloud on OpenStack using OPM

This section describes how to deploy UnityCloud on OpenStack using Operations and Policy Manager (OPM). OPM is an application that allows administrators to deploy Kubernetes clusters on various cloud platforms. OPM is packaged as a Virtual Machine (VM) and deployed using OpenStack Heat templates.

After the OPM is deployed, an administrator can deploy a Kubernetes cluster. The Kubernetes cluster is where OPM will deploy a group of microservices that together perform a Cloud-Native Network Function (CNF). A group of microservices is also referred to as a federation.

The following table provides an example of the VMs for deploying a Kubernetes cluster.

Note: For the OPM VM, the minimum disk space is 50G. However, if you plan to use OPM's Docker Registry and Helm Chart Repository, you will need enough disk space to hold all the container images and charts.

Table 1. Example: VMs to Deploy a Kubernetes Cluster (Full Production)

VM	IP Address	vCPU	Memory	Disk Needed if VM ISSU disabled	Disk Needed if VM ISSU enabled
OPM	10.165.5.10	4	16G	200G	400G
Master-1	10.165.5.11	4	8G	150G	300G
Node-1	10.165.5.12	16	78G	150G	300G
Node-2	10.165.5.13	16	78G	150G	300G
Node-3	10.165.5.14	16	78G	150G	300G
Node-4	10.165.5.15	16	78G	150G	300G
Node-5	10.165.5.16	16	78G	150G	300G
Node-6	10.165.5.17	16	78G	150G	300G
Node-7	10.165.5.18	16	78G	150G	300G
Node-8	10.165.5.19	16	78G	150G	300G
Node-9	10.165.5.20	16	78G	150G	300G
Node-10	10.165.5.21	16	78G	150G	300G
Node-11	10.165.5.22	16	78G	150G	300G

Deploy UC on OpenStack using OPM

VM	IP Address	vCPU	Memory	Disk Needed if VM ISSU disabled	Disk Needed if VM ISSU enabled
Node-12	10.165.5.23	16	78G	150G	300G
Node-13	10.165.5.24	16	78G	150G	300G

The following table lists the steps for deploying UnityCloud on OpenStack using OPM. Be sure to perform the steps in the following order.

Note: For information about the Platform VM and OPM VM versions supported with this release, see the *Software Release Notes for UnityCloud*.

Table 2. Steps to Deploy UnityCloud on Openstack Using OPM

Step	Description
1.	Set Up OpenStack Environment
2.	Install OPM into OpenStack Select one of the following methods for installing OPM VMs into the OpenStack Environment. Deploy OPM Using Local Server Disks (Limited lab usage only) Deploy OPM Using Cinder Volumes (Recommended)
3.	Deploy Kubernetes Cluster
4.	Download Affirmed Networks Images
5.	Private Docker Registry
6.	Affirmed Networks, Inc.Helm ChartMuseum
7.	Manage Federations with OPM

Set Up OpenStack Environment

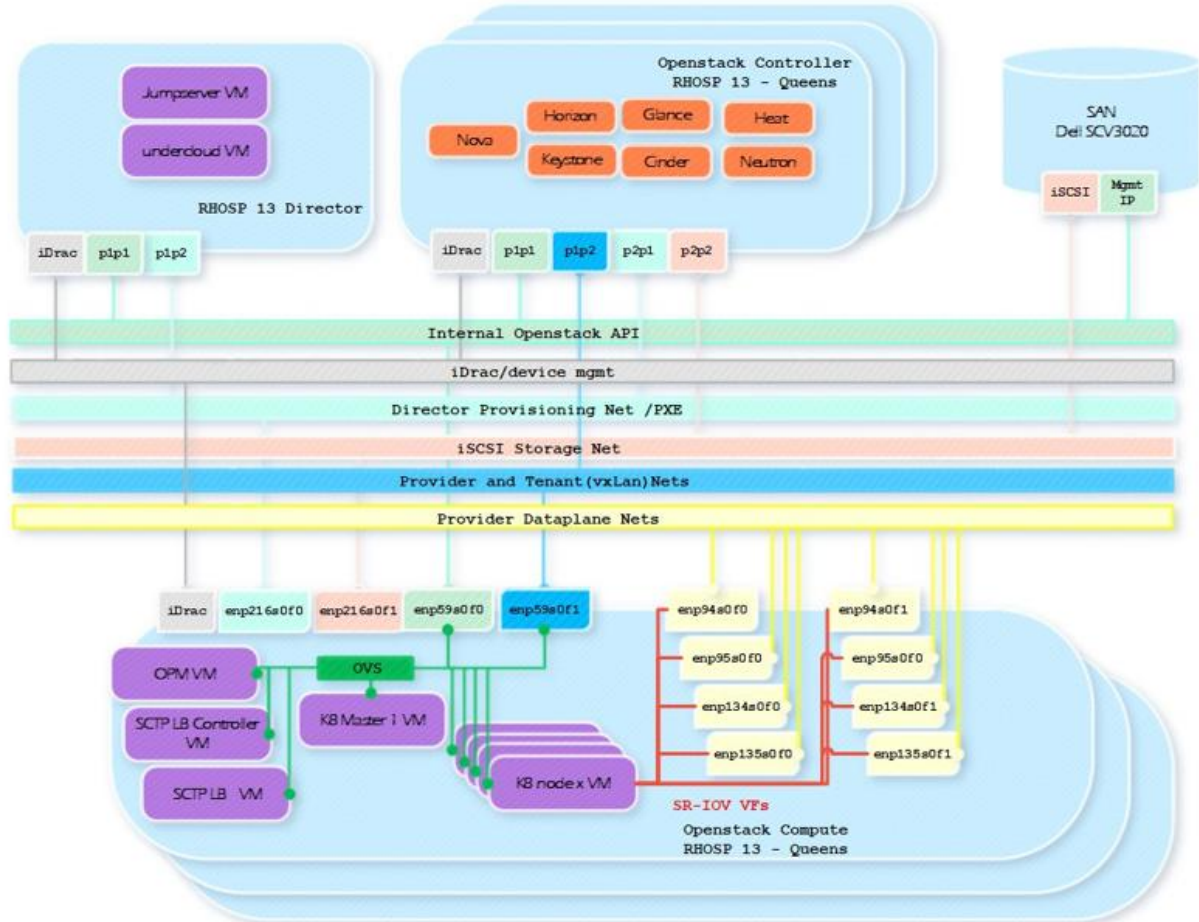
This deployment example assumes that OpenStack is used.

Set Up BIOS

To set up BIOS:

1. Enable PXE boot on the Ethernet port that will be used for the RHEL OpenStack Director provisioning network or other OpenStack installers networks.
2. Enable Single Root I/O Virtualization (SR-IOV) on all ports that will use SR-IOV.
3. Enable performance setting for CPU and memory.
4. Enable hyper threading.
5. Enable virtualization.
6. Configure the disk to use some type of RAID.
7. Change the boot order so that PXE boot happens first.

The following figure illustrates the recommended network topology for OpenStack and OPM.

Figure 1. Recommended Network Topology for OpenStack and OPM

Server Groups

The parameter `Os Server Group Id` may be defined via [Deploy Kubernetes Cluster](#). This parameter allows for the Compute VMs (such as Kubernetes Masters VMs) to be linked to a Server Group in OpenStack. What then follows is the ability to attach policies to these groups within these groups to enforce capabilities, such as anti-affinity and affinity.

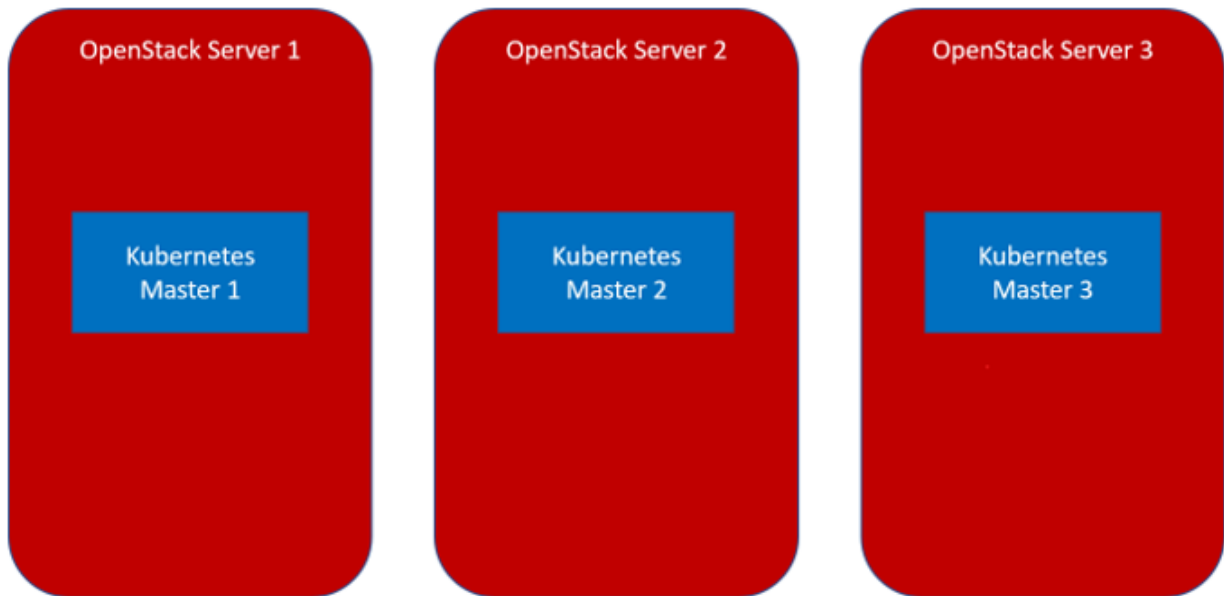
Anti-Affinity User Case and Setup

For example, if you want every Compute VM to run on different hardware for higher availability of the Kubernetes Control Plane, you can attach an anti-affinity policy to a server group.

This can be done by creating the server group initially using the OpenStack CLI:

```
# openstack server group create --policy anti-affinity <SERVER GROUP NAME>
```

Figure 2. Anti-Affinity Policy

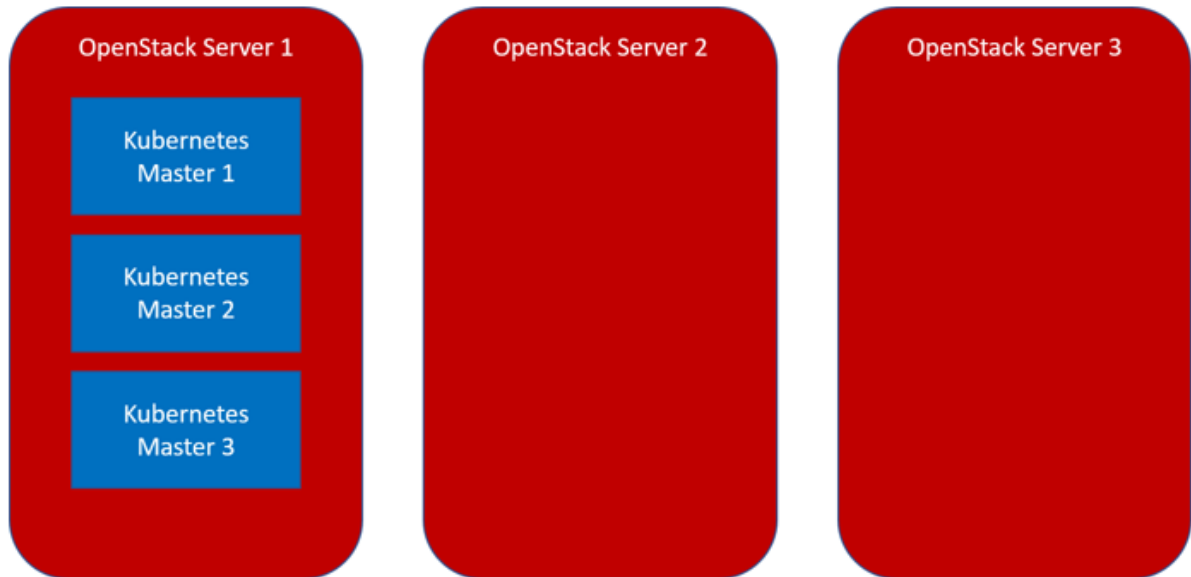


Affinity Use Case and Setup

Another example, if you want every Compute VM to run on the same hardware for strict management or tighter communication among Control Plane components, you can attach an affinity policy to a server group.

This can be done while creating the server group initially using the OpenStack CLI:

```
# openstack server group create --policy affinity <SERVER GROUP NAME>
```

Figure 3. Affinity Policy

UnityCloud Configuration

Once a server group is created in OpenStack with a policy attached, it can be specified during Kubernetes cluster creation.

To specify the server group when creating a cluster with OPM, add the server group ID to the **Os Server Group Id** field.

Alternately, you may set the value by Heat template by providing the `os_server_group_id` parameter:

```
os_server_group_id: <server group id>
```


Download Affirmed Networks Images

After the OPM VM is up and running, you can download all the files required to run Affirmed Networks microservices using the install script (cna_deploy_inventory). The script takes input from JSON inventory file(s), which specify the Helm charts and Docker images.

You can download the cna_deploy_inventory script and JSON inventory file(s) as described in [Affirmed Networks Public Repository - Push Charts](#).

Affirmed Networks Public Repository – Push Charts

Perform the following steps to download and push images to the local Docker Registry and Helm Chart Museum from Affirmed Networks to your local repository. These steps require public Internet access.

1. Issue the following commands to download the inventory files to your local OPM system.

```
# ssh root<opm ip>
```

```
# mkdir -p /an_release/inventory
```

2. Login to your customer account: From a web browser:

<https://uc.a4opacketcore.microsoft.com>

3. In the left panel, click on Artifactory and then Artifacts. In the right panel, double click on UC and then releases. Navigate to the release of interest, for example UC/releases/UC-4.3.0-VMIMAGES-47/vmimages/

4. Download the qcow2 images to the OPM's /an_release/images folder.

5. Navigate to the inventory folder for your release. For example: UC/releases/UC-4.3.0-VMIMAGES-47/inventories/

6. Download the prd json files to the OPM's /an_release/inventory folder.

7. Run the cna_deploy_inventory script to pull all the images from the Affirmed Networks public repository and push them to your Docker Registry and Helm Chart Museum.

```
# cna_deploy_inventory \
```

```
--inventory /an_release/inventory/*.json \
```

```
--dest_docker_url https://<OPM dreg FQDN>:5005 \
```

```
--dest_helm_url https://<OPM FQDN>/chartm \
```

```
--docker_yaml_path <OPM dreg FQDN>:5005 \
```

```
--login <username>:<password> \
--dest_docker_login <username>:<password> \
--dest_helm_login <username>:<password>
```

Note: If you are using multi-node OPM, instead of OPM IP use OPM FQDN.

The following table describes the `cna_deploy_inventory` parameters.

Table 3. cna_deploy_inventory parameters

Parameter	Description
<code>--inventory</code>	The name of the inventory file(s) to use for deployment. Multiple blank separated files may be specified.
<code>--dest_docker_url</code>	The URL of the destination Docker Registry path to which to push files (include <code>http://</code> or <code>https://</code>).
<code>--dest_helm_url</code>	The URL of the Helm ChartMuseum path to which to push files (include <code>http://</code> or <code>https://</code>).
<code>--docker_yaml_path</code>	The Docker Registry path that should be set in each Helm Chart's <code>values.yaml</code> file. Otherwise, it will be set to empty.
<code>--login</code>	The username and password to use if the source Docker Registry requires a login to pull images. Specify as username and password separated by a colon and no spaces. For example, <code><username>:<password></code> . Note: If only the username is specified (with or without a trailing colon), you will be prompted for the password.
<code>--dest_docker_login</code>	The username and password to use if the destination Docker registry requires a login. Specify as username and password separated by a colon and no spaces. For example, <code><username>:<password></code> .
<code>--dest_helm_login</code>	The username and password to use if the destination Helm ChartMuseum requires a login. Specify as username and password separated by a colon and no spaces. For example, <code><username>:<password></code> .

Special Consideration Pushing Charts to Multi-node OPM 4.3

In Unity Cloud 4.3 release, due to performance issues observed with rook ceph serving docker images, multi-node OPM storage has been moved to use local disk rather than rook ceph storage. As a result, the following MOP should be used when pushing docker images and charts to local docker registry and chart museum on multi-node OPM:

1. Scale dreg and chartm pods on multi-node OPM to 1 and pay special attention to the worker nodes the chartm and dreg pods are running:

```
kc scale -n vnfmgr deployments vnfmgr-chartm --replicas 1
```

```
kc scale -n vnfmgr deployments vnfmgr-dreg --replicas 1
```

2. Run can deploy_inventory and it will populate one of the disk that the dreg and chartm are running on.

```
./cna_deploy_inventory --inventory prd-paas.rel_4_3_0-31.a4oprodaf.json --dest_docker_url https://opm.unitycloud.com:5005 --dest_helm_url https://opm.unitycloud.com/chartm0/ --use_nerdctl
```

3. Copy docker registry data to the other nodes:

- a. Ssh to the node where dreg is running. Run the below command to get the node details:

```
kc get po -n vnfmgr -l app=vnfmgr-dreg -o custom-columns=NODE:spec.nodeName,IP:status.hostIP
```

- b. Get the node details to SCP the data:

```
kc get node --selector='!node-role.kubernetes.io/master' -o wide | grep -v $(kc get po -n vnfmgr -l app=vnfmgr-dreg -o jsonpath='{.items[0].spec.nodeName}')
```

- c. SCP the data to all the node (from the above command) IPs using the below command:

```
scp -r /var/storage/vnfmgr/dreg root@<NODE-IP>:/var/storage/vnfmgr/
```

4. Copy chartm data to other worker nodes.

- d. SSH to the node where chartm is running. Run the below command to get the node details:

```
kc get po -n vnfmgr -l app=vnfmgr-chartm -o custom-columns=NODE:spec.nodeName,IP:status.hostIP
```

- e. Get the node details to SCP the data:

```
kc get node --selector='!node-role.kubernetes.io/master' -o wide
| grep -v $(kc get po -n vnfmgr -l app=vnfmgr-chartm -o
jsonpath='{.items[0].spec.nodeName}')
```

- f. SCP the data to all the node (from the above command) IPs using the below command:

```
scp -r /var/storage/vnfmgr/chartm root@<NODE-IP>:/var/storage/vnfmgr/
```

5. Scale back the dreg and chartm pods back to 3

```
kc scale -n vnfmgr deployments vnfmgr-chartm --replicas 3
```

```
kc scale -n vnfmgr deployments vnfmgr-dreg --replicas 3
```

Special Consideration Pushing Charts to Multi-node OPM 5.1

In Unity Cloud 5.1 release, multi-node OPM simplifies the steps to push images and charts to local docker registry and chart museum. `cna_deploy_inventory` should be run 3 times to the specific destination ports (5006-5008) for docker registry and (/chartm0-/chartm2) for each of the charm pods as shown below:

1. `./cna_deploy_inventory --inventory prd-paas.rel_4_3_0-31.a4oprodaf.json --dest_docker_url https://opm.unitycloud.com:5006 --dest_helm_url https://opm.unitycloud.com/chartm0/ --use_nerdctl`
2. `./cna_deploy_inventory --inventory prd-paas.rel_4_3_0-31.a4oprodaf.json --dest_docker_url https://opm.unitycloud.com:5007 --dest_helm_url https://opm.unitycloud.com/chartm1/ --use_nerdctl`
3. `./cna_deploy_inventory --inventory prd-paas.rel_4_3_0-31.a4oprodaf.json --dest_docker_url https://opm.unitycloud.com:5008 --dest_helm_url https://opm.unitycloud.com/chartm2/ --use_nerdctl`

Special Consideration Pushing Charts to Multi-node OPM 5.2

In Unity Cloud 5.2 release, multi-node OPM uses fed-infra-dreg to support the local docker registry. Fed-infra-dreg requires Cinder CSI as shared storage and exposes the service through its own FQDN and port 5005. `cna_deploy_inventory` should still be run 3 times to populate the 3 chartm pods.

1. `./cna_deploy_inventory --inventory prd-smf.rel_5_1_0-259.tk5af.json --dest_docker_url https://registry.unitycloud.com:5005 --dest_helm_url https://opm.unitycloud.com/chartm0/ --use_nerdctl`
2. `./cna_deploy_inventory --inventory prd-smf.rel_5_1_0-259.tk5af.json --dest_docker_url https://registry.unitycloud.com:5005 --dest_helm_url https://opm.unitycloud.com/chartm1/ --use_nerdctl`

```
3. ./cna_deploy_inventory --inventory prd-smf.rel_5_1_0-259.tk5af.json--dest_docker_url  
https://registry.unitycloud.com:5005 --dest_helm_url https://opm.unitycloud.com/chartm2/ --  
use_nerdctl
```

Verify Images in Local Docker Registry

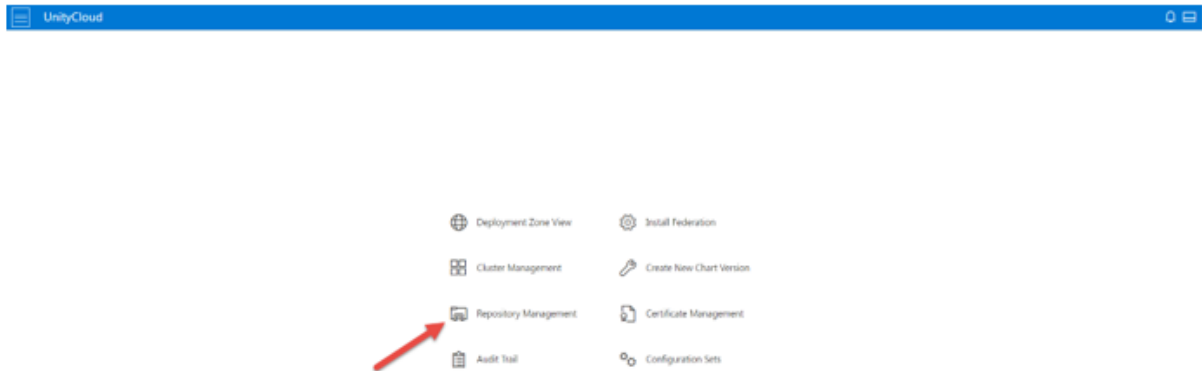
Run the following commands:

```
# curl -ik https://<OPM dreg FQDN>:5005/v2/_catalog  
  
# curl -ik https://<OPM dreg FQDN>:5005/v2/<repository name>/tags/list
```

Associate Local Docker Registry with OPM GUI

1. From the Main OPM screen, Click **Repository Management**.

Figure 4. Main OPM Screen



2. From the Repository Management screen, click **Docker Images** and click the Plus icon (+).

Figure 5. Repository Management – Docker Images

The screenshot shows a 'Repository Management' dialog box with a close button (X) in the top right corner. It has two tabs: 'Charts' and 'Docker Images', with 'Docker Images' being the active tab. The form contains the following fields and controls:

- Name ***: A dropdown menu with 'Select...' and a downward arrow, followed by a plus sign (+).
- Host ***: A text input field.
- Port ***: A text input field.
- Path**: A text input field.
- Protocol ***: Two radio buttons, 'HTTP' (selected) and 'HTTPS'.
- Username**: A text input field.
- Password**: A text input field.

At the bottom of the dialog are three buttons: 'Save', 'Delete', and 'Close'.

3. When prompted, enter the following information.

Table 4. Repository Management – Docker Image parameters

Parameter	Description
Name	Name of the local Docker registry.
Host	OPM IP address.
Port	Port of the local Docker registry. In this case, 5005.
Path	The local Docker registry path. In this case, enter /.
Username	The username to use if the local Docker registry requires a login.

Parameter	Description
Password	The password to use if the local Docker registry requires a login.

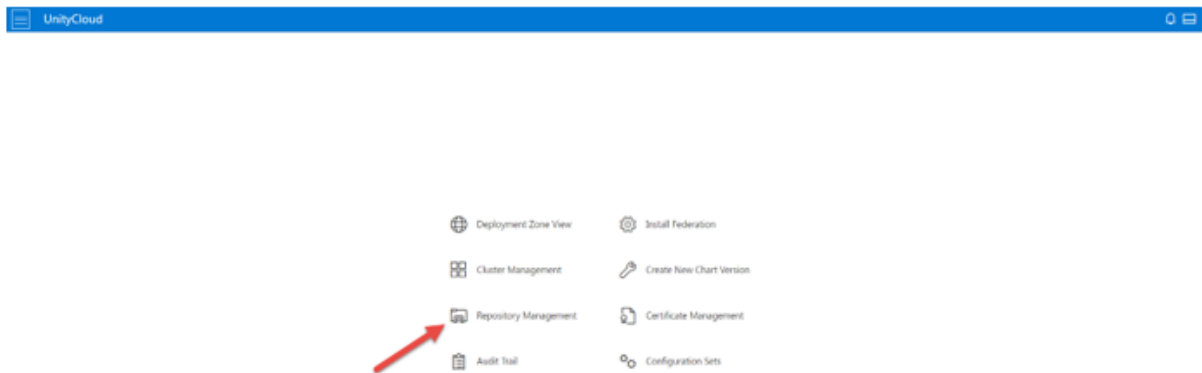
4. Click Save.

Associate Local Helm Chart Museum with OPM GUI

To associate the Local Helm ChartMuseum with the OPM GUI:

1. From the Main OPM screen, Click Repository Management.

Figure 6. Main OPM Screen



2. From the Repository Management screen, click Charts and click the Plus icon (+).

Figure 7. Repository Management – Charts

The screenshot shows a 'Repository Management' dialog box with a close button (X) in the top right corner. It has two tabs: 'Charts' (selected) and 'Dock Images'. The 'Charts' tab contains the following fields:

- Name ***: A dropdown menu with 'Select...' and a downward arrow, followed by a plus sign (+).
- Host ***: A text input field.
- Port**: A text input field.
- Type ***: A dropdown menu with 'Select...' and a downward arrow.
- Path**: A text input field.
- Username**: A text input field.
- Password**: A text input field.

At the bottom of the dialog are three buttons: 'Save', 'Delete', and 'Close'.

3. When prompted, enter the following information.

Table 5. Repository Management – Chart parameters

Parameter	Description
Name	Name of the local Helm ChartMuseum.
Host	Name of the local chartm service: vnfmgr-chartm
Port	Port of the local Helm ChartMuseum. In this case, 8081.
Type	Repository type.
Path	The local Helm ChartMuseum path.
Username	The username to use if the local Helm ChartMuseum requires a login.

Download Affirmed Networks Images

Parameter	Description
Password	The password to use if the local Helm ChartMuseum requires a login.

4. Click Save.

Private Docker Registry

When the OPM VM is deployed, a local Docker Registry is started. The registry uses the same IP address as the OPM management IP address. The port is 5005. You may choose to use either the OPM local docker registry or another private docker registry for image management.

If you are using OPM's private Docker repository, you must follow the steps in this section. If you are using your own private Docker repository with certificates, modify the following steps with your private Docker repository information.

Note: If using containerd as a result of a Kubernetes upgrade, make sure to replace the **docker** command with **nerdctl** where indicated below.

Obtain Local Docker Registry Certificate

To push images to the local Docker registry from another Docker registry such as the Affirmed public Docker registry, you must first copy the certificate from the local Docker registry.

To obtain the OPM local Docker Registry Certificate:

1. From the Docker server for which you want the certificate, issue the following command.

```
# scp root@<opm ip>:vnfmgr/certs/dreg.crt
```

2. Issue the following commands to install this certificate on your Docker client. OPM dreg FQDN is the OPM IP of single-node OPM.

```
# mkdir /etc/containerd/certs.d/<OPM dreg FQDN>:5005/
```

```
# cp dreg.crt /etc/containerd/certs.d/<OPM dreg FQDN>:5005/
```

Note: For multi-node OPM in 5.1, the docker registry certificate should be installed in the following directories in /etc/containerd/certs.d:

opm.unitycloud.com:5005 , opm.unitycloud.com:5006 , opm.unitycloud.com:5007 ,
opm.unitycloud.com:5008

Note: For multi-node OPM in 5.2, the docker registry certificate should be installed in the following directories in /etc/containerd/certs.d:

registry.unitycloud.com:5005

You can now pull and push images to the local Docker Registry.

Add Docker Images to Local Docker Registry

If your OPM has IP connectivity to another Docker Registry, you can pull images from that Registry. To do this, you must first install the Docker certificate as described in [Obtain Local Docker Registry Certificate](#). If there is no connectivity, you must copy a tar file to the Docker Registry and load the file into the registry.

SSH to the Docker registry that has the image and issue the following commands:

```
# nerdctl save testImage > testImage.tar

# scp testImage.tar root@<opm ip>:/vnfmgr/dreg # ssh to root@<opm ip>

# nerdctl load -i testImage.tar

# nerdctl tag <test image id>:<tag> <OPM dreg FQDN>:5005/myImage:TestImage

# nerdctl push <OPM dreg FQDN>:5005/myimages: TestImag
```

The image should now be available on the local Docker Registry.

Read Docker Imager from Local Docker Registry

To view a list of images on the local Docker Registry:

1. Login to the OPM via SSH

```
# ssh root@<opm ip>
```

2. List all the docker images

```
# nerdctl images
```

3. Issue the following command

```
# nerdctl pull <OPM dreg FQDN>:5005/myimages: TestImage
```

Delete All Docker Images

Perform the following steps to delete all Docker images before a specific date from the local Docker Registry.

1. Log in to the OPM via SSH.

```
# ssh root@<opm ip>
```

2. Issue the following command.

```
# cd /vnfmgr/dreg
```

3. Edit registryCleaner.sh. This deletes repository images before a date or number of old images per repository.

- `--d < MM-DD-YYYY >` – Delete everything in this repository up to this date.
- `--n < N >` – Keep the last N images in this repository.
 - -Change to the path/name you gave your repository.
 - -Change the date to the required date.
 - -You could use `--n` option instead.

4. Run registryCleaner.sh.

Delete One Docker Image from Local Docker Registry

Perform the following steps to delete one Docker image from the local Docker Registry.

Example

Delete OPM/cna-vnfmgr_k8_resmgr:1.0.0.0-26.rel from the local Docker Registry.

1. Log in to the OPM via SSH.

```
# ssh root@<opm ip>
```

2. Issue the following command:

```
# /vnfmgr/dreg/delete_docker_registry_image.py -I OPM/cna-  
vnfmgr_k8_resmgr:1.0.0.0-26.rel
```

Docker Registry Certificate Management

To ensure the OPM and its managed clusters can communicate with the associated Docker registry securely, use Certificate Management to install the local Docker registry certificates on OPM and its deployed clusters' nodes.

From the OPM main screen:

1. Click **Certificate Management**.

2. Fill in the details and then click **Import**:
 - **Target IP:** Enter the local Docker Registry IP where OPM will use to pull images from.
 - **Port:** 5005
 - **Attachment:** Docker Registry certificate file to import.
3. A confirmation message appears, indicating all the nodes where Docker Registry Certificate was pushed successfully.

***Note:** If the OPM has clusters that are attached, the certificates will need to be pushed to the nodes of these clusters manually. Follow instructions in [Obtain Local Docker Registry Certificate](#) to do that*

Affirmed Networks, Inc.Helm ChartMuseum

By default, OPM deploys a Helm ChartMuseum microservice that can be used by OPM, the deployed Kubernetes cluster or any other Kubernetes cluster. The Helm ChartMuseum has the same IP address as OPM with a suffix of /chartm.

The Helm ChartMuseum uses OPM's Network File System (NFS) storage by default.

Example:

Helm ChartMuseum: `https://10.165.5.10/chartm`

Actions

- Push a chart:

```
# curl --data-binary "@fed-example-<chart-version>.tgz"
https://10.165.5.10/chartm/api/charts
```

- Display a list of charts:

```
# curl https://10.165.5.10/chartm/api/charts | jq
```

- Display all federations available in the chartm:

```
# curl -k https://10.165.5.10/chartm/api/charts | jq 'keys'
```

- Delete a chart:

```
# curl -X DELETE https://10.165.5.10/chartm/api/charts/fed-
example/0.170.0
```

Install OPM into OpenStack

Installing OpenStack is outside the scope of this document. After OpenStack is set up, it should have the same network topology as shown in [Figure 1 Recommended Network Topology for OpenStack and OPM](#).

Note: Once a single-node OPM has been installed, you may use `fed-opm_multi` to install multi-node OPM to a deployed Kubernetes cluster using the [Deploy UnityCloud Network Function Applications with OPM](#) process.

Upload VM Images to Glance

1. Ensure you have downloaded the following files from the Affirmed Networks software release site.

Note: For information about the Platform VM and OPM VM versions supported with this release, see the [Software Release Notes for UnityCloud™](#).

- `vm-opm-<version>-main-int.qcow2`
- `vm-platform-<version>-main-int.qcow2`

2. Upload the OPM VM image.

```
# glance image-create --name vm-opm-<version> --disk-format qcow2 --
container-format bare --progress --visibility public -- file vm-opm-
<version>-main-int.qcow2
```

```
# openstack image set --property hw_disk_bus=virtio --property
hw_cdrom_bus=ide
```

3. Upload the Platform VM image.

```
# glance image-create --name vm-platform-<version> --disk-format qcow2 --
container-format bare --progress --visibility public --file vm-platform-
<version>-main-int.qcow2
```

```
# openstack image set --property hw_disk_bus=virtio --property
hw_cdrom_bus=ide
```

(Optional) Enable Multiqueue

```
# openstack image set --property hw_vif_multiqueue_enabled=true <image>
```

Note: If using Ceph for Glance, you must convert *qcow2* to *raw* using `qemu-img convert -f qcow2 -O raw vm-opm-<version>-main-int.qcow2 vm-opm-<version>-main-int.raw`. You must also change `--disk-format` from *qcow2* to *raw*.

Create Provider Networks in OpenStack

To create provider networks in OpenStack:

1. Use this network for management (mgmt) and East/West traffic. The following example uses a VLAN of 1900.

```
# openstack network create CNA-MGMT --share --provider- physical-network
base --provider-network-type vlan --provider- segment 1900
```

```
# neutron subnet-create CNA-MGMT 10.165.5.0/24 --name cna_mgmt_sub --no-
gateway --disable-dhcp --allocation-pool start=10.165.5.1,end=10.165.5.240
```

2. Use this network for high speed data plane traffic. Since the data plane uses SR-IOV, you must create a flat provider network. The data plane microservices will add and remove the VLAN tag, not OpenStack.

```
# openstack network create cna-SR-IOV-1 --share --provider- physical-
network SR-IOV1 --provider-network-type flat
```

```
# openstack subnet create --subnet-range 20.20.21.0/24 --no- dhcp --
network cna-SR-IOV-1 cna_SR-IOV1_21_subnet
```

Deploy OPM VMs

Select one of the following methods for installing OPM VMs into the OpenStack Environment:

- [Deploy OPM using Local Server Disks](#)
- [Deploy OPM using Cinder Volumes](#)

Deploy OPM using Local Server Disks

Note: Affirmed Networks recommends that you use Cinder and not local storage for deploying OPM.

To deploy OPM using the local server disks:

1. Create flavors matching the deployment example in Table 1 Example: VMs to Deploy a Kubernetes Cluster (Full Production).

Note: The disk must be 50G

a. OPM Flavor

```
# openstack flavor create --public opm.local.8.4.400 --ram 8192 --disk 400
--vcpus 4
```

b. K8 Master Flavor

```
# openstack flavor create --public platform.local.8.4.300 -- ram 8192 --
disk 300 --vcpus 4
```

(Optional) Enable Multiqueue

```
# openstack flavor set <flavor name> --property hw:vif_
multiqueue_enabled='true'
```

c. K8 Worker Node Flavor

```
# openstack flavor create --public platform.local.78.16.300 --ram 79872 --
disk 300 --vcpus 16
```

(Optional) Enable Multiqueue

```
# openstack flavor set <flavor name> --property hw:vif_
multiqueue_enabled='true'
```

2. Create the OpenStack Heat environment file.

- a. Ensure you have downloaded CNA_OPM_VM_local-<version>-main-int.yaml (local disk) from the Affirmed Networks software release site.

b. Create the Heat environment file to be used with this Heat template. For example: CNA- OPM-local.env.

Table 6. Heat parameters for installing OPM using local disks

Parameter	Description	Example
name	Name of VM	CNA-OPM-local-1
image	Glance image	dbeb5be6-3b20-480b-8df6-780d13f10fa2
flavor	OpenStack flavor used to deploy VM	opm.local.8.4
zone	OpenStack zone	platform-zone
mgmt_net_id	OpenStack networks used for OPM	8c417049-038c-4348-a5cc-6b020bd1dd80
mgmt_net_ip	IP address	10.165.5.10
mgmt_subnet_id	Management subnet ID	
k8_master_ip	This value should be the same as mgmt_net_ip.	10.165.5.10
k8_master_token	Kubernetes bootstrap token Bootstrap tokens take the form of abcdef.0123456789abcdef. Must match the regular expression [a-z0-9]{6}\.[a-z0-9]{16}. This must be different between all OPMs deployed on the same network.	14ced3.6933cb3b4e428a92
k8_master_port	Kubernetes master port	6443
k8_master_token	Kubernetes master token	
k8_master_base_pod_0_image	cnfmgr database image	
vnfmgr_dreg_external_ip	This value should be the same as mgmt_net_ip.	10.165.5.10
vnfmgr_dreg_external_port	cnfmgr docker reg external port	5005
k8_network	Kubernetes network plugin	calico
k8_dashboard	Kubernetes master dashboard enable	true
k8_rbac	Kubernetes master RBAC (superuser or none)	superuser
k8_cidr		192.167.0.0/16

Install OPM into OpenStack

Parameter	Description	Example
vm_type	Type of VM to configure	cnfmgr
host_gw	Network gateway	
host_dns	DNS for this network	
host_dns2	Second DNS for this network	
host_domain	Domain	
host_mtu	OpenStack and outside networks' MTU must be set equal to or greater than the VM MTU	9000
host_netmask	Network netmask	
security_group_id	Use none if your OpenStack setup is not using port security, otherwise enter the security group ID.	
opm_helm_chart	OPM Helm chart	/home/kuberadmin/fed-opm-1.4.0-47.tgz
opm_ext_docker_path	OPM Helm chart Docker path	cnreg-rel:5000
timezone	Time zone for the guest OS. The default is UTC.	UTC
ntp_server_ip	NTP server IP. If none then use public pools	none
root_domain_ca_key_	Per user-domain certificate authority public keys that can issue user auth certificates	
admin_domain_ca_key	Per user-domain certificate authority public keys that can issue user auth certificates	
liadmin_domain_ca_key	Per user-domain certificate authority public keys that can issue user auth certificates	
maint_domain_ca_key	Per user-domain certificate authority public keys that can issue user auth certificates	
host_certificate	Host certificate to supplement host key authentication	
enable_local_passwd_auth	Use of local password user authentication is disabled by default except in lab environments	
enable_root	Root login is disabled by default except in lab environments	
disable_maint	Maint group (non-root bash user(s)) are enabled by default	
root_passwd	Replace default root sha512 password	

Install OPM into OpenStack

Parameter	Description	Example
root_pubkey	Define public key for root user authentication	

The following table describes user parameters.

Caution: In order to allow SSH access to the VM, you must add desired "maint", "romaint" (read-only), or "limaint" (lawful intercept) group users with either public keys, certificate authority keys, SHA-512 encrypted passwords, or configure one or more TACACS servers.

Table 7. User-specific Heat parameters for installing OPM using local disks

Parameter	Description	Example
user_count	Number of defined users	"6"
user_1	Name of user 1	"maint"
passwd_1	Password for user 1	
pubkey_1	Pubkey for user 1	
group_1	User group	"maint"
user_2	Name of user 2	"romaint"
passwd_2	Password for user 2	
pubkey_2	Pubkey for user 2	
group_2	User group	"romaint"
user_3	Name of user 3	"limaint"
passwd_3	Password for user 3	
pubkey_3	Pubkey for user 3	
group_3	User group	"limaint"
user_4	Name of user 4	
passwd_4	Password for user 4	
pubkey_4	Pubkey for user 4	
group_4	User group	
user_5	Name of user 5	

Parameter	Description	Example
passwd_5	Password for user 5	
pubkey_5	Pubkey for user 5	
group_5	User group	
user_6	Name of user 6	
passwd_6	Password for user 6	
pubkey_6	Pubkey for user 6	
group_6	User group	

The following table describes the TACACS service parameters.

Table 8. TACACS-specific Heat parameters for installing OPM using local disks

Parameter	Description	Example
tserver_count	Number of defined TACACS servers	"1"
tacacs_service	TACACS service name presented to the tacacs servers	"tacacs server 1"
tacacs_protocol	A subset of primary service.	
tserver_1_ip_address	TACACS+ server IP	"0.0.0.0"
tserver_1_auth_type	Server authentication type	"login"
tserver_1_port	Server port	"49"
tserver_1_key	Server secret key	
tserver_1_auth_timeout	Server timeout	"5"

3. Deploy the OPM VM.

```
# openstack stack create CNA-OPM -t ./CNA_OPM_VM_local- <version>-main-
int.yaml -e ./CNA-OPM-local.env
```

Deploy OPM using Cinder Volumes

To deploy OPM using Cinder volumes:

1. Create flavors matching the deployment example in [Table 1](#) Example: VMs to Deploy a Kubernetes Cluster (Full Production).

Install OPM into OpenStack

Cinder does not need a flavor to have disk information. In fact, this has a negative effect because it skews OpenStack's view on how much local disk is available.

a. OPM Flavor

```
# openstack flavor create --public opm.cinder.8.4 --ram 8192 --vcpus 4
```

b. K8 Master Flavor

```
# openstack flavor create --public platform.cinder.8.4 --ram 8192 --vcpus 4
```

(Optional) Enable Multiqueue

```
# openstack flavor set <flavor name> --property hw:vif_multiqueue_enabled='true'
```

c. K8 Worker Node Flavor

```
# openstack flavor create --public platform.cinder.78.16 -- ram 79872 -- vcpus 16
```

(Optional) Enable Multiqueue

```
# openstack flavor set <flavor name> --property hw:vif_multiqueue_enabled='true'
```

2. Create a Cinder volume snapshot. Creating a Cinder Volume snapshot allows for a faster VM deployment. Without this, each time a VM is created you must install the Glance image onto the new volume, which can take a long time.

a. Get the Glance image: vm-opm-<version>-main-int.qcow2.

```
# openstack image list
```

```
+ +
```

```
-+ +
```

```
| ID | Name | Status |
```

```
+ +
```

```
-+ +
```

```
| 71dec84b-15fe-41e8-8228-a1736f93a94d | vm-opm-1.1.74 | active |
```

```
| 5c48eba2-ae53-45f3-8914-20defe15c107 | vm-platform-1.1.71
```

```
| active |
```

b. Create a Cinder volume based on the Glance ID.

Install OPM into OpenStack

```
# openstack volume create --image vm-opm-<version> --size 50 vm-opm-  
<version>-snapbase
```

c. Create a Cinder volume snapshot to be used with the deployed VM.

```
# openstack volume snapshot create --volume vm-opm- <version>-snapbase vm-  
opm-<version>-snap
```

d. Repeat these steps for vm-platform-<version>-main-int.qcow2, with a --size of 20.

For example:

```
# openstack volume create --image vm-platform- -- size 20 vm-  
opm--snapbase
```

e. Get the Cinder volume snapshot list.

```
# openstack volume snapshot list
```

```
+ + +-  
+ + +  
| ID | Name | Description | Status | Size |  
+ + +-  
+  
| 2f957833-cc3a-4880-8249-4b575c7a6cbf | vm-platform-1.1.71-snap | None |  
| 39138dfb-056e-4fa1-80e3-1fd287543407 | vm-opm-1.1.74-snap | None |  
+ + +-  
+
```

3. Create the OpenStack Heat environment file.

a. Ensure you have downloaded CNA_OPM_VM_cinder-<version>-main-int.yaml (Cinder volume) from the Affirmed Networks software release site.

b. Create the Heat environment file to be used with this Heat template. For example, CNA_OPM-cinder.env.

The following table describes the parameters.

Table 9. Heat parameters for installing OPM using Cinder volumes

Parameter	Description	Example
name	Name of VM	cna-OPM-cinder-1
image	Cinder volume snapshot image	77311671-ad3a-40c4-8f60-7cce0f1d2465

Install OPM into OpenStack

Parameter	Description	Example
cinder_zone	Cinder availability zone	"nova"
cinder_size	Cinder volume size	"50"
cinder_name	Name of cinder volume	cna-OPM-cinder-1
cinder_type	Cinder type	"ceph"
flavor	OpenStack flavor used to deploy VM.	opm.cinder.8.4
zone	OpenStack zone	platform-zone
mgmt_net_id	Openstack Network ID	8c417049-038c-4348-a5cc-6b020bd1dd80
mgmt_net_ip	IP address	10.165.5.10
mgmt_subnet_id	Management subnet id	
k8_master_ip	This value should be the same as mgmt_net_ip.	10.165.5.10
k8_master_port	Kubernetes master port	6443
k8_master_token	Kubernetes bootstrap token Bootstrap tokens take the form of abcdef.0123456789abcdef. Must match the regular expression [a-z0-9]{6}\.[a-z0-9]{16}. This must be different between all OPMs deployed on the same network.	14ced3.6933cb3b4e428a92
k8_master_base_pod_0_image	cnfmgr database image	mongo-4-0-28.tar
vnfmgr_dreg_external_ip	This value should be the same as mgmt_net_ip.	10.165.5.10
vnfmgr_dreg_external_port	cnfmgr docker reg external port	5005
k8_network	Kubernetes network plugin	calico
k8_dashboard	Kubernetes master dashboard enabled	true
k8_rbac	Kubernetes master RBAC (superuser or none)	superuser
k8_cidr		192.167.0.0/16
vm_type	Cnfmgr	cnfmgr
host_gw	Network gateway	
host_dns	DNS for this network	
host_dns2	Second DNS for this network	
host_domain	Domain	

Install OPM into OpenStack

Parameter	Description	Example
host_mtu	OpenStack and outside networks' MTU must be set equal to or greater than the VM MTU	9000
host_netmask	Network netmask	
security_group_id	Use none if your OpenStack setup is not using port security, otherwise enter the security group ID.	
opm_helm_chart	OPM helm chart	/home/kuberadmin/fed-opm- 1.4.0-47.tgz
opm_ext_docker_path	OPM Helm chart Docker path	"cnreg-rel:5000"
timezone	Time zone for guest OS. Default is UTC.	UTC
ntp_server_ip	NTP server IP. If none then use public pools.	none
root_domain_ca_key_	Per user-domain certificate authority public keys that can issue user auth certificates	
admin_domain_ca_key	Per user-domain certificate authority public keys that can issue user auth certificates	
liadmin_domain_ca_key	Per user-domain certificate authority public keys that can issue user auth certificates	
maint_domain_ca_key	Per user-domain certificate authority public keys that can issue user auth certificates	
host_certificate	Host certificate to supplement host key authentication	
enable_local_passwd_auth	Use of local password user authentication is disabled by default except in lab environments	
enable_root	Root login is disabled by default except in lab environments	
disable_maint	Maint group (non-root bash user(s)) are enabled by default	
root_passwd	Replace default root sha512 password	
root_pubkey	Define public key for root user authentication	

The following table describes user parameters.

Caution: In order to allow SSH access to the VM, you must add desired "maint", "romaint" (read-only), or "limaint" (lawful intercept) group users with either public keys, certificate authority keys, SHA-512 encrypted passwords, or configure one or more TACACS servers.

Table 10. User-specific Heat parameters for installing OPM using Cinder volumes

Parameter	Description	Example
user_count	Number of defined users	"6"
user_1	Name of user 1	"maint"
passwd_1	Password for user 1	
pubkey_1	Pubkey for user 1	
group_1	User group	"maint"
user_2	Name of user 2	"romaint"
passwd_2	Password for user 2	
pubkey_2	Pubkey for user 2	
group_2	User group	"romaint"
user_3	Name of user 3	"limaint"
passwd_3	Password for user 3	
pubkey_3	Pubkey for user 3	
group_3	User group	"limaint"
user_4	Name of user 4	
passwd_4	Password for user 4	
pubkey_4	Pubkey for user 4	
group_4	User group	
user_5	Name of user 5	
passwd_5	Password for user 5	
pubkey_5	Pubkey for user 5	
group_5	User group	
user_6	Name of user 6	

Parameter	Description	Example
passwd_6	Password for user 6	
pubkey_6	Pubkey for user 6	
group_6	User group	

The following table describes the TACACS service parameters.

Table 11. TACACS-specific Heat parameters for installing OPM using Cinder Volumes

Parameter	Description	Example
tserver_count	Number of defined TACACS servers	"1"
tacacs_service	TACACS service name presented to the tacacs servers	"tacacs server 1"
tacacs_service	A subset of primary service.	
tserver_1_ip_address	TACACS+ server IP	"0.0.0.0"
tserver_1_auth_type	Server authentication type	"login"
tserver_1_port	Server port	"49"
tserver_1_key	Server secret key	
tserver_1_auth_timeout	Server timeout	"5"

4. Deploy the OPM VM.

```
# openstack stack create CNA-OPM -t ./CNA_OPM_VM_cinder- <version>-main-
int.yaml -e ./CNA_OPM-cinder.env
```

Download MongoDB used in OPM

You must download the MongoDB container for microservices to operate. Affirmed Networks cannot distribute the mongoDB container image. Therefore, the OPM VM must have access to the Internet to allow this container to download automatically.

If your OPM VM does not have an Internet connection, you must perform the following steps before OPM microservices will run.

1. Log in to the OPM via SSH.

```
# ssh root@<opm ip>
```

2. Obtain this image on the server that has Docker and access to the public Internet.

- a. # `docker pull mongo:4.0.28`
- b. Verify that the Digest displayed after the docker pull command exactly matches with the following: `sha256:4ca81c89ad08f4cfa9906005126112bffe8fb363800466ef5e50f6238f6f6af`
- c. # `docker save mongo:4.0.28 > mongo.tar`
- d. # `scp mongo.tar root@<opm-ip>:/vnfmgr/dreg`
- e. # `ssh to root@<opm ip>`
- f. # `docker load -i mongo.tar`

Verify OPM Microservices are running

Make sure all pods are in the running state before moving to the next step. Issue the following commands to verify that OPM microservices are running.

```
# ssh kuberadmin@<opm ip>
# kubectl get pods -n vnfmgr
```

Note: The `vnfmgr-osrm` pod does not exist until the OPM is associated with at least one OpenStack controller. For detailed documentation on this, see [Associating OPM With an OpenStack Controller](#) section in [Deploy Kubernetes Cluster](#).

The system displays output similar to the following. The pods with the asterisk (*) might take some time before they are in a running state.

```
NAME READY STATUS RESTARTS AGE
ambassador-86c68f9b89-5tbt7 1/1 Running 0 49m vnfmgr-apigw-889f87575-j28qt
1/1 Running 0 49m
vnfmgr-chart-editor-7df688556-pschv 1/1 Running 0 49m vnfmgr-chartm-
56db55f9dc-nzx45 1/1 Running 0 49m vnfmgr-chartm-web-7bd5cd9986-sz9xp 1/1
Running 0 49m
*vnfmgr-db-6686f4c9df-48slg 1/1 Running 0 49m vnfmgr-dreg-5c8bcc4c9b-42ttc
2/2 Running 0 49m
*vnfmgr-master-rm-8448cfc677-fvbkd 1/1 Running 0 49m vnfmgr-osrm-1-
7b6b7dc75b-8lpbc 1/1 Running 0 41m vnfmgr-web-7879b59855-mp7j9 1/1 Running
0 49m
```

Upgrade the OPM Helm Chart

Use the following procedure to upgrade the OPM Helm chart when a new version of a chart is released. For instructions, see [Download Affirmed Networks Images](#).

To upgrade the OPM Helm chart:

1. Download the OPM JSON inventory file(s) from the Affirmed Networks software release site.
2. SSH into the deployed OPM federation.

```
# ssh root@<opm ip>
```

3. Change the current working directory to root.

```
# cd /root
```

4. Issue the following command to download the fed-opm-values.yaml file.

```
# helm3 get values fed-opm -n vnfmgr -o yaml > fed-opm-values.yaml
```

5. In the fed-opm-values.yaml file, change `repoPath` so that it points to where the docker images are stored, and make sure that `mongoServerImage` is set to `mongo:4.0.28`.

Note: In earlier versions, `registry_docker_path` was used to define where the docker images are stored. If `registry_docker_path` exists in your `fed-opm-values.yaml` file, remove it and replace it with `repoPath` as shown in the following hierarchy:

```
global: registry:
  docker:
    repoPath: <docker_images_path>
```

6. Issue the following command to execute the upgrade.

```
# helm3 upgrade fed-opm -n vnfmgr <opm helm chart url> -f fed-opm-values.yaml
```

Manage Emissary Fallback Certificate

1. Fetch fallback secret name <FALLBACK-SECRET-NAME>

```
# kubectl get hosts wildcard-host -n vnfmgr -o jsonpath="{.spec.tlsSecret.name}"
```

2. Fetch the fallback certificate

```
# kubectl get secret <FALLBACK-SECRET-NAME> -n vnfmgr -o
jsonpath="{.data.tls\.crt}" | base64 --decode > fallback-cert.pem
```

3. Check fallback certificate's expiry date

```
# openssl x509 -enddate -noout -in fallback-cert.pem
```

4. Generate a new certificate if the expiry date is < 30 days.

```
# openssl req -x509 -newkey rsa:4096 -keyout ambassadorKey.pem -out
ambassadorCert.pem -subj '/CN=ambassador-cert' -nodes -days <number_of_days> -
addext "subjectAltName=IP:<OPM_IP>"
```

<number_of_days> ==> Validity Period for certificate.

If you don't specify -days <number_of_days>, the certificate will be generated with a default validity of 30 days.

5. Modify the fallback secret with the newly generated certificate and key

```
# cat ambassadorCert.pem | base64 -w0
# cat ambassadorKey.pem | base64 -w0
```

Once fetched, update the fallback secret with the certificate and key.

```
# kubectl edit secret -n vnfmgr <FALLBACK-SECRET-NAME>
```

Install an Official Certificate for Bootstrap OPM

By default, OPM is installed with a self-signed certificate. This results in a security warning when connecting to OPM. To install an official certificate for OPM, use the following procedure:

1. Fetch fallback secret name <FALLBACK-SECRET-NAME>

```
# kubectl get hosts wildcard-host -n vnfmgr -o jsonpath="{.spec.tlsSecret.name}"
```

2. Modify the fallback secret with the official cert.pem and key.pem

```
# cat cert.pem | base64 -w0
# cat key.pem | base64 -w0
```

Once fetched, update the fallback secret with the certificate and key.

```
# kubectl edit secret -n vnfmgr <FALLBACK-SECRET-NAME>
```

Enable OPM NFS for Non-OPM Kubernetes Cluster Deployments

If you plan to deploy one or more Kubernetes clusters without using OPM, but wish to use the NFS server provided by OPM, make the following changes to your OPM server:

1. Log into your OPM server using SSH.

2. Run the following command:

```
# vi /vnfmgr/nfsexport
```

3. Add your server IP to the file in the following format:

```
/filestore <server_ip>(rw, sync, no_root_squash)
```

4. Save the file and exit vi.

5. Allow a few seconds for the changes to propagate, then verify that `/etc/exports` includes the server you have added.

Install Multi Node OPM

OPM can be deployed as a multi node Kubernetes cluster where the services run on worker nodes. This improves resiliency and scalability of the management solution by providing more resources for OPM-related services and enabling resilient deployment features like those available for UnityCloud network functions.

Multi node OPM requires a bootstrap OPM for deployment. The bootstrap OPM is used to create the cluster required for the multi node OPM and to deploy the multi node OPM with all its dependencies.

1. Deploy a bootstrap OPM if you do not have an existing OPM deployment:

— [Install OPM into OpenStack](#)

Table 12. Bootstrap OPM Size Requirements

VM	vCPU	Memory
Bootstrap OPM	4	8G

2. Using the bootstrap OPM, Deploy a Kubernetes cluster for multi master OPM:

— [Multi-Master Kubernetes](#)

Table 13. Multi Master OPM Size Requirements

VM	vCPU	Memory
Master-1	8	32G
Master-2	8	32G
Master-3	8	32G
Node-1	16	64G
Node-2	16	64G
Node-3	16	64G

3. Create the prerequisite services for multi master OPM:
 - [Enable OPM NFS for Non-OPM Kubernetes Cluster Deployments](#)
4. Deploy multi master OPM:
 - [Deploy UnityCloud Network Function Applications with OPM](#)
 - Use the chart `fed-opm_multi` for deployment.

Note: Refer to the [MNOPM HashiCorp Vault Integration](#) section for enabling HashiCorp Vault during multi-node OPM deployment.

Table 14. fed-opm multi parameters

Parameters	Location	Expected values
Username	global.dbNoSqlRam.mongo.username	root or any other valid string
Password	global.dbNoSqlRam.mongo.password	affirmed or any other valid string
Init Username	global.dbNoSqlRam.mongo.initUsername	admin or any other valid string
Init Password	global.dbNoSqlRam.mongo.initPassword	affirmed or any other valid string
Connection String	global.dbNoSqlRam.mongo.connectionString	For AKS deployment provide valid CosmosDB connection string
Username	global.opm.username	admin or any other valid string
Password	global.opm.password	admin or any other valid string
Mode	global.opm.mode	multiNode or singleNode

Install OPM into OpenStack

Parameters	Location	Expected values
Capacity	global.fileserver.local.capacity	1000Gi or any other valid size string
Repo Path	global.registry.docker.repoPath	Docker repository path to fetch the container images
Upstream Repo Path	global.registry.docker.upstreamRepoPath	Upstream docker repository path to fetch the UC Infra Docker Registry images
Enable OPM Dreg	global.registry.opmDreg.enabled	True to enable OPM Docker Registry
Enable Infra Dreg	global.registry.infraDreg.enabled	True to enable UC Infra Docker Registry
Infra Dreg IP	global.registry.infraDreg.loadBalancerIP	IP address where the UC Infra Docker Registry will be exposed
Infra Dreg Port	global.registry.infraDreg.port	Port of the UC Infra Docker Registry, set to 5005
Infra Dreg URL	global.registry.infraDreg.registryUrl	URL of the UC Infra Docker Registry
Ingress Gw IP	global.mesh.istio.ingressGwIP	IP address where the OPM GUI and NB APIs will be exposed
Ingress Gw TLS	global.mesh.istio.ingressGwTLS.userProvided	If using cert-manager then "enabled" flag in this section should be set to false
Mongo Server Image	pod-opm_master_rm.deployment.mongoServerImage	Full container image tag for Mongo DB

Required parameters for fed-redis_cluster

Redis Primaries	pod-redis_cluster.numberOfPrimaries	Should be set to 1 for fed-opm_multi deployment
Redis Replicas	pod-redis_cluster.replicationFactor	Should be set to 2 for fed-opm_multi deployment
Redis Resources	resources.limits.cpu	Should be set to 500m for fed-opm_multi deployment
	resources.limits.memory	Should be set to 1Gi for fed-opm_multi deployment
	resources.requests.cpu	Should be set to 100m for fed-opm_multi deployment
	resources.requests.memory	Should be set to 512Mi for fed-opm_multi deployment

Parameters for fed-kube_prom

Install OPM into OpenStack

Parameters	Location	Expected values
Type	pod-kube_prom.prometheus.storage.type	Should be set to local
Storage Class Name	pod-kube_prom.prometheus.storage.storageClassName	Should be set to kube-prom
Parameters for fed-prometheus		
thanosQuery Resources	thanosQuery.resources.limits.cpu	Should be set to 500m for fed-opm_multi deployment
	thanosQuery.resources.limits.memory	Should be set to 1Gi for fed-opm_multi deployment
	thanosQuery.resources.requests.cpu	Should be set to 100m for fed-opm_multi deployment
	thanosQuery.resources.requests.memory	Should be set to 256Mi for fed-opm_multi deployment
Prometheus Resources	prometheus.resources.limits.cpu	Should be set to 500m for fed-opm_multi deployment
	prometheus.resources.limits.memory	Should be set to 1Gi for fed-opm_multi deployment
	prometheus.resources.requests.cpu	Should be set to 100m for fed-opm_multi deployment
	prometheus.resources.requests.memory	Should be set to 256Mi for fed-opm_multi deployment

— Verify that the multi node OPM is running successfully by checking its status in the bootstrap OPM Cluster View UI.

— Verify the self monitoring in the multi node OPM by checking its status in the multi node OPM UI.

5. Generate and install CA certificate for Multi-Node OPM. This is a required step to ensure the FQDN is set up properly for the certificate used in multi-node OPM.

Generate the root CA certificates with a wildcard CN.

CERTNAME=unitycloud

openssl genrsa -out \$CERTNAME.key 2048

**openssl req -x509 -new -nodes -key \$CERTNAME.key -sha256 -subj
"/C=US/ST=MA/L=Acton/O=Microsoft/OU=CNI/CN=*.unitycloud.com/
emailAddress=kabhijeet@microsoft.com" -days 730 -out \$CERTNAME.crt**

Install OPM into OpenStack

The `unitycloud.crt` and `unitycloud.key` are generated. Provide these as values for `tlsrct` and `tlskey` respectively. The following image shows the `tlsrct` and `tlskey` boxes where the newly generated values should be entered.

Figure 8. Values for `tlsrct` and `tlskey`

Life Cycle Management : Create Configuration Set

Cluster: `tr-48-master-1`

Shared values: Select...

Add Chart +

- fed-opm_multi-4.3.0-98-main-int
 - fed-crds-4.3.0-134-main-int
 - fed-rbac-4.3.0-141-main-int
 - fed-kube_addons-4.3.0-155-main-int
 - fed-opa-4.3.0-150-main-int
 - fed-metalb-4.3.0-146-main-int
 - fed-opa_policies-4.3.0-26-main-int
 - fed-istio-4.3.0-150-main-int
 - fed-paas_helpers-4.3.0-154-main-int
 - fed-service_reg-4.3.0-142-main-int
 - fed-redis_operator-4.3.0-125-main-int
 - fed-grafana-4.3.0-143-main-int
 - fed-prometheus-4.3.0-149-main-int
 - fed-redis_cluster-4.3.0-156-main-int
 - fed-kube_prom-4.3.0-112-main-int
 - fed-opm_multi-4.3.0-98-main-int

General

docker-reg * image-pull-secrets *

unique-name-suffix *

OPM

opm-password * admin opm-platform * openstack

opm-username * admin

Load Balancer

grafana-lb-ip * istio-lb-ip *

kargo-lb-ip * kube-prom-lb-ip *

oam-lb-subnet * none opm-lb-ip *

thanos-lb-ip *

PaaS

admin-password * elasticsearch-host *

grafana-url * istio-gw-jaeger-hostname * fqdn.com

istio-proxy-include-ip-ranges * 10.96.0.0/12,192.168.0.1 jaeger-host *

leader-election-renew-deadline * 20s storage-type *

tlsrct **tlskey**

Users must provide/input the FQDNs from the following table for OPM, Grafana and Prometheus.

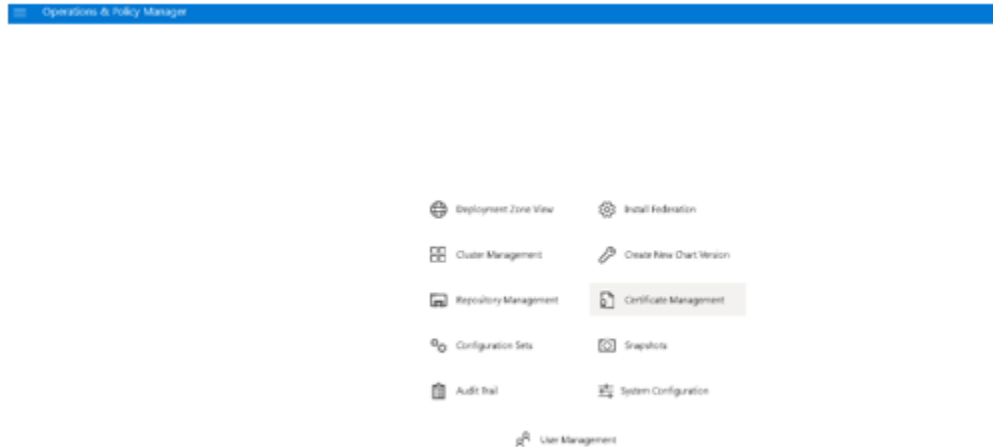
Table 15. FQDN values

FQDN	Fed Name	Path of the field in values.yaml	Value to use
OPM FQDN	fed-opm_multi	global.mesh.istio.ingressGwFQDN	opm.unitycloud.com
UC Infra Dreg FQDN	fed-opm_multi	global.regisry.infraDreg.registryUrl	registry.unitycloud.com
Grafana FQDN	fed-grafana	global.mesh.istio.ingressGwFQDN	grafana.unitycloud.com
Prometheus FQDN	fed-prometheus	global.mesh.istio.ingressGwFQDN	thanos.unitycloud.com

Active/Standby OPM

Active and Standby is supported only with multi-node OPM, also known as fed-opm_multi. fed-opm_multi can be in one of the two modes. Upon deployment, OPM will be configured as Active.

Figure 9. OPM Configured as Active



A standby OPM can be employed to ensure docker resiliency. Clusters hosting the network functions should be attached, or managed by both the active and standby OPM's. In case the active OPM is no longer available to serve the images, docker registry residing on the standby OPM will be used.

In addition, by default, OPM uses the certmgr generated certificate as both the ingress and docker registry certificates. Intelligence is built in OPM to ensure the same certificate is used by both Active and Standby OPM. When the certificate rotates, the client certificates used by the network functions will be updated automatically.

Deploy Standby OPM

Prerequisites

Push the Charts to the Local Repository

Perform the steps in [Affirmed Networks Public Repository – Push Charts](#) to download and push images to the local Docker Registry and Helm ChartMuseum from the Affirmed Networks public repository.

Deploy Standby OPM Cluster

If the current deployment doesn't contain second multi-master OPM cluster, see Deploying Multi Master OPM section in [Install Multi Node OPM](#) to create using the multi-node OPM.

Install Fed-OPM Multi on Standby OPM

Install OPM into OpenStack

- To install Fed-OPM Multi on Standby OPM, see Deploying Federations with OPM.
- Verify that the multi node OPM is running successfully by checking its status in the bootstrap OPM Cluster View UI.
- Verify the self monitoring in the multi node OPM by checking its status in the multi node OPM UI.

Procedure

1. To have a standby OPM assigned for an active OPM, toggle the Global OPM mode switch and enter details about active OPM.

IP Address : <Active OPM' Master IP> M2M Keys: <Active OPM's m2mkeys >

Figure 10. Global Standby Mode Values

Global Standby Mode Values

IP Address * 10.163.66.111

M2M Keys *

```
2zRWWY+7/IAJOTDg
VRUotbD/hZzR4S+yIA6+AAAAJnJ
vb3RAadm5mbWdyLW9zcm0tMTAw
MC01ZDZkN2RmODQ2LT
U1NnhuAQIDBAUGBw==
-----END OPENSSH PRIVATE KEY-----
```

Update Global Standby Mode

Note: Active OPM should be using the same FQDN as the standby OPM, otherwise error message will appear.

By Default FQDN is set as `opm.unitycloud.com` in `fed-opm_multi values.yaml`

Figure 11. Ingress GW FQDN Value

mesh

istio

Ingress Gw *

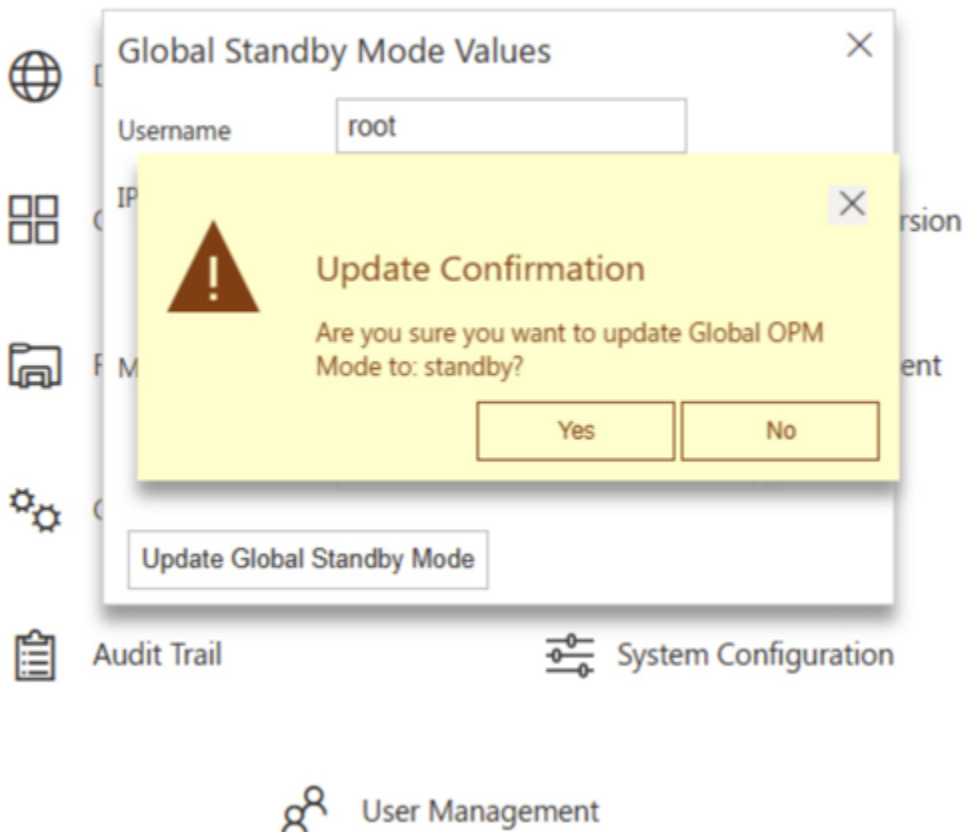
Ingress Gw IP *

Ingress Gw FQDN *

Mtls Enabled * ☐

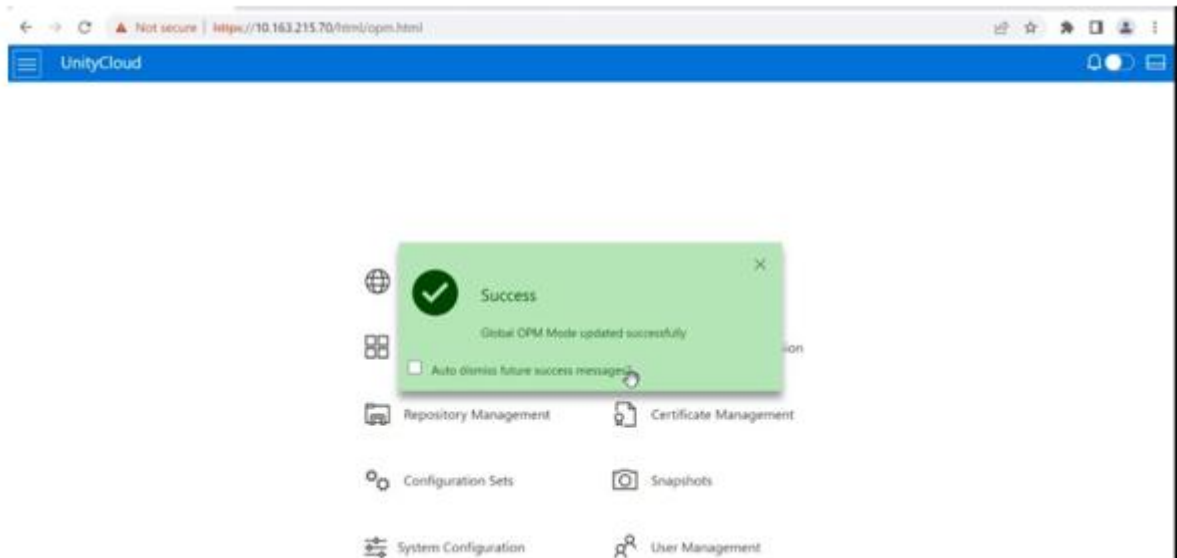
- Once given the required values, click on Update Global Standby Mode and click Yes to confirm.

Figure 12. Global Standby Mode Values – Updated Confirmation



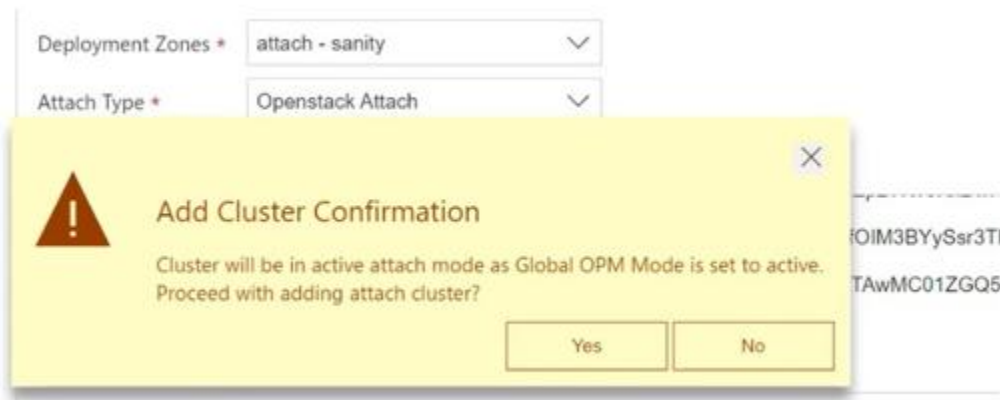
- Once confirmed, the OPM switches to standby mode and the toggle on top right for Global OPM Mode Switch will toggle **OFF**.

Figure 13. Global OPM Mode Updated Success Screen



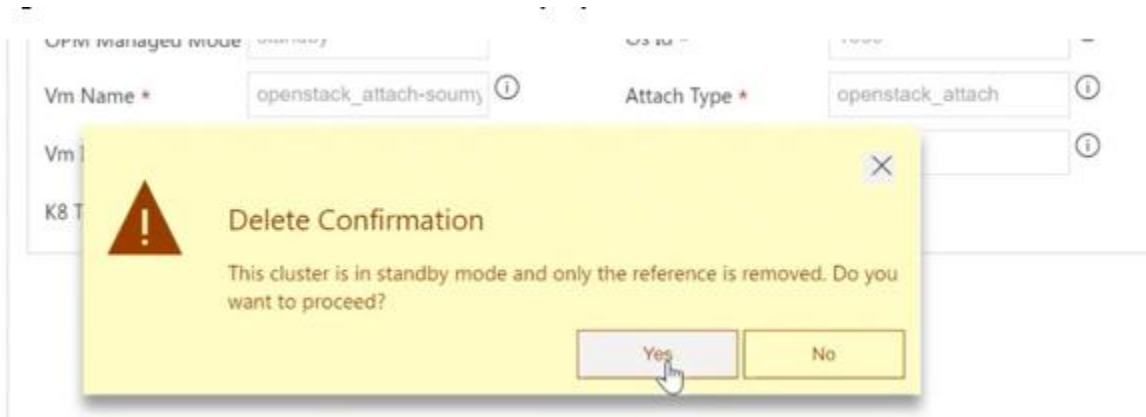
User is responsible to use the standby OPM to attach to all the network function clusters. When attaching a cluster to OPM in active/standby mode, it will give a pop-up message confirming the same, that the cluster will be in active/standby mode respectively.

Figure 14. Adding Cluster Confirmation Pop-up



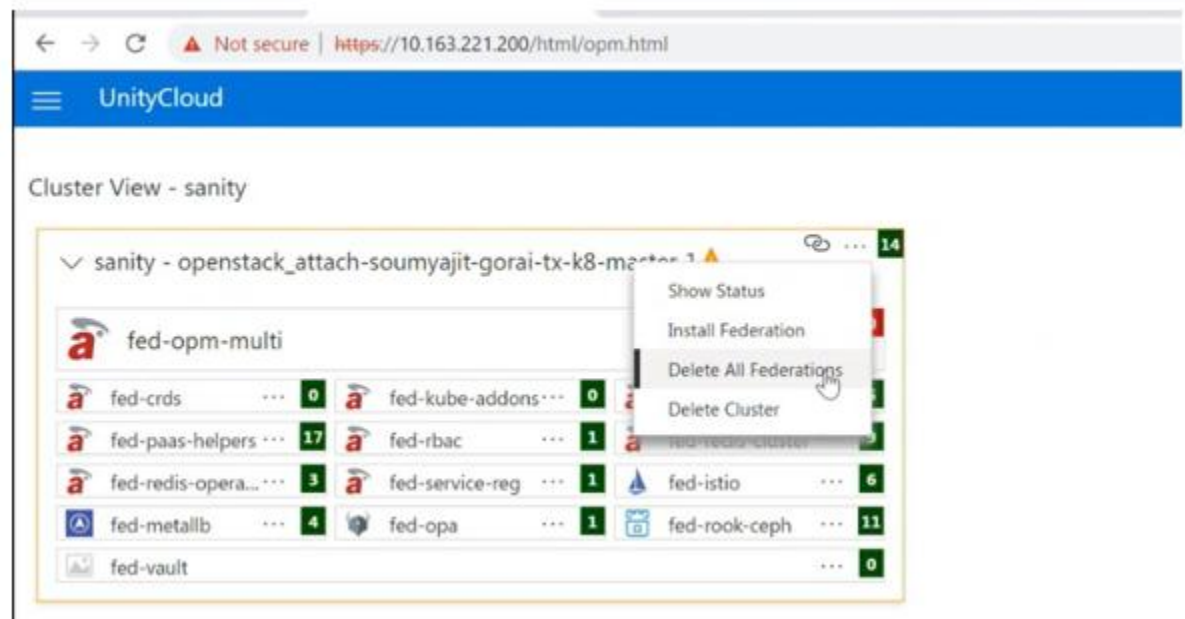
4. Similarly, when deleting a cluster to the active/standby OPM, it will give a pop-up message confirming the same.

Figure 15. Delete Cluster Confirmation Pop-up



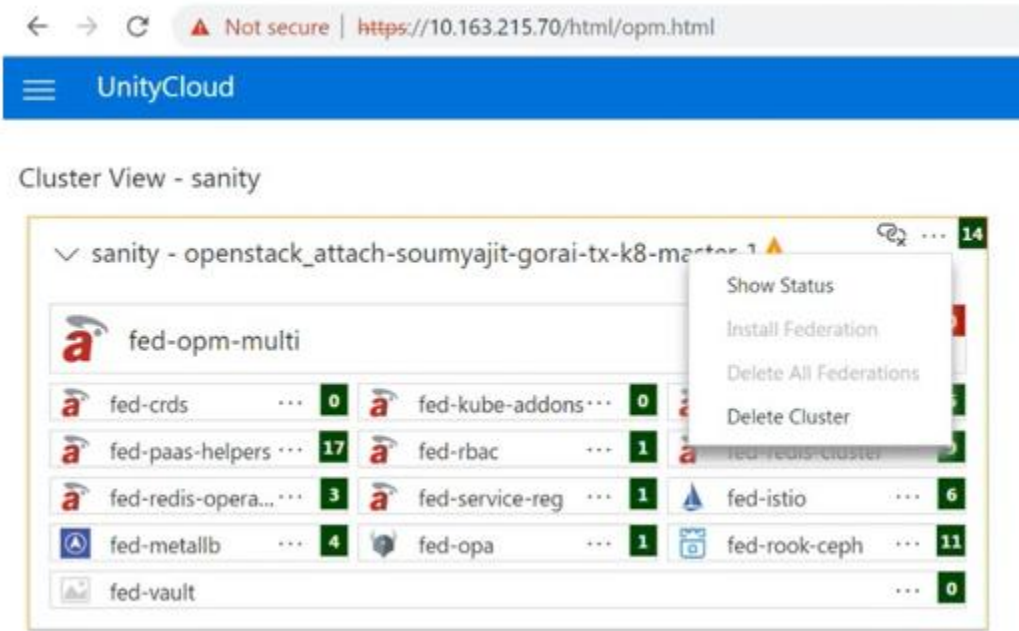
Note: User can use the active OPM to add/update/delete federations on the attached clusters.

Figure 16. Cluster View – Sanity



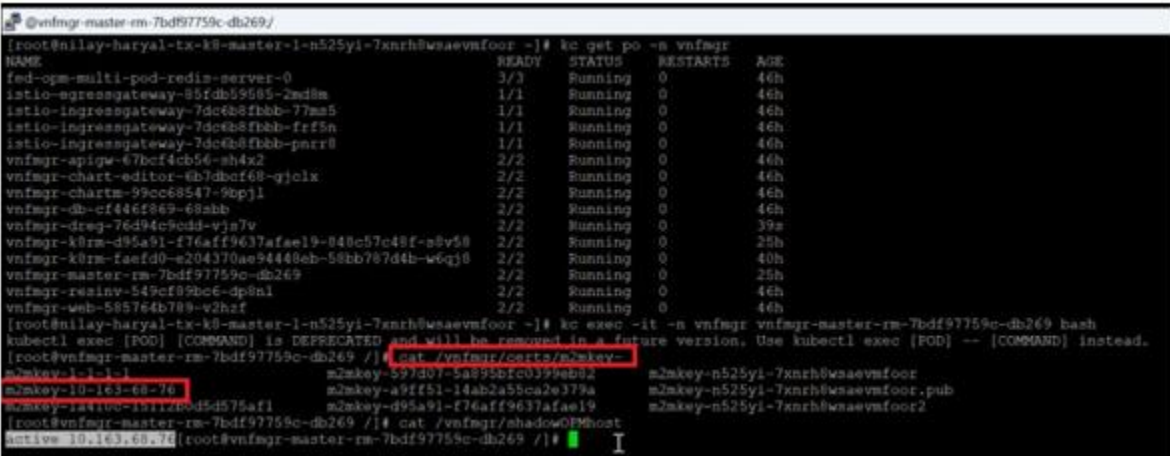
- Whereas on the standby OPM, user can only use OPM to view cluster information.

Figure 17. Standby OPM – View Cluster Information



6. The active OPM's m2mkeys are stored in the standby OPM and the active OPM details are maintained using /vnfmgr/shadowOPMhost file is present inside any vnfmgr-master-rm pods or in Mongo (in vnfmgrDB):

Figure 18. Active OPM m2m keys



7. The active OPM will also have the standby OPM's m2mkeys and details maintained in /vnfmgr/shadowOPMhost file is present inside any vnfmgr-master-rm pods or in Mongo (in vnfmgrDB):

Figure 19. Standby OPM m2m keys

```

[root@alex-xia-ia-k8-master-1-6p4pk2-kqnslzwc2opx0uft ~]# kubectl exec -it -n vnfmgr vnfmgr-master-rm-f457f5c74-8tn8n bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
[root@vnfmgr-master-rm-f457f5c74-8tn8n /]# cat /vnfmgr/shadowOPMhost
standby 10.167.66.111
[root@vnfmgr-master-rm-f457f5c74-8tn8n /]# cat /vnfmgr/certs/m2mkey-
m2mkey-0417ed-1ef916e97129bcc2 m2mkey-10-163-66-111 m2mkey-974c3e-a3abe0e40e41fff787
[root@vnfmgr-master-rm-f457f5c74-8tn8n /]# cat /vnfmgr/certs/m2mkey-

```

Manage Cluster in Standby Mode

After you have deployed a Kubernetes cluster, you can then deploy it in standby mode on another instance of OPM. In the event that the primary OPM becomes unavailable, the secondary OPM can be used to switch deployed clusters from standby mode to active mode and assume management of those clusters.

Prerequisites

Before deploying a cluster in standby mode, you must collect the following information from the active cluster on the primary OPM:

1. Export a template of the active cluster on the primary OPM. See [Export Cluster and Deployment Zone Settings](#) for details.
2. Log into the primary OPM using SSH and retrieve the public and private m2m keys of the active cluster. They will be used in setting up the standby cluster. Use the following commands:

```

# cat /vnfmgr/certs/m2mkey-[K8_Token_of_Active_Cluster]

# cat /vnfmgr/certs/m2mkey-[K8_Token_of_Active_Cluster].pub

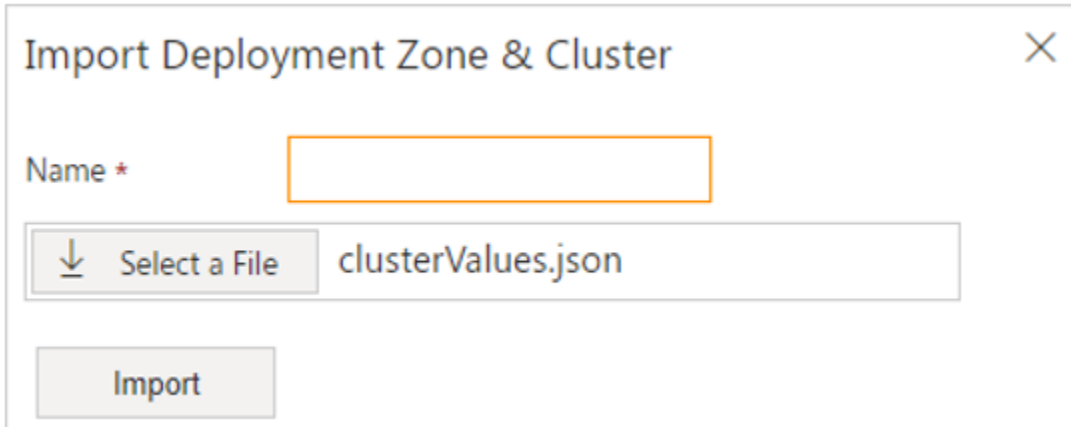
```

Deploy Cluster in Standby Mode

To deploy a cluster in standby mode, log into a separate instance of OPM and perform the following:

1. Click **Cluster Management**.
2. Click **Import Deployment Zone or Cluster** (down arrow).
3. Click **Select a File** and choose the saved template from the active cluster.
4. Click **Import**.

Figure 20. Import Saved Template



Import Deployment Zone & Cluster

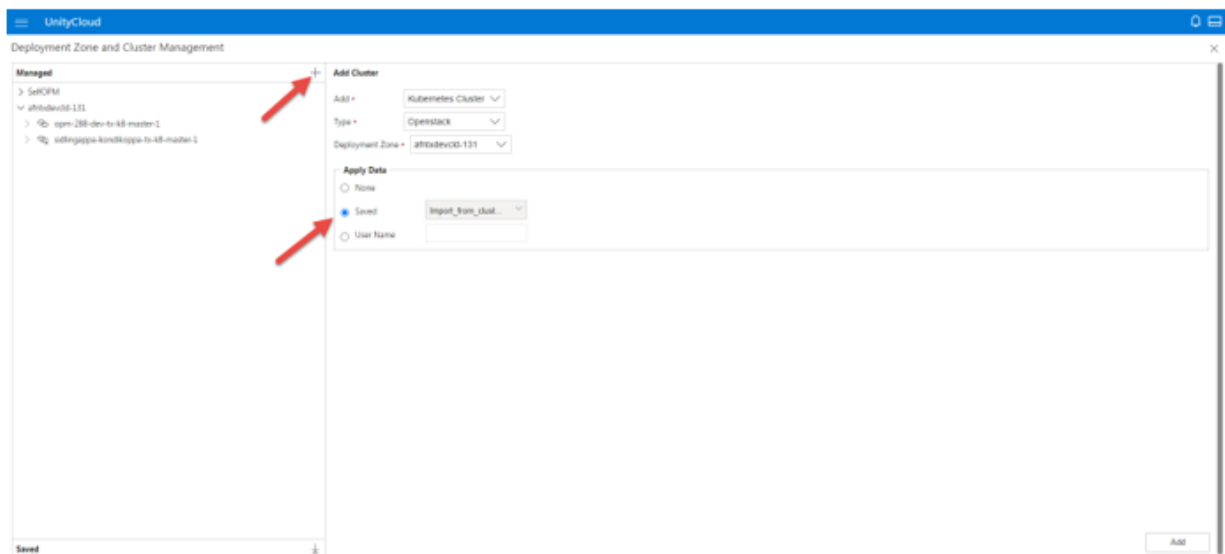
Name *

Select a File clusterValues.json

Import

5. Click **Create Deployment Zone or Cluster** (plus).
6. Under the **Add** dropdown, choose **Kubernetes Cluster**.
7. Under **Type** and **Deployment Zone**, choose values that match the active cluster.
8. Under **Apply Data**, click **Saved** and choose your imported template from the dropdown, then click **Add**.

Figure 21. Add a Cluster



UnityCloud

Deployment Zone and Cluster Management

Managed

- Set OPM
- athdev00-131
 - opm-280-dev-to-k8-master-1
 - ollapappa-kondappa-to-k8-master-1

Add Cluster

Add +

Kubernetes Cluster

Type +

Openstack

Deployment Zone +

athdev00-131

Apply Data

None

☒ Saved

Import_from_cluster...

User Name

Save

Add

9. On the Create Cluster Form, switch **OPM Managed Mode** from Active to **Standby**.
10. Under **M2m Pubkey** and **M2m Privkey**, provide the public and private keys saved from the active cluster.

- Fill out any remaining required fields, making sure to match the values in the active cluster exactly.
- Click **Create** to deploy the cluster in standby mode.

Figure 22. Required Fields to Deploy in Standby Mode

[illegible]

Certificate Rotation

The CertMgr managed certificates get auto renewed based on the duration set in `fed-opm_multi values.yaml`. By default, it is 8760 hours (1 year).

Figure 23. CertManager Screen

certManager			
Enabled	☒	Cert Duration	8760h
Renew Before	10h	Organization	affirmednetworks
Issuer Name	affirmed-clusterissuer		

The active OPM docker certificate then gets synced across all OPM and attached Clusters. This would ensure docker resiliency across active OPM, standby OPM and all the attached clusters.

Docker Registry Failover

In case the docker registry is not available on active OPM, the attached clusters switches to use the standby OPM docker certificate.

This is made possible using a cronjob set to run every hour on the on NF clusters to verify docker access. In case the active OPM IP is not resolvable by the FQDN, it will try the same with standby OPM details. If that works, it will update the /etc/hosts file accordingly to ensure access persists.

This cronjob is running on each node of the attached clusters, to ensure docker registry is accessible from all the clusters' nodes.

Logs can be fetched using `journalctl | grep dregMonitorCron` command after SSH to any node.

OPM Switchover

User can switch over the active and standby OPM in cases of docker registry failover or other failure or maintenance scenarios.

First, the standby OPM needs to be toggled back using the Global OPM mode switch to active OPM.

Then, the earlier active OPM can now be attached as standby OPM to now active OPM.

This would be critical as we cannot have a standby OPM attached to another standby OPM, hence need to make standby as active first before attaching another OPM to it as its standby.

Modify SSH banner text

Follow these steps to change the SSH banner text for the following UC VMs:

- vm-opm
- vm-platform

Tip: The steps for both are the same.

1. SSH into the VM
2. Edit /etc/issue.net add text customization
3. Save

After saving the changes, the next time the user executes ssh into the VM, the SSH banner displays the customized banner text.

Apply Kubernetes Timer Optimization in UnityCloud

This section provides step by step instruction on how to update existing or add a new Kubernetes timer, which can improve performance, robustness and recovery time of the application.

kubelet

The kubelet updates the node conditions to API server based on the configured `--node-status-updatefrequency`, which is 10 seconds by default.

We recommend updating the value of `node-status-update-frequency` to 5 seconds to match the period of syncing Node Status in NodeController. This parameter should be set using the config file specified by the Kubelet's `--config`.

1. The file containing the current kubelet's ComponentConfig is located at `/var/lib/kubelet/config.yaml`. Validate the current configuration by running the following command:

```
# cat /var/lib/kubelet/config.yaml | grep nodeStatusUpdateFrequency
```

2. Update `nodeStatusUpdateFrequency` by running the following commands to update the value to 5 seconds.

```
# Var1='nodeStatusUpdateFrequency: 10s'
```

```
# Var2='nodeStatusUpdateFrequency: 5s'
```

```
# sed -i "s/${Var1}/${Var2}/g" /var/lib/kubelet/config.yaml
```

3. Validate the config by running the following command:

```
# cat /var/lib/kubelet/config.yaml | grep nodeStatusUpdateFrequency
```

4. Run the following command to restart kubelet service and load the new configuration into the kubelet.

```
# systemctl daemon-reload && systemctl restart kubelet
```

5. Make sure to log into all the nodes and repeat all the above steps.

kube-controller-manager

By default, a running Node is allowed to be unresponsive for 40 seconds before marking it as unhealthy. This value must be N times more than kubelet's `nodeStatusUpdateFrequency`, where N is the number of retries allowed for kubelet to post node status.

We recommend updating **node-monitor-grace-period** to 20 seconds, so kubelet will retry four times before marking the node as unhealthy.

1. Locate the kube-controller-manager pod in the cluster by running the following command:

```
# kc get po -n kube-system | grep kube-controller-manager
```

2. Keep a note of current configuration of controller manager by running the following command:

```
# kc describe po [kube-controller-manager_name] -n kube-system
```

3. Add the timer `node-monitor-grace-period` to the manifest file located at `/etc/kubernetes/manifests` in file `kube-controller-manager.yaml` in the master node:

- a. Ensure that you are logged in on master node. Run the following command to add `node-monitor-graceperiod` and update the value to 20s. Please ensure the command is run just once.

```
# sed -i '/ - kube-controller-manager/ a \ \ \ \ - --node-monitor-grace-period=20s' /etc/kubernetes/manifests/kube-controller-manager.yaml
```

- b. Validate that `node-monitor-grace-period` is added only once by running the following command:

```
# cat /etc/kubernetes/manifests/kube-controller-manager.yaml
```

4. After the manifest file is updated, kube-controller-manager pod will be re-created loaded with the new configuration. To validate, perform step 1 and 2.

kube-apiserver

default-not-ready-toleration-seconds indicates the `tolerationSeconds` of the toleration for `notReady:NoExecute` that is added by default to every pod that does not already have such a toleration.

default-unreachable-toleration-seconds indicates the `tolerationSeconds` of the toleration for `unreachable:NoExecute` that is added by default to every pod that does not already have such a toleration.

We recommend updating the `default-not-ready-tolerationseconds` and `default-unreachable-tolerationseconds` from default value of 300 seconds to 170 seconds.

1. Locate the kube-apiserver pod in the cluster by running the following command:

```
# kc get po -n kube-system | grep kube-apiserver
```

2. Keep a note of current configuration of apiserver by running the following command:

```
# kc describe po [kube-apiserver_name] -n kube-system"
```

3. Timers `default-not-ready-toleration-seconds` and `default-unreachable-toleration-seconds` need to be added in the manifest file located at `/etc/kubernetes/manifests` in file `kube-apiserver.yaml` in the master node:

a. Ensure that you are logged in on master node. Run the following 2 commands to add `default-not-readytoleration-seconds` and `default-unreachable- toleration-seconds` and update the values to 170. Please ensure the commands are run just once each:

```
# sed -i '/ - kube-apiserver/ a \\ \\ - --default-not- ready-toleration-seconds=170' /etc/kubernetes/manifests/kube-apiserver.yaml
```

```
# sed -i '/ - kube-apiserver/ a \\ \\ - --default- unreachable-toleration-seconds=170' /etc/kubernetes/manifests/kube-apiserver.yaml
```

4. Validate that `default-not-ready-toleration-seconds` and `default-unreachable-toleration-seconds` is added only once by running the following command:

```
# cat /etc/kubernetes/manifests/kube-apiserver.yaml
```

5. After the manifest file is updated, `kube-apiserver` pod will be re-created loaded with the new configuration. To validate, perform step 1 and 2.

Cluster Administration

Deploy Kubernetes Cluster

This section uses the information in the following table to demonstrate how to deploy a Kubernetes cluster using OPM.

Note: For the OPM VM, the minimum disk space is 50G. However, if you plan to use the OPM's Docker Registry and Helm Chart Repository, you will need enough disk space to hold all the container images and charts.

With the new VM ISSU feature, the disk space needed for the VM , must be doubled in order to create upgrade partition in VM.. The VM ISSU feature is on by default.

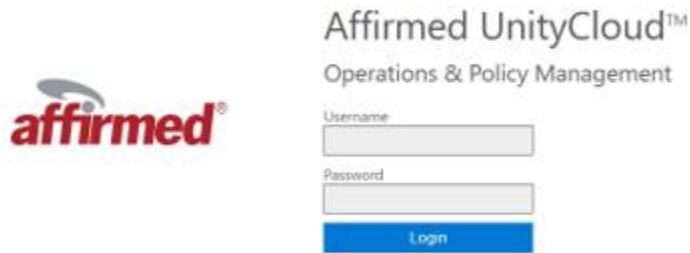
Table 16. Example: VMs for Deploying Kubernetes Cluster

VM	IP Address	vCPU	Memory	Disk Space	With ISSU feature enabled
OPM	10.165.5.10	4	8G	200G	400G
Master-1	10.165.5.11	4	8G	150G	300G
Node-1	10.165.5.12	16	78G	150G	300G
Node-2	10.165.5.13	16	78G	150G	300G

Associate OPM with OpenStack Controller

OPM communicates with OpenStack to deploy VMs (platform VM). These VMs make up the Kubernetes cluster.

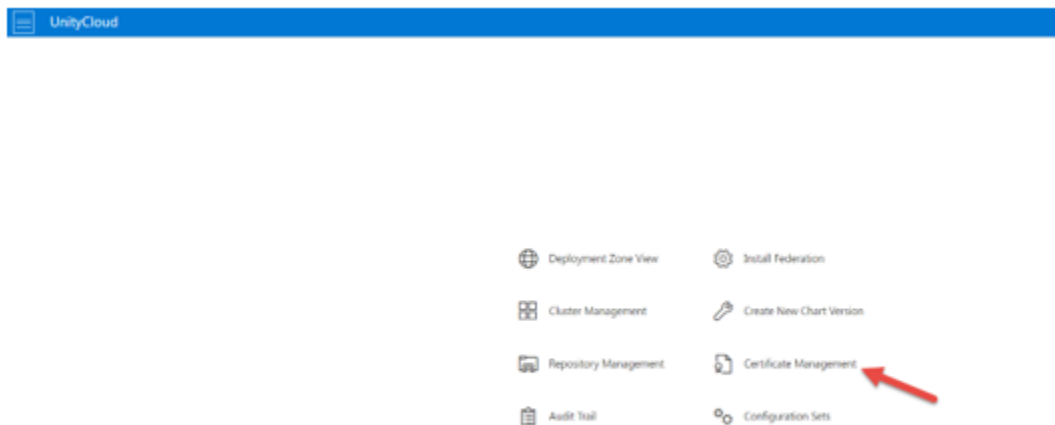
1. To log in to the OPM GUI, use a web browser to go to `https://<opm ip>`.

Figure 24. OPM GUI Login

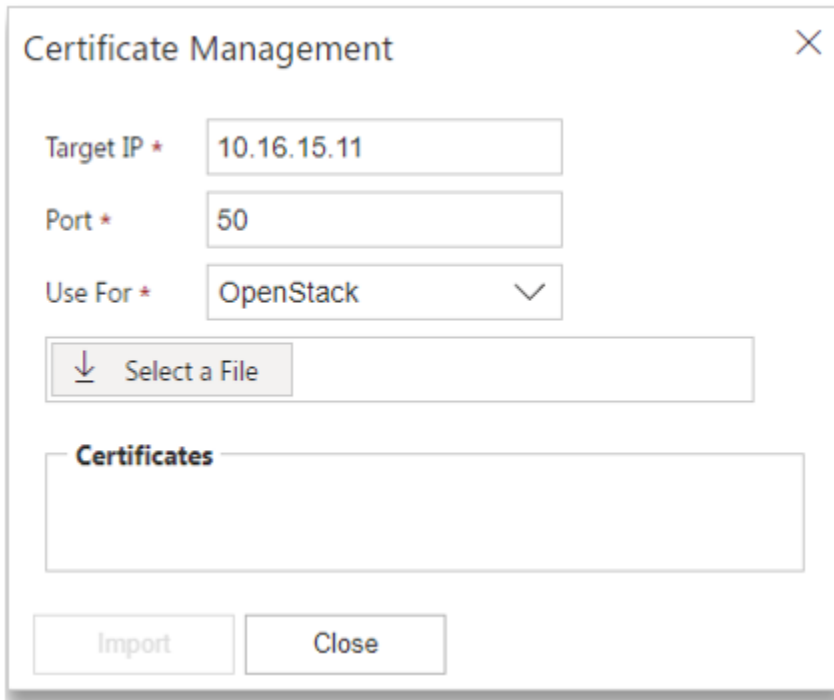
2. Enter your username and password.

Note: If you are *not* using a certificate file with OpenStack, **proceed to step 7.**

3. From the main OPM screen, click **Certificate Management**.

Figure 25. Main OPM Screen

4. Enter the Certificate Management parameters.

Figure 26. Certificate Management


The image shows a 'Certificate Management' dialog box with a close button (X) in the top right corner. It contains the following fields and controls:

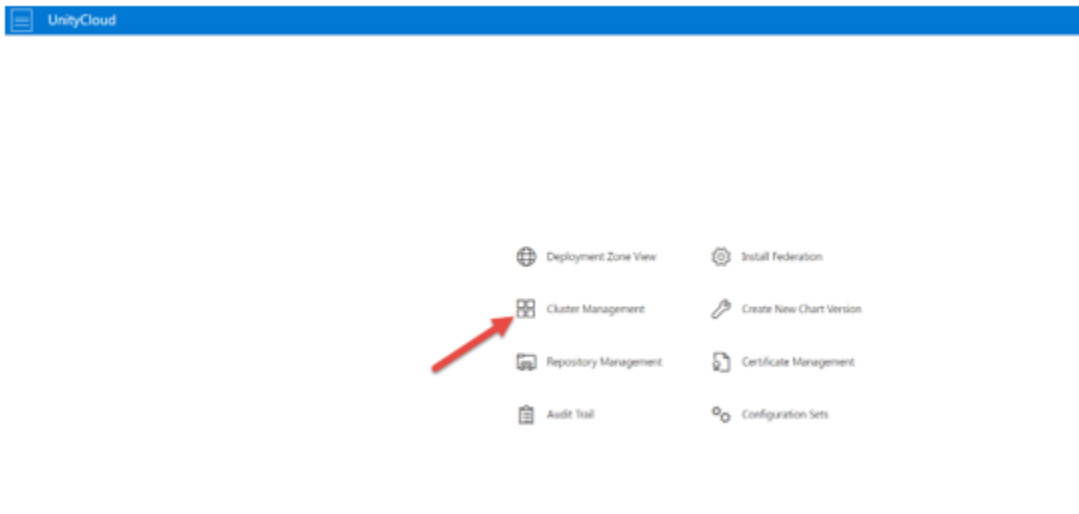
- Target IP ***: A text input field containing '10.16.15.11'.
- Port ***: A text input field containing '50'.
- Use For ***: A dropdown menu with 'OpenStack' selected and a downward arrow.
- Select a File**: A button with a downward arrow icon.
- Certificates**: A large empty text area.
- Import** and **Close**: Two buttons at the bottom.

Table 17. Certificate management parameters

Parameter	Description
Target IP*	Enter the IP address of the OpenStack server. If using Docker, enter the IP address of the Docker Registry.
Port	Port
Use for:	Select OpenStack from the drop-down menu.

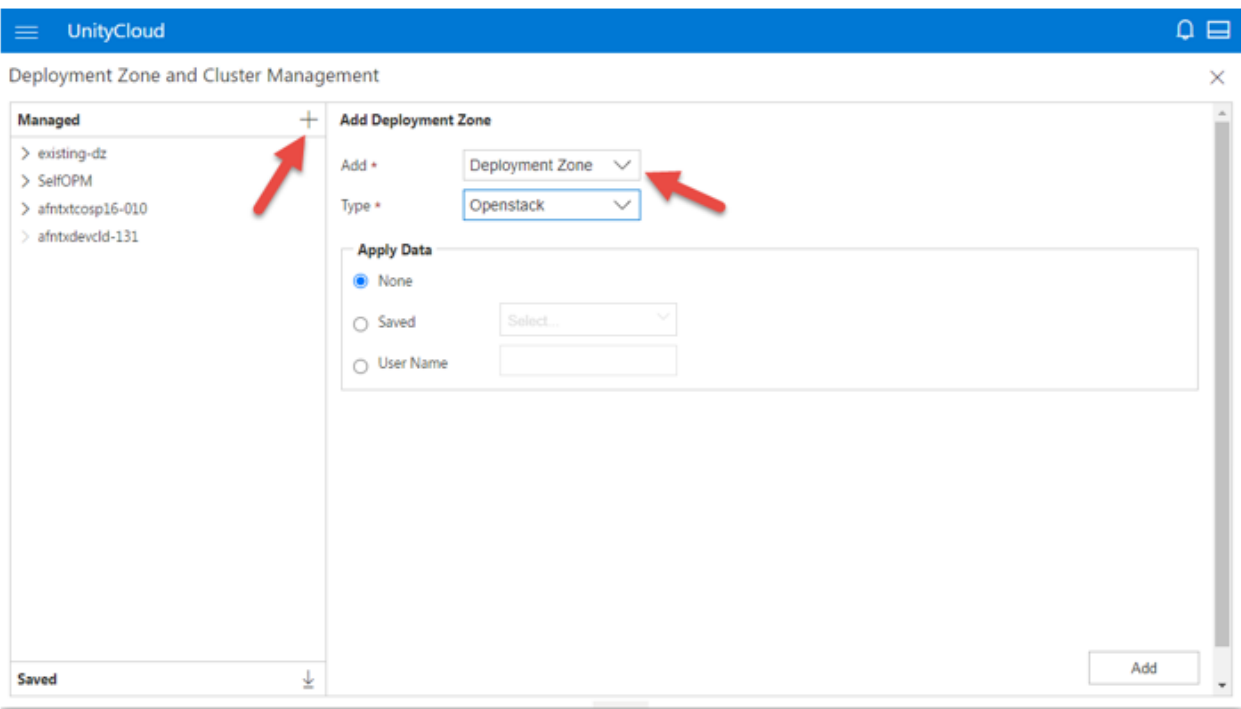
5. Click **Select a File** and insert the OpenStack certificate file.
6. Click **Import**. This file is now copied to OPM at path /vnfmgr/certs.
7. From the main OPM screen, click **Cluster Management**.

Figure 27. Main OPM Screen



8. Click the Plus icon (+) and choose **Deployment Zone** from the **Add** dropdown menu.

Figure 28. Deployment Zones



- 9. Select **Openstack** as the deployment zone type and click **Add**.
- 10. Enter the OpenStack deployment zone information and click **Deploy**.

Figure 29. OpenStack Deployment Zine Information

The screenshot shows the 'Create Deployment Zone Form' in the UnityCloud interface. The form is titled 'Deployment Zone and Cluster Management'. On the left, a 'Managed' sidebar lists existing deployment zones: 'existing-dz', 'SelfOPM', 'afntxtcosp16-010', and 'afntxdevcd-131'. The main form area contains the following fields:

- Os Auth Url Version *
- Os Id *
- Os Name *
- Os Port *
- Os Project Id (set to 'none')
- Os User *
- Os Location *
- Os Project Domain Name
- Local DNS (toggle)
- Os Cert Path
- Os Ip *
- Os Pass * (password field)
- Os Project Name *
- Os Protocol Type *
- Type * (set to 'openstack')
- Os User Domain Name
- Os Region Name

At the bottom right, there are 'Deploy' and 'Save Values' buttons. The bottom left shows a 'Saved' status.

Table 18. OpenStack deployment zone parameters

Parameter	Description
OS Pass*	OpenStack API password.
OS Name*	Name associated with this OpenStack instance.
OS Ip*	OpenStack API IP address.
OS Protocol Type*	This is either HTTP or HTTPS. If you are importing a certification file for use with this OpenStack connection, enter HTTPS.
Os Project Domain Name	OpenStack project domain name. The default value is Default.
Os User Domain Name	OpenStack user domain name. The default value is Default.
Cluster Ids	Read-only list of clusters.
OS Id*	Unique numeric ID to identify this OpenStack instance.
OS User*	OpenStack API user.
OS Location*	String to identify location.
Os Region Name	OpenStack region name. The default value is regionOne.
OS Project Name*	OpenStack API project name.

Parameter	Description
OS Cert Path	Path to where the OpenStack certification file is located. If you are importing a certification file for use with this OpenStack connection, enter <code>/vnfmgr/certs/<name of cert file added with Certificate Management></code> .
Type*	Indicates how the cluster will be created. In this case, it is using OpenStack.
OS Auth Url Version*	OpenStack Auth URL version and API version. Select V2 or V3.
OS Port	OpenStack API port.

* You can obtain this information by downloading the OpenStack RC File V3 or V2 on the OpenStack Horizon GUI.

11. (Optional) After you have created a deployment zone, you can save the deployment zone configuration information by clicking **Save Values**. This you to recreate the deployment zone without having to type in all the information again.

Figure 30. Deployment Zone Save Template

The screenshot shows the 'Create Deployment Zone Form' in the UnityCloud interface. On the left, there is a 'Managed' list with items like 'existing-dz', 'SelfOPM', 'afntxcosp16-010', and 'afntxdevcid-131'. The main area contains a form with two columns of input fields. A red arrow points to the 'Save Values' button at the bottom right of the form.

Field	Value
Os Auth Url Version *	Example
Os Id	
Os Name *	Example
Os Port *	Example
Os Project Id	Example
Os User *	Example
Os Location *	Example
Os Project Domain Name	Example
Local DNS	☐
Os Cert Path	Example
Os Ip *	Example
Os Pass *	*****
Os Project Name *	Example
Os Protocol Type *	Example
Type *	openstack
Os User Domain Name	Example
Os Region Name	Example

Buttons: Deploy, Save Values

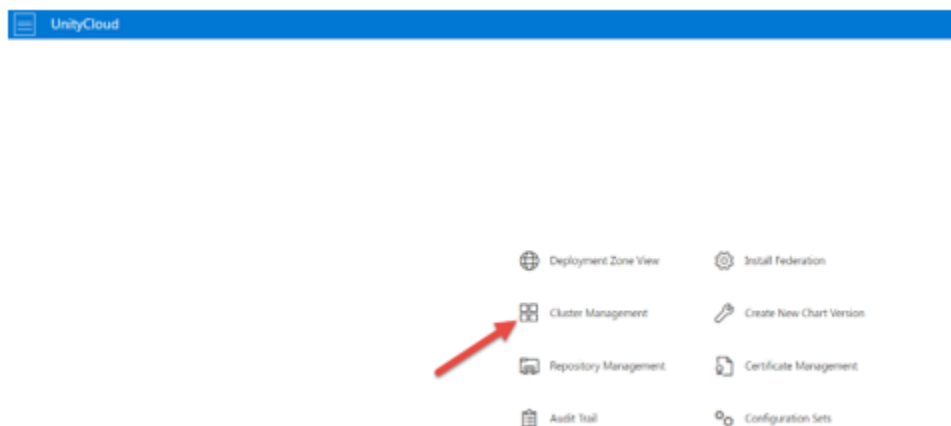
The next time you need to create this deployment zone, from the **Add Deployment Zone** dialog box, select the **Saved** name before clicking **Add**.

Create Kubernetes Cluster

To create a Kubernetes Cluster:

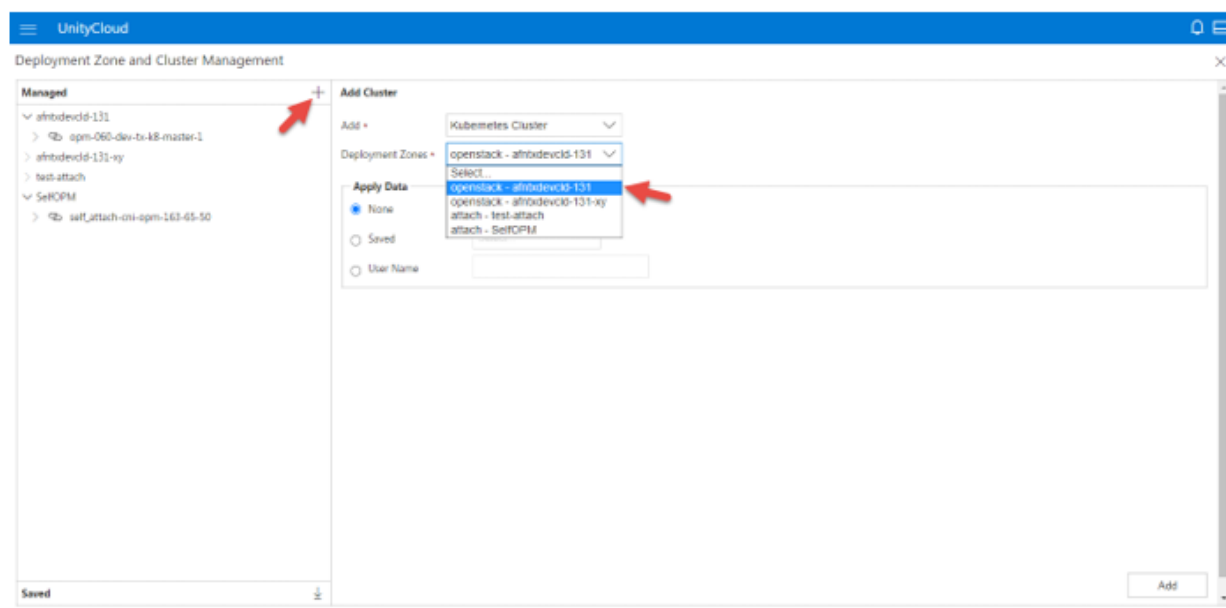
1. From the main OPM screen, click **Cluster Management**.

Figure 31. Main OPM Screen



2. Click the Plus icon (+) and then select **Kubernetes Cluster** from the **Add** dropdown.
3. Select the Openstack deployment zone created in the previous step from the **Deployment Zones** dropdown.

Figure 32. Create New Cluster



4. Click **Add** and enter the Kubernetes cluster information.

- By default, only fields that are essential for cluster creation are shown. To view all available fields, change the **Form Filter** dropdown from **Basic** to **Advanced**.

Figure 33. Kubernetes Cluster Information

Table 19. Kubernetes Cluster Properties

Parameter	Description
General	
OPM Managed Mode	Enable this flag to enable OPM to deploy new clusters for cold standby scenarios. See Deploy Cluster in Standby Mode .
M2m Pubkey	Public M2m Key for cluster managed by OPM in standby mode.
M2m Privkey	Private M2m Key for cluster managed by OPM in standby mode.
OS Id	Deployment Zone ID used to identify where the Kubernetes cluster will live. Currently, there is only OpenStack.
Vm DNS	DNS IP
Vm Domain	Domain
Vm Mtu	Sets MTU in both vm-platform OpenStack and outside networks' MTU must be set equal to or greater than the VM MTU.
Vm GW	Gateway IP
Vm Image	OpenStack Platform VM image ID. Use Glance for local disk and Cinder volume snapshot ID for cinder.

Cluster Administration

Parameter	Description
Vm Mgmt Net Id	OpenStack Network ID
Vm Netmask	Netmask
Ntp Service Ip	NTP server IP address.
Timezone	Timezone.
Vm Security Group ID	OpenStack Security Group ID. Use none if OpenStack is using port security, otherwise enter the security group ID.
Ready Percent	The minimum percentage of worker nodes to be ready apart from the master, for cluster to be considered in created state.
Os Server Group Id	OpenStack server group ID.
Vm Cinder	Toggle on to use Cinder Volumes. Toggle off to use local disk.
VM Disk Type	Only available if VM Cinder is set to <code>True</code> .
Cloud Provider=OpenStack	Toggle on if deploying the cluster using OpenStack Cloud Provider.
Project Id	Cloud Provider Tenant ID. Use the following steps to retrieve the Cloudp Tenant ID: 1. In the OpenStack Horizon GUI, click Project. 2. Click API Access. 3. Click View Credentials. You will find the tenant ID under Project ID.
Subnet Id	Cloud Provider Subnet ID. If using OpenStack, this is the OpenStack network subnet ID used for the master management IP address. To retrieve the OpenStack network subnet ID, enter the following command on OpenStack: <pre># openstack subnet list</pre>
ARC Enable	[PB] Transform your kubernetes clusters into a Managed Azure object. This toggle will load the required microservices for ARC.
ARC Name	Name your cluster in Azure.
ARC Resource Group	Create or link the Azure Arc enabled cluster to a Azure Resource Group.
ARC Log Workspace	Name log analytics resource created in Azure.
ARC SP Client Id	Service principal can be retrieved from Azure Portal.
ARC SP Secret Key	Service key to be used with Service Principal ID.
ARC Tenant Id	Tenant ID from your Azure Active Directory.
ARC Subscription	Azure subscription where you want to deploy Azure Arc.
ARC Object Id	ObjectID from your Azure Active Directory.

Cluster Administration

Parameter	Description
AAA	
Tacacs Service	TACACS service name provided to the TACACS servers.
Tacacs Protocol	A subset of primary service.
Tacacs Servers	Enable to specify Tacacs server(s)
Ip Address	TACACS + server IP
Auth Type	Authentication type
Port	Server port. Default 49.
Key	Server secret
Auth Timeout	Server timeout. Default 5.

Table 20. Kubernetes Cluster Security Properties

Parameter	Description
Security Parameters	Enable to set general security parameters.
Root Domain Ca Key	Per user-domain certificate authority public keys that can issue user auth certificates
Maint Domain Ca Key	Per user-domain certificate authority public keys that can issue user auth certificates
Admin Domain Ca Key	Per user-domain certificate authority public keys that can issue user auth certificates
Liadmin Domain Ca Key	Per user-domain certificate authority public keys that can issue user auth certificates
Host Certificate	Host certificate to supplement host key authentication
Enable Local Passwd Auth	Use of local password user authentication is disabled by default except in lab environments
Enable Root	Root login is disabled by default except in lab environments
Disable Maint	Maint group (non-root bash user(s)) are enabled by default
Root Passwd	Replace default root sha512 (\$6) passwd
Root Pubkey	Define public key for root user authentication
Users	Enable to specify user parameters.
User	Name of user
Passwd	Password for user
Pubkey	Pubkey for user
Group	User group

Caution: In order to allow SSH access to the VM, you must add desired "maint", "romaint" (read-only), or "limaint" (lawful intercept) group users with either public keys, certificate authority keys, SHA-512 encrypted passwords, or configure one or more TACACS servers.

5. If using multi-master Kubernetes, enable Multi Master. For details on multi-master Kubernetes and its configuration, please refer to [Multi-Master Kubernetes](#).
6. Enter the Kubernetes node information.

Figure 34. Kubernetes Node Information

The following table describes the Kubernetes node parameters.

Caution: OPM has no visibility to check the available number of VFs automatically. An OpenStack failure occurs when the number of VFs requested exceeds the number of VFs available. This can be difficult to diagnose. Before provisioning the sriovNetwork list, it is recommended to verify the number of VFs available.

Table 21. Kubernetes Master Properties

Parameter	Description
K8 Master Port	Listening port of Kubernetes API server.
K8 Pass	Kubernetes master node password.
K8 Token	Kubernetes unique identifier of this cluster.
K8 User	Kubernetes master node user name.
Topology Manager Policy	Topology manager policy.

Cluster Administration

Parameter	Description
Multi Master	Enabling HA will create at least 3 master nodes and API server will be reachable through HA Proxy VIP. See Multi-Master Kubernetes for more information.
K8 Master Lb Vip	When enabling multimaster, the worker nodes will reach masters using this IP.
K8 Cert Key	K8 cert key.
K8 Public Vip	K7 Public Vip.
Vip Mgmt Net Id	Vip Mgmt net ID.
Vip Ext Net Id	Vip ext net ID.
Haproxy Combine	Combine by default unless otherwise set in Advanced mode.
Master	
Vm Name	Name of the node VM. VM names must not include uppercase letters, and typically consist of lowercase letters, numerals, and dashes.
Vm Disk Name (cinder only)	Cinder volume name.
Vm Ip	HA proxy VM AM IP address. If using a single interface this will also be one of the VRRP addresses.
Vm Flavor	OpenStack flavor for the Kubernetes node.
Vm Zone	OpenStack zone to deploy the VM.
Disk Size (cinder only)	Size of the disk to create. Minimum disk size is 100G. When VM ISSU feature is on it should be at least 200G
Add Net	Enable to specify additional networks.
Additional Network	
Net Name	Network name.
Net Id	Network ID.
Net Ip	Network IP address.
Net Gw	Network gateway.
SR-IOV	Toggle on to configure VF ports on platform VMs. To add additional SR-IOV interfaces, click the + button.
Net Id (SR-IOV only)	OpenStack Network ID that is associated with an SR-IOV PF. If the value is none, do not configure on VM.
Label (SR-IOV only)	Label associated with an SR-IOV PF.
Vlan (SR-IOV only)	OpenStack VLAN ID that is associated with an SR-IOV PF. If the value is none, there is no VLAN tag.
Dpdk (SR-IOV only)	If enabled, SR-IOV port is bound to VFIO. If disabled, SR- IOV port is bound to Linux.
Trunk (SR-IOV only)	Enable configuration of trunk subports.
Subports (SR-IOV only)	Subport parameters to use when Trunk is enabled. Click the Plus Button (+) to add additional subports:

Parameter	Description
	▪ Net Id - OpenStack subnet net ID. ▪ Net Ip - IP address within the range defined by the net ID. ▪ Vlan Id - OpenStack VLAN ID.

Table 22. Kubernetes Worker Node Properties

Parameter	Description
Disk Size	Size of disk to create, cannot be less than 100G. When VM ISSU feature is on it should be at least 200G
Ceph Disk Size	Size of Ceph disk to create.
K8 Node Ip	IP address of Kubernetes node. If single interface, it will be used for OAM and Kubernetes traffic.
Vm Disk Name	Cinder volume name.
Vm Flavor	OpenStack flavor for the Kubernetes node.
Vm Name	Name of the node VM. Lower case only.
Vm Zone	OpenStack Zone to deploy the VM.
Reserve CPU	Reserve CPU.
Hugepages	Enable to configure Hugepages options.
Hugepages Size	Hugepages size.
Hugepages Number	Hugepages number.
SR-IOV	Toggle on to configure VF ports on platform VMs. To add additional SR-IOV interfaces, click the + button.
SRIOV Intf	
Net Id (SR-IOV only)	OpenStack Network ID that is associated with an SR-IOV PF. If the value is none, do not configure on VM.
Label (SR-IOV only)	Label associated with an SR-IOV PF.
Vlan (SR-IOV only)	OpenStack VLAN ID that is associated with an SR-IOV PF. If the value is none, there is no VLAN tag.
Dpdk (SR-IOV only)	If enabled, SR-IOV port is bound to VFIO. If disabled, SR-IOV port is bound to Linux.
Trunk (SR-IOV only)	Enable configuration of trunk subports.
Subports (SR-IOV only)	Subport parameters to use when Trunk is enabled. Click the Plus Button (+) to add additional subports: Subport parameters to use when Trunk is enabled. Click the Plus Button (+) to add additional subports:

Parameter	Description
	▪ Net Id - OpenStack subnet net ID. ▪ Net Ip - IP address within the range defined by the net ID. ▪ Vlan Id - OpenStack VLAN ID.

7. Click **Create**. Three VMs will be deployed using OpenStack:

- One Kubernetes master
- Two Kubernetes nodes

8. (Optional) After you have created a cluster, you can save the cluster configuration information as a template by clicking **Save Values**. A template allows you to recreate the cluster without having to type in all the information again.

The next time you need to create this cluster, from the **Add Cluster** dialog box, select the **Saved** name before clicking **Add**.

In addition, you can export the saved cluster configuration to a file using the **Export Cluster Template** function.

Verify Kubernetes Cluster

To verify that the Kubernetes cluster is ready, SSH into the deployed Kubernetes cluster.

Example:

Issue the following commands to verify the Kubernetes cluster.

```
# ssh kuberadmin@<kubernetes master ip>
```

```
# kubectl get nodes
```

The system displays output like the following:

```
NAME STATUS ROLES AGE VERSION
cna-k8-master-165-5-11 Ready master 4m v1.14.2
cna-k8-node-165-5-12 Ready <none> 4m v1.14.2
cna-k8-node-165-5-13 Ready <none> 4m v1.14.2
```

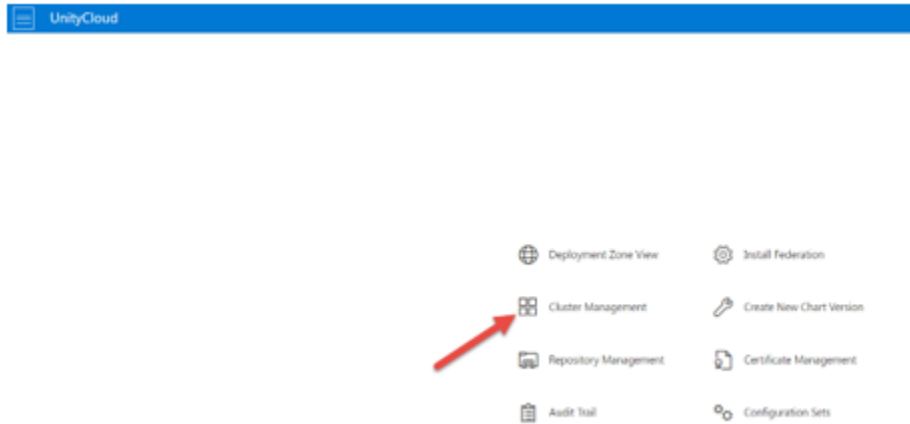
Note: Before proceeding to the next step, ensure each node has a status of Ready.

Add Nodes to Kubernetes Cluster

To add nodes to a Kubernetes Cluster:

1. From the main OPM screen, click **Cluster Management**.

Figure 35. Main OPM Screen



2. Select the Kubernetes cluster to which you want to add a node.
3. Scroll down to a Master or Worker node and click the **Add Master** or **Add Worker Nodes** button.

Figure 36. Kubernetes Clusters – Add Node

4. Scroll down to the bottom of the form and fill in the node information.

- Click **Update**.
- After OpenStack has deployed the VM, issue the following commands to verify that the new node is in the Kubernetes cluster.

```
# ssh kuberadmin@<kubernetes master ip>
```

```
# kubectl get nodes
```

The system displays output like the following:

```
NAME STATUS ROLES AGE VERSION
```

```
cna-k8-master-165-5-11 Ready master 4m v1.14.2
```

```
cna-k8-node-165-5-12 Ready <none> 4m v1.14.2
```

```
cna-k8-node-165-5-13 Ready <none> 4m v1.14.2
```

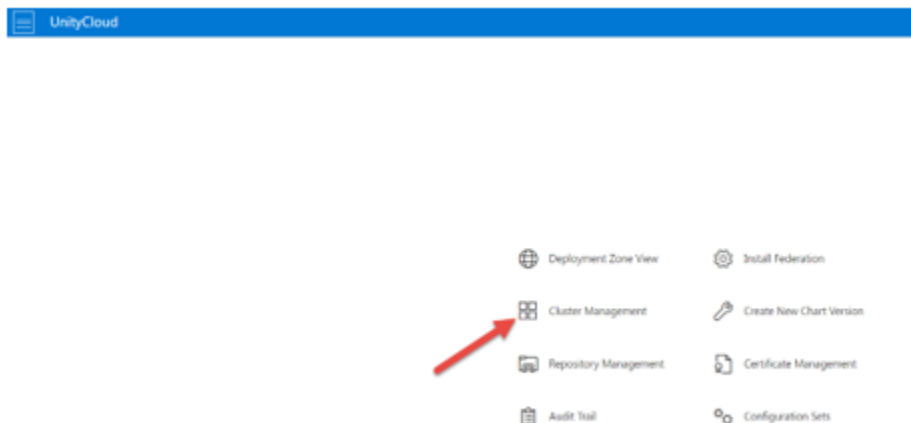
```
cna-k8-node-165-5-14 Ready <none> 4m v1.14.2
```

Delete Kubernetes Node

To delete a Kubernetes node:

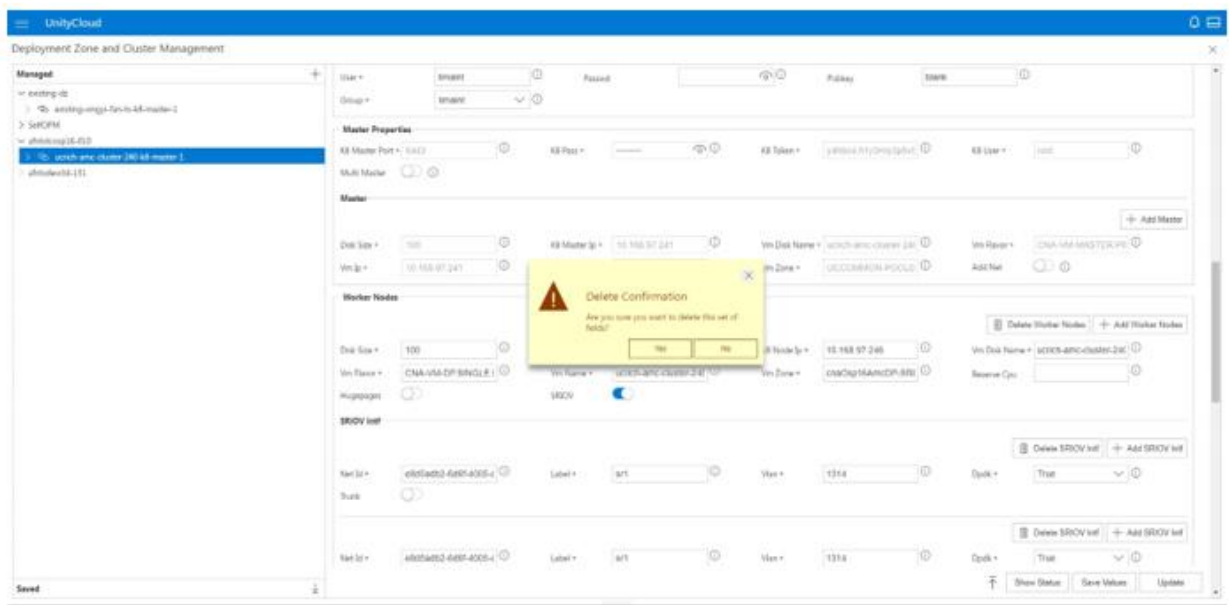
- From the main OPM screen, click **Cluster Management**.

Figure 37. Main OPM Screen



- Select the cluster and click on the Trash Can icon or **Delete Worker Nodes** next to the node you want to delete.

Figure 38. Delete Kubernetes Node



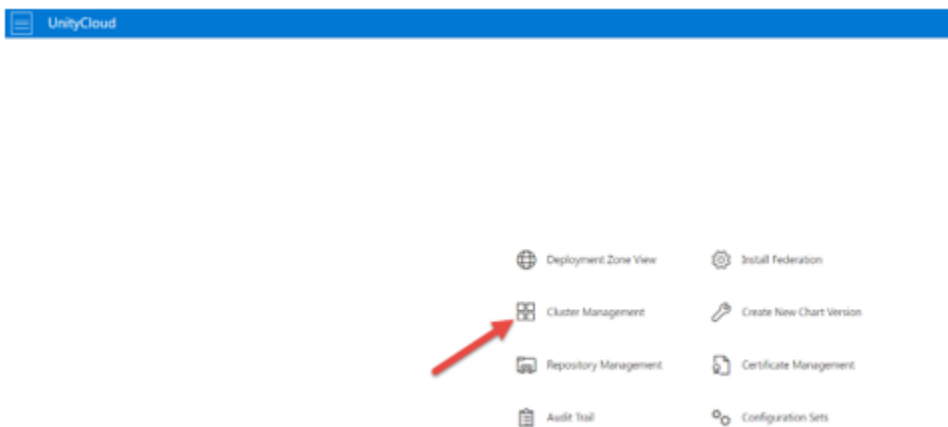
3. Enter **Yes** in the text field and then click **Yes** to confirm the deletion.

Delete Kubernetes Cluster

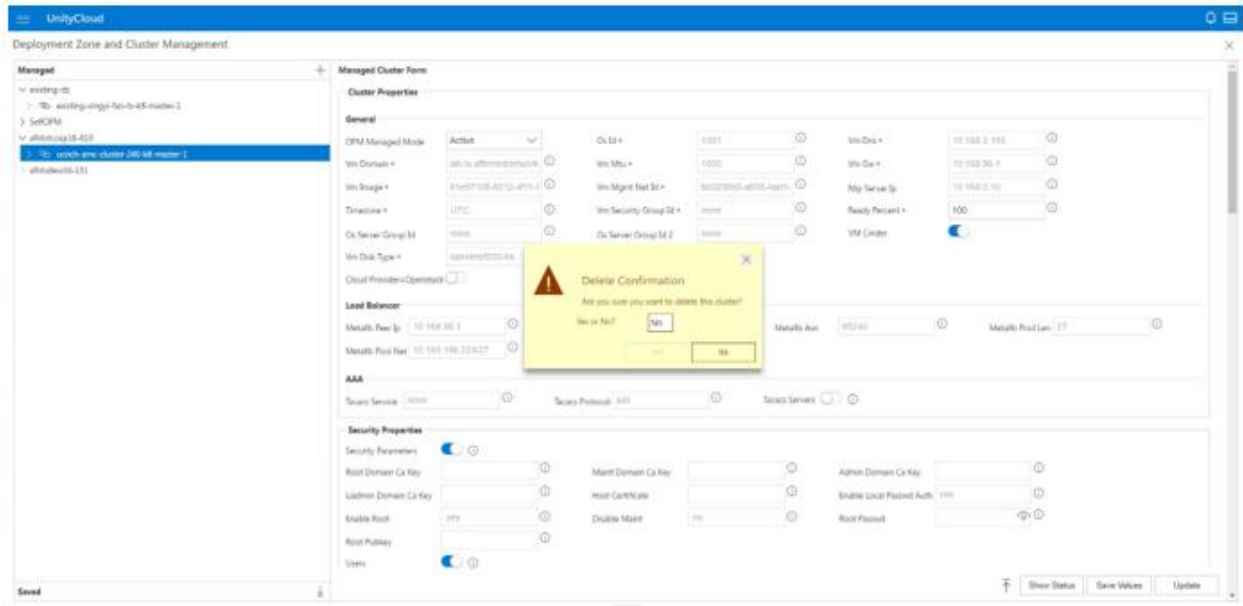
To delete a Kubernetes cluster:

1. From the main OPM screen, click **Cluster Management**.

Figure 39. Main OPM Screen



2. Move the mouse cursor over the cluster name that you want to delete.
3. Click on the Trash Can icon next to the cluster name that you want to delete.

Figure 40. Delete Kubernetes Cluster

4. Click **Yes** to confirm the deletion.

Attach to an Existing Kubernetes Cluster

OPM can also connect to a Kubernetes cluster that was created without OPM. This allows OPM to install Affirmed Networks federations (Helm charts and Docker images) as if it created the Kubernetes cluster itself.

Note: OPM supports attaching both Azure and Non-Azure clusters.

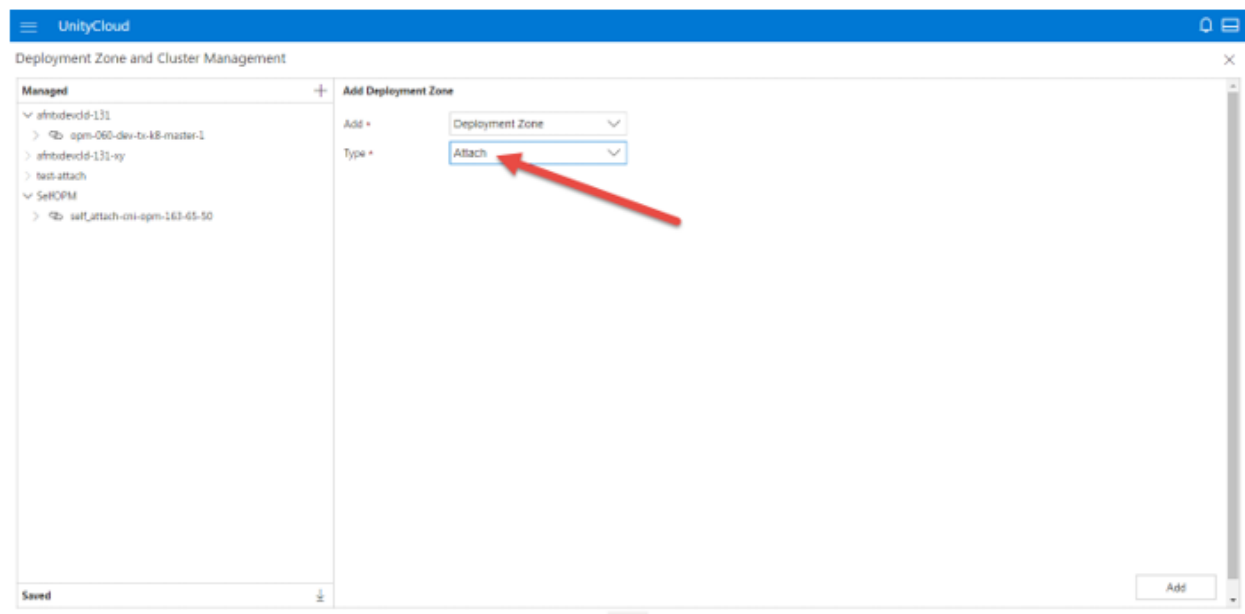
1. To log in to the OPM GUI, use a web browser to go to `https://<opm ip>`.

Figure 41. OPM GUI Login



2. Enter your username and password.
3. From the main OPM screen, click **Cluster Management** and click the Plus icon (+).

Figure 42. Add Attach Deployment Zone



4. Select **Attach** as the deployment zone **Type** and click **Add**.

Figure 43. Attach Deployment Zine Parameters

5. Enter the deployment zone information and click **Deploy**.

The following table describes the existing deployment zone parameters.

Table 23. Deployment zone parameters

Parameter	Description
Id*	Unique numeric ID to identify this deployment zone instance.
Type*	Value must be set to attach . This value is read-only.
Name*	Name of the deployment zone.
Location*	Select the location of the deployment zone.
Chart Repo*	Select the Helm chart repository.

6. Click the Plus icon (+) and select **Kubernetes Cluster** in the **Add** dropdown.
7. Choose the previously-created attach deployment zone from the **Deployment Zones** dropdown.
8. To Attach Non Azure Cluster, click **Non Azure Attach** for the **Attach Type** field.
 - a. Enter the **IP Address** of the master node.
 - b. **M2M Keys** of the cluster that is to be attached. The M2M private key can be obtained from the OPM where the cluster was created, under /vmfmgr/certs. See [Generating M2M Key](#) section.
 - c. Enter the username in **Username** field.

Figure 44. Attach Non-Azure Cluster

The screenshot shows the 'Operations & Policy Manager' interface. On the left, under 'Managed', there are two entries: 'attach-a4orchdevcid-025' and 'SelfOPM'. On the right, the 'Add Cluster' form is displayed. The 'Add' field is set to 'Kubernetes Cluster'. The 'Deployment Zones' field is set to 'attach - attach-a4orchdevcid-025'. The 'Attach Type' field has two buttons: 'Non Azure Attach' (which is selected and highlighted in blue) and 'Azure Attach'. Below these are input fields for 'IP Address', 'Username', and 'M2M Keys'.

9. To Attach Azure Cluster, click **Azure Attach** for the **Attach Type** field.

Prerequisite

You will need the kubeconfig file of the Kubernetes cluster to which you want to attach.

- a. To obtain the kubeconfig file, go to the home directory of the user who installed Kubernetes. For example, user kuberadmin.

```
# cd /home/kuberadmin/.kube/
```

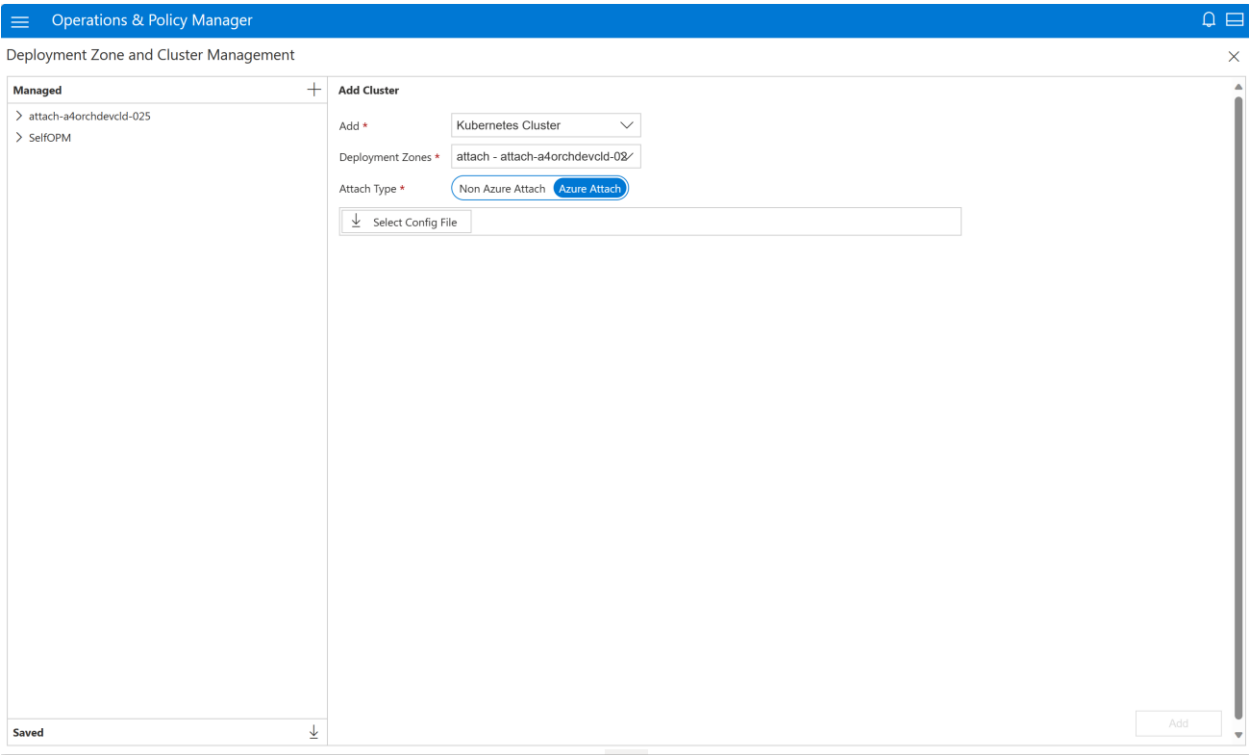
```
# cat config
```

When getting this file for a Kubernetes cluster created using OPM, the config file is located here: /home/kuberadmin/admin.conf.

- b. Save this file to the system on which you will run the web browser for the OPM GUI.

Note: For AKS clusters, run the command to get the config file: **az aks getcredentials --resource-group <resource-group-name> --name <aks-clustername> -f admin.conf**

Figure 45. Attach Azure Cluster



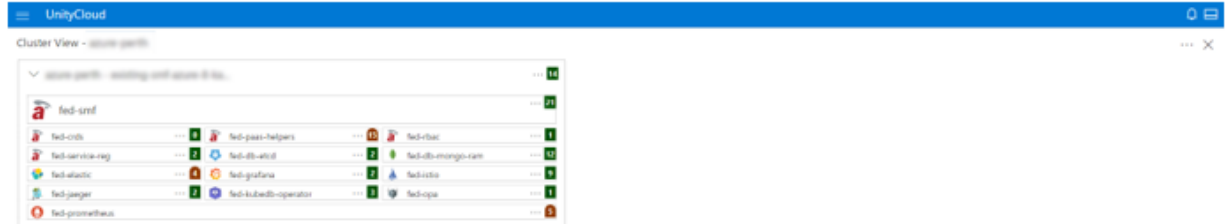
10. Click **Add**.

Note: The VM name of an attached cluster changes its prefix to match the attach type.

11. From the deployment zone view, select the cluster to view the attached cluster.

Figure 46. Deployment Zone View

The system displays the cluster view and shows all federations running on the attached cluster.

Figure 47. Cluster View

Add SSH M2MKeys to a Cluster

To generate the keys and install them onto all nodes in the cluster:

1. From any Linux server, create a keypair:

```
$ ssh-keygen -N "" -t ed25519 -f mykeypair
```

2. Add the public key to the cluster (master and all worker you want to access):

if username used to attach the cluster is not root: `$ cat mykeypair.pub >> /home/<username>/.ssh/authorized_keys`

if username used to attach the cluster is root: `$ cat mykeypair.pub >> /root/.ssh/authorized_keys`

3. To test if the m2mkey generated is successful, use the private key to connect:

```
$ ssh -i mykeypair -o StrictHostKeyChecking=no <username>@<IP ADDRESS>
```

Add Location to Deployment Zone Geographic View

When using OPM to view the geographic map of deployment zones, you may add or edit the list of locations. The location parameter is required when creating new deployment zones and must match a location name from this list.

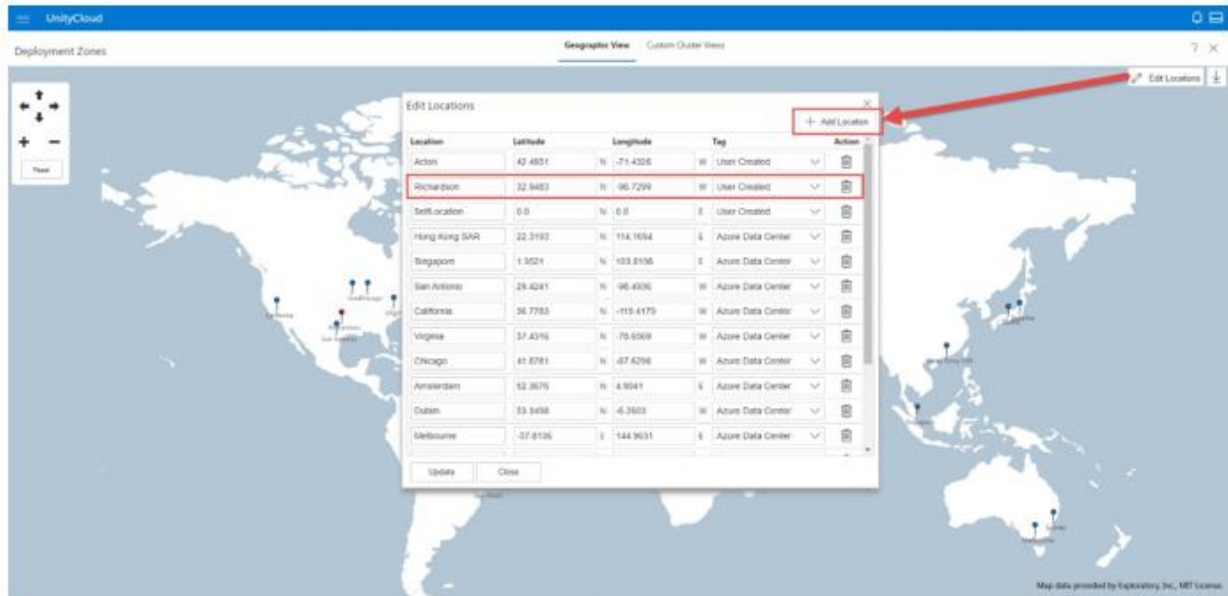
To add or edit a location on the geographic map:

1. Log into OPM using a web browser.
2. From the OPM main screen, choose **Deployment Zone View**. The Geographic View appears, showing a map of the world with locations marked by icons.
3. Click **Edit Locations**.
4. Click **Add Location** to add a new line to the list, or edit fields in an existing Location entry.

Provide a **Location** name, global **Latitude** in degrees, global **Longitude** in degrees, and a **Tag** of either `User Created` or `Azure Data Center`.

If you wish to delete a location, click the **Trash icon** in the Action column.

5. Click **Update** to update the list of locations and the Geographic View.

Figure 48. Add Location in Geographic View

Enable server access to NFS server

1. Connect to OPM using SSH.
2. Run the following command:

```
# vi /vnfmgr/nfsexport
```

3. Add the following for each server in the cluster, where <server_ip> is the server that needs access:

```
/filestore <server_ip>(rw, sync, no_root_squash)
```

4. Save the file.

Manage Cluster Configuration Templates

While configuring a Kubernetes cluster, you may save the cluster configuration for future use.

Save Cluster Configuration

On the Deployment Zone and Cluster Management screen:

1. In the cluster tree, expand the Managed section.
2. Select or create a cluster. Confirm the configuration values you wish to save.
3. Click the **Save Values** button.

Figure 49. Save Values

Operations & Policy Manager

Deployment Zone and Cluster Management

Managed Cluster Form

Form Filter: Basic

Cluster Properties

General

Create Image Pull Secret ☐ ⓘ

Vm Dns * 172.23.1.70 ⓘ

Vm Domain * a4opacketcare.micros ⓘ

Vm Gw * 10.161.48.1 ⓘ

Vm Image * f3dc1fce-7633-4281-8 ⓘ

Vm Mgmt Net Id * 09524cc0-3638-4d12- ⓘ

Vm Netmask * 255.255.248.0 ⓘ

Ntp Server Ip 172.23.1.70 ⓘ

Vm Security Group Id * none ⓘ

VM Cinder * True

Vm Disk Type * oephHcsAllFlash ⓘ

Cloud Provider=Openstack ☐

Load Balancer

AAA

Security Properties

Security Parameters ☒ ⓘ

Enable Local Passwd Auth yes ⓘ

Enable Root yes ⓘ

Root Passwd ⓘ ⓘ

Master Properties

K8 Master Port * 6443 ⓘ

K8 Pass * ⓘ ⓘ

K8 Token * ins9na.jr6ltczni0shwfvsi ⓘ

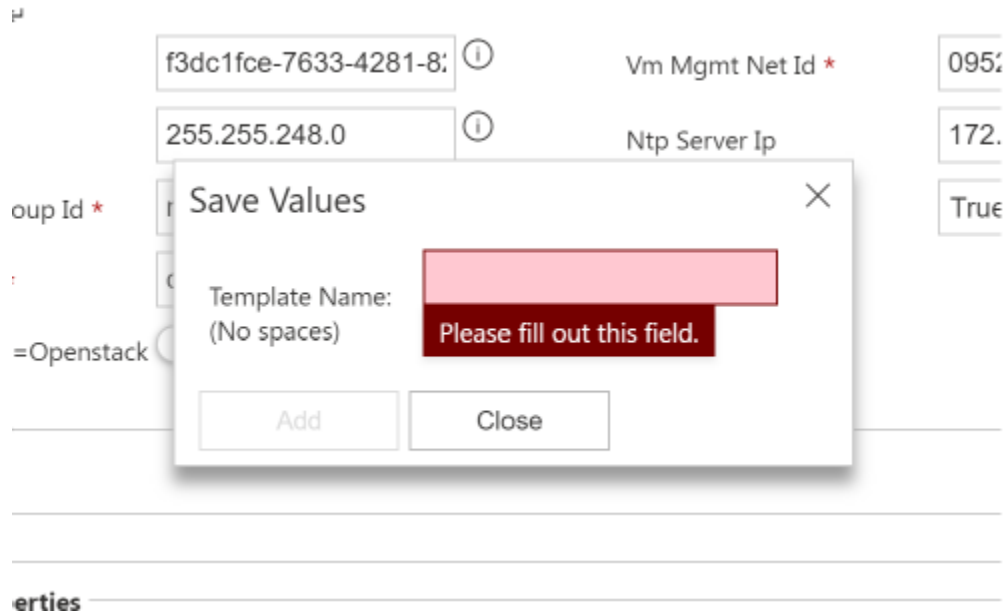
K8 User * root ⓘ

Multi Master ☐ ⓘ

↑ Show Status Save Values Update

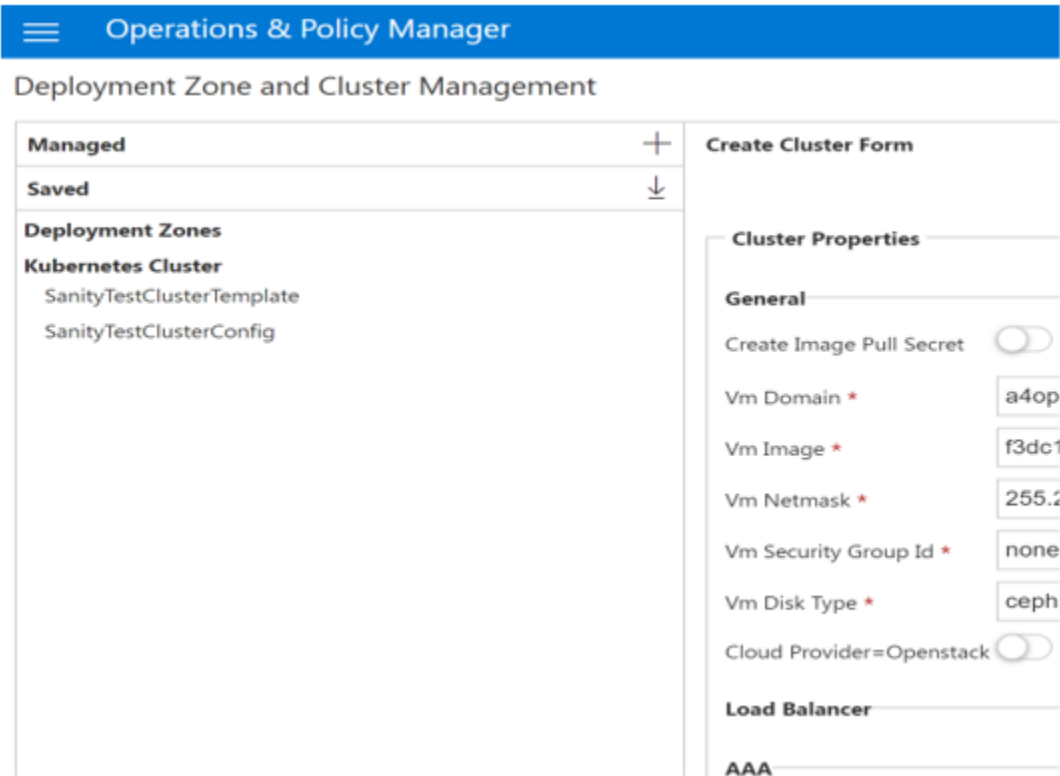
4. In the Save Values dialog, enter a template name and then click **Add** to save the values to the named template.

Figure 50. Save Values



- 5. The saved cluster configuration template will show up in the Saved section of the cluster tree.

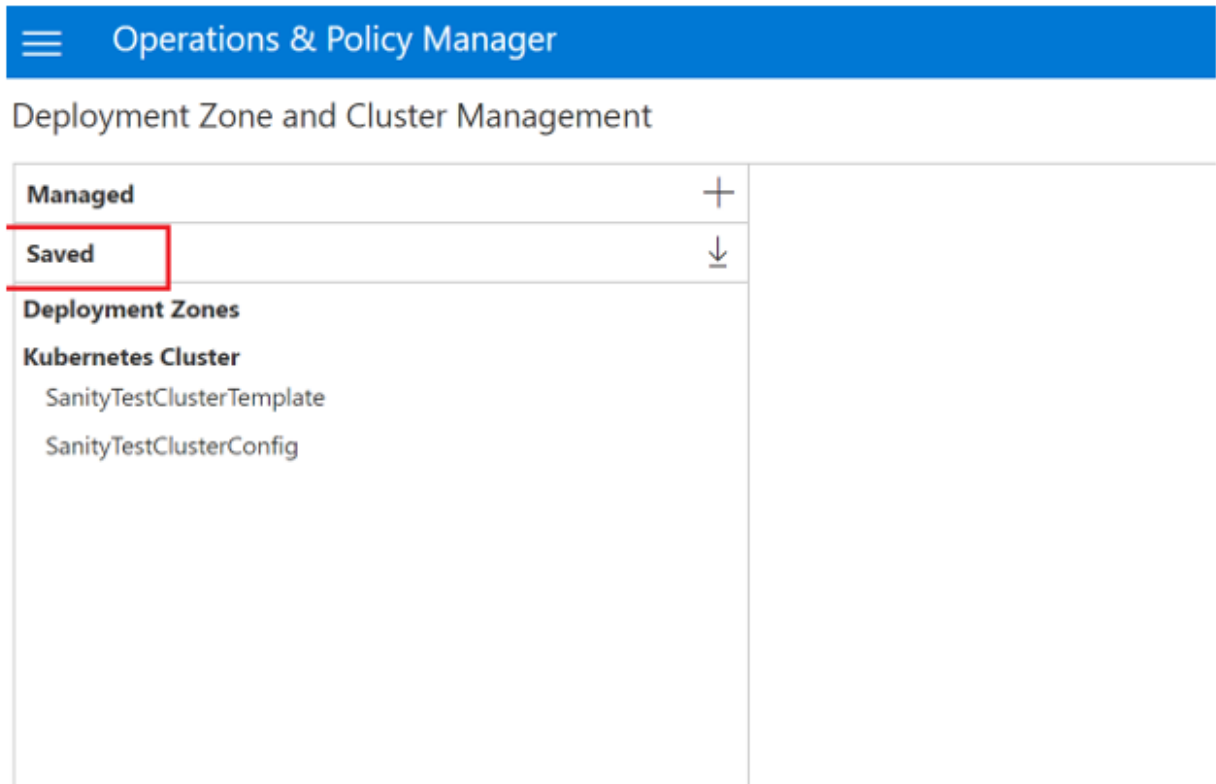
Figure 51. Saved cluster configuration



Copy/Update Saved Cluster Template

On the Deployment Zone and Cluster Management screen:

1. In the cluster tree, expand the **Saved** section.

Figure 52. Expand the Saved section

2. Select the saved cluster template to show the cluster configuration.
3. Click **Clone** to copy the values to a new template, or **Update** to save any changes you make to the template. Use **Export Saved Values** to export the saved values to a file.

Figure 53. Export Saved Values

Operations & Policy Manager

Deployment Zone and Cluster Management

Managed

Saved

Deployment Zones

Kubernetes Cluster

SanityTestClusterTemplate

SanityTestClusterConfig

Saved Cluster Form

Form Filter: Basic

Cluster Properties

General

Create Image Pull Secret: ☐

Vm Dns: 172.23.1.70

Vm Gw: 10.161.48.1

Vm Image: fd9c1f5e-7635-4281-8f-...

Vm Network: 255.255.248.0

Vm Security Group Id: none

Vm Disk Type: openHicsAllFlash

Vm Mgmt Nat Id: 00524e08-3636-4d12-1...

Http Server Ip: 172.23.1.70

Vm Cidr: True

Cloud Provider: Openstack

Load Balancer

AAA

Security Properties

Security Parameters: ☒

Enable Local Password Auth: yes

Enable Root: yes

Root Password:

Master Properties

K8s Master Port: 6443

K8s User: root

K8s Pass:

K8s Token: inslra_j8tcmOsfefsp

Multi Master: ☐

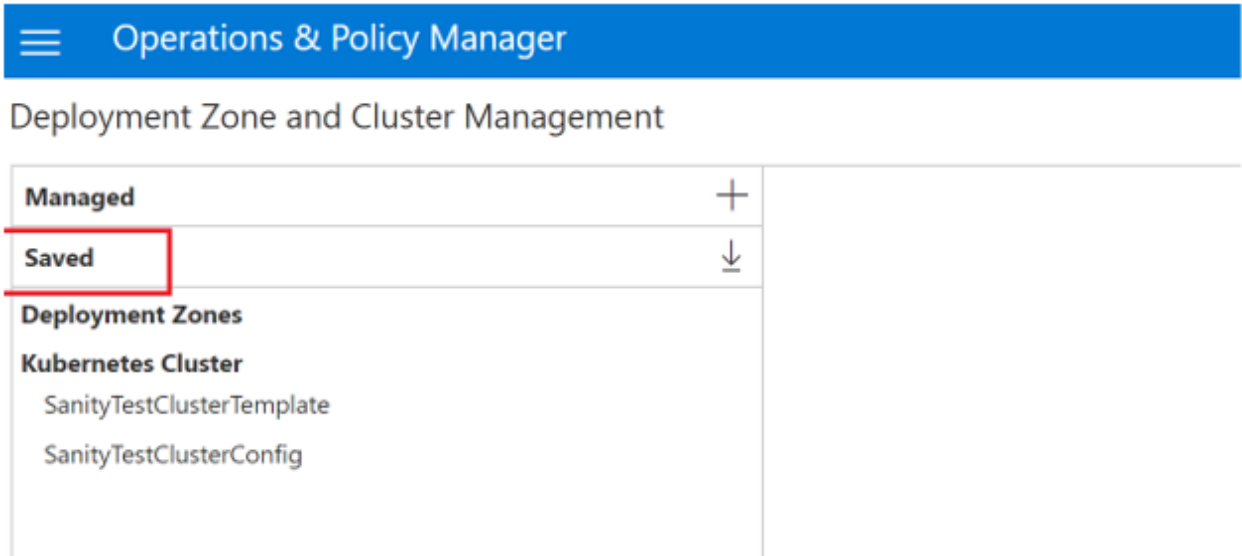
Clone Update

Export Cluster Template

On the Deployment Zone and Cluster Management screen:

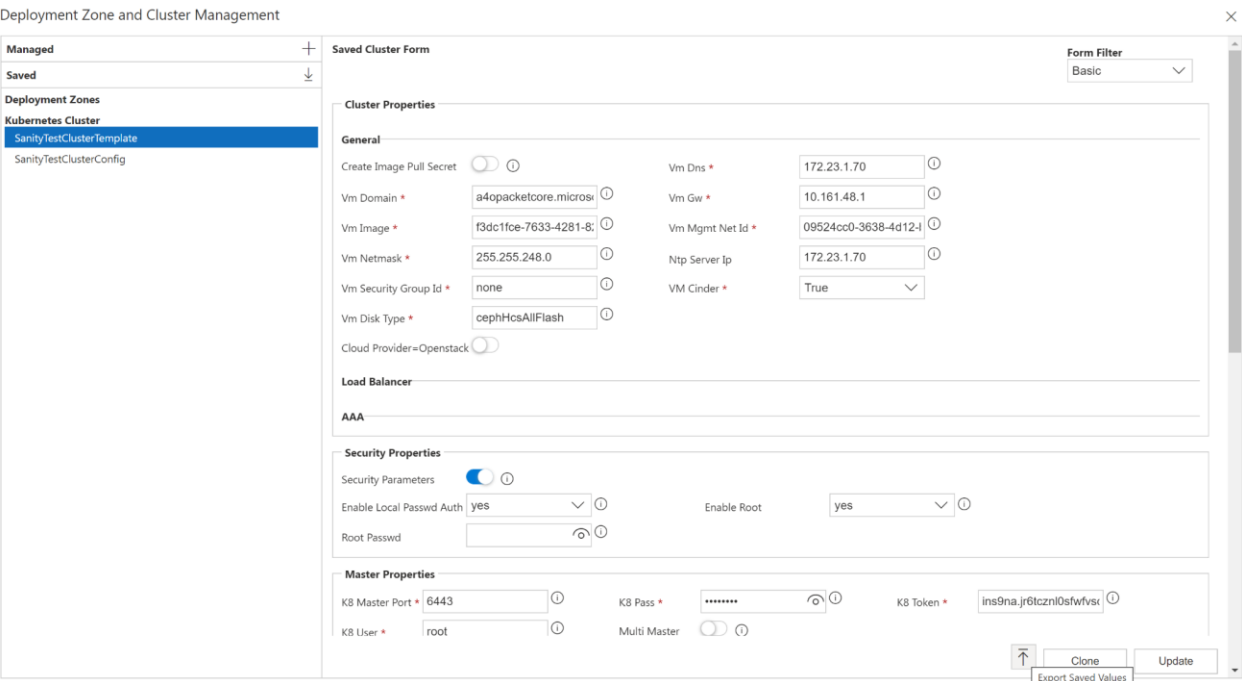
1. In the cluster tree, expand the **Saved** section.

Figure 54. Expand Saved section



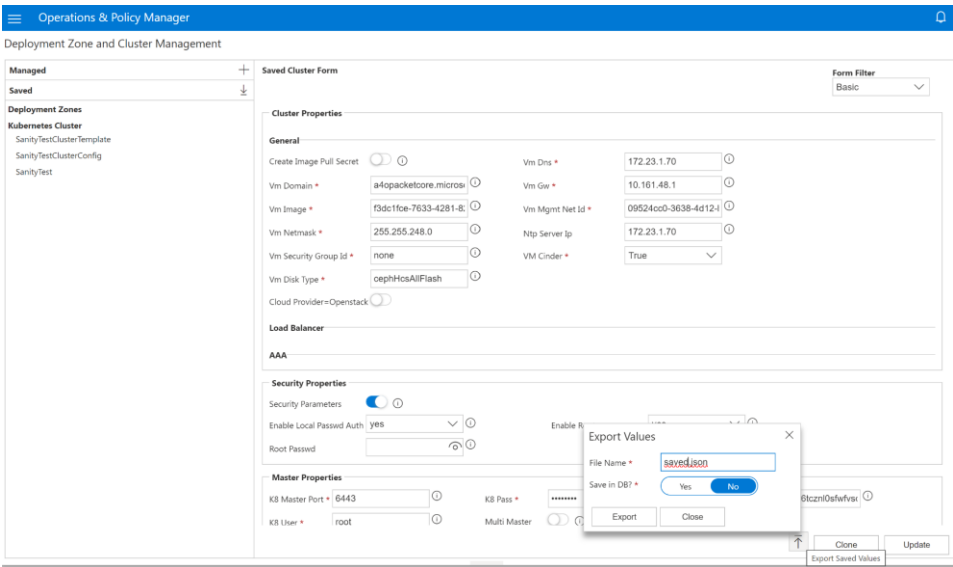
2. Select the saved template to show the cluster configuration.
3. Click the **Export Saved Values** button.

Figure 55. Export Saved Values



4. In the Export Template dialog, choose a file name and then click **Export** to save the configuration to a JSON-formatted file on local computer.

Figure 56. Export



Import Cluster Template

On the Deployment Zone and Cluster Management screen:

1. In the cluster tree, next to the Saved section header, Click the **Import Deployment Zone or Cluster** button.

Figure 57. Import Deployment Zone or Cluster

Operations & Policy Manager

Deployment Zone and Cluster Management

Managed

Saved

Deployment Zones

Kubernetes Cluster

SanityTestClusterTemplate

SanityTestClusterConfig

+

↓

Import Deployment Zone or Cluster

General

Create Image Pull Secret

Vm Domain *

Vm Image *

Vm Netmask *

Vm Security Group Id *

Vm Disk Type *

Cloud Provider=Openstack

Load Balancer

AAA

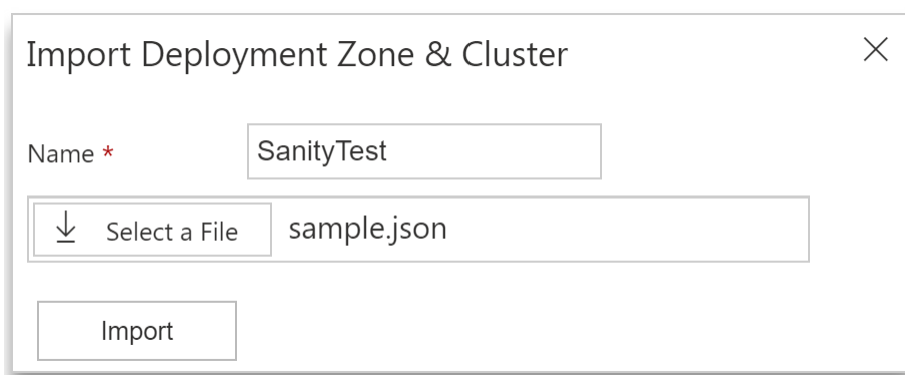
2. On the Import Deployment Zone & Cluster popup, click **Select a File** to select a previously exported cluster template JSON file from your local computer.

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Figure 58. Select a File


Import Deployment Zone & Cluster

Name *

3. Click **Import**.
4. Saved values will be imported successfully and show in the Saved section of the cluster tree.

Figure 59. Saved values imported

Deployment Zone and Cluster Management

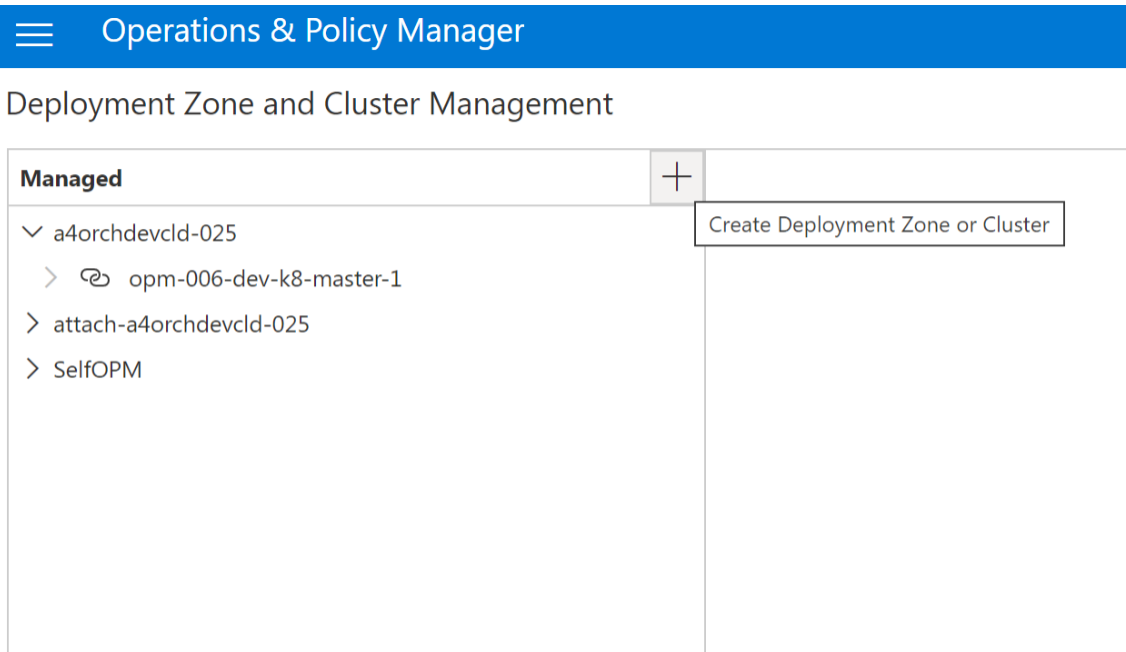
Managed	+
Saved	↓
Deployment Zones	
Kubernetes Cluster	
SanityTestClusterTemplate	
SanityTestClusterConfig	
SanityTest	

Deploy Saved Cluster Template

On the Deployment Zone and Cluster Management screen:

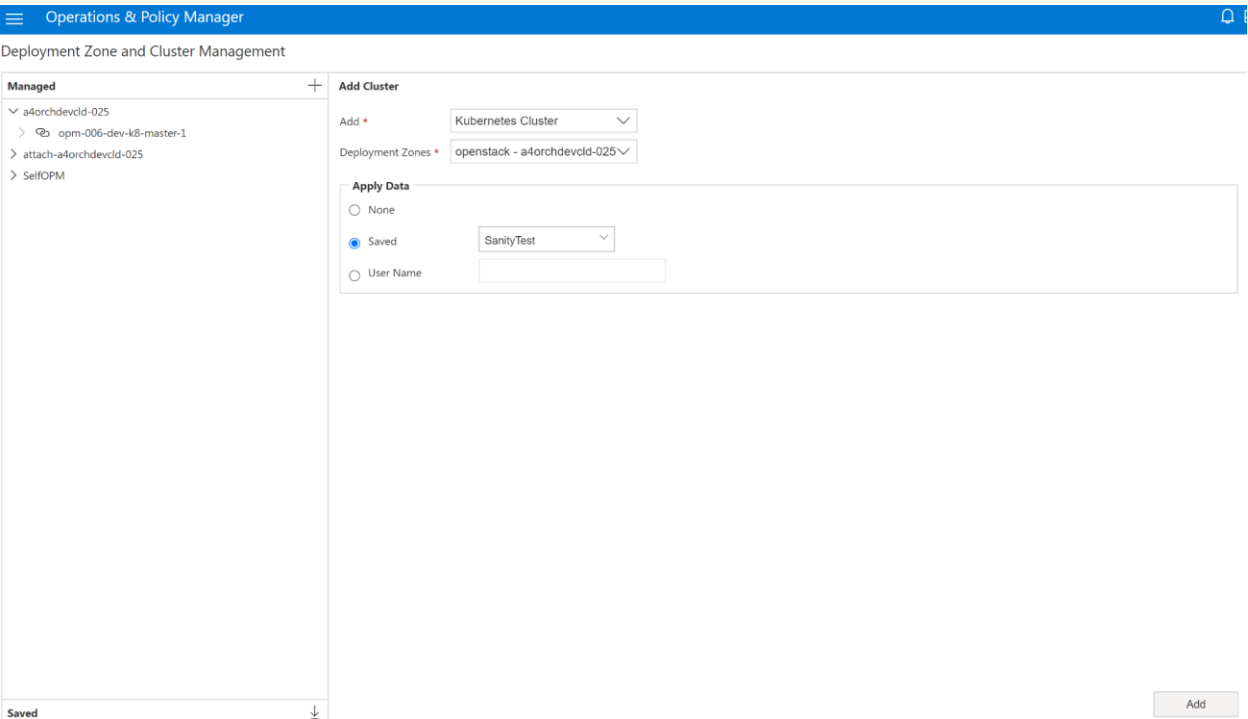
1. In the cluster tree, next to the Managed section header, Click the Create Deployment Zone or Cluster button.

Figure 60. Create Deployment Zone or Cluster



2. Select Add Kubernetes Cluster and Apply Data using saved template. Then click the Add button at the bottom right.

Figure 61. Add Kubernetes Cluster and Apply Data



3. Click the Create button to create the cluster using the saved configuration.

Figure 62. Click Create button

The screenshot shows the 'Create Cluster Form' in the 'Operations & Policy Manager' interface. The form is divided into three main sections: General, Security Properties, and Master Properties. The General section includes fields for Vm Domain, Vm Image, Vm Netmask, Vm Security Group Id, Vm Disk Type, Cloud Provider, Vm Dns, Vm Gw, Vm Mgmt Net Id, Ntp Server Ip, and VM Cinder. The Security Properties section includes Security Parameters, Enable Local Passwd Auth, Root Passwd, Enable Root, and K8 Token. The Master Properties section includes K8 Master Port, K8 User, K8 Pass, Multi Master, and K8 Token. At the bottom right, there are buttons for 'Show Status', 'Save Values', and 'Create'.

Note: For backward compatibility, OPM supports importing previously saved CSV configuration files. All new configurations are now exported in JSON format.

Switch Cluster from Standby to Active Mode

In the event that the primary OPM fails or becomes unavailable, a secondary OPM with clusters deployed in standby mode can switch those clusters to active mode and assume management of those clusters. To switch a cluster from standby to active mode, perform the following:

1. Log into the secondary OPM.
2. Click Cluster Management.
3. Select the cluster that you wish to switch from standby to active mode.
4. Switch **OPM Managed Mode** from **Standby** to **Active**.
5. Click **Update**.

Caution: Only one OPM should manage a cluster in active mode at a time. There is no active synchronization between the two instances of OPM, so using both to manage a cluster in active mode should be avoided.

Saved	
-------	--

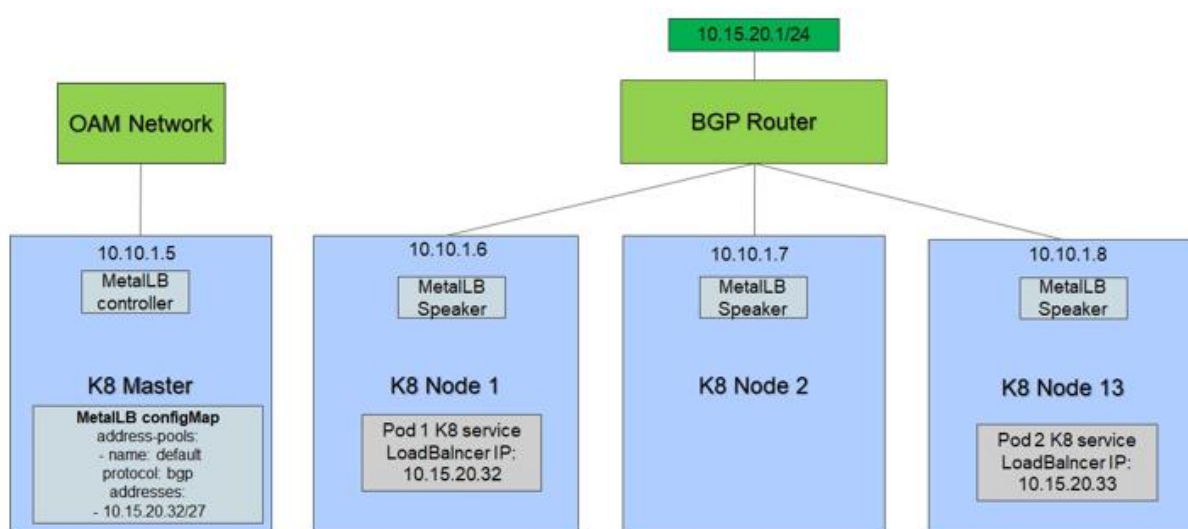
Install MetalLB Federation

MetalLB(fed-metallb) is a load balancer implementation for bare metal Kubernetes clusters, using standard routing protocols. It enables you to create Kubernetes services of type “LoadBalancer” in clusters that do not run on a cloud provider.

Border Gateway Protocol (BGP) is used by MetalLB to advertise the IPs of external cluster services to a BGP enabled router. Assuming the router is configured to support multipath, this enables load balancing across the nodes.

For more information about MetalLB and how it uses BGP, see <https://metallb.universe.tf/>

Figure 64. BGP Implementation in MetalLB Installation



For installation of fed-metallb using OPM, follow the steps below:

1. Log into the OPM UI.
2. Navigate to the Install Federation and double-click on **fed-metallb**.
3. The chart can be configured using OPM's basic or YAML expert mode.

Table 24. Customizable HELM parameters for fed-metallb

Parameter	Description	Default Value
Release Name - (string)	Specifies the name of the federation.	None
Namespaces -(string)	Specifies the namespace where fed- metallb will run.	None
Global Parameters		

Install MetalLB Federation

Parameter	Description	Default Value
enableSCC (boolean)	Toggles whether OpenShift SCC resources are deployed.	False
registry.docker.repoPath (string)	Specifies the docker registry path to get the docker images.	cnreg-rel:5000
pod-metallb Parameters		
rbacEnabled (boolean)	Toggles whether fed-metallb's RBAC resources are deployed or not.	True
ipAddressPools[]	List of named IP address pools that are advertised by MetalLB speakers via BGP.	By default, a pool named default- pool is configured. To add more, add a new named pool to the list. Do not remove default-pool.
ipAddressPools[].name (string)	Name of this IP address pool.	default-pool. If adding a new pool, name sure to set a new name for it.
ipAddressPools[].addressesList (list of strings) OPM Bulk install parameter: metallb- pool-addresses.metallb- ip-address-range	Required. List of IPV4/prefix address pool ranges	None.
bgp.peers[]	List of named BGP routers that MetalLB speakers will establish a BGP session with.	By default, a peer is already added that needs to be configured.
bgp.peers[].name (string) OPM Bulk install parameter: metallb- peer-name		
bgp.peers[].myASN(string) OPM Bulk install parameter: metallb- my-asn	Specifies the BGP ASN number of the MetalLB BGP speakers. Note this is a string.	"64500" (needs to be changed)
bgp.peers[].peerASN (string) OPM Bulk install parameter: metallb- peer-asn	Specifies the BGP ASN number of the peer router. Note this is a string.	"64501" (needs to be changed)
bgp.peers[].peerAddress (string) OPM Bulk install parameter: metallb- peer-ip	Specifies the IPV4 address of the peer router	10.0.0.1 (needs to be changed)
bgp.advertisements[]	List of named BGP advertisements that can optionally be assigned to a	By default, an advertisement named default-bgp- advertisement is configured.

Install MetalLB Federation

Parameter	Description	Default Value
	named address pool. BGP advertisements can also be associated with BGP communities. By default an advertisement is applied to all available IP address pools.	To add more, add a new named advertisement to the list.
bgp.advertisements[].name (string)	Specifies name of the BGP advertisement	default-bgp-advertisement
bgp.advertisements[].communities[] (array of strings)	List of BGP community names associated with this advertisement. By default, the well-known NO_ADVERTISE BGP community is configured with default-bgp-advertisement. This prevents routes from being re-advertised by the BGP peer.	NO_ADVERTISE
bfd.enabled (boolean)	NOT YET SUPPORTED. Toggles whether BFD support is enabled with the MetalLB BGP speakers.	False
bfd.profiles[]	NOT YET SUPPORTED. Specifies a list of BFD profiles with configurable BFD parameters.	By default, a profile named default-profile is configured. To add more, add a new named BFD profile to the list. Do not remove default-profile.
bfd.profiles[].name (string)	NOT YET SUPPORTED. Specifies a name for the BFD profile	default-profile
prometheus	NOT YET SUPPORTED. Specifies Prometheus monitoring and alerting configuration.	
controller	Do not modify unless needed. This optionally configures the MetalLB controller deployment.	
speaker	Do not modify unless needed. This optionally configures the MetalLB speaker deployment.	

Install MetalLB Federation

Parameter	Description	Default Value
speaker.disabled (boolean)	If set to true, the MetalLB speaker daemonset is not deployed.	false
speaker.logLevel (string)	Specifies speaker log levels. Valid values are all, debug, info, warn, error or none.	info
speaker.tolerateMaster (boolean)	If set to true, MetalLB speaker pods are run on Kubernetes control plane nodes.	false
speaker.frr.enabled (boolean) OPM Bulk install parameter: metallb- enable-frr	NOT YET SUPPORTED. If enabled, MetalLB FRR mode is enabled. This is needed to support BFD and IPV6 enabled BGP speakers.	false
speaker.frr.image.repoPath (string) OPM Bulk install parameter: metallb- frr-repo	NOT YET SUPPORTED. Required if FRR mode is enabled. This specifies the image repository where the FRR image can be pulled. Note that the FRR image is non- redistributable and needs to be manually downloaded from the Internet. Please refer to the section on Downloading the FRR image Used in Federations section.	None
speaker.frr.image.name (string) OPM Bulk install parameter: metallb- frr-image	NOT YET SUPPORTED. Required if FRR mode is enabled.	frr:v7.5.1

- Click **Select Cluster** and select the cluster to deploy the **fed-metallb** federation.
- Click **Install Federation**.

Note: After all the pods in the **fed-metallb** namespace are running, the Kubernetes Load Balancer service uses IP addresses from the Addresses defined in the **fed-metallb** federation.

(Optional) Downloading the FRR Image Used by Fed-metallb

Affirmed Networks cannot distribute the FRR container image that is used by MetalLB when it is running in FRR mode. Before you install fed-metallb, you must pull the Docker image from the internet and install it in the OPM Docker Registry. If your OPM VM does not have an internet connection, you must perform the following steps to install the container prior to installing fed-metallb. If there is internet connection, skip these steps and install the fed-metallb federation specifying 'frrouting' in the speaker.frr.image.repoPath parameter.

1. Obtain this image on the server that has Docker and access to the public Internet.

- a. `# docker pull frrouting/frr:v7.5.1`

- b. Verify that the Digest displayed after the docker pull command exactly matches with the following:

```
sha256:cf9cc84733a3494ea2682f36300b41484877840b3e6a61f820955 54e991d3671
```

- c. `# docker save frrouting/frr:v7.5.1 > frr-v7.5.1.tar`

- d. `# scp frr-v7.5.1.tar root@<opm ip>:/tmp`

2. Copy the tarball to the OPM VM and untar the packages.

- a. `# ssh to root@<opm ip>`

- b. `# docker load -i /tmp/ frr-v7.5.1.tar`

3. Issue the following commands to get the image ID, tag it locally, and push it to the registry. Do not change the FRR image and version since Rook checks for the official image name and version.

- a. `# docker images frrouting/frr:v7.5.1`

- b. `# docker tag <image id> <opm ip>:5005/frr:v7.5.1`

- c. `# docker push <opm ip>:5005/frr:v7.5.1`

If the registry used in the steps above matches the registry configured in the global registry docker repo path parameter, you may proceed to install the fed-metallb federation. Otherwise, when installing fed-metallb, configure the speaker.frr.image.repoPath parameter with `<opm ip>:5005`, or whichever repo was used to upload the FRR image.

Configure additional addressesList items, IP address pools or BGP peers

The fed-metallb chart can be configured with multiple IP addresses in a pool, multiple IP address pools or multiple BGP peers. Please refer to the Helm parameters table above. To add new addresses to an existing pool, or to add additional IP address pools or BGP peers when fed-metallb

is already installed, perform an upgrade of the same fed-metallb chart with the updated Helm parameters.

1. Navigate to **Deployment Zone View** and select the site's deployment zone from the map in the **Geographic View**.
2. Select **Upgrade Federation** in the fed-metallb menu
3. Toggle **YAML Expert Mode** and configure the additional addressList item(s), additional IP address pool(s) or BGP peer(s). Refer to the Helm parameters table above and example YAML below for more details.

Below is an example YAML snippet configuring multiple IP addresses in a pool, multiple IP address pools and multiple BGP peers.

```
pod-metallb:
ipAddressPools:
# The first pool uses addresses variable
- name: default-pool poolLabels: {} addressesList:
- 10.163.205.64/27
# The second pool uses the addressesList array variable to configure
multiple IP address ranges in a pool
- name: second-pool
poolLabels: {} addressesList:
- 10.163.105.64/27
- 10.163.5.64/27
bgp:
# The example shows 2 peers peers:
- name: leaf-1 myASN: "64563"
peerASN: "65202"
peerAddress: 10.163.64.1
- name: leaf-2 myASN: "64563"
peerASN: "65203"
peerAddress: 10.163.65.1
```

4. Click the **rolling upgrade** button to apply changes to the live network.
5. Verify that the IP address pool(s) resources have been created by running the following command:

```
# kubectl describe ipaddresspools.metallb.io -n fed-metallb
```

6. Verify that the BGP peer resource(s) have been created by running the following command:

```
# kubectl describe bgppeers.metallb.io -n fed-metallb
```

7. Verify that external services of type LoadBalancer have External IPs set.

```
# kubectl get services -A
```

8. Verify that the BGP session(s) with the configured peer(s) are established by inspecting any one of the MetalLB speaker pod's log.

```
# kubectl logs -n fed-metallb <speaker pod name> | grep "peerAdded "
```

Example output:

```
{"caller":"bgp_
controller.go:214","event":"peerAdded","level":"info","msg":"pe er
configured, starting BGP session","peer":"10.165.0.1","ts":"2022-10-
28T13:55:55Z"}
```

Note: Removing, updating or adding IP address pool(s) require a restart of the fed-metallb controller pod.

Add Multiple IP Address Pools

To configure multiple IP address pools for the MetalLB load balancer, you must use the expert mode when deploying with OPM GUI or modify the values file when deploying manually with the helm install command.

By default, a typical fed-metallb values file looks similar to the following example:

```
global:

registry_docker_path: cnreg-rel:5000

pod-metallb:

config:

addresses: 10.15.20.32/27

aggregationLength: "27"

myAsn: "65500"

peerAddress: 10.15.10.1

peerAsn: "65550"
```

The above example shows one address pool which is the default pool. The default name is "default".

If you want to add more address pools, add the following lines to the values file:

```
add_pool:

pool1:

name: "mypool_1"
```

```
addresses: 10.15.27.180/27
```

```
aggregationLength: "27"
```

The following output shows an example of three pools:

```
global:
```

```
registry_docker_path: cnreg-rel:5000
```

```
pod-metallb:
```

```
config:
```

```
addresses: 10.15.20.32/27
```

```
aggregationLength: "27"
```

```
myAsn: "65500"
```

```
peerAddress: 10.15.10.1
```

```
peerAsn: "65550"
```

```
add_pool:
```

```
pool1:
```

```
name: "mypool_1"
```

```
addresses: 10.15.27.180/27
```

```
aggregationLength: "27"
```

```
pool2:
```

```
name: "mypool_2"
```

```
addresses: 10.15.30.200/27
```

```
aggregationLength: "27"
```

To use these pools, you must modify your K8 service yaml and specify the pool name. If you do not include a name, then the default pool will be used.

```
apiVersion: v1
```

```
kind: Service metadata:
```

```
annotations:
```

```
metallb.universe.tf/address-pool: mypool_1
```

```
name: myappsrv
```

```
namespace: mynamespace

spec:

  selector: app: myapp

  ports:

    name: udp

    protocol: UDP

    port: 5202

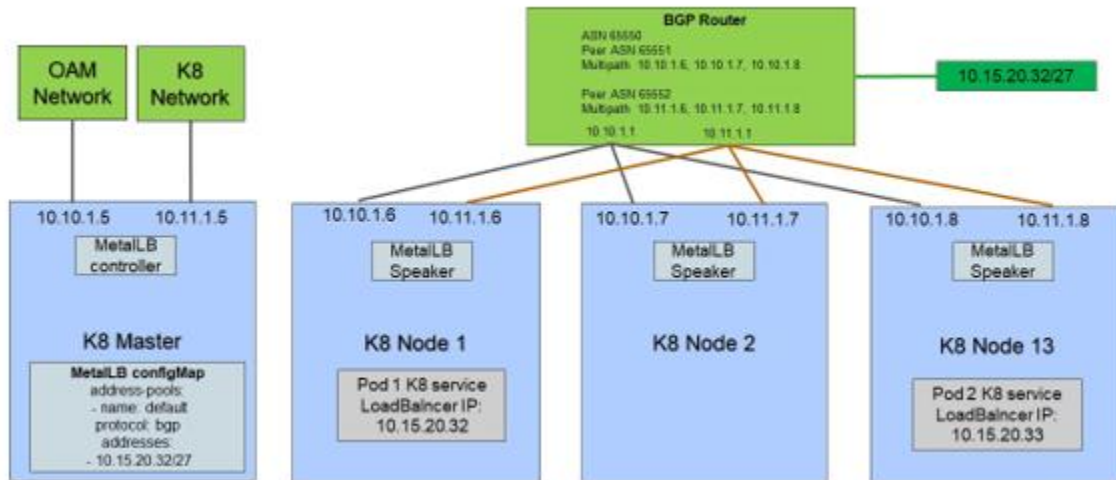
    targetPort: 5202

  type: LoadBalancer
```

Add Multiple BGP Peers

The MetalLB load balancer (fed-metallb) supports having multiple BGP peers. Once configured, all peers will get all the advertised BGP routes.

Note: The BGP router determines the correct path. This can be achieved in a few ways by rejecting routes for different peers. However, this BGP config is outside the scope of this document

Figure 65. Multiple BGP Peer implementation

To configure multiple BGP peers for the MetalLB load balancer, you must use the expert mode when deploying with OPM GUI or modify the values file when deploying manually with the helm install command.

By default, the typical fed-metallb values file looks similar to the following:

```
global:
registry_docker_path: cnreg-rel:5000

pod-metallb:

config:
addresses: 10.15.20.32/27
aggregationLength: "27"
myAsn: "65501"
peerAddress: 10.10.1.1
peerAsn: "65550"
```

The above example shows one peer which implies that all advertised routes go to this peer. If you want to add more peers, add the following:

```
add_peer:

peer1:

myAsn: "65502"
peerAddress: 10.11.1.1
```

```
peerAsn: "65550"
```

The following output shows an example of three peers:

```
global:
```

```
registry_docker_path: cnreg-rel:5000
```

```
pod-metallb:
```

```
config:
```

```
addresses: 10.15.20.32/27
```

```
aggregationLength: "27"
```

```
myAsn: "65501"
```

```
peerAddress: 10.10.1.1
```

```
peerAsn: "65550"
```

```
add_peer:
```

```
peer1:
```

```
myAsn: "65502"
```

```
peerAddress: 10.11.1.1
```

```
peerAsn: "65550"
```

```
peer2:
```

```
myAsn: "65503"
```

```
peerAddress: 10.12.1.1
```

```
peerAsn: "65550"
```


Third-Party Docker Images used in UnityCloud

Affirmed Networks, Inc. cannot distribute certain third-party container images. Therefore, you must pull the Docker images from the Internet and install them in the OPM Docker Registry prior to installing. The following is a list of qualified third-party Docker images used by UnityCloud for this release:

- ceph/ceph:v18.2.4
 - OPA version: 0.68.0
 - Kubemgmt: 8.5.10
- Rook: 1.15.7
 - cert-manager: 1.15.3
 - Istio: 1.16.1
 - nfvpe/multus:v3.7
 - nfvpe/sriov-device-plugin:v3.3.2

Note: These image versions have been qualified to work with UnityCloud current release. Other versions may not support proper operation.

Deploy fed-rook_ceph

Note: If selected, `fed-rook_ceph` must be installed after `fed-rbac`, but before any other federation.

If you choose to install `fed-rook_ceph`, the local or remote NFS server is replaced with a Ceph Storage Cluster using CephFS or Ceph block storage. CephFS is a mountable, POSIX-compliant file system layered on top of a Ceph Storage Cluster.

CephFS provides the following advantages over NFS:

- Supports authentication
- Clustered storage
- Replication of data at the Ceph layer instead of using RAID

`fed-rook_ceph` deploys CephFS using Rook. Rook is a Kubernetes operator for Ceph that automates the tasks of a storage administrator, including deployment, configuration, scaling, monitoring, and disaster recovery.

Note: For more information about Ceph, CephFS, and Rook, please refer to the official [Ceph documentation](#) and [Rook documentation](#).

Prerequisites to Deploy fed-rook_ceph on OpenStack

There are new requirements for the UnityCloud OpenStack infrastructure to support running a Ceph cluster on Kubernetes.

Note: These requirements assume that all Kubernetes worker nodes are also storage nodes.

1. Ceph requires at least 3 worker nodes.
2. When the Kubernetes cluster is provisioned in OPM, you may define the Ceph disk size in the **Ceph Disk Size** field, in gigabytes. If this value is larger than zero, the Ceph disk is added on the worker node.
3. By default the Ceph replication factor is 3, which means that the useable storage available for all federations to use is based on the following formula:

$$\text{Useable size} = (\text{Ceph Disk Size} \times \text{No of worker nodes}) / 3$$
4. The required useable size should be enough to meet the storage requirements of federations that will use fed-rook_ceph storage, taking into consideration data retention for a given number of active users.
5. If selected, fed-rook_ceph must be the first federation installed.

Download Ceph Image used in Federations

Affirmed Networks cannot distribute the Ceph container image, so before you install fed-rook_ceph, you must pull the Docker image from the internet and install it in the OPM Docker Registry. If your OPM VM does not have an internet connection, you must perform the following steps to install the container prior to installing fed-rook_ceph. If there is internet connection, skip these steps and install the fed-rook_ceph federation specifying 'ceph' in the Ceph Image Repo parameter.

1. Obtain this image on the server that has Docker and access to the public Internet.
 - a. `# docker pull quay.io/ceph/ceph:v18.2.4`
 - b. Verify that the Digest displayed after the docker pull command exactly matches with the following:
`sha256:b96f1d0c606bb72ae4980f3a9348f75a1b7001320689c3aca33cd282c25b4610`
 - c. `# docker save quay.io/ceph/ceph:v18.2.4 > ceph-18.2.4.tar`
 - d. `# scp ceph-18.2.4.tar root@<opm ip>:/tmp`
2. Copy the tarball to the OPM VM and untar the packages.
 - a. `# ssh to root@<opm ip>`
 - b. `# docker load -i /tmp/ceph-18.2.4.tar`
3. Issue the following commands to get the image ID, tag it locally, and push it to the registry. Do not change the Ceph image and version since Rook checks for the official image name and version.
 - a. `# docker images quay.io/ceph/ceph:v18.2.4`
 - b. `# docker tag <image id> <opm ip>:5005/ceph:v18.2.4`

Deploy fed-rook_ceph

c. # **docker push <opm ip>:5005/ceph:v18.2.4**

If the registry used in the steps above matches the registry configured in the global registry docker repo path parameter, you may proceed to install the fed-rook_ceph federation. Otherwise, when installing fed-rook_ceph, configure the Ceph Image Repo parameter with **<opm ip>:5005**, or whichever repo was used to upload the Ceph image.

Deployment

Deploy fed-rook_ceph using OPM as described in [Install Federations Individually](#). You may use the following configuration parameters:

Table 25. fed-rook_ceph parameters

Parameter	Description	Default Value
Release Name (Required)	Enter the release name.	
Namespaces (Required)	Enter the namespace for this federation.	
pod-rook_ceph		
cephImageRepoPath (Required)	The repository for the Ceph image. Affirmed Networks is not able to distribute the Docker image due to licensing restrictions.	
deviceFilter (Required)	A Golang regular expression to match the short kernel names of devices that allows Rook to select which block devices to set up Ceph on.	vdb
dashboardEnabled (Required)	Boolean control on whether Rook deploys the Ceph Dashboard.	true
storageClassName (Required)	Name of the storage class that will be used to dynamically create a persistent volume with CephFS or block storage. Use rook-cephfs for CephFS, and rook-cephbp for block storage.	rook-cephfs

Deploy fed-rook_ceph

Parameter	Description	Default Value
journalSizeMB	For environments where the disk size is less than 20G, set this to 1024. Otherwise, leave blank.	
databaseSizeMB	For environments where the disk size is less than 100G, set this to 1024. Otherwise, leave blank.	
service		
externalIpAddress	The address where the Ceph dashboard is accessible	
loadBalancer.enabled (Required)	Boolean control on whether an external load balancer is used to expose the Ceph dashboard.	true
loadBalancer.loadBalancerFQDN	If using load balancer, this is the FQDN where the Ceph dashboard is accessible.	
loadBalancer.loadBalancerIP	If using load balancer, this is the IP address where the Ceph dashboard is accessible.	
loadBalancer.loadBalancerSharingKey	If using a shared load balancer, this is the key shared with other services.	paas
cephConfig		
global.monOsdDownOutSubtreeLimit	The smallest CRUSH unit type that Ceph will not automatically mark out. For instance, if set to host and if all OSDs of a host are down, Ceph will not automatically mark out these OSDs. If set to root it means that any failure can mark OSDs as out and hence trigger rebalancing (which is not a problem if amount of data stored in the ceph cluster is small).	root

Deploy fed-rook_ceph

Parameter	Description	Default Value
failureDomain.fs.data	CephFilesystem pool failure domain for data pool, translates to chooseleaf_firstn setting in CRUSH	host
failureDomain.fs.metadata	CephFilesystem pool failure domain for metadata pool, translates to chooseleaf_firstn setting in CRUSH	host
failureDomain.block	CephBlockPool for data failure domain	host
failureDomain.mgr	CephBlockPool for Ceph Manager failure domain	host
metadataServer.preferredTopologyKey	topologyKey for preferredDuringSchedulingIgnoredDuringExecution for MDS pods	topology.kubernetes.io/zone
metedataServer.requiredTopologyKey	topologyKey for requiredDuringSchedulingIgnoredDuringExecution for MDS pods	kubernetes.io/hostname

You may verify the installation by checking that all the Rook and Ceph pods are running. Make sure that `rook-ceph-osd-0`, `rook-ceph-osd-1`, `rook-ceph-osd-2` (up to the number of worker nodes present) are running. These are the object storage daemons running on each node, providing access to the Ceph disks. Enter the following command to verify the installation:

```
$ kubectl -n fed-rook-ceph get pods
```

Resilient fed-rook_ceph deployment

Default fed-rook_ceph helm values do not consider deployments on clusters which are comprised of hosts deployed across multiple racks/zones etc. For a deployment where a cluster is deployed across multiple zones and amount of data in a ceph cluster is relatively small there are the following resiliency requirements:

- Placement Group replicas and Metadata server pods are spread over zones
- Every OSD failure is accounted for by Ceph (when OSD is down it may be marked as **out** which may trigger Placement Group rebalancing)

Helm values for relevant parameters need to look as follows:

```

pod-rook_ceph:
  cephConfig:
    global:
      monOsdDownOutSubtreeLimit: root
  failureDomain:
    fs:
      data: zone
      metadata: zone
      block: zone
      mgr: zone
  metadataServer:
    preferredTopologyKey: topology.kubernetes.io/zone
    requiredTopologyKey: topology.kubernetes.io/zone

```

Ceph dashboard

The Ceph dashboard is accessible by external IP or external load balancer, depending on which was configured. After the external load balancer (fed-metallb) is deployed, the dashboard should be accessible at the configured load balancer IP address.

Dashboard credentials

The default dashboard user's name is `admin`.

To obtain the admin user's password for the dashboard, enter the following command:

```
$ kubectl -n fed-rook-ceph get secret rook-ceph-dashboard-password -o
jsonpath="{['data']['password']}" | base64 --decode && echo
```

Log into the Ceph dashboard

You can view the Ceph dashboard by using a web browser to access

https://<configured_name_or_ip>:8443. Log in using the user name and password for admin.

Cleanup for re-install

To clean up fed-rook_ceph and all its data, perform the following procedures:

Warning: Uninstalling fed-rook_ceph erases CephFS data. Back up your data before uninstalling.

1. Uninstall fed-rook_ceph:

```
# helm uninstall --namespace fed-rook-ceph fed-rook-ceph
```

2. Delete the fed-rook_ceph namespace:

```
# kubectl delete namespace fed-rook_ceph
```

3. Re-initialize the disks on each worker node by running the following bash script:

Warning: Any data on the Ceph disks will be lost when you execute this script. Update the script if Ceph disk is not vdc! If using SSDs, you can use 'blkdiscard' instead of 'dd' to cleanup the disks. See below.

```
#!/usr/bin/env bash

rm -rf /var/lib/rook

DISK="/dev/vdb"

# Zap the disk to a fresh, usable state (zap-all is important, b/c MBR has
to be clean)

# You will have to run this step for all disks. sgdisk --zap-all $DISK

# If sgdisk is unresponsive or if it hangs, you can use wipefs instead

# wipefs --all --force $DISK
```


Deploy fed-rook_ceph

```
# Clean hdds with dd

dd if=/dev/zero of="$DISK" bs=1M count=100 oflag=direct,dsync

# Clean disks such as ssd with blkdiscard instead of dd

#blkdiscard $DISK

# These steps only have to be run once on each node

# If rook sets up osds using ceph-volume, teardown leaves some devices
mapped that lock the disks.

ls /dev/mapper/ceph-* | xargs -I% -- dmsetup remove %

# ceph-volume setup can leave ceph-<UUID> directories in /dev and
/dev/mapper (unnecessary clutter)

rm -rf /dev/ceph-*
```

Troubleshoot Cleanup

Namespace deletion stuck at 'Terminating'

If the namespace deletion is stuck at 'Terminating', you may need to forcefully terminate it by patching the namespace's finalizer to null. See steps below:

1. Run the following command:

```
# kubectl get namespace fed-rook-ceph -o json > fed-rook- ceph.json
```

2. Edit the file and delete "kubernetes" in the spec.finalizers array:

```
"spec": { "finalizers": [
]
},
```

3. Save the file.

4. In same terminal run the kubeproxy. You can also run as a foreground process in another terminal:

```
# kubectl proxy &
```

5. Send request to API server to forcefully delete the namespace:

```
# curl -k -H "Content-Type: application/json" -X PUT --data-binary @fed-rook-ceph.json http://127.0.0.1:8001/api/v1/namespaces/fed-rook-ceph/finalize
```

6. Verify that namespace is deleted.
7. Terminate proxy by executing `# fg` and pressing ctrl-c.

Rook CRDs not deleted when using cna stopfed

If using cna stopfed, the Rook CRDs may not get deleted so you need to manually delete the CRDs. Execute the following commands:

```
# kubectl delete crd cephblockpools.ceph.rook.io --ignore-not-found #
kubectl delete crd cephclients.ceph.rook.io --ignore-not-found
# kubectl delete crd cephclusters.ceph.rook.io --ignore-not-found
# kubectl delete crd cephfilesystemmirrors.ceph.rook.io --ignore-not-found
# kubectl delete crd cephfilesystems.ceph.rook.io --ignore-not-found
# kubectl delete crd cephnfses.ceph.rook.io --ignore-not-found
# kubectl delete crd cephobjectrealms.ceph.rook.io --ignore-not-found #
kubectl delete crd cephobjectstores.ceph.rook.io --ignore-not-found
# kubectl delete crd cephobjectstoreusers.ceph.rook.io --ignore-not-found
# kubectl delete crd cephobjectzonegroups.ceph.rook.io --ignore-not-found
# kubectl delete crd cephobjectzones.ceph.rook.io --ignore-not-found
# kubectl delete crd cephrbdmirrors.ceph.rook.io --ignore-not-found
# kubectl delete crd objectbucketclaims.objectbucket.io --ignore-not-found
# kubectl delete crd objectbuckets.objectbucket.io --ignore-not-found
# kubectl delete crd
volumereplicationclasses.replication.storage.openshift.io -- ignore-not-found
# kubectl delete crd volumereplications.replication.storage.openshift.io -
-ignore-not-found
# kubectl delete crd volumes.rook.io --ignore-not-found
```

Manage Ceph Nodes

Recover Ceph OSD

This procedure describes how to recover the Ceph Object Storage Daemon (OSD) if it becomes corrupted or gets stuck in a crash loop state. The old Ceph OSD can be removed from the Ceph Cluster so the new one can initialize properly.

Prerequisites and Notes

- Confirm you will have enough space on your cluster after removing your OSDs to properly handle the deletion
 - Log into the ceph-toolbox container and check space is enough:

```
# kubectl -n fed-rook-ceph exec -it $(kubectl -n fed-rook-ceph get pod -l
"app=rook-ceph-tools" -o jsonpath='{.items [0].metadata.name}') bash
```

```
# ceph df
```

```
# ceph osd df
```

- Confirm the remaining OSDs and their placement groups (PGs) are healthy in order to handle the rebalancing of the data

→ Log into the ceph-toolbox container and check that the remaining OSDs and PGs are healthy:

```
# kubectl -n fed-rook-ceph exec -it $(kubectl -n fed-rook-ceph get pod -l
"app=rook-ceph-tools" -o jsonpath='{.items [0].metadata.name}') bash
```

```
# ceph health detail
```

```
# ceph -s
```

- Do not remove too many OSDs at once
- Wait for rebalancing between removing multiple OSDs

Recovery Procedure

1. Disable the Rook Ceph Operator:

```
$ kubectl -n fed-rook-ceph scale deployment rook-ceph-operator --
replicas=0
```

2. Take note of the crashing OSD to be recovered:

```
$ kubectl -n fed-rook-ceph get pods -o wide | grep osd
```

3. Disable the OSD deployment that was crashing:

```
$ kubectl -n fed-rook-ceph scale deployment rook-ceph-osd-N -- replicas=0
Where N is the OSD number.
```

4. Log into the Ceph toolbox to make sure the OSD is down:

```
$ kubectl -n fed-rook-ceph exec -it $(kubectl -n fed-rook-ceph get pod -l
"app=rook-ceph-tools" -o jsonpath='{.items [0].metadata.name}') bash
```

```
# ceph osd down osd.1
```

5. Purge the OSD using the following Rook purge utility:

a. **wget -N https://raw.githubusercontent.com/rook/rook/v1.6.2/cluster/examples/kubernetes/ceph/osd-purge.yaml**

b. Edit file and change namespace to `fed-rook-ceph`.

c. Change OSD ID in args to match OSD number to purge.

d. Change the rook-ceph image to the deployed version. For example, `<registry>/cna-rook_ceph-rook_ceph:<release_version>`

6. Run the job and verify in job's pod logs that OSD is purged successfully:

```
$ kubectl create -f osd-purge.yaml
```

7. Delete the OSD deployment

```
$ kubectl delete deployment -n fed-rook-ceph rook-ceph-osd-N
```

Where N is the OSD number.

8. Delete the underlying data on the node by logging onto the node where the OSD is scheduled. Run the cleanup script on the node to re-initialize the Ceph disk. Make sure to update the DISK variable to the configured Ceph disk:

```
#!/usr/bin/env bash
rm -rf /var/lib/rook
DISK="/dev/vdb"
# Zap the disk to a fresh, usable state (zap-all is important, b/c MBR has
to be clean)
# You will have to run this step for all disks.
sgdisk --zap-all $DISK
# If sgdisk is unresponsive or if it hangs, you can use wipefs instead
# wipefs --all --force $DISK
# Clean hdds with dd
dd if=/dev/zero of="$DISK" bs=1M count=100 oflag=direct,dsync
# Clean disks such as ssd with blkdiscard instead of dd
#blkdiscard $DISK
# These steps only have to be run once on each node
# If rook sets up osds using ceph-volume, teardown leaves some devices
mapped that lock the disks.
ls /dev/mapper/ceph-* | xargs -I% -- dmsetup remove %
# ceph-volume setup can leave ceph-<UUID> directories in /dev and
/dev/map- per (unnecessary clutter)
rm -rf /dev/ceph-*
```

9. Enable the Rook Ceph Operator and wait for the new OSD to be re-created by Rook operator:

```
$ kubectl -n fed-rook-ceph scale deployment rook-ceph-operator --
replicas=1
```

Remove OSDs Gracefully When Removing a Worker Node

To gracefully remove an OSD on a worker node that is scheduled to be decommissioned:

1. Disable the operator:

```
$ kubectl -n fed-rook-ceph scale deployment rook-ceph-operator --replicas=0
```

2. Disable OSD that is on the node to be removed:

```
$ kubectl -n fed-rook-ceph scale deployment rook-ceph-osd-N -- replicas=0
```

Where **N** is the OSD number of the pod running on the node to be removed.

3. Purge the OSD using the following Rook purge utility:

- a. Edit file and:

- i. Change namespace to fed-rook-ceph.

- ii. Change OSD ID in args to match OSD number to purge.

- iii. Change the rook-ceph image to the deployed version.

For example, `<registry>/cna-rook_ceph-rook_ceph:<release_version>`

4. Run the job and verify in job's pod logs that OSD is purged successfully:

```
$ kubectl create -f osd-purge.yaml
```

5. Delete the OSD deployment

```
$ kubectl delete deployment -n fed-rook-ceph rook-ceph-osd-N
```

Where **N** is the OSD number.

6. Remove the node / Add new node if needed.

7. Delete the underlying data on the node by logging onto the node where the OSD is scheduled. Run the cleanup script on the node to re-initialize the Ceph disk. Make sure to update the DISK variable to the configured Ceph disk:

```
#!/usr/bin/env bash
rm -rf /var/lib/rook
DISK="/dev/vdb"
# Zap the disk to a fresh, usable state (zap-all is important, b/c MBR has
to be clean)
# You will have to run this step for all disks.
sgdisk --zap-all $DISK
# If sgdisk is unresponsive or if it hangs, you can use wipefs instead
# wipefs --all --force $DISK
# Clean hdds with dd
```

Deploy fed-rook_ceph

```
dd if=/dev/zero of="$DISK" bs=1M count=100 oflag=direct,dsync # Clean
disks such as ssd with blkdiscard instead of dd #blkdiscard $DISK
# These steps only have to be run once on each node
# If rook sets up osds using ceph-volume, teardown leaves some devices
mapped that lock the disks.
ls /dev/mapper/ceph-* | xargs -I% -- dmsetup remove %
# ceph-volume setup can leave ceph-<UUID> directories in /dev and
/dev/map- per (unnecessary clutter)
rm -rf /dev/ceph-*
```

8. Enable the operator again and wait for new OSD to get created:

```
$ kubectl -n fed-rook-ceph scale deployment rook-ceph-operator --
replicas=1
```

Fail Over a Ceph Monitor

If a Ceph Monitor (mon) is unhealthy and the Kubernetes pod restart or liveness probe are not sufficient to bring a mon back up, the operator makes the decision to terminate the unhealthy monitor deployment and bring up a new monitor with a new identity. By default, the operator waits 10 minutes before failing over an unhealthy mon.

For detailed information about failing over a Ceph Monitor, see the Rook documentation here: [Monitor Health](#)

Install Istio

This section describes special considerations when installing Istio version 1.11.2 and later.

Additional Concepts Affecting Installation

CNI - Container Network Interface, which is basically the code that K8s shall call to setup network interface(s) in pods (for example, calico).

Istio-CNI - Istio in the new architecture introduces istio-cni which is a container network interface (CNI) plugin that sets up the pod network traffic redirection to/from the Istio sidecar proxy (read more at Istio / Install Istio with the Istio CNI plugin)

Multus - A meta-CNI plugin, i.e. a CNI that calls other CNIs. Multus is invoked by K8s for attaching multiple network interfaces to pods, e.g. in UC, multus is installed by fed-sriovdp and it's used for making multiple SRIOV interfaces available in the required pods.

Important note for multus

Please note that since multus is a meta-CNI (a CNI that invokes other CNIs) plugin, once it's installed, kubelet (K8)s will always invoke multus for any container networking, so even though UnityCloud uses multus to enable multiple SRIOV interfaces, multus is not conceptually coupled with SRIOV. Multus may be used to enable any type of multiple interfaces, even two calico interfaces in the POD for example.

Note: For full instructions on deploying Multus and SR-IOV Device Plugin, refer to "Prerequisites for UPF Deployment" in the User Plane Function Setup Guide.

Istio and Multus Interworking

The introduction of istio-cni by istio 1.11.2 means that the fed-istio installation procedure needs to be aware of the presence or absence of Multus. Here is a brief explanation:

When the K8s cluster does NOT have multus installed

In this case, kubelet (K8s) will directly invoke the Calico CNI plugin for pod networking. When fed-istio triggers the Istio installation, then istio-cni is set up as a chained plugin, which means changes are made directly to existing Calico configuration. Whenever the Calico cni is invoked, istio-cni also gets invoked and the pods that register for istio-injection will get the istio-proxy side car.

When the K8s cluster does have multus installed

In this case, kubelet (K8s) invokes the Multus CNI plugin for pod networking. Now, multus will further invoke required CNI plugins such as Calico, SRIOV (depends on use-case), and istio-cni. So, in this case, when fed-istio triggers the Istio installation, then the istio-cni plugin files will be independently set up as a non-chained plugin so that Multus can parse and process them.

Install Istio

The installation of fed-istio is directly linked to the presence of absence of Multus.

How to determine if Multus will be installed on the Kubernetes cluster

If Multus will be installed on the cluster, additional values must be set when installing fed-istio. Multus is installed by fed-sriovdp, so any federation that uses fed-sriovdp will require additional configuration.

At this time, the following federations require fed-sriovdp:

- fed-upf
- fed-amf - Not in all cases. Only set the flag when Multus is used.
- fed-mme - Not in all cases. Only set the flag when Multus is used.
- fed-agf
- fed-n3iwf

Multus not installed on the Kubernetes cluster

If Multus is not installed on the cluster, no specific helm variables need to be passed to the installation.

Multus installed on the Kubernetes cluster

Because fed-istio must be installed before fed-sriovdp (which installs Multus), it is not possible to automate the following Helm values for installing fed-istio:

`global.multusDeployed=true` value should be set in fed-istio's `values.yaml` or `--set global.multusDeployed=true` can be passed in the fed-istio's helm install command.

Caution: If you are installing any of the above-mentioned fed's helm chart in a namespace which is different from the default namespace, such as installing the fed-upf helm chart in the fed-upf-1 namespace then you need to configure that namespace in the `values.yaml` of fed-upf helm chart as follows :

```
global:
```

```
  istioEnabledFeds:
```

```
  - fed-upf-1
```

If this is not done then the installation will fail with the following error:

```
Warning NoNetworkFound 0s multus cannot find a network- attachment-definition
(istio-cni) in namespace (fed-upf-1): network-attachment-
definitions.k8s.cni.cncf .io "istio-cni" not found
```

Install Multiple Federations on Same Cluster

When installing multiple application federations on the same cluster, any federations that depend on fed-sriovdp must be installed first.

fed-istio will be configured for the needs of the first federation that installs it, meaning that if fed- istio is installed without the `global.multusDeployed=true` flag (such as for fed-smf), subsequent installations that require fed-sriovdp (such as fed-upf) will fail because fed-istio was not configured properly for Multus. If this happens, you must re-create the cluster.

Example scenario

For example, when there is a requirement to install fed-smf and fed-upf in the same cluster, fed- upf must be installed first. As a prerequisite to fed-upf installation, fed-istio is installed and the helm variable `global.multusDeployed` for fed-istio must be set to `true`.

Once fed-upf installation completes, then fed-smf and any other subsequent federations can be installed regardless of whether they depend upon fed-sriovdp.

Manage Federations with OPM

Affirmed Networks Cloud Native applications are structured as collections of microservices known as federations. Federations can be true applications, such as Access and Mobility Management Function (AMF), Session Management Function (SMF), User Plane Function (UPF), Network Repository Function (NRF), or a Platform as a Service (PaaS), such as Elastic Search, Grafana, or Prometheus.

Microservices are described by Helm charts that are stored in a Helm repository and specific Docker images, stored in registries. The federations are managed with OPM by selecting the appropriate Helm chart from the repository and then configuring the required parameters.

Bulk Deploy Network Functions using Configuration Sets

Note: In addition to network functions, this process allows an existing single-node or multi- node OPM to deploy `fed-opm_multi` (multi-node OPM) to a Kubernetes cluster.

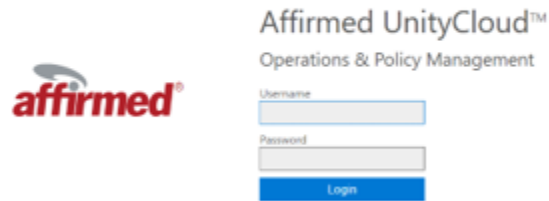
You can use OPM to bulk deploy UnityCloud network functions on a Kubernetes cluster. OPM automatically detects and deploys prerequisite federations during the deployment procedure.

For information on required Helm variables for individual federations, please refer to [Install Federations Individually](#).

To bulk deploy a network function to an existing cluster with OPM:

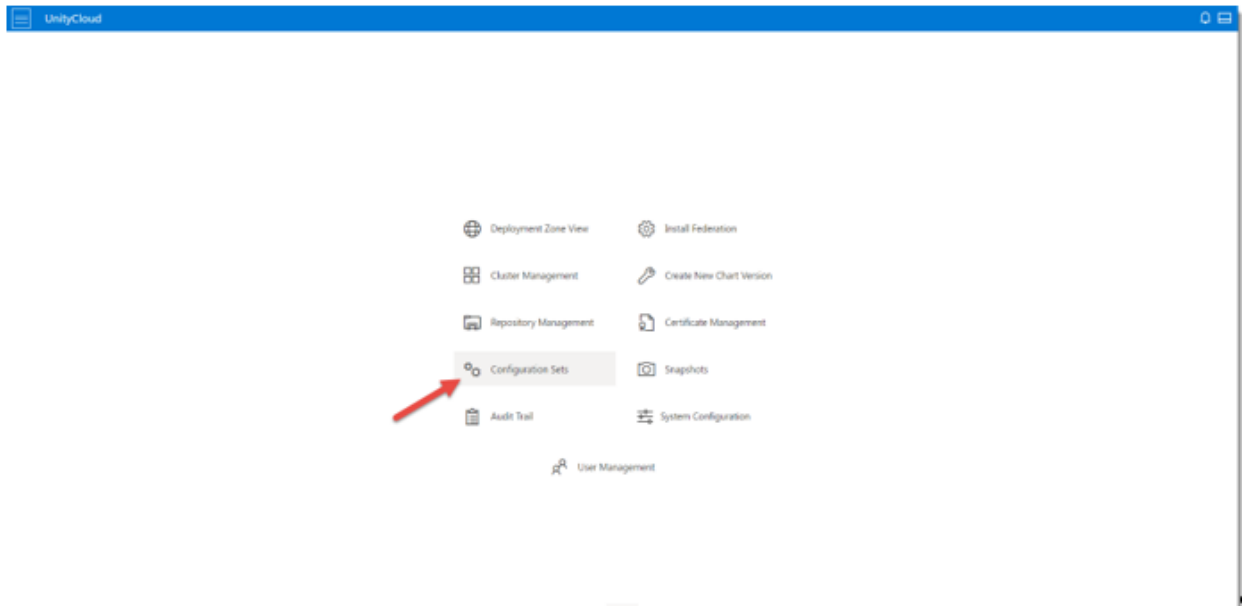
1. Log into the OPM GUI by using a web browser to visit `https://<opm_ip>`.

Figure 66. OPM Login



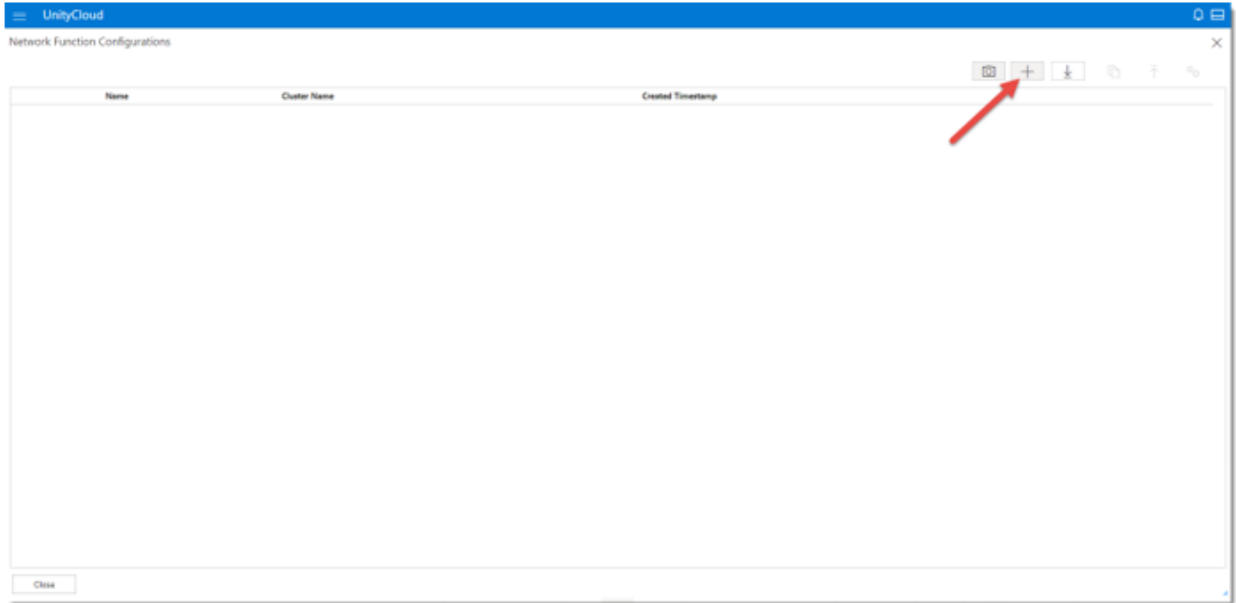
2. Enter your user name and password.
3. From the main OPM screen, click **Configuration Sets**.

Figure 67. Main OPM Screen



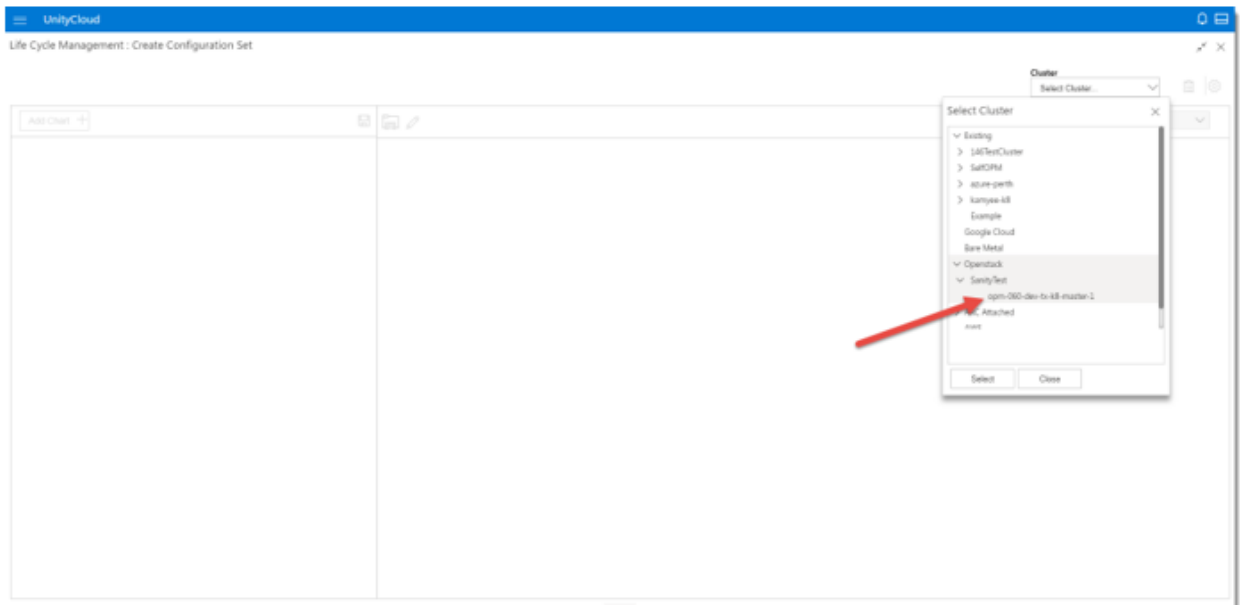
4. On the Network Function Configurations screen, click the **Plus (+) button** to begin creating a new configuration set.

Figure 68. Create Configuration



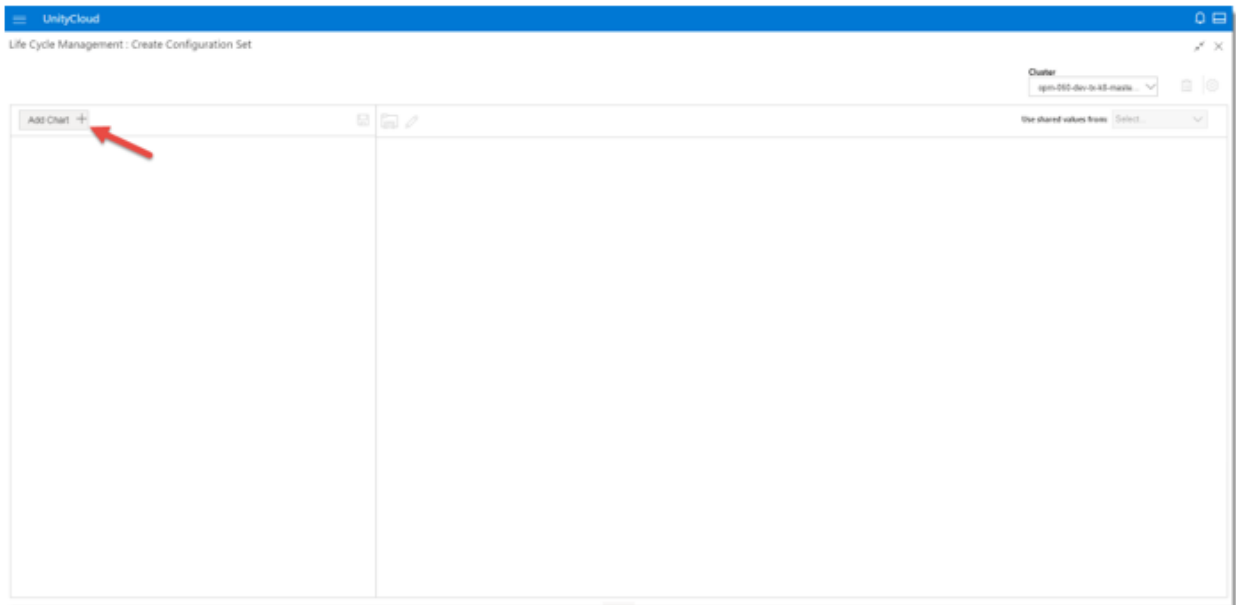
5. In the **Cluster** dropdown, select your existing Kubernetes cluster, then click **Select**.

Figure 69. Select Cluster



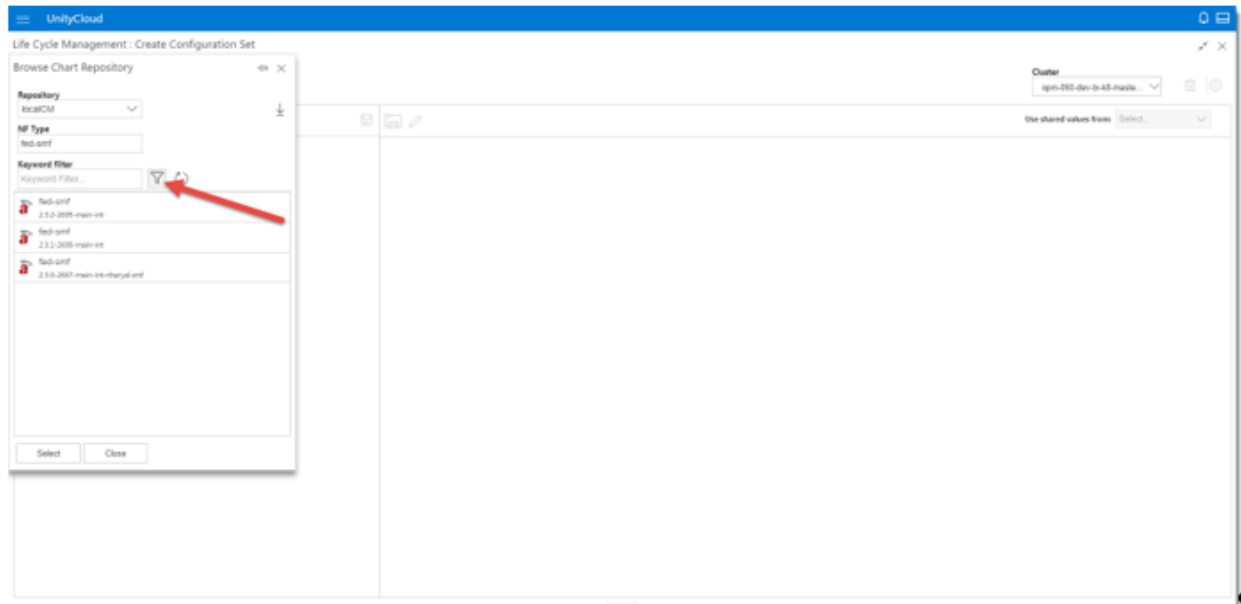
6. Click **Add Chart +**.

Figure 70. Add Chart



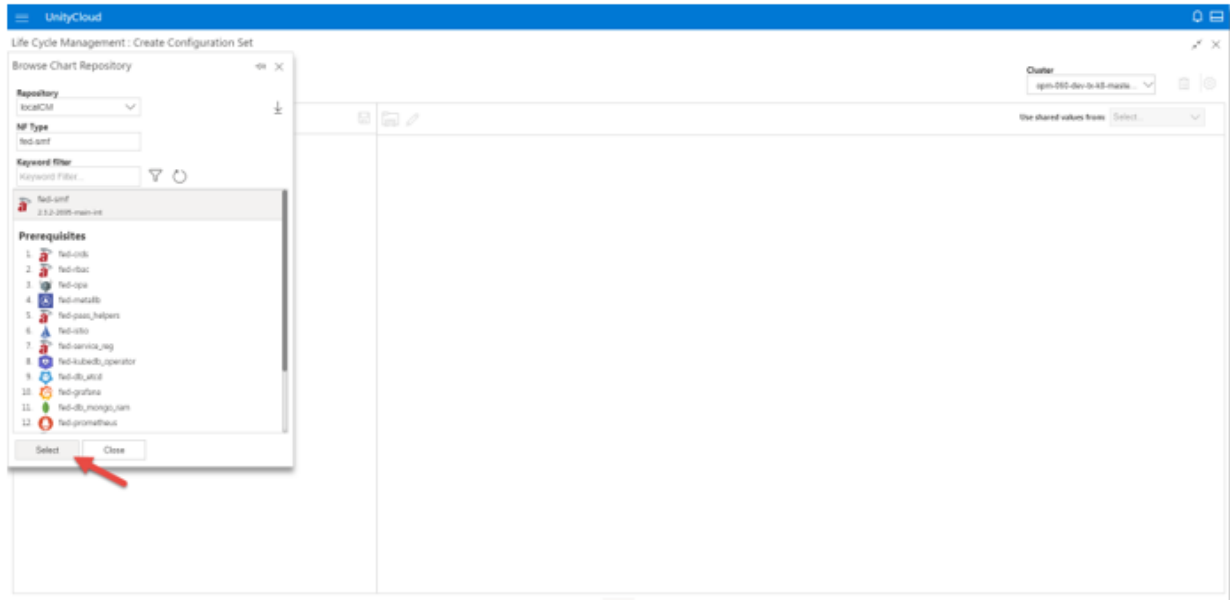
7. In the Browse Chart Repository window, filter by **Repository**, **NF Type**, and **Keyword filter**, then click the **Render Charts** button to show the results

Figure 71. Render Charts



8. Select from the list of available federations and then click **Select**.

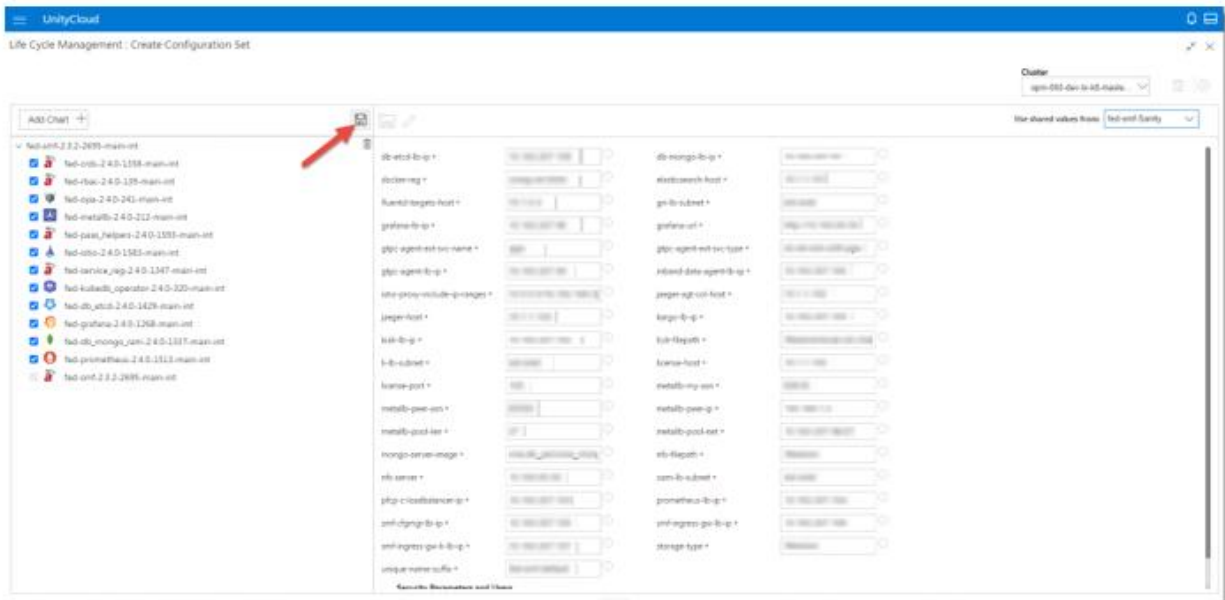
Figure 72. Select Chart



9. Select from the Use shared values from: dropdown if you wish to copy values from a previously saved NF configuration set.

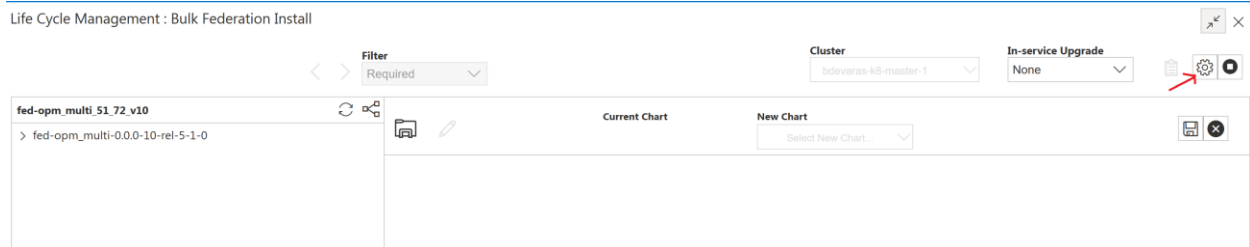
- Expand the NF config tree. The latest version of PaaS components in the chart repository are selected for installation.
- By default, all PaaS components will be installed. Un-checking a PaaS component will skip its installation.
- Click the **Create Configuration Set** button and specify a name to create a NF configuration set.

Figure 73. Create Configuration Set and Begin Bulk Federation Install



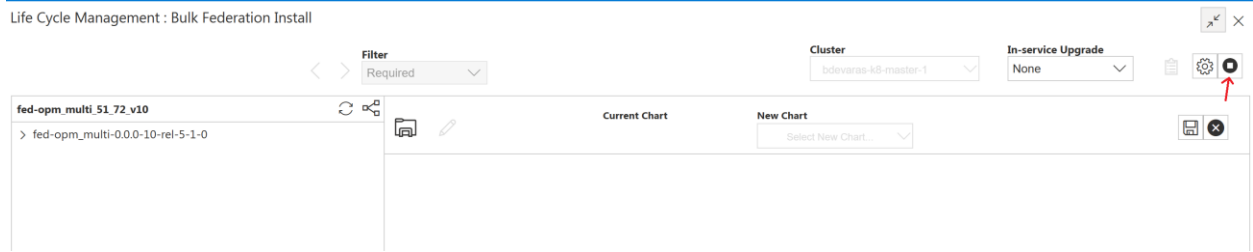
10. Expand the list of requisite components in the list-side pane and click components to review configuration values.

11. Click the **Bulk Install Federations** to kick off the bulk deployment of the network function.



12. You may monitor the deployment status by clicking the **Notifications icon (Bell)**.

Note: To stop the bulk installation process, you may click the **Stop Bulk Installation** button.



Bulk Restore or Upgrade Network Functions using Snapshots

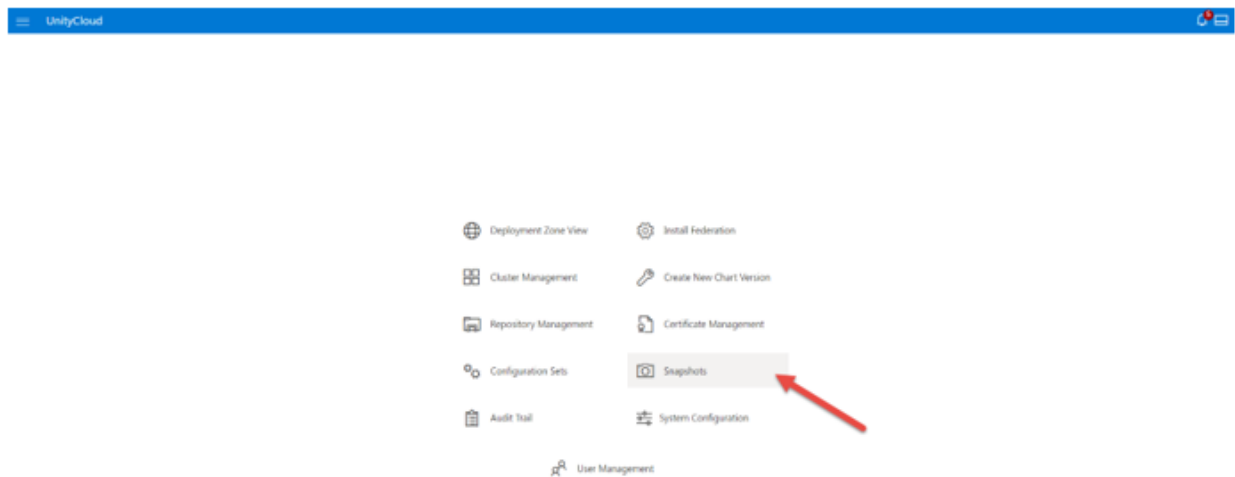
A snapshot of a cluster can be used to restore or upgrade federations in a cluster. This section demonstrates how to take a snapshot of a cluster and how to use the snapshot.

Take a Cluster Snapshot

Taking a cluster snapshot captures currently deployed federations in the cluster. From the OPM main screen:

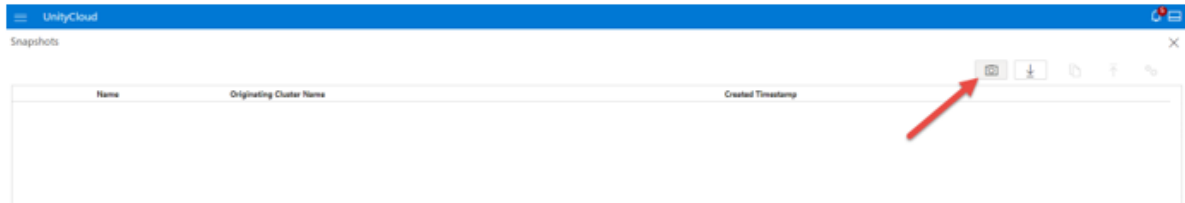
1. Click **Snapshots**.

Figure 74. OPM Main Screen



2. In the Snapshots window, click the **Create a snapshot of live instance's configurations** button.

Figure 75. Create snapshot of live instance configuration



3. In the Network Function Snapshot pop-up, select the target cluster ID and then click the **Get** button. Currently deployed federations in the cluster are displayed in the pop-up window.

Figure 76. Network Federation Snapshot Window

The screenshot shows the 'Network Function Snapshot' pop-up window. It has a title bar with a close button. The form includes the following fields and sections:

- Cluster ID ***: A dropdown menu showing 'sidlingappa-kondikoppa-t...' and a 'Get' button.
- Name ***: A text input field containing 'Test_Snapshot'.
- Description**: A text area containing 'Test snapshot for documentation.'
- Network Functions**: An empty section for listing network functions.
- PaaS Components**: A section containing four items, each with a checkbox and a logo:
 - ☒ fed-opa - 2.4.0-283-main-int
 - ☒ fed-paas_helpers - 2.4.0-1645-main...
 - ☒ fed-rbac - 2.4.0-175-main-int
 - ☒ fed-crds - 2.4.0-1394-main-int
- Snapshot**: A button at the bottom left.

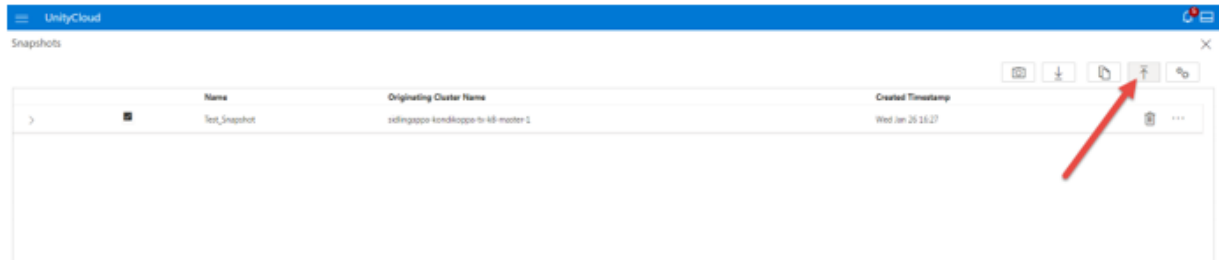
4. Provide a name and description for the snapshot. By default, all federations deployed on the cluster will be included in the snapshot. Unselect any federations to skip, Then click the **Snapshot** button.
5. A snapshot of selected federations on the cluster is saved under the specified name and is displayed in Snapshots.

Import/Export Snapshots

After a snapshot is saved, it can be exported to the file system for later retrieval. To export a saved snapshot:

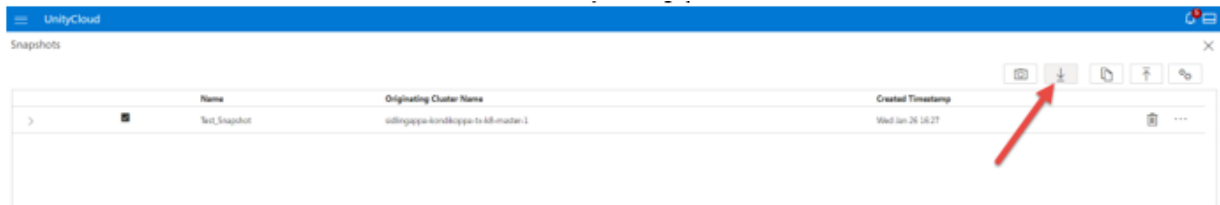
1. From Snapshots, check the box next to the target snapshot and click the **Export selected configurations** button.

Figure 77. Export selected configurations



2. In the Export Configuration Sets dialog box, click the **Export** button and the snapshot is saved to your file system.
3. If the snapshot in your database is corrupted or deleted, you can import your saved copy from file system using the **Import configuration ZIP file** button.

Figure 78. Import configuration ZIP file



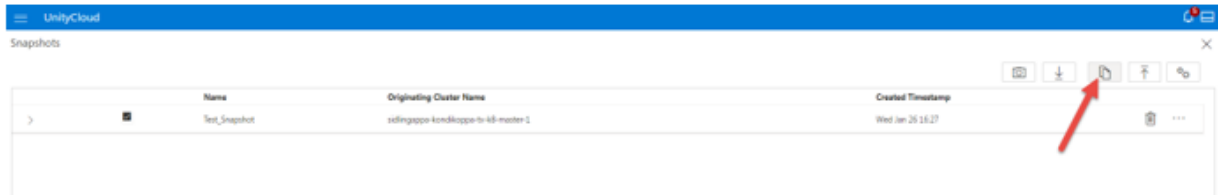
4. If the cluster is re-created, the snapshot can be used to re-deploy all federations.

Install or Upgrade Federations using Snapshots

A snapshot can be used to bulk install or upgrade federations. One possible use of this feature is to quickly restore a set of federations after a cluster has been deleted and re-created:

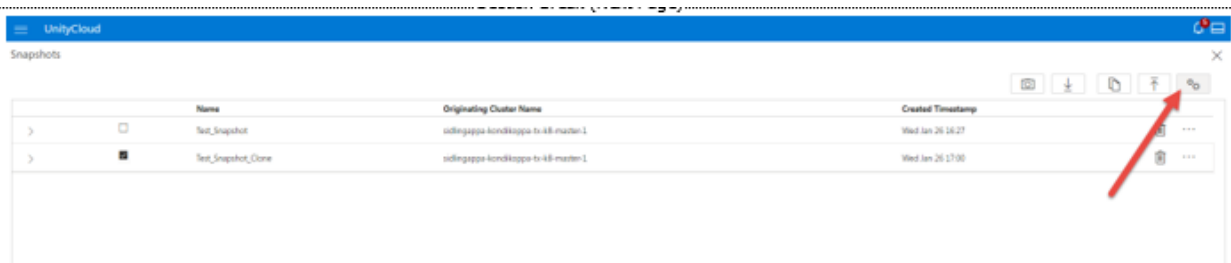
1. To avoid corrupting the snapshot, always make a clone of the snapshot using the **Clone selected configurations** button.

Figure 79. Clone selected configurations

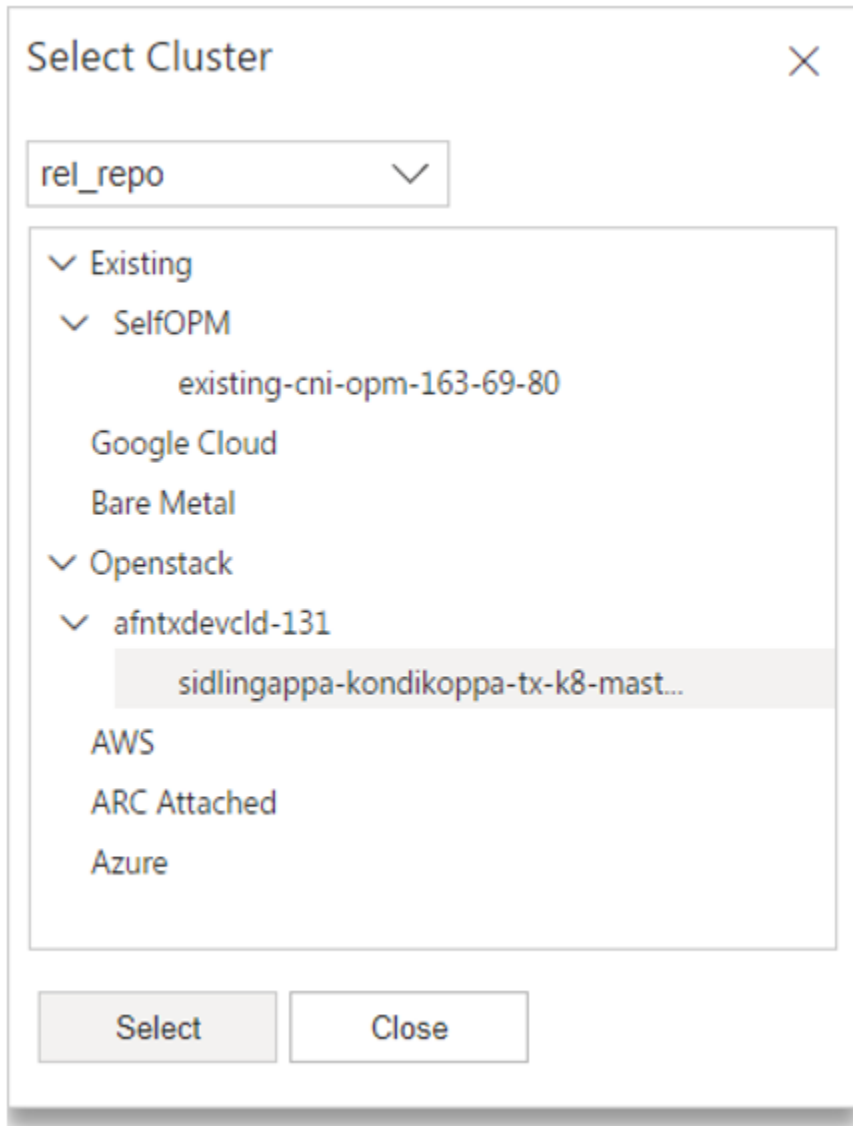


2. Provide a new snapshot name and click the Clone button. The cloned snapshot is saved to Snapshots.
3. Select the cloned snapshot and click the **Launch Bulk Installation View** button.

Figure 80. Launch Bulk Installation View

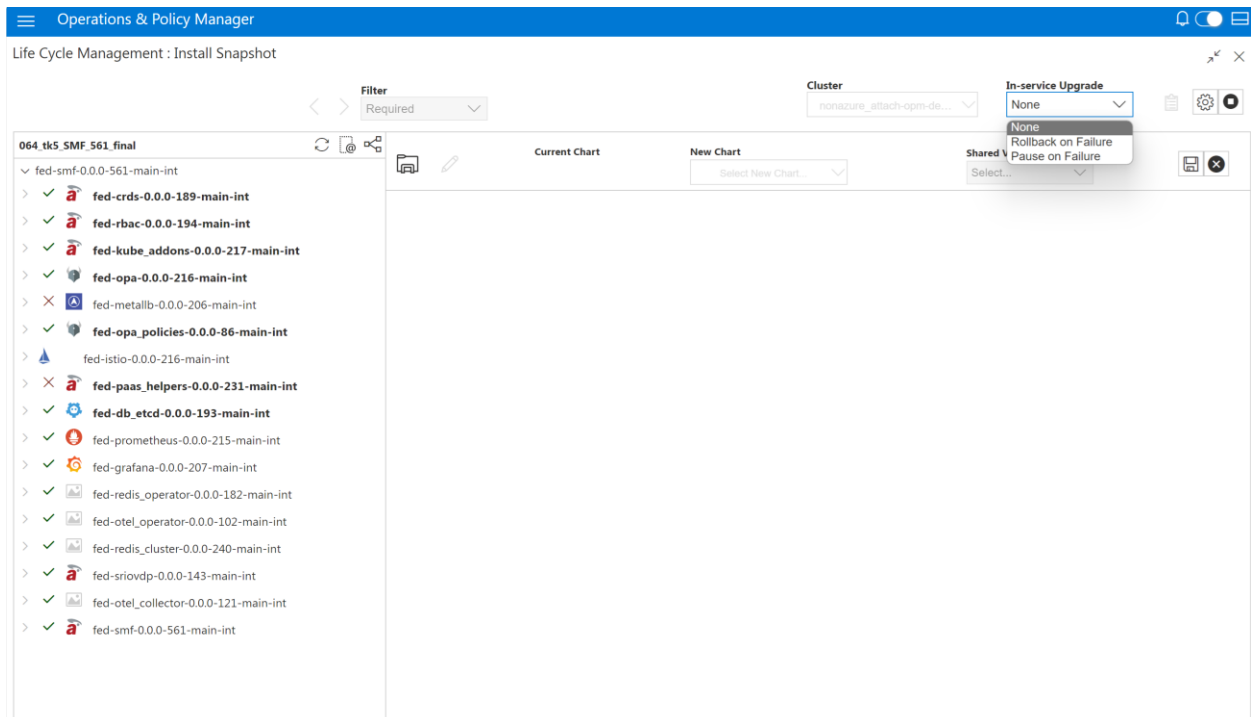


4. In the Select Cluster dialog, provide a chart repository, select the target cluster where the charts will be upgraded, and click **Select**.

Figure 81. Select Cluster

5. The Install Snapshot window is displayed, with the list of federations and their pre-requisites displayed on the left-hand pane in deployment order. The snapshot name (highlighted) is displayed above the NF tree.

Figure 82. Install Snapshot Window

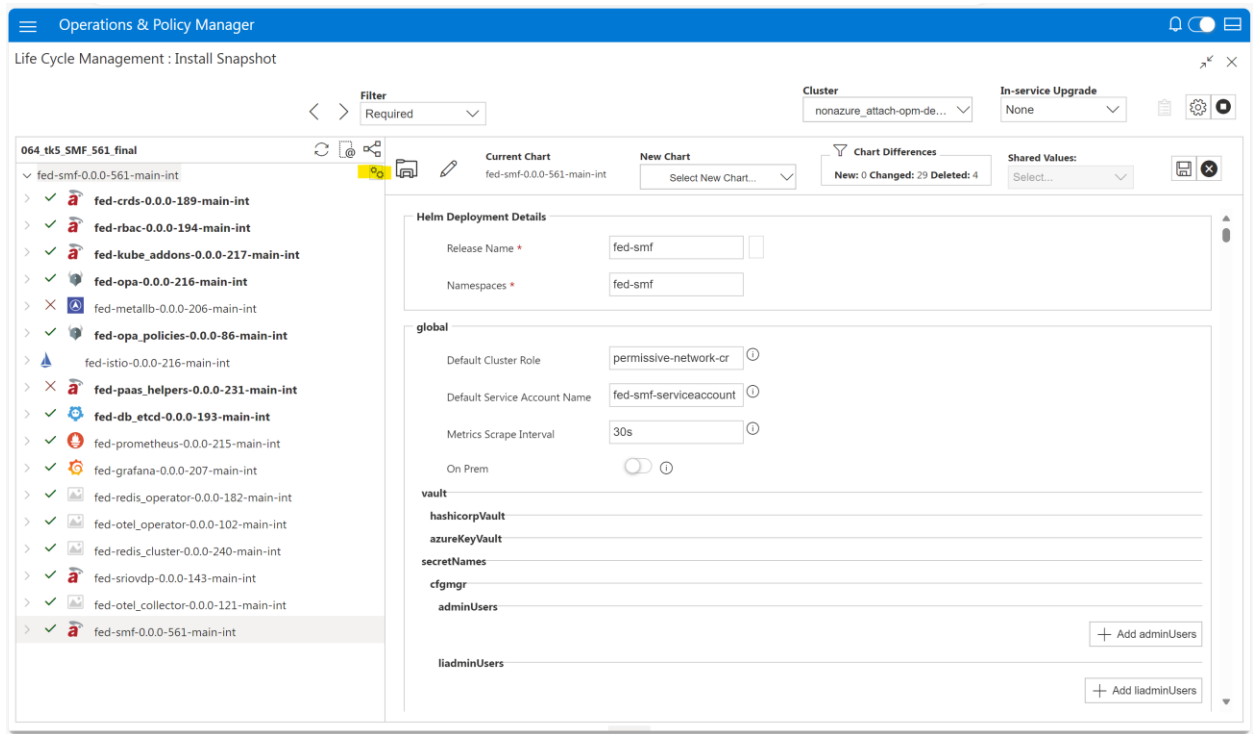


6. Select the desired In-service Upgrade option:

- **None** – Upgrade will follow startorder.json with no In-service Upgrade hooks to ensure minimal impact to services. This is the equivalent of existing snapshot upgrade support in pre-5.1.0 releases.
- **Pause on Failure** – Upgrade will follow upgradeorder.json with in-service upgrade support. If a federation fails to upgrade, the upgrade will roll back the failed federation and pause for user intervention.
- **Rollback on Failure** – Upgrade will follow upgradeorder.json with in-service upgrade support. If a federation fails to upgrade, the upgrade will roll back the failed federation as well as any already upgraded federations to the last installed versions.

7. If upgrading a network function federation, select it from the tree. Click **Update icon (highlighted)** and then select a new chart from your desired repository. The dependent chart versions will be updated based on the selected new network function chart version. Values of the new network function chart are displayed on the right-hand pane.

Figure 83. Update icon



8. Review and update values where applicable by clicking federations in the tree. All and Required tabs view all values and significant fields respectively in a form. **Toggle YAML Expert Mode** shows data in expert mode.

9. Make sure to click **Save updated chart values** icon to accept any edits to any chart, and acknowledge the confirmation. After the changes have been accepted, the new chart is highlighted on the tree. This is also an indication that the changes have been accepted successfully.

10. Click the **Bulk Install Federations** icon to trigger upgrade or install. The charts will be upgraded according to the start order or upgrade order dependent on the In-service Upgrade option selected.

11. Check Notifications pane on the right-hand side on bulk upgrade progress. There will also be an icon in the tree next to the federation to indicate the various status of the federation during the upgrade.

Table 26. Tree Status icon and description

Tree Status Icon	Federation Status Description
	InProgress
	Deployed
	Complete
	Failed / Rollback Failed
	Rollback Complete
	Warning

12. Check Console pane on the bottom for detailed status of each federation, including helm command and responses as well as pod status.

Figure 84. Console pane

The screenshot shows the 'Operations & Policy Manager' console. The main panel displays 'Helm Deployment Details' for a snapshot named 'fed-snap-0.0.0-123-main-int'. Below this, the 'Helm Responses' section shows the output of a Helm upgrade command, indicating that the snapshot was successfully deployed. The 'Pods' table at the bottom shows two pods in a 'Running' state.

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE
istio-cni-node-bongd	1/1	Running	0	18h	192.168.4.9	aks-worker-39185284-vmss000004	<none>
istio-cni-node-dbdw	1/1	Running	0	18h	192.168.0.6	aks-worker-39185284-vmss000001	<none>

The right sidebar contains a 'Notifications' pane with several messages, including a warning about a config group upgrade and several success messages for various components being upgraded successfully.

Update Site Specific Shared Parameters

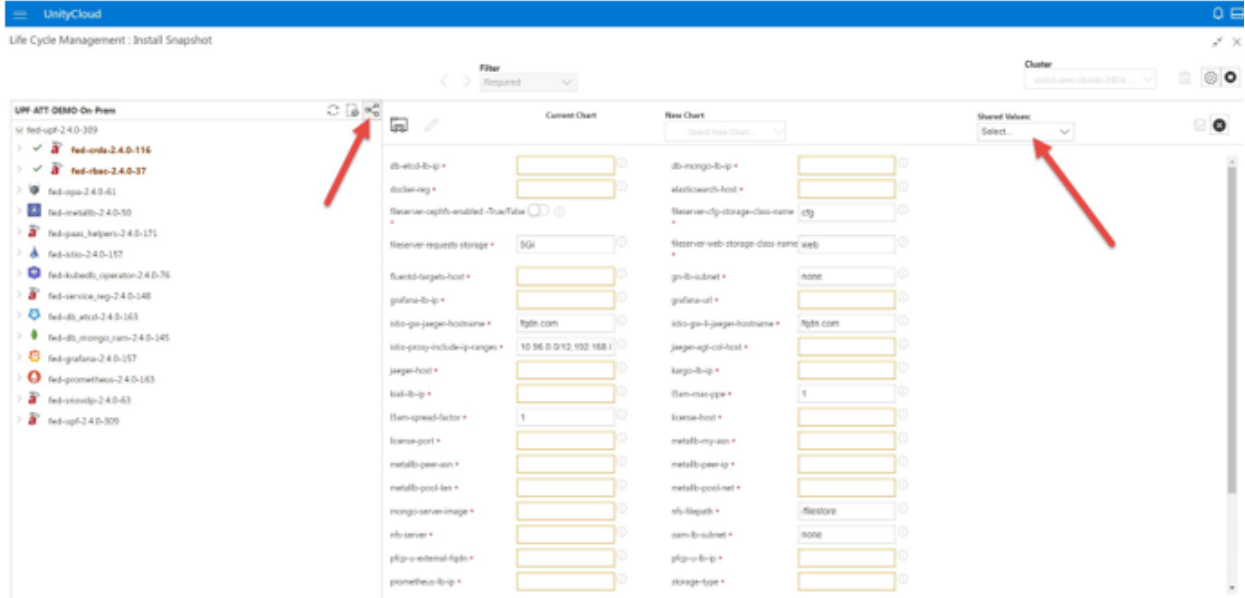
When installing a snapshot, you may update site specific parameters shared across multiple federations. This is helpful when deploying a standard snapshot over multiple sites.

Refer to **Shared Parameters for Network Function Deployment** for details of the shared parameters.

To edit site-specific shared parameters while installing a snapshot:

1. Click the **Site specific values** button on the Install Snapshot screen. The list of shared values appears.
2. Edit the shared values by entering them manually, or by selecting a set of values from an existing configuration set or snapshot under the **Shared Values** dropdown.
3. When finished editing, click **Save updated shared parameter values (floppy disk icon)**, and then proceed with bulk installation.

Figure 85. Site Specific Parameters



ISSU Per-Federation Helm Options Override

When installing a snapshot with ISSU, user can override default options used in helm operations for each federations in the snapshot. The following options can be overridden:

1. **timeout**: the default timeout is loaded from upgradeOrder.json of chart. If not specified, OPM system configuration runtime timeout (default 1200s) is loaded.
2. **debug**: debug is disabled by default.
3. **force**: fed-crds has force enabled by default. Other feds have it disabled by default.
4. **wait**: wait is enabled by default.

Figure 86. Launch per-federation options configuration page

<
>

Filter
 Required ▼

testing
↻ ⌵

▼
fed-kafka-0.0.0-47-rel-5-1-0

>
✓
fed-crds-0.0.0-43-rel-5-1-0

>
✓
fed-rbac-0.0.0-17-rel-5-1-0

>
✓
fed-kube_addons-0.0.0-47-rel-5-1-0

>
✓
fed-opa-0.0.0-44-rel-5-1-0

>
✓
fed-metallb-0.0.0-46-rel-5-1-0

>
✓
fed-opa_policies-0.0.0-47-rel-5-1-0

>
✓
fed-istio-0.0.0-48-rel-5-1-0

>
✓
fed-paas_helpers-0.0.0-50-rel-5-1-0

>
✓
fed-db_etcd-0.0.0-50-rel-5-1-0

>
✓
fed-prometheus-0.0.0-52-rel-5-1-0

>

fed-zookeeper-0.0.0-41-rel-5-1-0

>

fed-kafka-0.0.0-47-rel-5-1-0

Toggle helm chart

force

fed-istio

timeout

force

fed-paas-helpers

timeout

force

fed-db-etcd

timeout

force

Figure 87. Helm Chart options form

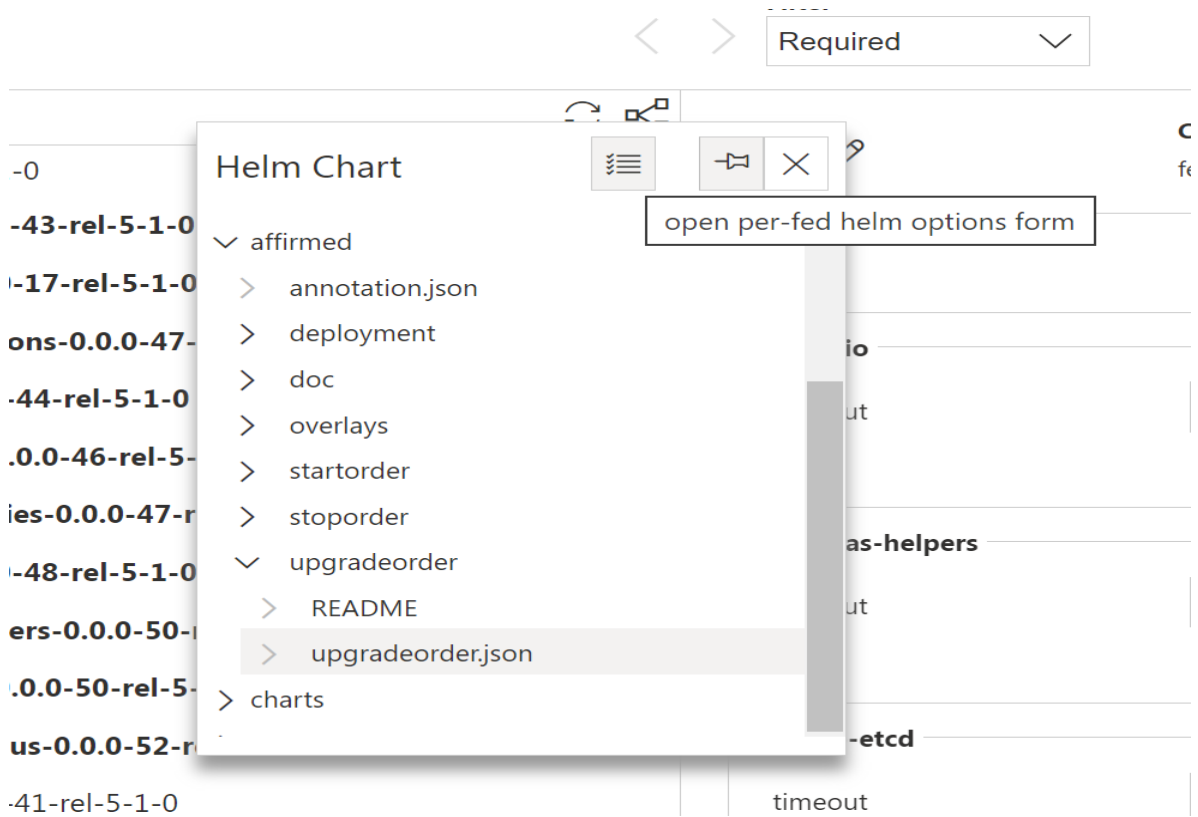


Figure 88. Bulk Fed Install

Life Cycle Management : Bulk Federation Install

Follow the screen shots above to launch the per-fed options page for configuration. Make sure to hit the Save button to save any changes. The saved per-fed options will be used for helm upgrade and helm rollback command for pause-on-failure or rollback-on-failure ISSU invocation.

ISSU Upgrade Dry Run

Before running ISSU using a snapshot, use Dry Run button to test run the ISSU process. Dry Run will follow the same ISSU upgrade process and add --dry-run to the helm upgrade command.

Install Federations Individually

Federation Deployment Order

The following tables list the Affirmed Networks federations in the recommended deployment order, where PaaS components are deployed first, followed by Application federations. Some PaaS

components can run on remote clusters (Elastic, Kibana, Elastalert), but are shown as if being started on the same cluster.

During ISSU, the federations that are grouped together will be upgraded in parallel. In the new installation scenario, they will be serially deployed.

After you have successfully started the application federations, you can add your 3GPP network functions, depending on which functions you are deploying.

Table 27. UnityCloud Deployment Order

Order	PaaS Component/Application
1.	fed-rbac fed-crds
2.	fed-kube_addons
3.	fed-opa
4.	fed-metallb fed-opa_policies
5.	fed-istio
6.	fed-db_etcd
7.	fed-paas_helpers
8.	fed-prometheus fed-grafana fed-kubedb_operator fed-otel_operator fed-redis_operator
9.	fed-infra_dreg fed-otel_collector fed-redis_cluster fed-sriovdp
10.	fed-elastic
11.	fed-rook_ceph
12.	fed-kube_prom
13.	fed-elastalert
14.	fed-kafka_strimzi
15.	fed-kibana
16.	fed-timer

Order	PaaS Component/Application
17.	fed-jaeger
18.	fed-nrf
19.	fed-amf_mme fed-smf fed-upf

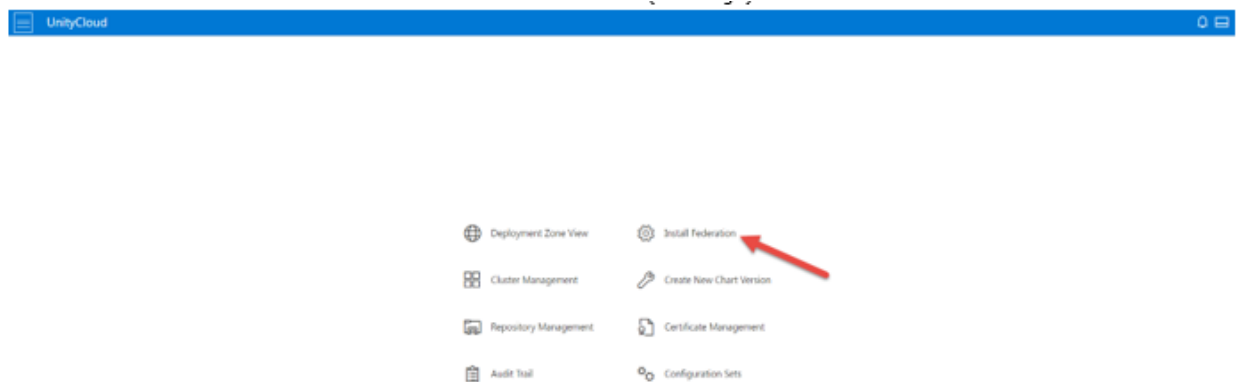
Install Individual Federations

Perform the following steps to install a federation.

1. From the main OPM screen, click **Install Federation**.

Note: Federations can also be managed from a Geographical Map. This section assumes a deployment zone and cluster have already been created.

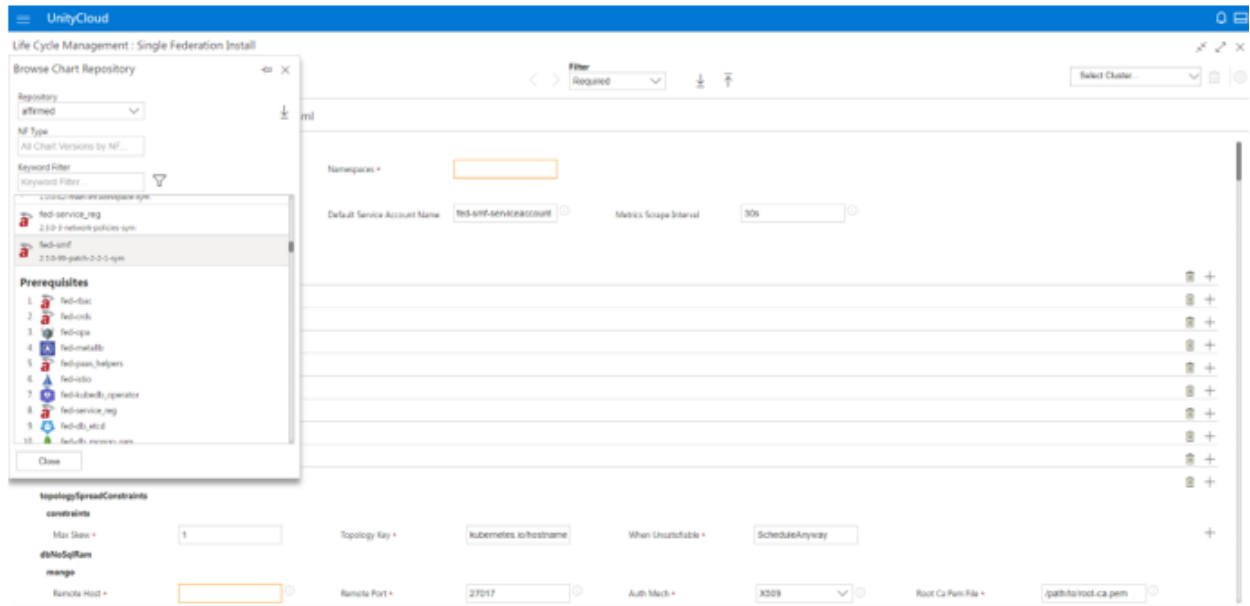
Figure 89. Main OPM Screen



2. From the **Install Federation** screen, select the Helm repository from the drop-down to display a list of charts. To filter the list, enter a filter condition (for example, version number) and click the Render Charts button.

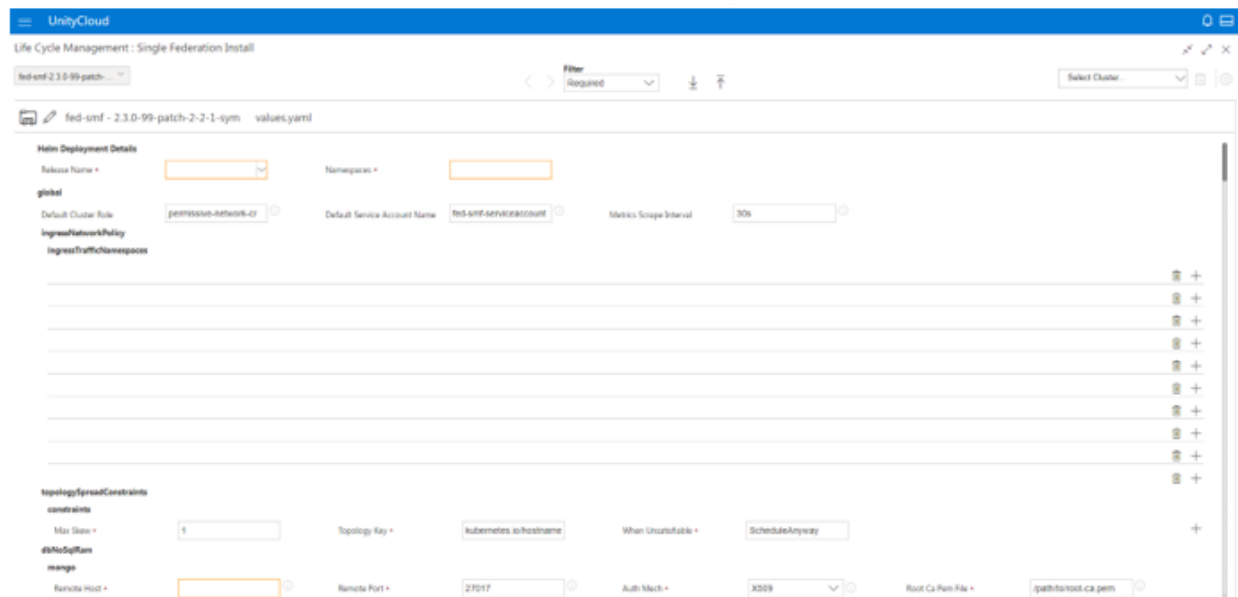
Note: To manage Helm repositories, select Repository Management from the main OPM page. See [Associate Local Helm ChartMuseum with OPM GUI](#).

Figure 90. Install Federation



3. Select a chart from the list and configure the parameters. By default, the system displays the Required parameters from the values.yaml file, which contains the default configuration values for this chart. To display all the configurable values, select the **All** tab.

Figure 91. Helm Chart Selection



The following tables list the required parameters and values for each PaaS component and Application federation. See [Manage Federations with OPM](#) for more information on the recommended deployment order for federations. When installing federations, note the following dependencies:

- `fed-rook_ceph` is optional, but must be installed immediately after `fed-rbac` if it is selected.
- `fed-rbac` must be installed first and deleted last.
- `fed-paas_helpers` must be deleted second to last.
- `fed-postgres` must be installed before `fed-alerta`.
- For all federations that have Grafana enabled (`global.grafana.enabled=true`), the Grafana URL (`global.grafana.url`) must be defined. If the URL is not defined, the federation deployment fails and displays an indicative error.
 - The format for the URL is: `http://[paas-lb-ip]:[grafana-port]`
 - If Grafana is deployed with TLS, the format for the URL is: `https://[paas-lb-ip]:[grafana-port]`

Caution: In order to access the network function applications with the CLI, you must define "admin", "oper", "readonly", "liadmin", and "lioper" group users in `pod-cfgmgr` for each application, with either public keys and/or certificate authority keys and/or SHA-encrypted passwords, and/or configure one or more TACACS servers.

Table 28. fed-rbac Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.

Table 29. fed-crdns Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.

Table 30. fed-rook_ceph Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.

pod-rook_ceph

Parameter	Value
cephImageRepoPath	The repository for the Ceph image.
cephImage	The Ceph image. (Example: ceph:v17.2.6)
deviceFilter	A Golang regular expression to match the short kernel names of devices that allows Rook to select which block devices to set up Ceph on.
dashboardEnabled	Boolean control on whether Rook deploys the Ceph Dashboard.
storageClassName	Name of the storage class that will be used to dynamically create a persistent volume with CephFS.
service > loadBalancer > enabled	Boolean control on whether an external load balancer is used to expose the Ceph dashboard.

Table 31. fed-kube_addons Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.

Table 32. fed-opa Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global > envVarsFedUniqueName	Enter a globally unique name.

Table 33. fed-metallb Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
pod-metallb > config	
peerAddress	Enter BGP router IP address.
peerAsn	Enter BGP router ASN.
MyASN	Enter cluster BGP ASN number.

Parameter	Value
Addresses	Enter pool of addresses to be used for LB.
AggregationLength	Must match prefix for Addresses.

Table 34. fed-paas_helpers Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
fileserver > ui > loadBalancerIP	IP address for optional load balancer service.
fileserver > ui > loadBalancerPort	Port number for optional load balancer service. Default is 85.
fileserver > nfs > Remote Host	Enter OPM IP address.
fileserver > nfs > Remote Path	Input desired path for the chart. Default is/filestore. Path must already exist on remote server.
grafana > enabled	Toggle On.
grafana > url	http://<metalLB_PaaSIP>:3000
envVarsFedUniqueName	Enter a globally unique name.
logging > enabled	Toggle On.
logging > elastic > IP	<metalLB_PaaSIP>
pod-fluentd_operator > fluentd > Number Of Workers	20
pod-jaeger-agent	
agent > collector > Host	<metalLB_PaaSIP>
service > loadBalancerIP	<metalLB_PaaSIP>
service > loadBalancerFQDN	<metalLB_PaaSIP>
pod-cores_mgr > coresStorage > Type	Enter storage type.

Table 35. fed-otel_operator Required Parameters

Parameter	Value
Release Name	Enter the release name.

Parameter	Value
Namespaces	Enter the namespace for this federation.

Table 36. fed-otel_collector Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.

Table 37. fed-istio Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
Outbound CIDR	When using non-default pod and services CIDR, this variable must set with the appropriate IP ranges. For any additional external traffic routing via Istio service mesh, the required IP range also must be appended to the existing range. Example: <pre>--set pod-istio.istioproxy.includeIPRanges="10.96.0.0/12,192.168.0.0/16,10.15.25.0/24"</pre> Where, 10.96.0.0/12 is the K8 service CIDR 192.168.0.0/16 is the pod CIDR 10.15.25.0/24 is the external peer IP, which requires routing configuration using external host API invocation by applications.
grafana > enabled	Toggle On.

global

grafana > url	http://<metalLB_PaaSIP>:3000
multusDeployed	Set to true if fed-istio is going to be deployed with fed-upf, fed-agf, or fed- n3iwf. Set to true if fed-istio is going to be deployed with fed-amf_mme and this fed is going to use multus for providing service. See Install Istio for additional details.

pod-istio

paasSelector > Jaeger Host Name	Enter the <Kubernetes master ip> if Jaeger is installed locally. Otherwise, enter the IP address of the Jaeger server.
---------------------------------	--

Parameter	Value
paasSelector > Grafana Host Name	Enter the <Kubernetes master ip> if Grafana is installed locally. Otherwise, enter the IP address of the Grafana server.
kialiService > loadBalancerIp	<metalLB_PaaSIP>
kialiService > loadBalancerFqdn	<metalLB_PaaSIP>
istioinit useMaistraUbi8Image	Set to true if deploying to a host running Red Hat Enterprise Linux 8.x or higher.

Table 38. fed-postgres Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
service > type	NodePort (or LoadBalancer for multi-node cluster)

Table 39. fed-kubedb_operator Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
pod-kubedb_catalog > Mongo Server Image	Image to be included.

Table 40. fed-redis_operator Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.

Table 41. fed-prometheus Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.

Manage Federations with OPM

Parameter	Value
global	
enVarsFedUniqueName	Enter a globally unique name.
logging > Enabled	Toggle On.
logging > elastic > IP	<metalLB_PaaSip>
grafana > Enabled	Toggle On.
grafana > Url	http://<<metalLB_PaaSip>>:3000
pod-prometheus	
service > External Ip Address	<metalLB_PaaSip>
service > loadBalancer > loadBalancerIp	<metalLB_PaaSip>
service > loadBalancer loadBalancerFqdn	<metalLB_PaaSip>
thanosQuery > service externalIpAddress	<metalLB_PaaSip>

Table 42. fed-alerta Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
pod-alerta	
Keyfile	Replace with the host.key file. Default is a self signed certificate key.
Certfile	Replace with the host.crt file. Default is a self signed certificate file.
Alerta Admin Password	Default password for all alerta admin users.
Alerta Admin Key	Default API key for admin users.
alertaAdminUsers	Use to create admin users.
postgresql > Host	Postgresql IP address.
postgresql > Postgresql Username	Postgresql username.
postgresql > Postgresql Password	Postgresql password.

Note: Ensure the `host.key` and `host.crt` files match the ones used for setting up `fed-elastalert`.

Table 43. fed-elastic Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
loadBalancerSourceRanges	
envVarsFedUniqueName	Enter a globally unique name.
logging > enabled	Toggle On.
grafana > enabled	Toggle On.
grafana > url	<metalLB_PaaSIP>
grafana > prometheusInfo > user	
grafana > prometheusInfo > password	
pod-elastic	
Pod JVMOpts	-Xms10240m -Xmx10240m
replicas > Master replicas > Client	3
replicas > Data	5
externalStorage > Storage Provider Type	Enter the storage provider type.
storageSize > Master storageSize > Client	20Gi
storageSize > Data	150Gi
service > Exposed IP	<<metalLB_PaaSIP>>
service > loadbalancer > enabled	Toggle On.
service > loadbalancer loadbalancerIp	<metalLB_PaaSIP>
service > loadbalancer loadbalancerIp	<metalLB_PaaSIP>

Table 44. fed-db-etcd Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
envVarsFedUniqueName	Enter a globally unique name.
logging > Enabled	Toggle On.
logging > elastic > IP	<<metalLB_PaaSIP>>
logging > elastic > port	9200
grafana > Enabled	Toggle On.
grafana > Url	http://<<metalLB_PaaSIP>>:3000
pod-etcd_operator	
service > LoadBalancer > enabled	Toggle On.
service > loadBalancer > loadBalancerIp	<metalLB_PaaSIP>
service > loadBalancer > loadBalancerFqdn	<metalLB_PaaSIP>
service > loadBalancer loadBalancerSharingKey	paas

Table 45. fed-elastalert Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
envVarsFedUniqueName	Enter a globally unique name.
logging > elastic > IP	<metalLB_PaaSIP>
Elastalert Ext IP	<<metalLB_PaaSIP>>
pod-elastalert	
service > loadBalancer > loadBalancerIp	<<metalLB_PaaSIP>>

Parameter	Value
service > loadBalancer loadBalancerFqdn	<metalLB_PaaSIP>

Table 46. fed-jaeger Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
envVarsFedUniqueName	Enter a globally unique name.
logging > enabled	Toggle On.
logging > elastic > IP	<metalLB_PaaSIP>
grafana > enabled	Toggle On.
grafana > url	http://<metalLB_PaaSIP>:3000
pod-jaeger	
storage > elasticsearch > Host	<metalLB_PaaSIP>
storage > collector > service exposedExternalIP	<metalLB_PaaSIP>
storage > collector > service loadBalancer > loadBalancerIp	<metalLB_PaaSIP>
storage > collector > service > loadBalancer > loadBalancerFqdn	<metalLB_PaaSIP>
storage > query > service exposedExternalIP	<metalLB_PaaSIP>
storage > query > service >loadBalancer loadBalancerIp	<metalLB_PaaSIP>
storage > query > service > loadBalancer loadBalancerFqdn	<metalLB_PaaSIP>

Table 47. fed-grafana Required Parameters

Parameter	Value
Release Name	Enter the release name.

Manage Federations with OPM

Parameter	Value
Namespaces	Enter the namespace for this federation.
global	
logging > enabled	Toggle On.
logging > elastic > IP	<metalLB_PaaSIP>
grafana > enabled	Toggle On.
grafana > url	http://<<metalLB_PaaSIP>>:3000
pod-grafana	
debug	Set to false. Setting to true may have a significant impact on performance.
service > External Ip Address	<metalLB_PaaSIP>
service > loadBalancer > loadBalancerIp	<metalLB_PaaSIP>
service > loadBalancer loadBalancerFqdn	<metalLB_PaaSIP>

Table 48. fed-kafka-strimzi Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
envVarsFedUniqueName	Enter a globally unique name.
grafana > url	http://<<metalLB_PaaSIP>>:<port>
pod-kafka_strimzi	
broker > externalLBip	<metalLB_PaaSIP>
broker > staticLBips	<metalLB_PaaSIP>
	<metalLB_PaaSIP>
	<metalLB_PaaSIP>
Mirrormaker > enabled	<boolean>

Table 49. fed-kibana Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
envVarsFedUniqueName	Enter a globally unique name.
logging > Enabled	Toggle On.
logging > elastic > ip	<metalLB_PaaSIP>
pod-kibana	
Kibana Ext IP	<metalLB_PaaSIP>
Elastic Ext IP	<metalLB_PaaSIP>
service > loadBalancer > loadBalancerIp	<metalLB_PaaSIP>
service > loadBalancer loadBalancerFqdn	<metalLB_PaaSIP>

Table 50. fed-kube_prom Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global > envVarsFedUniqueName	Enter a globally unique name.
pod-kube_prom	
service > loadBalancer > loadBalancerIp	<metalLB_PaaSIP>
service > loadBalancer loadBalancerFqdn	

Table 51. fed-redis_cluster Required Parameters

Parameter	Value
Release Name	Enter the release name.

Manage Federations with OPM

Parameter	Value
Namespaces	Enter the namespace for this federation.
Global > topologySpreadConstraints	Enter two constraints. 1. For Kubernetes node topology (example label: kubernetes.io/hostname) 2. For the Zone or Rack cluster topology (example label: topology.kubernetes.io/zone)

Table 52. fed-sriovdp Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global > Type	Select the deployment type. Possible values: vm-generic beremetal-generic. Default value: vm-generic. To view or edit the default sriovdp parameters, select None in the Filter dropdown list at the top of the OPM window.
global > registry > docker > Repo Path	Displays the Registry Path.
sriovdpVmConfigMap	
resourceList > Resource Name	Resource name. Default values: sr1 / sr2.
resourceList > selectors > drivers	SR-IOV device drivers. Default value: vfio-pci.
resourceList > selectors > pciAddresses	PCI addresses of the virtual function (VF). Note: This is required only for the Virtual Function (VF) deployment type (vm-generic).
sriovdpBareMetalConfigMap	
resourceList > Resource Name	Resource name. Default values: sr1 / sr2.
resourceList > selectors > drivers	SR-IOV device drivers. Default value: vfio-pci.
resourceList > selectors > pfNames	Physical function (PF) names for the device. Note: This is required only for the Physical Function (PF) deployment type (beremetal-generic) .

Note: The OPM Configuration screenshot and the configmap sinppets given below are only examples for illustrative purpose. The **pciAddress** values and **pfName** values must be adapted for the specific deployment scenario.

Figure 92. Required Parameters for sriovdpp Deployment Type - Virtual Function (vm-generic)

The screenshot shows the OPM web interface for configuring a sriovdpp deployment. The top bar indicates the current release is 'fed-sriovdp - 3.3.0-4-rel-3-3-0-cbhasin.rel-3-dbg' and the file is 'values.yaml'.

Helm Deployment Details

Release Name: Namespaces:

global

Type: **vm-generic** Sriov Port Prefix:

registry

docker

Repo Path:

sriovdppVmConfigMap

resourceList

Buttons:

Resource 1

Resource Name:

selectors

drivers

pciAddresses

Buttons:

Resource 2

Resource Name:

selectors

drivers

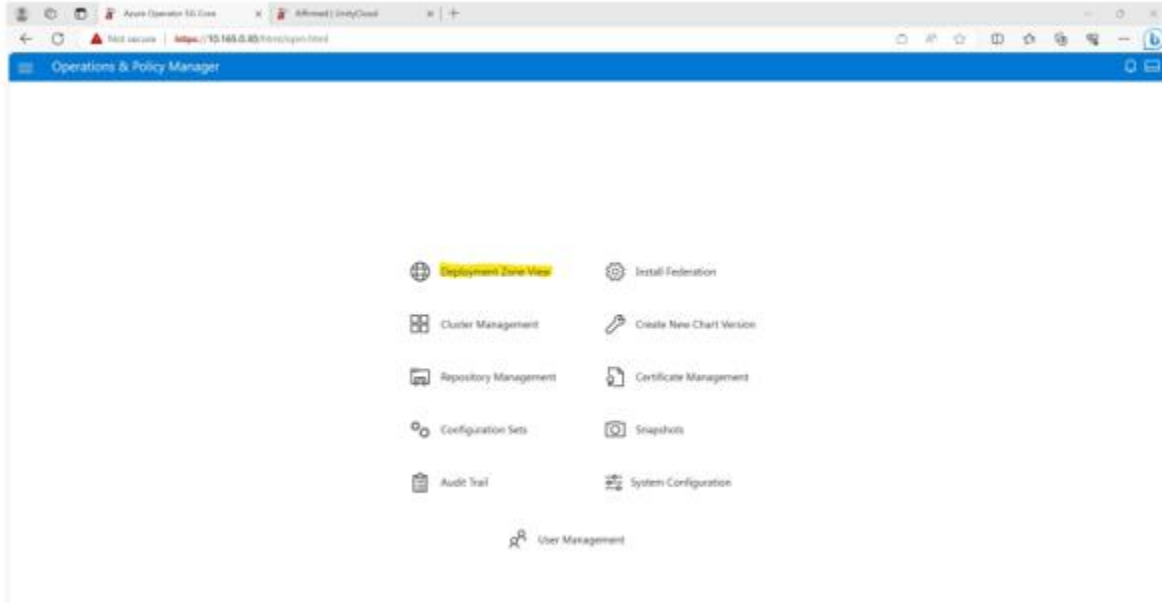
pciAddresses

Buttons:

Upgrade Individual Federations

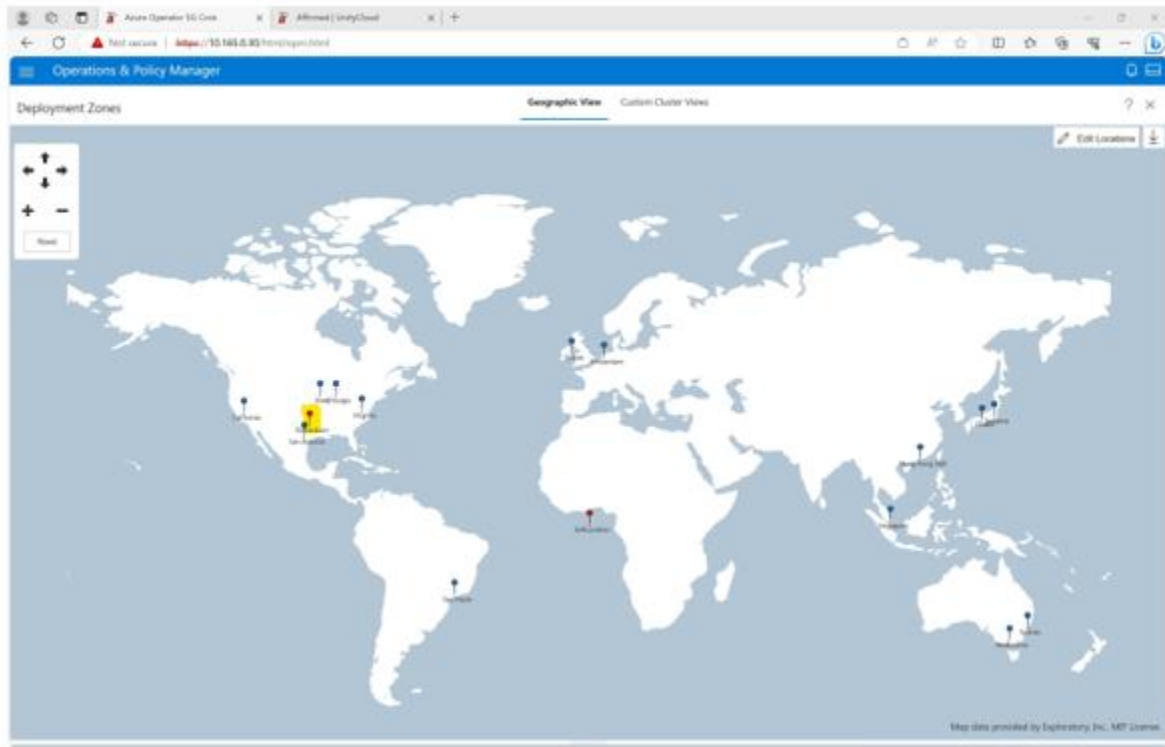
1. Login to OPM IP and click on **Deployment Zone View**.

Figure 93. Deployment Zone Review



2. Click on the deployment zone of the cluster.

Figure 94. Deployment Zone



3. Click on the federation to upgrade.

Figure 97. Select New Chart

Operations & Policy Manager

Life Cycle Management: Upgrade Federation

Repository: ADO Current Chart: fed-and-rmme-4.3.0-26-r1 New Chart: **Select New Chart...**

Filter: Required Cluster: Management Cluster - v1.0.0

Release Name: fed-and-rmme Namespace: fed-and-rmme

global

Default Cluster Role: permissive-network-cr Default Service Account Name: fed-and-rmme-serviceaccount Enable Anvil Function: On

Enable Anvil Function: On

Enable (32): On

deployment

IngressNetworkPolicy: Enabled

IngressTrafficNamespaces: Enabled

logging

Enabled: On

buffer

Buffer Limit: 512M Retry Wait: 180s Max Retries: 10

alerts

Add Targets: + Add addTargets

forward

Host: 10.165.178.227

alerts

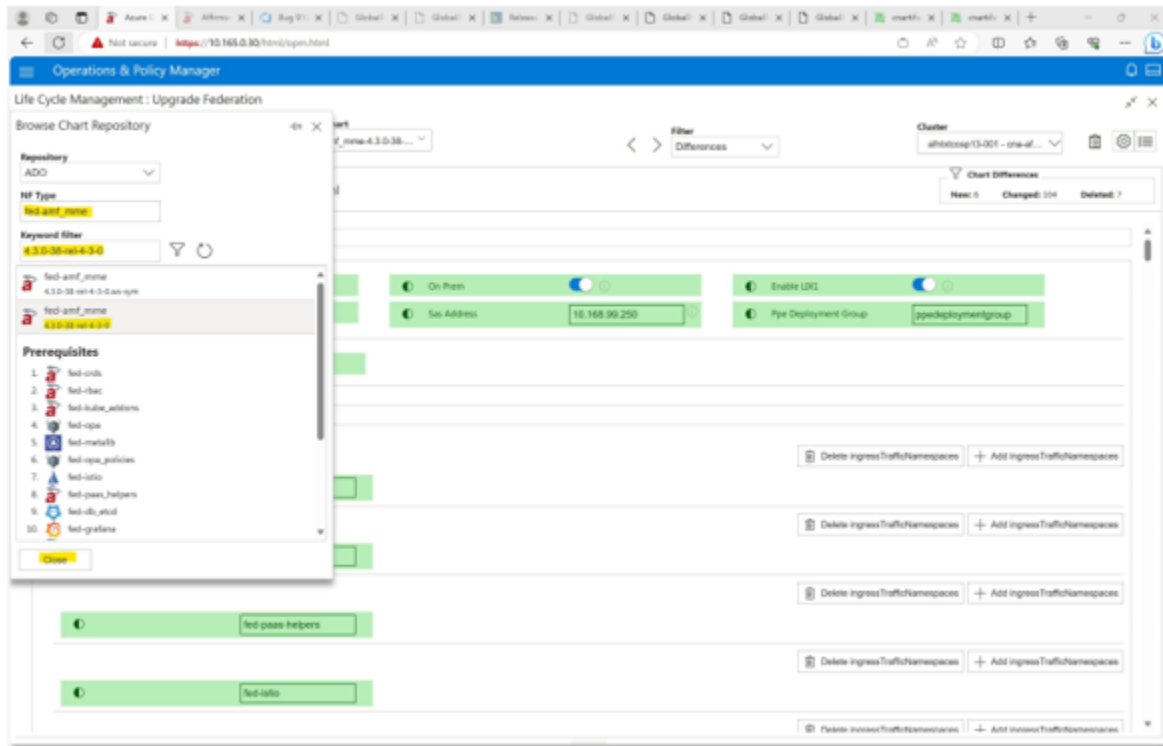
Host: 10.165.178.224

realtime

realtime

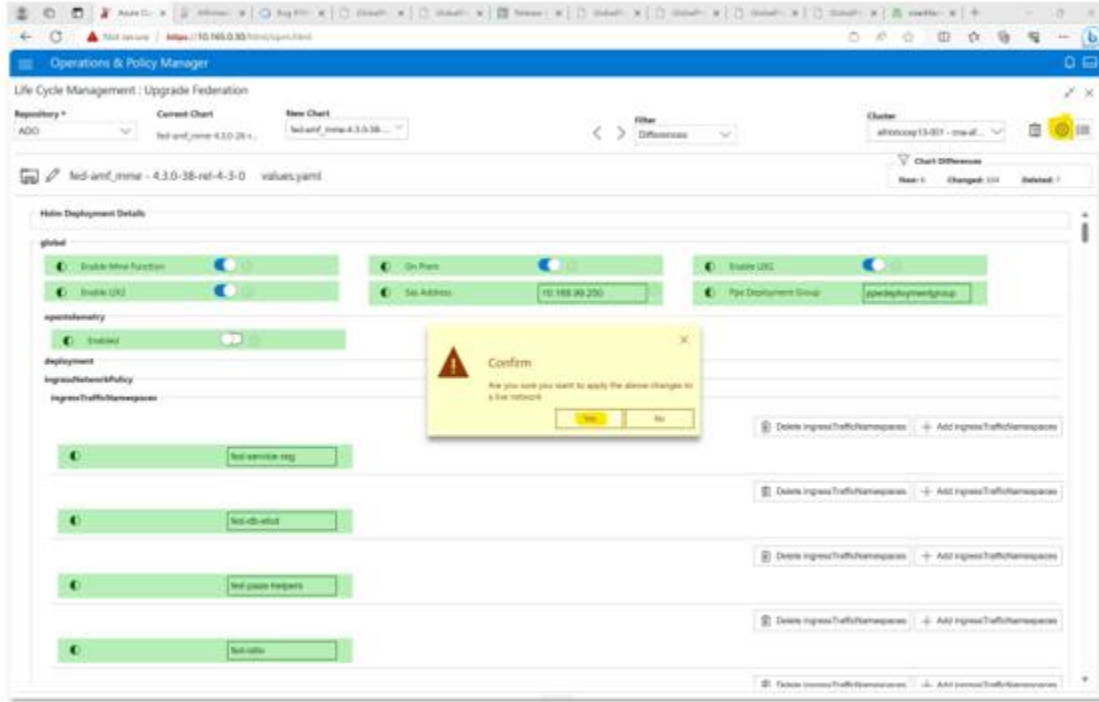
6. Fill in the NF type and the keyword filter to get the desired image.

Figure 98. NF Type

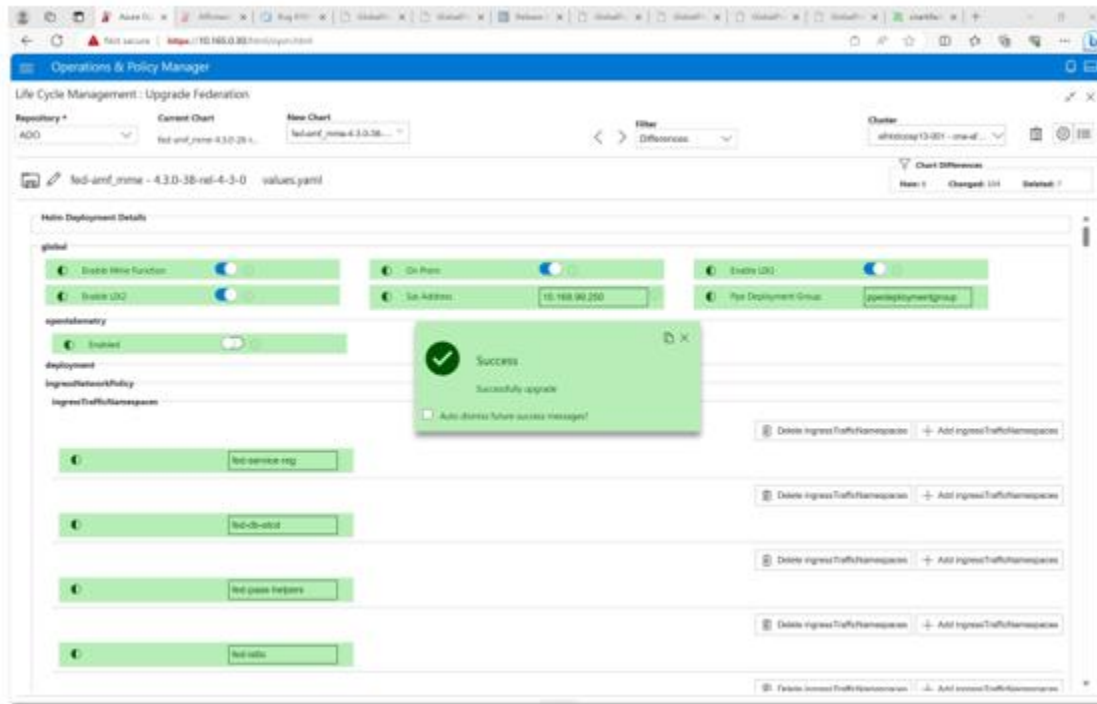


7. Click on the gear wheel icon to start the upgrade.

Figure 99. Start the Upgrade



8. Following message is seen on successful upgrade.

Figure 100. Upgrade Success Message

Delete a Federation

To delete a federation:

1. From the main OPM screen, click **Deployment Zone View**.
2. Select the deployment zone containing the federation you want to delete.
3. In Cluster View, find the federation you wish to delete and select ... > **Delete Federation** to delete the federation.

Delete All Federations in a Cluster

To delete all federations in a cluster:

1. From the main OPM screen, click **Deployment Zone View**.
2. Select the deployment zone containing the cluster with federations you want to delete.
3. In Cluster View, find the cluster and select ... > **Delete All Federations**.
4. In the Select Repository dialog that appears, select your chart repository and then click **Delete Federations** to delete all federations from the cluster.

Note: By default, `fed-rook_ceph`, `fed-rbac`, and `fed-crds` are not deleted by this operation. To delete them as well, you can disable **Skip deletion of rook-ceph, rbac, crds** in the Select Repository dialog.

Delete Persistent Volumes

If a deleted federation made use of `fed-rook_ceph`, you may also delete its persistent volumes (PVs):

1. Log into the OPM server using SSH.
2. For each persistent volume, run the following command:

`kubect1 delete pv <pv_name>`, where `<pv_name>` is the persistent volume to be deleted.

Caution: Deleting a PV is a destructive operation and will result in the loss of data associated with the federation. By default, these federations use Kubernetes Persistent Volumes with the "Retain" reclaim policy. When a federation is uninstalled, its PVCs are deleted but the corresponding PVs remain in a "Released" state and must be manually deleted.

Identify Persistent Volumes

Identifying persistent volumes (PVs) requires prior knowledge of which federations use `fed-rook_ceph`. To identify PVs bound to the PVCs of an existing federation, you can run the following command:

`kubect1 -n <federation_name> get pvc`

Multi-Master Kubernetes

One of the issues with first-stage deployments of Kubernetes clusters is that they contain a single master node. While workloads have been delegated to Kubernetes to ensure applications are up and versioned, the control plane itself has a single point of failure. To mitigate this risk, a multimaster setup can assist in ensuring the control plane continues to run in the event that a master node fails.

For UnityCloud, we have standardized the deployment of multi-master using Heat templates. This method uses an OpenStack VIP as the control plane endpoint, as well as haproxy service running in the Master VMs, to load-balance API calls among the multiple masters in round-robin fashion. Because the workers are joined to the cluster by the VIP address, a master node outage will not halt workloads.

Component Descriptions

VIP Address

The VIP address serves as the endpoint to access the Kubernetes control plane, rather than the IP address of the single Kubernetes master node.

Note: The OpenStack VIP facilities are not used; the VIP is handled through the HAProxy and keepalive messages on the Master VMs.

Master K8s VM(s)

We have chosen HAProxy as the standard load balancer for multi-master Kubernetes in UnityCloud. To standardize the configuration and management of this load balancer, we have added this service in the Master VMs. This load balancer is set up to forward API requests to each of the Kubernetes master nodes in round-robin fashion. The HAProxy service binds the VIP address with Keepalived. This entails that there is a primary HAProxy load balancer and a backup HAProxy load balancer. Load Balancer Health Check Behavior Health checks in HAProxy occur every 10 seconds. This timing avoids the misinterpretation of Kubernetes latency as an outage. When the Kubernetes Master is down, the health check occurs every 5 seconds. Two consecutive failed health checks separated by 10 seconds identify a Kubernetes master as down. Two consecutive successful health checks separated by 5 seconds identify a Kubernetes master as back up. The HAProxy Primary server and HAProxy Secondary server are configured in Active-Passive mode. This means that HAProxy Primary will take all requests as long as it is up, and HAProxy Secondary will only receive requests if HAProxy Primary is down.

All Kubernetes masters are configured in Active-Active-Active mode. This means that as long as each master node is up, requests will be sent to each master node in round-robin fashion.

Kubernetes Master(s) Instead of a single master node hosting the Kubernetes control plane, the control plane is shared among multiple master nodes. The method of etcd delivery within Kubernetes is currently stacked, meaning every Kubernetes master node has its own etcd instance.

Note: The recommended number of Master nodes are three.

Kubernetes Worker(s)

Rather than joining a single master node as the Kubernetes control plane, each worker node is joined to the VIP address which represents the Control Plane endpoint.

Allow VRRP Protocol in OpenStack Environment

In order to bind VMs to the VIP address chosen, the VRRP Protocol must be permitted in the default security group in OpenStack:

```
$ADMIN_PROJECTID=$(openstack project list|grep admin|awk {'print $2'})

$SEC_GROUPID=$(neutron security-group-list -c id -c tenant_id -c name -f
value|grep default|grep $ADMIN_PROJECTID|awk {'print $1'})

$openstack security group rule create $SEC_GROUPID --protocol 112
```

Generate Kubernetes Certificate Key

In single master node clusters, the default operation during Kubernetes cluster creation is to generate random, new certificates. However, with multiple master nodes, all must share the same certificate. Rather than copying the certificate files manually from one VM to another, Kubernetes provides a mechanism to specify a “Certificate Key”, in which Kubernetes will exchange the Certificate Files among the multiple Master Nodes automatically.

To generate this random 32-byte hexadecimal key, enter the following Bash command:

```
$ openssl rand -hex 32
```

This key should be set aside to be passed into the Heat deployment of the cluster. It will allow for Kubernetes to generate and share the necessary certificate files among the Master nodes.

Create Certificate Files in Predefined Path

For the cluster to be able to communicate with an internal Docker registry and with an OpenStack environment, the Heat templates use a predefined path for these certificates to be defined.

- `/vnfmgr/certs/dreg.crt` (Docker Registry Certificate for Kubernetes to pull AN Docker Images from.)
- `/vnfmgr/certs/osca.crt` (OpenStack CA Certificate for AN to run any needed OpenStack API Calls.)

If either of these are unneeded, the files must still both be created.

Deploy Multi-Master Kubernetes Using OPM

Follow the steps described in [Deploy Kubernetes Cluster](#). Enable Multi Master and configure the following parameters:

Table 53. Multi Master parameters

Parameter	Description
K8 Master Lb Vip	Virtual IP address to access load-balanced Kubernetes master node through the VMs.
K8 Cert Key	Enter the Kubernetes certificate key.
K8 Public Vip	Kubernetes public virtual IP. Only used if using more than one network.
Vip Mgmt Net Id	OpenStack management ID.
Vip Ext Net id	Only used if using more than one network.
Haproxy Combine	Set to True to enable deployment of HA [1]Proxy with the master nodes. Note: Setting to False is not supported.

Note: Haproxy Combine is set to True, setting to False has been Deprecated along with running HAproxy VMs.

Table 54. K8 Extra Master parameters

Parameter	Description
VM Name	A unique name for the VM.
VM Disk Name	A unique name for the VM disk. This may match the VM name.

Parameter	Description
Vm Ip	IP address.
K8 Master Ip	Kubernetes master IP address. When using a single network, this should match Vm Ip.
Vm Flavor	OpenStack flavor name.
Vm Zone	OpenStack zone to deploy VM.
Disk Size	Disk size. minimum 20G.
Add Net	Enable to add additional networks.

Note: The recommended number of Kubernetes Master Nodes is three. The first is already defined as part of the cluster parameters. Use Extra Masters to define two additional master nodes. Click the Plus Button (+) to configure additional master nodes.

Communicate with Multi-Master Cluster after Deployment

There are a few ways to run kubectl commands against a multi-master cluster after it has been created:

- **SSH into a Master node to execute the kubectl commands directly.**
- **Retrieve admin.conf file from a Master Node, copy to any server to run kubectl commands from:**
 - a. SSH as root user into a Master Node.
 - b. Copy `/home/kuberadmin/admin.conf` to any server to run kubectl commands against the cluster. This server only needs network access to the VIP address of the cluster.
- **Retrieve user Kubernetes Config File from a Master Node, copy to any server to run kubectl commands from:**
 - a. SSH as non-root user into a Master Node.
 - b. Copy file from `$KUBECONFIG` to any server to run kubectl commands against the

Rejoin master node to Multi-Master Cluster

In a multi-master cluster, there are usually three masters. If one of the masters is deleted or corrupt, follow these instructions to add a new master node into the existing cluster.

1. Make sure Master VM is deleted in Openstack

```
# openstack stack delete -y <masternode1>
```

2. SSH to the Kubernetes master, then execute the following:

```
# kubectl delete node <masternode1>
```

Where <masternode1> is the master which was removed.

```
# kubeadm init phase upload-certs --upload-certs -certificatekey <key>
```

The key can be found by looking at /var/log/configParser.log

```
# kube# kubectl exec -it etcd-opm-trunk-k8-master-1-bbc807-4b2c09f7b029c1b8 sh -n kube-system
```

```
# etcdctl --cacert /etc/kubernetes/pki/etcd/ca.crt -cert /etc/kubernetes/pki/etcd/peer.crt -key /etc/kubernetes/pki/etcd/peer.key member list
```

```
# etcdctl --cacert /etc/kubernetes/pki/etcd/ca.crt -cert /etc/kubernetes/pki/etcd/peer.crt -key /etc/kubernetes/pki/etcd/peer.key member remove <masternode1 id>
```

Where <masternode1 id> is the Master that was removed and shown in previous cmd

```
# exit
```

3. Recreate the master node. This can be done either by using OPM GUI, or by logging into the OPM VM with SSH and executing the following:

```
# kubectl get pods -n vnfmgr
```

```
# kubectl exec -it vnfmgr-osrm-1-xxxxxxx bash -n vnfmgr
```

```
# cd /root/heat
```

```
# source /root/heat/stack.rc;openstack stack create -f json demo-k8-master-1-dcc807-4b2c09f7b029c1b8 -t /root/heat/CNA_platform_all.yaml -e /root/heat/demo-k8-master-1-dcc807- 4b2c09f7b029c1b8.env
```

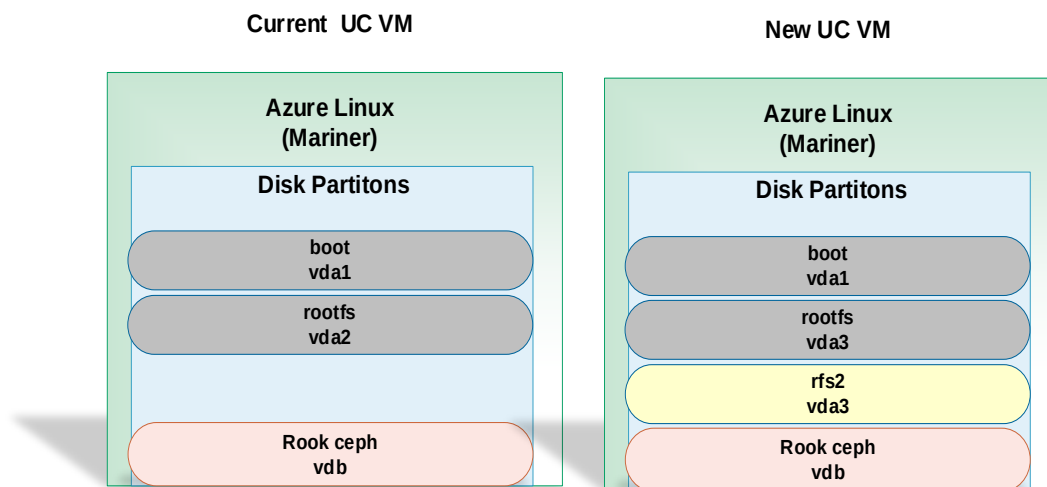
demo-k8-master-1 is replaced with your master name.

VM ISSU

Starting in UC 5.1 the VMs can now be upgraded using in-service software upgrade feature. This feature is on by default. This means, when the VM is created an extra disk partition is created which allows for future upgrades to be installed and switched to.

To achieve this, **the VM will now need twice as much disk space allocated to create a disk partition equal to the installed root disk partition desired.**

Figure 101. Disk Partition



This only applies to the root disk and not any additional disk added for rook-ceph in vm-platform or /filestore in vm-opm.

Once the VMs are deployed with this feature, future upgrade images (seed files) can be applied to the VM using the steps below. The example below will use a 5.1.1 upgrade to show the in-service upgrade will work.

For the OPM GUI, you would put in 200GB VM disk size if you wanted the VM to have a 100G root disk(vda2). This would also create an upgrade partition (vda3) of 100GB in size.

Download seed files

These files are tarballs that have all the files needed to upgrade the VM image from OS upgrades to Kubernetes upgrades.

From the same location that qcow2 images are downloaded, you can also download seed files tarballs.

For the example below you would download **ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar** and then copy to all the VMs that will be upgraded. The files needs to be copied to this dir / .

Upgrade VM-PLATFORM

To use in-service upgrade, your Kubernetes cluster should have at least 3 Master running.

This example show 3 masters and 3 workers.

Steps:

On Master 2

1. ssh to Master 2
2. # kubeadm init phase upload-certs --upload-certs --certificate-key <using same key as before>

The key can be found by looking at /var/log/configParser-<xxx>.log

This is one of the Heattemplate values passed in k8_cert_key.

You can also grep the configParser.log for certificate-key

When a Kubernetes cluster is created, this cert is created to allow nodes to join the cluster. This cert **expires in 2 hours**. To renew this cert and restart the 2-hour window,

The cmds above needs to be done once per cluster.

3. # kubectl drain --ignore-daemonsets --delete-emptydir-data <master 1 name>
4. # kubectl delete node <master 1 name>
5. # kubectl get pods -n kube-system | grep etcd
6. # kubectl exec -it <etcd pod for master 2 from cmd above> sh -n kube-system
7. # etcdctl --cacert /etc/kubernetes/pki/etcd/ca.crt --cert /etc/kubernetes/pki/etcd/peer.crt --key /etc/kubernetes/pki/etcd/peer.key member list
8. # etcdctl --cacert /etc/kubernetes/pki/etcd/ca.crt --cert /etc/kubernetes/pki/etcd/peer.crt --key /etc/kubernetes/pki/etcd/peer.key member remove <master 1 id from cmd above>

On Master1

9. SSH to Master 1
10. Download ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar to / , if you haven't already.

11. # tar -xvf **ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar** -C /

Produces:

- seed- platform-5.1.1-1-rel-5-1-1..tgz
- ucUpgrade.py
- ucUpgraderConfig.json

12. Modify /ucUpgraderConfig.json to configure how you want your VM to be upgraded.

13. # /ucUpgrader.py -f /ucUpgraderConfig.json

a. Wait until “**Successfully Upgraded VM**” is display on console.

14. # reboot

15. # tail -f /var/log/configParser-2024-xx-xx-xx-xx-xx.log

On Master2

16. SSH to Master 2

17. # kubectl get nodes -o wide

Make sure nodes have a Ready status

18. # kubectl get pods -o wide | grep master-1

Make sure pods have a Running status and all containers are ready per pod.

On Master1

19. SSH to Master 1

20. # kubectl drain --ignore-daemonsets --delete-emptydir-data <master 2 name>

21. # kubectl delete node <master 2 name>

22. # kubectl get pods -n kube-system | grep etcd

23. # kubectl exec -it <etcd pod for master 1 from cmd above> sh -n kube-system

24. # etcdctl --cacert /etc/kubernetes/pki/etcd/ca.crt --cert /etc/kubernetes/pki/etcd/peer.crt --key /etc/kubernetes/pki/etcd/peer.key member list

25. # etcdctl --cacert /etc/kubernetes/pki/etcd/ca.crt --cert /etc/kubernetes/pki/etcd/peer.crt --key /etc/kubernetes/pki/etcd/peer.key member remove <master 2 id from cmd above>

On Master2

26. SSH to Master 2

27. Download ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar to / , if you haven't already.

28. # tar -xvf **ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar** -C /

Produces:

- seed- platform-5.1.1-1-rel-5-1-1..tgz
- ucUpgrade.py
- ucUpgraderConfig.json

29. Modify /ucUpgraderConfig.json to configure how you want your VM to be upgraded.

30. # /ucUpgrader.py -f /ucUpgraderConfig.json

- a. Wait until **"Successfully Upgraded VM"** is display on console

31. # reboot

32. # tail -f /var/log/configParser-2024-xx-xx-xx-xx-xx.log

On Master1

33. SSH to Master 1

34. # kubectl get nodes -o wide

- a. Make sure nodes have a Ready status

35. # kubectl get pods -o wide | grep master-2

- a. Make sure pods have a Running status and all containers are ready per pod.

36. # kubectl drain --ignore-daemonsets --delete-emptydir-data <master 3 name>

37. # kubectl delete node <master 3 name>

38. # kubectl get pods -n kube-system | grep etcd

39. # kubectl exec -it <etcd pod for master 3 from cmd above> sh -n kube-system

40. # etcdctl --cacert /etc/kubernetes/pki/etcd/ca.crt --cert /etc/kubernetes/pki/etcd/peer.crt --key /etc/kubernetes/pki/etcd/peer.key member list

41. # etcdctl --cacert /etc/kubernetes/pki/etcd/ca.crt --cert /etc/kubernetes/pki/etcd/peer.crt --key /etc/kubernetes/pki/etcd/peer.key member remove <master 3 id from cmd above>

On Master3

42. SSH to Master 3

43. Download ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar to / , if you haven't already.

44. # tar -xvf **ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar** -C /

Produces:

- seed- platform-5.1.1-1-rel-5-1-1..tgz
- ucUpgrade.py
- ucUpgraderConfig.json

45. Modify /ucUpgraderConfig.json to configure how you want your VM to be upgraded.

46. # /ucUpgrader.py -f /ucUpgraderConfig.json

- a. Wait until **"Successfully Upgraded VM"** is display on console

47. # reboot

48. # tail -f /var/log/configParser-2024-xx-xx-xx-xx-xx.log

On Master1

49. SSH to Master 1

50. # kubectl get nodes -o wide

- a. Make sure nodes have a Ready status

51. # kubectl get pods -o wide | grep master-3

- a. Make sure pods have a Running status and all containers are ready per pod.

52. # kubectl drain --ignore-daemonsets --delete-emptydir-data --disable-eviction=true **<worker 1 name>**

53. # kubectl delete node **<worker 1 name>**

On worker 1

54. SSH to worker 1

55. Download ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar to / , if you haven't already.

56. # tar -xvf **ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar** -C /

Produces:

- seed- platform-5.1.1-1-rel-5-1-1..tgz

- ucUpgrade.py
- ucUpgraderConfig.json

57. Modify /ucUpgraderConfig.json to configure how you want your VM to be upgraded.

58. # /ucUpgrader.py -f /ucUpgraderConfig.json

- a. Wait until **“Successfully Upgraded VM”** is display on console.

59. # reboot

60. # tail -f /var/log/configParser-2024-xx-xx-xx-xx-xx.log

On Master1

61. SSH to Master 1

62. # kubectl get nodes -o wide

- a. Make sure nodes have a Ready status

63. # kubectl get pods -o wide | grep worker-1

- a. Make sure pods have a Running status and all containers are ready per pod.

64. # kubectl drain --ignore-daemonsets --delete-emptydir-data --disable-eviction=true <worker 2 name>

65. # kubectl delete node <worker 2 name>

On Worker 2

66. SSH to worker 2

67. Download ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar to / , if you haven't already.

68. # tar -xvf **ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar** -C /

Produces:

- seed- platform-5.1.1-1-rel-5-1-1..tgz
- ucUpgrade.py
- ucUpgraderConfig.json

69. Modify /ucUpgraderConfig.json to configure how you want your VM to be upgraded.

70. # /ucUpgrader.py -f /ucUpgraderConfig.json

- a. Wait until **“Successfully Upgraded VM”** is display on console

71. # reboot

72. # tail -f /var/log/configParser-2024-xx-xx-xx-xx-xx.log

On Master1

73. SSH to Master 1

74. # kubectl get nodes -o wide

a. Make sure nodes have a Ready status

75. # kubectl get pods -o wide | grep worker-2

a. Make sure pods have a Running status and all containers are ready per pod.

76. # kubectl drain --ignore-daemonsets --delete-emptydir-data --disable-eviction=true <worker 3 name>

77. # kubectl delete node <worker 3 name>

On Worker 3

78. SSH to worker 3

79. Download ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar to / , if you haven't already.

80. # tar -xvf **ucUpgrader-vm-platform-5.1.1-1-rel-5-1-1.tar** -C /

Produces:

- seed- platform-5.1.1-1-rel-5-1-1..tgz
- ucUpgrade.py
- ucUpgraderConfig.json

81. Modify /ucUpgraderConfig.json to configure how you want your VM to be upgraded.

82. # /ucUpgrader.py -f /ucUpgraderConfig.json

a. Wait until **"Successfully Upgraded VM"** is display on console

83. # reboot

84. # tail -f /var/log/configParser-2024-xx-xx-xx-xx-xx.log

On Master1

85. SSH to Master 1

86. # kubectl get nodes -o wide

a. Make sure nodes have a Ready status

87. # kubectl get pods -o wide | grep worker-3

ucUpgrader.py

ucUpgrader.py script will not only install upgrade but also:

- Get status of what partition is active root.
- Allow reverting to the old version (other partition)
- Allow debug to boot into either partition.

ucUpgraderConfig.json sets the configuration parameters for ucUpgrader.py.

* You get vmVersion by running `cat /var/log/ver.txt` on VM

** You can get K8s version by running `kubectl get nodes` on VM

```
{
  "vmType" : "platform",
  "vmVersion" : "vm-platform-0.0.0-3-issu-3.qcow2",
  "seedFile" : "seed-vm-platform-0.0.0-3-issu-3.tgz",
  "upgradeVM" : "True",
  "ignoreConfigparserOverride" : "True",
  "k8sVersionChange" : "1.29.1",
  "multiBootDebug" : "False",
  "forceErase" : "False"
}
```

Table 55. Key Descriptions

Key	Description
vmType	What VM are we upgrading (vm-opm or vm-platform) If this check fails script exits with error
vmVersion	What version of VM are we upgrading. If this check fails script exits with error
seedFile	What seed file are we using. If this file does not exist, then script exits with error
upgradeVM	False means just a new install on VM not upgrade (Same thing as if you just redeploy the VM with new image). If this is a cluster with only one Master then this should be False .

Key	Description
ignoreConfigparserOverride	If True, when VM reboots after upgrade, configParser will run as normal and will not stop after IP address is configured.
k8sVersionChange	What K8s version to upgrade to. Same version means do not upgrade K8s. However, everything else will; upgrade OS, kernel bug fixes, etc if there are any. With the caveat that the new seed file must have the same K8s version you are upgrading to. If it doesn't then the script exits with error
multiBootDebug	If True then the multiboot menu will stay up for 4 minutes , then go to default partition. This allows user to select non default partition to do any type of debugging. By default, this is False which has only 30 second timeout for multiboot menu.
forceErase	If True, script will not ask user if it is ok to delete old partition. Default is False

OPM MongoDB Backup and Restore

Mongo uses a mounted directory /data/db for both single node and multi-node opm, all the data is typically stored in that directory. For creating backups, save all backups in that directory before copying the contents of the backup to the host directory.

Backup

1. To SSH to any of the master node, execute below command:

```
ssh root@<SSH-ip>
```

2. To save database contents, execute the below command:

```
kc exec -n vnfmgr -it $(kc get po -n vnfmgr -l app=vnfmgr-db -o  
jsonpath='{.items[0].metadata.name}') -- mongodump -- username admin --  
password affirmed --archive=/data/db/mongo- backup
```

3. To store the backup, copy the backup file to the master node to store them locally:

```
kc cp -n vnfmgr $(kc get po -n vnfmgr -l app=vnfmgr-db -o  
jsonpath='{.items[0].metadata.name}') :/data/db/mongo-backup mongo-backup
```

Restore Backup

The restore operation can be performed on the same OPM where the backup was taken or it can be restored to any other OPM to replicate the database contents.

1. To SSH to any master node, execute below command:

```
ssh root@<SSH-ip>
```

2. To find out the primary database instance for fed-opm_multi, execute below command:

```
echo 'rs.isMaster().primary' | kubectl -n vnfmgr exec -ti vnfmgr-db-0 -c  
vnfmgr-db -- mongo --shell --quiet
```

3. To load the backup file, first get the backup file to the master node (if not already present):

```
scp mongo-backup root@<SSH-ip>:/root/mongo-backup
```

4. To copy the backup file inside database pod: For fed-opm, execute the below command:

```
kc cp -n vnfmgr mongo-backup $(kc get po -n vnfmgr -l app=vnfmgr-db -o  
jsonpath='{.items [0].metadata.name}') :/data/db/mongo-backup
```

5. To copy the file to primary database instance: For fed-opm_multi, execute below command:

```
kc cp -n vnfmgr mongo-backup <vnfmgr-db-pod-name>:/data/db/mongo-backup
```

Restore Contents

1. For fed-opm, execute below command:

```
kc exec -n vnfmgr -it $(kc get po -n vnfmgr -l app=vnfmgr-db -o
jsonpath='{.items[0].metadata.name}') -- mongorestore -- username admin --
password affirmed --drop -- excludeCollection=image_collection --
excludeCollection=audittrail --archive=/data/db/mongo-backup
```

2. For fed-opm_multi, execute below command on the current primary instance:

```
kc exec -n vnfmgr -it <vnfmgr-db-pod-name> -- mongorestore - --username
admin --password affirmed --drop -- excludeCollection=image_collection --
excludeCollection=audittrail --archive=/data/db/mongo-backup
```

Things to do after restore

Re-attach the clusters (if any):

Once the data is restored, login to the GUI and delete already attached clusters. After deleting, re-attach them again.

The restore operation only restores the database contents, the associated k8rm pods are not recreated automatically. To get them created again, re-attach the clusters.

Re-attach the created clusters and deployment zones (if any)

Once the data is restored, attach the previously created clusters and associated deployment zones. This is to recreate the associated osrm and k8rm pods.

Miscellaneous Information

Kubernetes Certificate Renewal – Single or HA Multi-Master Deployment

Note: If the certificate has already expired, perform the procedure for all master nodes else a certificate error still generates.

To renew the Kubernetes certificates for one year, use the following procedure on both the OPM VM and Platform VM masters. You must log in as the root user:

1. Check the current certificate expiration:

```
# ssh root@<vmip>
# kubectl get nodes
```

2. Renew all the Kubernetes certificates:

```
# kubectl get nodes
```

3. Verify that the certificates expire a year from now:

```
# kubectl get nodes
```

4. Restart Kubernetes system pods:

```
# mv /etc/kubernetes/manifests/kube-controller-manager.yaml
/tmp
```

```
# mv /etc/kubernetes/manifests/kube-scheduler.yaml /tmp
```

```
# mv /etc/kubernetes/manifests/kube-apiserver.yaml /tmp
```

```
# mv /etc/kubernetes/manifests/etcd.yaml /tmp
```

5. Wait 30 seconds, then:

```
# mv /tmp/kube-apiserver.yaml /etc/kubernetes/manifests
```

```
# mv /tmp/kube-controller-manager.yaml
```

```
/etc/kubernetes/manifests
```

```
# mv /tmp/kube-scheduler.yaml /etc/kubernetes/manifests
# mv /tmp/etcd.yaml /etc/kubernetes/manifests
```

6. Verify kube-system pods come back up:

```
# kubectl get pods -n kube-system
```

Note: If this is a multi-master Kubernetes cluster, restarting kube system pods, needs to be done for the other two master nodes. Only do one master node at a time and make sure all pods come back up before proceeding to the next master node.

Note: If Kubernetes version is 1.18.3 then “alpha” is needed in kubeadm cmd line. For example:

```
# kubeadm alpha certs check-expiration
# kubeadm alpha certs renew all
```

Fed-sriovdp VM Configmap Example

```
# kc get configmap -n kube-system sriovdp-config -oyaml
apiVersion: v1
data:
  config.json: '{"resourceList":
[{"resourceName":"sr1","selectors":{"drivers":["vfio-
pci"],"pciAddresses":["0000:00:06.0","0000:00:07.0"]}},
{"resourceName":"sr2","selectors":{"drivers":["vfio-
pci"],"pciAddresses":["0000:00:08.0","0000:00:09.0"]}}]}'
kind: ConfigMap metadata:
  annotations:
    meta.helm.sh/release-name: fed-sriovdp
    meta.helm.sh/release-namespace: fed-sriovdp
    creationTimestamp: "2022-08-23T15:43:57Z"
  labels:
    app.kubernetes.io/managed-by: Helm
    name: sriovdp-config
    namespace: kube-system
    resourceVersion: "1644485"
    uid: 868c871a-ed63-45e5-b9e3-aba43cb499d4
```

Figure 102. Required Parameters for sriovdp Deployment Type - Physical Function (baremetal-generic)

fed-sriovdp - 3.3.0-4-rel-3-3-0-cbhasin.rel-3-dbg values.yaml

Helm Deployment Details

Release Name * Namespaces *

global

Type * Sriov Port Prefix

registry

docker

Repo Path *

pod-sriovdp

Sriovdp3party Image Multus3party Amd64Image Rbac Enabled ☒

Cluster Role Service Account Name

multus

Enabled ☒

sriovdpBaremetalConfigMap

resourceList

Resource Name

selectors

drivers

pfNames

Resource Name

selectors

drivers

pfNames

Fed-sriovdp Baremetal Configmap Example

```
# kubectl get cm -n kube-system sriovdp-config -o yaml
apiVersion: v1
data:
config.json: '{"resourceList":
[{"resourceName":"sr1","selectors":{"drivers":["vfio-
pci"],"pfNames":["ens1f0","ens7f0"]}},
{"resourceName":"sr2","selectors":{"drivers":["vfio-
```

Misc Information

```
pci"],"pfNames":["ens1f1","ens7f1"]}]}}}'
kind: ConfigMap
metadata: annotations:
meta.helm.sh/release-name: fed-sriovdp
meta.helm.sh/release-namespace: fed-sriovdp
creationTimestamp: "2022-08-22T05:18:49Z"
labels:
app.kubernetes.io/managed-by: Helm
name: sriovdp-config
namespace: kube-system
resourceVersion: "50028174"
uid: ab407816-5d0c-43b3-8d05-0383b83d943f
```

Caution: In order to access the following function with the CLI, you must define "admin", "oper", "readonly", "liadmin", and "lioper" group users in pod-cfgmgr with either public keys and/or certificate authority keys and/or SHA-encrypted passwords, and/or configure one or more TACACS servers.

Table 56. fed-nrf-Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
registry > docker > repoPath	<OPM_IP>:5005
logging > enabled	Toggle On.
logging > IP	<metalLB_PaaSIP>
grafana > enabled	Toggle On.
grafana > url	http://<<metalLB_PaaSIP>>:3000
grafana > prometheusinfo > user	
grafana > prometheusinfo > password	
fileserver > ui > loadBalancerIP	IP address for optional load balancer service.
fileserver > ui > loadBalancerPort	Port number for optional load balancer service. Default is 85.
Fileserver Nfs Remote Host	Enter the <opm ip address> if you are using the OPM Network file System (NFS) server. Otherwise, enter the IP address of the remote NFS server.
Fileserver Nfs Remote Path	Input desired path for the chart. Default is/filestore. Path must already exist on remote server.
mesh > istio > ingressGw	<metalLB_nrfIstioIP>

Misc Information

Parameter	Value
envVars > FedUniqueName	Enter a globally unique name.
pod-cfgmgr	
service > loadBalanceIp	<metalLb_nrfCfgMgrIp>
service > loadBalanceFqdn	<metalLb_nrfCfgMgrIp>
security > enable_local_passwd_auth	"1"
security > enable_root	"1"
security > local_admin_users > user	Enter a name for the admin user
security > local_admin_users > password	Enter a SHA-512 encrypted password
security > local_admin_users > pubkey	
security > local_oper_users > user	Enter a name for the oper user
security > local_oper_users > password	Enter a SHA-512 encrypted password
security > local_oper_users > pubkey	
security > local_readonly_users > user	Enter a name for the read-only user
security > local_readonly_users password	Enter a SHA-512 encrypted password
security > local_readonly_users > pubkey	
security > local_liadmin_users > user	Enter a name for the lawful intercept admin user
security > local_liadmin_users > password	Enter a SHA-512 encrypted password
security > local_liadmin_users > pubkey	
security > local_lioper_users > user	Enter a name for the lawful intercept oper user
security > local_lioper_users > password	Enter a SHA-512 encrypted password
security > local_lioper_users > pubkey	
userProvided > tacacs_protocol	TACACS protocol name
userProvided > tacacs_service	TACACS service name presented to the TACACS servers

Misc Information

Parameter	Value
userProvided > enable_tacacs_accounting	Enable pushing TACACS accounting data up to the servers: 1-enable, 0-disable and 0-default.
userProvided > tacacs_servers > Auth Timeout	TACACS server authentication timeout.
userProvided > tacacs_servers > Auth Type	TACACS server authentication type.
userProvided > tacacs_servers > Ip Address	TACACS server IP address.
userProvided > tacacs_servers > Key	TACACS server secret.
userProvided > tacacs_servers > Port	TACACS server port.
pod-istio_gw > paasSelector > jaegerHostName	<metalLB_PaaSIP>
pod-slice_inv_mgr	
deployment > apiServiceHost	<metalLB_nrfIstioIP>
deployment > apiServicePort	80
upgrade > stableInfralImage	
upgrade > stableWorkerImage	
pod-nf_selector	
upgrade > stableInfralImage	
upgrade > stableWorkerImage	
deployment > apiServiceHost	<metalLB_nrfIstioIP>
deployment > apiServicePort	80
pod-nrf_monitor	
deployment > apiServiceHost	<metalLB_nrfIstioIP>
deployment > apiServicePort	80
pod-oauth_token	
deployment > apiServiceHost	<metalLB_nrfIstioIP>
deployment > apiServicePort	80

Caution: In order to access the following function with the CLI, you must define "admin", "oper", "readonly", "liadmin", and "lioper" group users in pod-cfgmgr with either public keys and/or certificate authority keys and/or SHA-encrypted passwords, and/or configure one or more TACACS servers.

Table 57. fed-amf_mme Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
logging > enabled	Toggle On.
logging > fluentd > buffer > Buffer Limit	Total buffer size. Maximum amount of buffer space each fluentd thread would use to keep logs. After this space is exhausted, fluentd starts deleting the oldest logs to make room for new logs.
logging > fluentd > buffer > Buffer Limit	Interval in seconds at which the buffer flush is retried in case of failure.
logging > fluentd > buffer > Buffer Limit	Maximum number of retries to flush a buffer chunk. After this is exhausted, fluentd deletes the chunk.
grafana > enabled	Toggle On.
grafana > url	http://<<metalLB_PaaSIP>>:3000
grafana > prometheusinfo > Url	Prometheus server URL.
tabdb > etcd > Name	Etcd database instance name.
tabdb > etcd > Svc	Tab-DB service path.
tabdb > etcd > Port	Tab-DB service port.
tabdb > etcd > Target Port	Tab-DB service target port.
Fileserver > Nfs > Remote Host	Enter the <opm ip address> if you are using the OPM Network file System (NFS) server. Otherwise, enter the IP address of the remote NFS server.
Fileserver > Nfs > Remote Path	Input desired path for the chart. Default is /filestore. Path must already exist on remote server.
Fileserver > Nfs > Remote Port	Fileserver remote host port.
mesh > istio > ingress Gw	Service mesh ingress gateway.
mesh > Ingress Gw IP	Specific IP requested for Ingress Gateway LoadBalancer service.

Misc Information

Parameter	Value
mesh > Ingress Gw FQDN	FQDN to be added as annotation to the Ingress Gateway LoadBalancer Service.
mesh > Mtls Enabled	Enable or disable istio mutual TLS.
monitoring > prometheus Namespace	Monitoring namespace.
registry > docker > repoPath	<OPM_IP>:5005
envVars > Fed Type	Federation type.
envVars > Fed Unique Name	Enter a globally unique name.
envVars > Service Registry Namespace	Service registry namespace.
envVars > Encrypt Paas Data for LI	Enable or disable LI PaaS content encryption/decryption.
externalIps > N2external Ips	List of FQDN and external IP pairs for N2 interface. The FQDNs must be configured under amf-n2.
sctpServerPorts > N2sctp Server Ports	List of Server Ports for N2 sctp interface. The Port must be configured under amf-n2.
pod-fileserver	
Enabled	Enable/disable fileserver.
deployment > replicas	Deployment replica count.
pvc > cfg > Requests Storage	Requests storage for pvc.
pvc > cfg > Storage Class Name	Storage class name for pvc.
pvc > web > Requests Storage	Requests storage for pvc.
pvc > web > Storage Class name	Storage class name for pvc.
pod-cfgmgr	
Enabled	Enable/disable cnfmgr.
confd > Aes256cfb128Key	Encoded confd aes256cfb128Key.
service > Port Prefix	Service port prefix
deployment > replicas	Deployment replica count.
security > enable_local_passwd_auth	"1"
security > enable_root	"1"
security > local_admin_users > user	Enter a name for the admin user

Misc Information

Parameter	Value
security > local_admin_users password	Enter a SHA-512 encrypted password
security > local_admin_users pubkey	
security > local_oper_users > user	Enter a name for the oper user
security > local_oper_users password	Enter a SHA-512 encrypted password
security > local_oper_users pubkey	
security > local_readonly_users user	Enter a name for the read-only user
security > local_readonly_users password	Enter a SHA-512 encrypted password
security > local_readonly_users pubkey	
security > local_liadmin_users user	Enter a name for the lawful intercept admin user
security > local_liadmin_users password	Enter a SHA-512 encrypted password
security > local_liadmin_users pubkey	
security > local_lioper_users > user	Enter a name for the lawful intercept oper user
security > local_lioper_users password	Enter a SHA-512 encrypted password
security > local_lioper_users pubkey	
pod-istio_gw	
Enabled	Enable/disable istio-gw
ingressGateway > Enabled	Enable/disable ingress gateway
podPriority > hpa > Enabled	Enable/disable ingress gateway HPA
podPriority > hpa > Worker Cpu Requests	CPU request ingress application
podPriority > hpa > Min Replicas	Specify minimum replica count for ingress gateway

Misc Information

Parameter	Value
podPriority > hpa > Max Replicas	Specify maxium replica count for ingress gateway
podPriority > hpa > Average Utilization	Specify the target CPU utilization level when HPA would trigger
egressGateway > Enabled	Enable/disable egress gateway
egressGateway > hpa > Enabled	Enable/disable egress gateway HPA
egressGateway > hpa > Worker Cpu Requests	CPU request egress application
egressGateway > hpa > Min Replicas	Specify the minimum replica count for egress gateway
egressGateway > hpa > Max Replicas	Specify the maxium replica count for egress gateway
egressGateway > hpa > Average Utilization	Specify the target CPU utilization level when HPA would trigger
pod-istio_gw_li	
Enabled	Enable/disable istio-gw-li
Istio Nspace	Istio namespace
ingressGateway > Enabled	Enable/disable LI ingress gateway
egressGateway > Enabled	Enable/disable LI egress gateway
egressGateway > hpa > Enabled	Enable/disable egress gateway HPA
egressGateway > hpa > Worker Cpu Requests	CPU request egress application
egressGateway > hpa > Min Replicas	Specify minimum replica count for egress gateway
egressGateway > hpa > Max Replicas	Specify maximum replica count for egress gateway
egressGateway > hpa > Average Utilization	Specify the target CPU utilization level when HPA would trigger
ppsSelector > Jaeget Host Name	Jaeger host (IP or FQDN)
pod-amf_cc	
Cna Infra Thread Pool Size Db Client	DB client thread pool size
upgrade > Canary	Enable/disable canary sequence
upgrade > Canary Vers Weight	Canary pod weight percentage

Misc Information

Parameter	Value
upgrade > Stable Vers Weight	Stable pod weight percentage
upgrade > Stable Version	Stable pod image version
deployment > Replicas	Deployment replica count
deployment > Trace Log Enabled	Enable/disable tracelog
deployment > Inject Istio Proxy	Enable/disable istio-proxy
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
tracing > Enabled	Enable/disable tracing
tracing > Sampler type	Jaeger sampler type
tracing > Sampler Param	Jaeger sampler parameter
tracing > Agent Port	UDP port exposed by jaeger-agent
tracing > Log Enabled	Enable/disable Jaeger logging.
pod-amf_ta	
upgrade > Canary	Enable/disable canary sequence
upgrade > Canary Vers Weight	Canary pod weight percentage
upgrade > Stable Vers Weight	Stable pod weight percentage
upgrade > Stable Version	Stable pod image version
deployment > Replicas	Deployment replica count
deployment > Inject Istio Proxy	Enable/disable istio-proxy
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas

Misc Information

Parameter	Value
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
pod-amf_monitor	
Enabled	Enable/disable AMF monitor
deployment > Replicas	Deployment replica count
deployment > API Service Host	Virtual service host (IP or FQDN)
deployment > API Service Port	Ingress GW node port
Inject Istio Proxy	Enable/disable istio-proxy
pod-amf_n2	
deployment > Replicas	Deployment replica count
deployment > Trace Log Enabled	Enable/disable tracelog
deployment > Lb Tunnel Ip Family	LB tunnel IP family
Inject Istio Proxy	Enable/disable istio-proxy
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
pod-timer	
Enabled	Enable/disable timer microservice
Role	clusterrole is created as a part of fed-rbac
Service Account Name	service AccountName configuration is used to create serviceaccount per pod
deployment > Replicas	Deployment replica count
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests

Misc Information

Parameter	Value
hpa > Worker Cpu Limits	Worker container HPA CPU limits
Worker Memory Requests	Worker container HPA memory requests
Worker Memory Limits	Worker container HPA memory limits
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
pod-nrf-agent	
deployment > Replicas	Deployment replica count
deployment > Inject Istio Proxy	Enable/disable istio-proxy
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
pod-sub-analyzer	
Enabled	Enable/disable sub-analyzer
upgrade > Canary	Enable/disable canary sequence
upgrade > Canary Vers Weight	Canary pod weight percentage
upgrade > Stable Vers Weight	Stable pod weight percentage
upgrade > Stable Version	Stable pod image version
deployment > Replicas	Deployment replica count
deployment > Inject Istio Proxy	Enable/disable istio-proxy
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas

Misc Information

Parameter	Value
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
pod-gtpc_agent	
deployment > Replicas	Deployment replica count
deployment > Inject Istio Proxy	Enable/disable istio-proxy
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
tracing > Enabled	Enable/disable tracing
tracing > Sampler type	Jaeger sampler type
tracing > Sampler Param	Jaeger sampler parameter
tracing > Agent Port	UDP port exposed by jaeger-agent
tracing > Log Enabled	Enable/disable Jaeger logging.
pod-dns_agent	
Enabled	Enable/disable DNS Agent
deployment > Replicas	Deployment replica count
deployment > Inject Istio Proxy	Enable/disable istio-proxy
upgrade > Canary	Enable/disable canary sequence
upgrade > Canary Vers Weight	Canary pod weight percentage
upgrade > Stable Vers Weight	Stable pod weight percentage
upgrade > Stable Version	Stable pod image version
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits

Misc Information

Parameter	Value
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
pod-ulb	
Cluster Role	clusterrole and PSP for is created as a part of fed-rbac. clusterrole options are restricted-cr, permissive-network-cr, permissive-full-cr
Service Account Name	pod-ulb service account name
deployment > config deploymentGroups > Name	Name of the deployment group
deployment > config deploymentGroups > endpoints ips >	IP addresses
deployment > config deploymentGroups > endpoints > sctp > Enabled	Enable SCTP protocol for the endpoint
deployment > config > frontEnd externalLb > Empty	Empty field to select externalLb mode
pod-li_x1	
Cluster Role	clusterrole and PSP for is created as a part of fed-rbac. clusterrole options are restricted-cr, permissive-network-cr, permissive-full-cr
Service Account Name	serviceName configuration is used to create serviceaccount per pod
deployment > Replicas	Deployment replica count
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
upgrade > Canary	Enable/disable canary sequence

Misc Information

Parameter	Value
pod-li_x2	
Cluster Role	clusterrole and PSP for is created as a part of fed-rbac. clusterrole options are restricted-cr, permissive-network-cr, permissive-full-cr
Service Account Name	serviceAccountName configuration is used to create serviceaccount per pod
deployment > Replicas	Deployment replica count
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
upgrade > Canary	Enable/disable canary sequence
pod-mme_cc	
External Host	
upgrade > Canary	Enable/disable canary sequence
upgrade > Canary Vers Weight	Canary pod weight percentage
upgrade > Stable Vers Weight	Stable pod weight percentage
upgrade > Stable Version	Stable pod image version
deployment > Replicas	Deployment replica count
deployment > Api Service Host	Virtual service host (IP or FQDN)
deployment > Trace Log Enabled	Enable/disable tracelog
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas

Misc Information

Parameter	Value
hpa > Average Utilization	Pod average utilization
tracing > Enabled	Enable/disable tracing
tracing > Sampler type	Jaeger sampler type
tracing > Sampler Param	Jaeger sampler parameter
tracing > Agent Port	UDP port exposed by jaeger-agent
tracing > Log Enabled	Enable/disable Jaeger logging.
destinationRule > Tls Enabled	Enable/disable TLS.
pod-mme_ta	
deployment > Replicas	Deployment replica count
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
pod-mme_s1	
deployment > Replicas	Deployment replica count
deployment > Trace Log Enabled	Enable/disable tracelog
deployment > Lb Tunnel Ip Family	LB tunnel IP family
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
pod-diameter_base > diameterGroups	

Misc Information

Parameter	Value
Group Name	Name of diameter-base group
Enabled	Enable/disable diameter-base group
Transport	diameter-base transport protocol
Sctp Encap Type	SCTP encapsulation type
Role	Role of diameter-base group
External Ip Service Id	Service ID
ratelimit > Enabled	Enable/disable ratelimiting
ratelimit > Quota	Number of incoming requests allowed
ratelimit > Interval	Configure time interval
ratelimit > Source App	Source app
deployment > Replicas	Deployment replica count
hpa > Enabled	Enable/disable HPA
hpa > Infra Cpu Requests	Infra container HPA CPU requests
hpa > Worker Cpu Requests	Worker container HPA CPU requests
hpa > Worker Cpu Limits	Worker container HPA CPU limits
hpa > Max Replicas	Pod maximum replicas
hpa > Min Replicas	Pod minimum replicas
hpa > Average Utilization	Pod average utilization
tracing > Enabled	Enable/disable tracing
tracing > Sampler type	Jaeger sampler type
tracing > Sampler Param	Jaeger sampler parameter
tracing > Agent Port	UDP port exposed by jaeger-agent
tracing > Log Enabled	Enable/disable Jaeger logging.
pod-routing	
Enabled	Enable/disable routing microservice
deployment > Replicas	Deployment replica count
pod-interface_mgr	
Enabled	Enable/disable interface manager microservice
deployment > Replicas	Deployment replica count

Misc Information

Parameter	Value
pod-lb_ppe	
Enabled	Enable/disable LB PPE microservice
Cluster Role	clusterrole and PSP for is created as a part of fed-rbac. clusterrole options are restricted-cr, permissive-network-cr, permissive-full-cr
Service Account Name	serviceAccountName configuration is used to create serviceaccount per pod
Ppe First Worker	First core dataplane that will be allowed to use. This is not a physical core number; it is a core number within the range allocated to the pod.
Ppe Num Workers	Number of cores allocated to dataplane, including the first core.
Ppe Io Scaling	Dataplane I/O resource distribution
Vlan Stripping	Enable/disable vlan stripping on dataplane ports
Enable Jumbo Frames	Enable/disable jumbo frame support
deployment > Replicas	Deployment replica count
deployment > Deploy Capture Agent	Enable/disable capture agent deployment
resources > Hugepages Type	Huge pages type
resources > Ppe Cpu Requests	LB PPE container CPU usage threshold
resources > Ppe Cpu Limits	LB PPE container CPU usage maximum
resources > Ppe Memory	Allocated memory for LB PPE container
resources > Ppe Hugepages Size	Huge pages size
guaranteedScheduling > Enabled	Enable/disable guaranteedScheudling for LB PPE microservice
guaranteedScheduling > Infra Cpu Limit	Infra container CPU usage maximum limit
guaranteedScheduling > Infra emory Limit	Infra container memory usage maximum limit
guaranteedScheduling > Ulb Cpu Limit	ULB container CPU maximum limit
guaranteedScheduling > Ulb Memory Limit	ULB container memory usage maximum limit
guaranteedScheduling > Ppe Cpu Limit	LB PPE container CPU usage maximum limit

Misc Information

Parameter	Value
guaranteedScheduling > Ppe Memory Limit	LB PPE container memory usage maximum limit
netAttachDefs > Enabled	Enable/disable non-default network driver
tracing > Enabled	Enable/disable tracing
tracing > Sampler type	Jaeger sampler type
tracing > Sampler Param	Jaeger sampler parameter
tracing > Agent Port	UDP port exposed by jaeger-agent
tracing > Log Enabled	Enable/disable Jaeger logging.

Caution: In order to access the following function with the CLI, you must define "admin", "oper", "readonly", "liadmin", and "lioper" group users in pod-cfgmgr with either public keys and/or certificate authority keys and/or SHA-encrypted passwords, and/or configure one or more TACACS servers.

Table 58. fed-nssf Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
registry > docker > repoPath	<OPM_IP>:5005
logging > enabled	Toggle On.
logging > IP	<metalLB_PaaSIP>
grafana > enabled	Toggle On.
grafana > url	http://<<metalLB_PaaSIP>>:3000
grafana > prometheusinfo > user	
grafana > prometheusinfo > password	
fileserver > ui > loadBalancerIP	IP address for optional load balancer service.
fileserver > ui > loadBalancerPort	Port number for optional load balancer service. Default is 85.
Fileserver Nfs Remote Host	Enter the <opm ip address> if you are using the OPM Network file System (NFS) server. Otherwise, enter the IP address of the remote NFS server.
Fileserver Nfs Remote Path	Input desired path for the chart. Default is /filestore. Path must already exist on remote server.

Misc Information

Parameter	Value
mesh > istio > ingressGw	<metalLB_nssfIstioIp>
envVars > FedUniqueName	Enter a globally unique name.
pod-cfgmgr	
service > loadBalanceIp	<metalLb_nssfCfgMgrIp>
service > loadBalanceFqdn	<metalLb_nssfCfgMgrIp>
security > enable_local_passwd_auth	"1"
security > enable_root	"1"
security > local_admin_users > user	Enter a name for the admin user
security > local_admin_users > password	Enter a SHA-512 encrypted password
security > local_admin_users > pubkey	
security > local_oper_users > user	Enter a name for the oper user
security > local_oper_users > password	Enter a SHA-512 encrypted password
security > local_oper_users > pubkey	
security > local_readonly_users > user	Enter a name for the read-only user
security > local_readonly_users > password	Enter a SHA-512 encrypted password
security > local_readonly_users > pubkey	
security > local_liadmin_users > user	Enter a name for the lawful intercept admin user
security > local_liadmin_users > password	Enter a SHA-512 encrypted password
security > local_liadmin_users > pubkey	
security > local_lioper_users > user	Enter a name for the lawful intercept oper user
security > local_lioper_users > password	Enter a SHA-512 encrypted password
security > local_lioper_users > pubkey	
userProvided > tacacs_protocol	TACACS protocol name
userProvided > tacacs_service	TACACS service name presented to the TACACS servers
userProvided > enable_tacacs_accounting	Enable pushing TACACS accounting data up to the servers: 1-enable, 0-disable and 0-default.

Misc Information

Parameter	Value
userProvided > tacacs_servers > Auth Timeout	TACACS server authentication timeout.
userProvided > tacacs_servers > Auth Type	TACACS server authentication type.
userProvided > tacacs_servers > Ip Address	TACACS server IP address.
userProvided > tacacs_servers > Key	TACACS server secret.
userProvided > tacacs_servers > Port	TACACS server port.
pod-slice_inv_mgr	
deployment > apiServiceHost	<metalLB_nssf1stioIp>
deployment > apiServicePort	80
upgrade > stableInfralImage	
upgrade > stableWorkerImage	
pod-slice_selector	
deployment > Api Service Host	<metalLB_nssf1stioIp>
deployment > apiServicePort	80
upgrade > stableInfralImage	
upgrade > stableWorkerImage	
pod-nrf_agent	
upgrade > stableInfralImage	
upgrade > stableWorkerImage	
pod-nssf_monitor	
deployment > apiServiceHost	<mesh1stioIngressGWPoolIp>
deployment > apiServicePort	80

Caution: In order to access the following function with the CLI, you must define "admin", "oper", "readonly", "liadmin", and "lioper" group users in pod-cfgmgr with either public keys and/or certificate authority keys and/or SHA-encrypted passwords, and/or configure one or more TACACS servers.

Table 59. fed-smf Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
envVars > FedUniqueName	Enter a globally unique name.
registry > docker > repoPath	<OPM_IP>:5005
fileserver > ui > loadBalancerIP	IP address for optional load balancer service.
fileserver > ui > loadBalancerPort	Port number for optional load balancer service. Default is 85.
Fileserver Nfs Remote Host	Enter the <i><opm ip address></i> if you are using the OPM Network file System (NFS) server. Otherwise, enter the IP address of the remote NFS server.
Fileserver Nfs Remote Path	Input desired path for the chart. Default is <i>/filestore</i> . Path must already exist on remote server.
grafana > enabled	Toggle On
grafana > url	http://<metalLB_PaaSIP>:3000
grafana > prometheusinfo > user	
grafana > prometheusinfo > password	
logging > enabled	Toggle On
logging > ip	<metalLB_PaaSIP>
logging > port	9200
pod-cfgmgr	
service > loadBalancerIp	<metalLb_smfCfgMgrIp>
service > loadBalancerFqdn	<metalLb_smfCfgMgrIp>
security > enable_local_passwd_auth	"1"
security > enable_root	"1"
security > local_admin_users > user	Enter a name for the admin user
security > local_admin_users > password	Enter a SHA-512 encrypted password
security > local_admin_users > pubkey	
security > local_oper_users > user	Enter a name for the oper user

Misc Information

Parameter	Value
security > local_oper_users > password	Enter a SHA-512 encrypted password
security > local_oper_users > pubkey	
security > local_readonly_users > user	Enter a name for the read-only user
security > local_readonly_users > password	Enter a SHA-512 encrypted password
security > local_readonly_users > pubkey	
security > local_liadmin_users > user	Enter a name for the lawful intercept admin user
security > local_liadmin_users > password	Enter a SHA-512 encrypted password
security > local_liadmin_users > pubkey	
security > local_lioper_users > user	Enter a name for the lawful intercept oper user
security > local_lioper_users > password	Enter a SHA-512 encrypted password
security > local_lioper_users > pubkey	
userProvided > tacacs_protocol	TACACS protocol name
userProvided > tacacs_service	TACACS service name presented to the TACACS servers
userProvided > enable_tacacs_accounting	Enable pushing TACACS accounting data up to the servers: 1- enable, 0-disable and 0-default.
userProvided > tacacs_servers > Auth Timeout	TACACS server authentication timeout.
userProvided > tacacs_servers > Auth Type	TACACS server authentication type.
userProvided > tacacs_servers > Ip Address	TACACS server IP address.
userProvided > tacacs_servers > Key	TACACS server secret.
userProvided > tacacs_servers > Port	TACACS server port.
pod-sub_analyzer > apiServiceHost	<metalLB_smflstiulp>
pod-pfcp_c	
deployment > masterNodeIpAddress	<Kubernetes master ip>
deployment > service loadBalancerIp	<metalLB_smflstiulp>
deployment > service > externalFqdn	Must match the FQDN defined.

Parameter	Value
pod-smfcc	
deployment > masterNodeIpAddress	<metalLB_smflstiolp>
deployment > dynamicPccEnabled	<i>False</i>
pod-smf_monitor	
deployment > Master Node Ip Address	<metalLB_smflstiolp>
pod-inband_data_agt > service > loadBalancerIp	<metalLB_smflstiolp>
pod-gtpc_agent	
deployment > masterNodeIpAddress	<metalLB_smflstiolp>
externalSvc > loadBalancerIp	<metalLB_smflstiolp>
externalSvc > loadBalancerFqdn	<metalLB_smflstiolp>

fed-smf Parameters for ISSU (In-Service Software Upgrade)

SMF supports pre and post upgrade/rollback hooks to validate and track the status of ISSU.

Pre-Upgrade hook is used to report the state of SMF federation prior to upgrade. This includes NF monitor status, pod status and KPI metrics information.

Post Upgrade hook is used to validate the state of upgrade by checking the NF monitor status. Pod status and KPI metrics and the validation failures can result in action roll back/pause (full roll back or pause on single federation roll back as configured)

Pre rollback and Post rollback hooks report the state of SMF federation during rollback.

Table 60. fed-smf pre-upgrade Parameters for ISSU

Parameter	Value
reporting: nfMonitor pod Kpi	Report the state of pre-upgrade for defined parameters
Config; max_retry retry_interval kpi_min_metric_value	Define config to be used to generate report. number of retries to fetch nf monitor / pod status / KPI metrics. defines the retry interval. metric value to be used for kpi validation.

Misc Information

Parameter	Value
kpi_sampling_interval	sampling interval to fetch KPIs
Pod: name: validate value	Pod validation params name of the pod in SMF federation (example: smfcc) action on failure (default set to false to just report the state) percentage of pods to be up

Table 61. fed-smf post-upgrade Parameters for ISSU

Parameter	Value
Validate:	Enter True to take action on failures
Config; max_retry retry_interval kpi_min_metric_value kpi_sampling_interval	Define config to be used for retries and kpi scraping interval. number of retries to fetch nf monitor / pod status / KPI metrics. defines the retry interval. metric value to be used for kpi validation. sampling interval to fetch KPIs
Pod: name: validate value	Pod validation params name of the pod in SMF federation (example: smfcc) action on failure (default set to true to take action) percentage of pods to be up
KPI Metrics information: Kpi name: kpi_min_metric_value kpi_sampling_interval validate value	Call processing KPIs to be used for validation: name of the kpi: List of Kpis smfcc_nsmf_pdu_session_update_sm_context_acceptable_failures smfcc_gtpc_modify_bearer_request_acceptable_failures smfcc_service_request_acceptable_failures smfcc_xn_handover_acceptable_failures smfcc_ue_deregistration_acceptable_failures smfcc_pgw_ue_initiated_detach_acceptable_failures smfcc_ue_initiated_pdu_session_release_acceptable_failures smfcc_pgw_modify_bearer_request_acceptable_failures smfcc_pcf_initiated_policy_modification_acceptable_failures smfcc_pgw_pcf_initiated_policy_modification_acceptable_failures

Misc Information

Parameter	Value
	Fed SMF level - KPI will be calculated only if metric has minimum configured value. Fed SMF level - Duration for KPI monitoring in second during upgrade post PODS are in n/n state. per KPI - If false, then upgrade will not be failed even KPI does not match passing criteria. per KPI threshold - KPI to be compared against maximum value %.

Caution: In order to access the following function with the CLI, you must define "admin", "oper", "readonly", "liadmin", and "lioper" group users in pod-cfgmgr with either public keys and/or certificate authority keys and/or SHA-encrypted passwords, and/or configure one or more TACACS servers.

Note: After you have deployed the UPF federation, see the UPF Setup Guide for information on setting up the UPF federation in UnityCloud.

Table 62. fed-upf Required Parameters

Parameter	Value
Release Name	Enter the release name.
Namespaces	Enter the namespace for this federation.
global	
envVars > FedUniqueName	Enter a globally unique name.
fileserver > ui > loadBalancerIP	IP address for optional load balancer service.
fileserver > ui > loadBalancerPort	Port number for optional load balancer service. Default is 85.
Fileserver Nfs Remote Host	Enter the <i><opm ip address></i> if you are using the OPM Network file System (NFS) server. Otherwise, enter the IP address of the remote NFS server.
Fileserver Nfs Remote Path	Input desired path for the chart. Default is /filestore. Path must already exist on remote server.
mesh > istio > ingressGwlp	<metalLB_upflstiolp>
grafana > enabled	Toggle On

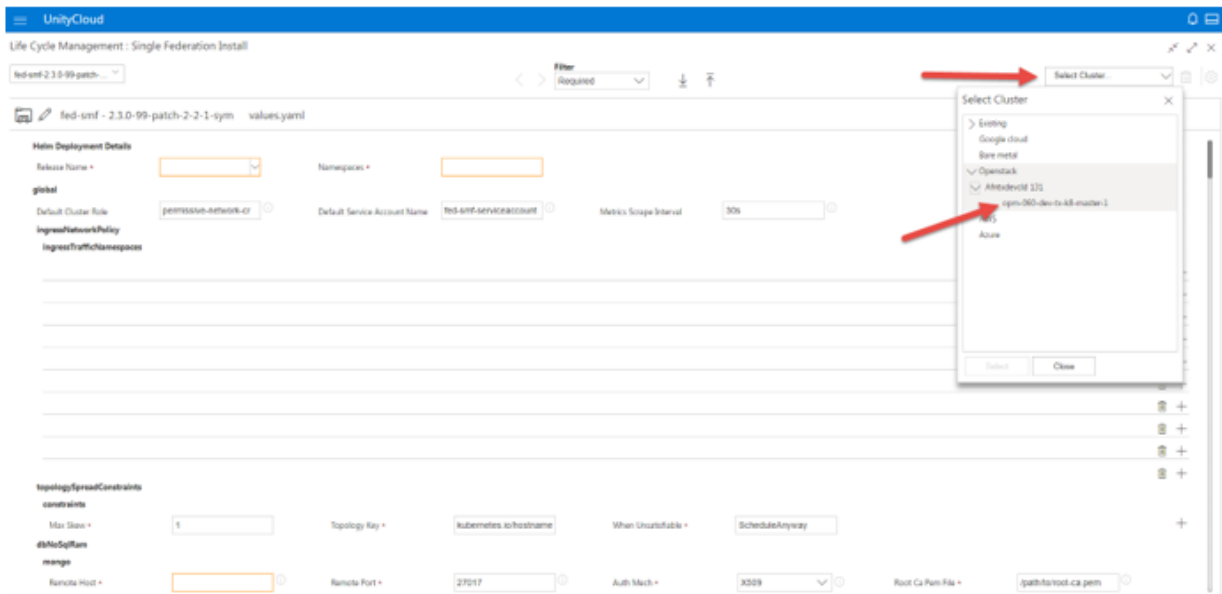
Misc Information

Parameter	Value
grafana > url	http://<<metalLB_PaaSIP>>:3000
grafana > prometheusinfo > user	
grafana > prometheusinfo > password	
logging > elastic > ip	<metalLB_PaaSIP>
logging > elastic > port	9200
registry > docker > repoPath	<OPM_IP>:5005
pod-cfgmgr	
service > loadBalanceIp	<metalLB_upfCfgMgrIp>
service > loadBalanceFqdn	upf.affirmed.com
security > enable_local_passwd_auth	"1"
security > enable_root	"1"
security > local_admin_users > user	Enter a name for the admin user
security > local_admin_users > password	Enter a SHA-512 encrypted password
security > local_admin_users > pubkey	
security > local_oper_users > user	Enter a name for the oper user
security > local_oper_users > password	Enter a SHA-512 encrypted password
security > local_oper_users > pubkey	
security > local_readonly_users > user	Enter a name for the read-only user
security > local_readonly_users > password	Enter a SHA-512 encrypted password
security > local_readonly_users > pubkey	
security > local_liadmin_users > user	Enter a name for the lawful intercept admin user
security > local_liadmin_users > password	Enter a SHA-512 encrypted password
security > local_liadmin_users > pubkey	
security > local_lioper_users > user	Enter a name for the lawful intercept oper user
security > local_lioper_users > password	Enter a SHA-512 encrypted password

Misc Information

Parameter	Value
security > local_lioper_users > pubkey	
userProvided > tacacs_protocol	TACACS protocol name
userProvided > tacacs_service	TACACS service name presented to the TACACS servers
userProvided > enable_tacacs_accounting	Enable pushing TACACS accounting data up to the servers: 1- enable, 0-disable and 0-default.
userProvided > tacacs_servers > Auth Timeout	TACACS server authentication timeout.
userProvided > tacacs_servers > Auth Type	TACACS server authentication type.
userProvided > tacacs_servers > Ip Address	TACACS server IP address.
userProvided > tacacs_servers > Key	TACACS server secret.
userProvided > tacacs_servers > Port	TACACS server port.
pod-pfcp_u	
deployment > masterNodeIpAddress	<Kubernetes master ip>
service > loadBalancerIp	<metalLB_upfPfcplp>
service > loadBalancerFqdn	upf.affirmed.com
pod-ppe	
ppeNbWorkers	2
deployment > masterNodeIpAddress	<Kubernetes master ip>
sriov > Labels	This value should match the value configured for the Sriovlabel1 parameter from the Cluster Management > Kubernetes Clusters screen. See Create Kubernetes Cluster .
sriov > Labels	This value should match the value configured for the Sriovlabel2 parameter from the Cluster Management > Kubernetes Clusters screen. See Create Kubernetes Cluster .
pod-upf-monitor > deployment masterNodeIpAddress	<Kubernetes master ip>

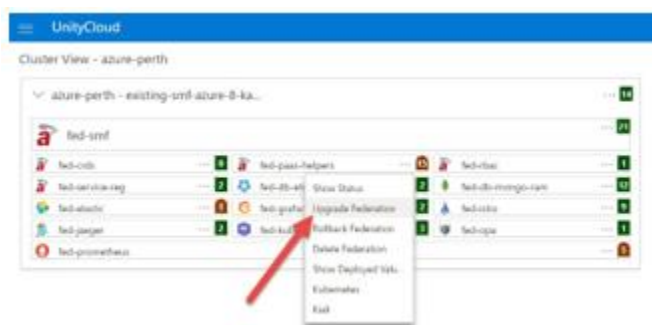
4. Click **Select Cluster** to display a dialog box from which you can select the cluster to which the federation will be deployed. After you have selected the cluster, the cluster is set for subsequent federation deployments.

Figure 103. Select Cluster for Federation Deployment

5. From the **Pick Cluster** dialog box, select the cluster and click Select.
6. (Optional) Click the Clipboard icon to verify that the chart will deploy correctly.
7. Click the Deploy Federation button (gear icon) to deploy the chart.

A pop-up window appears after the federation is installed successfully and the microservice status appears. Use the Ellipsis icon (...) to update the federation status. If a federation does not install successfully, you can roll back to a prior version from the **Install Federation** screen. For more information, see [Rollback Procedure](#).

8. To exit the **Install Federation** screen, click the X icon, which returns you to the geographical map and cluster view.
9. The cluster view shows all federations running on the cluster and each federation has a green indicator in the corner indicating the number of microservices. To manage each federation while in the cluster view, click the Ellipsis icon (...).

Figure 104. Cluster View of all Federations Running on Cluster

Jaeger Sampling and Debug Log Guidelines

- 1 (100%) Jaeger tracing supports 125K Simultaneous Active Users.
- When the load is greater than 125K SAU, it is recommended to lower Jaeger tracing to 0.001 and use the per SUPI/IMSI Jaeger tracing to get complete debug3 logs for specific SUPI/IMSI as needed.
- Default logging is Debug1. It is recommended not to enable Debug2 or Debug3 logging. Instead, use Jaeger tracing.

Shared Parameters for Network Function Deployment

The following references describe the shared parameters required when deploying a network function application using OPM.

NRF Shared Parameters

Table 63. fed-nrf Shared Parameters

Parameter	Value
docker-reg	Registry path.
elasticsearch-host	Elasticsearch IP address.
grafana-url	Exposed grafana URL.
nfs-filepath	Fileserver remote file path.
nfs-server	Fileserver remote host address.

Misc Information

Parameter	Value
nrf-cfgmgr-lb-ip	Load balancer IP address.
nrf-ingress-gw-lb-ip	Specific IP requested for Ingress Gateway LoadBalancer service.
oam-lb-subnet	Specify which Azure load balancer subnet to allocate from.
unique-name-suffix	Federation unique name suffix.
Security Parameters and Users	
Enable Local Password Auth	Use of local password for user authentication.
Enable Root	Root login is disabled by default.
Users	In order to access the network function with the CLI, you must define "admin", "oper", "readonly", "liadmin", and "lioper" group users in pod-cfgmgr with either public keys and/or certificate authority keys and/or SHA- encrypted passwords, and/or configure one or more TACACS servers.
user	User name.
passwd	User password.
pubkey	User public key.
group	User group.

AMF-MME Shared Parameters

Table 64. fed-amf_mme Shared Parameters

Parameter	Value
admin-password	Grafana password.
amf-cfgmgr-lb-ip	Load balancer IP address.
amf-ingress-gw-lb-ip	Specific IP requested for Ingress Gateway LoadBalancer service.
amf-ingress-gw-li-initial-protocol	Protocol for ingressGwLi port
amf-ingress-gw-li-lb-ip	Specific IP requested for LI Ingress Gateway LoadBalancer service.
db-etcd-lb-ip	Load balancer IP.
docker-reg	Registry path.

Misc Information

Parameter	Value
elasticsearch-host	Elasticsearch IP address.
fluentd-targets-host	Hostname or IP address of the forward server.
gn-lb-subnet	Specify which Azure load balancer subnet to allocate from.
grafana-lb-ip	Istio ingress gateway IP.
grafana-url	Exposed grafana URL.
gtpc-agent-ext-svc-name	Name of external gtpc agent service.
gtpc-agent-ext-svc-type	Interface type for gtpc_agent microservice.
gtpc-agent-lb-ip	https://kubernetes.io/docs/tasks/access-application-cluster/create-external-load-balancer/ .
image-pull-secrets	AS number to use for the local end of the session.
istio-gw-jaeger-hostname	Jaeger host (IP or FQDN).
istio-proxy-include-ip-ranges	Defines the subnets to be included within the Istio Service Mesh.
jaeger-agt-col-host	Collector host-configuration.
jaeger-host	The Jaeger host IP address.
kargo-lb-ip	Load-balancer IP.
kiali-lb-ip	Istio ingress gateway IP.
leader-election-renew-deadline	The timeout for the leader election of istio operator.
li-lb-subnet	Specify which Azure load balancer subnet to allocate from.
license-host	Flexera license server host.
license-port	Flexera license server port.
metallb-enable-frr	Enable speaker FRR mode.
metallb-frr-image	FRR image name.
metallb-frr-repo	FRR image repo path.
multusDeployed	Set to true if deploying along with fed-upf, fed-agf, and fed-n3iwf. In case of fed-amf, fed-mme nad fed-amf- mme, set to true if those feds use multus for providing services. In all other cases, set to false.

Misc Information

Parameter	Value
nfs-filepath	Fileserver remote file path.
nfs-server	Fileserver remote host address.
oam-lb-subnet	Specify which Azure load balancer subnet to allocate from.
prometheus-lb-ip	Istio ingress gateway IP.
redis-limit-cpu	CPU limit.
redis-limit-mem	Memory limit.
redis-primaries	Number of primaries.
redis-replicas	Number of replicas.
redis-request-cpu	CPU requests.
redis-request-mem	Memory requests.
storage-type	Storage type may be local, remote, or rook.
tlscrt	tlscrt configuration.
tlskey	tlskey configuration.
unique-name-suffix	Federation unique name suffix.
bgp-peer	
metallb-peer-name	BGP peer name.
metallb-peer-asn	AS number to expect from the remote end of the session.
metallb-peer-ip	Address to dial when establishing the session.
metallb-my-asn	AS number to use for the local end of the session.
ip-address-pool	
metallb-pool-name	IP address pool name.
metallb-pool-addresses	
metallb-ip-address-range	A single IP address range that can either be a CIDR prefix, or an explicit start-end range of IPs.
Security Parameters and Users	
Include security params and users	Enable to configure and include security parameters and users.
Enable Local Password Auth	Use of local password for user authentication.

Parameter	Value
Enable Root	Root login is disabled by default.
Users	In order to access the network function with the CLI, you must define "admin", "oper", "readonly", "liadmin", and "lioper" group users in pod-cfgmgr with either public keys and/or certificate authority keys and/or SHA- encrypted passwords, and/or configure one or more TACACS servers.
user	User name.
passwd	User password.
pubkey	User public key.
group	User group.

NSSF Shared Parameters

Table 65. fed-nssf Shared Parameters

Parameter	Value
admin-password	Grafana password.
db-etcd-lb-ip	Load balancer IP.
docker-reg	Registry path.
elasticsearch-host	Elasticsearch IP address.
grafana-lb-ip	Istio ingress gateway IP.
grafana-url	Exposed grafana URL.
image-pull-secrets	AS number to use for the local end of the session.
istio-gw-jaeger-hostname	Jaeger host (IP or FQDN).
istio-proxy-include-ip-ranges	Defines the subnets to be included within the Istio Service Mesh.
jaeger-host	The Jaeger host IP address.
kiali-lb-ip	Istio ingress gateway IP.
leader-election-renew-deadline	The timeout for the leader election of istio operator.
metallb-enable-frr	Enable speaker FRR mode.
metallb-frr-image	FRR image name.

Misc Information

Parameter	Value
metallb-frr-repo	FRR image repo path.
multusDeployed	Set to true if deploying along with fed-upf, fed-agf, and fed-n3iwf. In case of fed-amf, fed-mme nad fed-amf- mme, set to true if those feds use multus for providing services. In all other cases, set to false.
nfs-filepath	Fileserver remote file path.
nfs-server	Fileserver remote host address.
nrf-cfgmgr-lb-ip	Load balancer IP address.
nrf-ingress-gw-lb-ip	Specific IP requested for Ingress Gateway LoadBalancer service.
oam-lb-subnet	Specify which Azure load balancer subnet to allocate from.
prometheus-lb-ip	Istio ingress gateway IP.
storage-type	Storage type may be local, remote, or rook.
unique-name-suffix	Federation unique name suffix.
bgp-peer	
metallb-peer-name	BGP peer name.
metallb-peer-asn	AS number to expect from the remote end of the session.
metallb-peer-ip	Address to dial when establishing the session.
metallb-my-asn	AS number to use for the local end of the session.
ip-address-pool	
metallb-pool-name	IP address pool name.
metallb-pool-addresses	
metallb-ip-address-range	A single IP address range that can either be a CIDR prefix, or an explicit start-end range of IPs.
Security Parameters and Users	
Include security params and users	Enable to configure and include security parameters and users.
Enable Local Password Auth	Use of local password for user authentication.
Enable Root	Root login is disabled by default.

Misc Information

Parameter	Value
Users	In order to access the network function with the CLI, you must define "admin", "oper", "readonly", "liadmin", and "lioper" group users in pod-cfgmgr with either public keys and/or certificate authority keys and/or SHA- encrypted passwords, and/or configure one or more TACACS servers.
user	User name.
passwd	User password.
pubkey	User public key.
group	User group.

SMF Shared Parameters

Table 66. fed-smf Shared Parameters

Parameter	Value
admin-password	Grafana password.
db-etcd-lb-ip	Load balancer IP.
docker-reg	Registry path.
elasticsearch-host	Elasticsearch IP address.
fluentd-targets-host	Hostname or IP address of the forward server.
gn-lb-subnet	Specify which Azure load balancer subnet to allocate from.
grafana-lb-ip	Istio ingress gateway IP.
grafana-url	Exposed grafana URL.
gtpc-agent-ext-svc-name	Name of external gtpc agent service.
gtpc-agent-ext-svc-type	Interface type for gtpc_agent microservice.
gtpc-agent-lb-ip	https://kubernetes.io/docs/tasks/access-application-cluster/create-external-load-balancer/ .
image-pull-secrets	AS number to use for the local end of the session.
inband-data-agent-lb-ip	Load balancer IP address.
istio-gw-jaeger-hostname	Jaeger host (IP or FQDN).

Misc Information

Parameter	Value
istio-proxy-include-ip-ranges	Defines the subnets to be included within the Istio Service Mesh.
jaeger-agt-col-host	Collector host-configuration.
jaeger-host	The Jaeger host IP address.
kargo-lb-ip	Load-balancer IP.
kiali-lb-ip	Istio ingress gateway IP.
lcdr-filepath	lcdr filepath.
leader-election-renew-deadline	The timeout for the leader election of istio operator.
li-lb-subnet	Specify which Azure load balancer subnet to allocate from.
license-host	Flexera license server host.
license-port	Flexera license server port.
metallb-enable-frr	Enable speaker FRR mode.
metallb-frr-image	FRR image name.
metallb-frr-repo	FRR image repo path.
multusDeployed	Set to true if deploying along with fed-upf, fed-agf, and fed-n3iwf. In case of fed-amf, fed-mme nad fed-amf- mme, set to true if those feds use multus for providing services. In all other cases, set to false.
n4-lb-subnet	Specify which Azure load balancer subnet to allocate from.
nfs-filepath	Fileserver remote file path.
nfs-server	Fileserver remote host address.
oam-lb-subnet	Specify which Azure load balancer subnet to allocate from.
pfcp-c-loadbalancer-ip	Load balancer IP address.
pfcp-ext-svc-name	Name of external pfcp service.
pfcp-ext-svc-type	Interface type for pfcp microservice
pfcp-lb-ip	https://kubernetes.io/docs/tasks/access-application-cluster/create-external-load-balancer/ .
prometheus-lb-ip	Istio ingress gateway IP.

Misc Information

Parameter	Value
redis-limit-cpu	CPU limit.
redis-limit-mem	Memory limit.
redis-primaries	Number of primaries.
redis-replicas	Number of replicas.
redis-request-cpu	CPU requests.
redis-request-mem	Memory requests.
smf-cfgmgr-lb-ip	Load balancer IP address.
smf-ingress-gw-lb-ip	Specific IP requested for Ingress Gateway LoadBalancer service.
smf-ingress-gw-li-lb-ip	Specific IP requested for LI Ingress Gateway LoadBalancer service.
sshd-cipher-suite	Comma-separated ciphers to set the cipherSuiteString for sshd that terminates ssh connections (for CLI or NETCONF) on the cfgmgr pod sshd process.
storage-type	Storage type may be local, remote, or rook.
tls-cipher-suite	Colon-separated ciphers for TLS that terminates the RESTCONF interface on the cfgmgr pod confd process.
tlscrt	tlscrt configuration.
tlskey	tlskey configuration.
unique-name-suffix	Federation unique name suffix.
bgp-peer	
metallb-peer-name	BGP peer name.
metallb-peer-asn	AS number to expect from the remote end of the session.
metallb-peer-ip	Address to dial when establishing the session.
metallb-my-asn	AS number to use for the local end of the session.
ip-address-pool	
metallb-pool-name	IP address pool name.
metallb-pool-addresses	
metallb-ip-address-range	A single IP address range that can either be a CIDR prefix, or an explicit start-end range of IPs.

Misc Information

Parameter	Value
Security Parameters and Users	
Include security params and users	Enable to configure and include security parameters and users.
Enable Local Password Auth	Use of local password for user authentication.
Enable Root	Root login is disabled by default.
Users	In order to access the network function with the CLI, you must define "admin", "oper", "readonly", "liadmin", and "lioper" group users in pod-cfgmgr with either public keys and/or certificate authority keys and/or SHA-encrypted passwords, and/or configure one or more TACACS servers.
user	User name.
passwd	User password.
pubkey	User public key.
group	User group.

UPF Shared Parameters

Table 67. fed-upf Shared Parameters

Parameter	Value
admin-password	Grafana password.
db-etcd-lb-ip	Load balancer IP.
docker-reg	Registry path.
elasticsearch-host	Elasticsearch IP address.
fileserver-cephfs-enabled-True/False	Enable/disable CephFS.
fileserver-cfg-storage-class-name	Storage class name for pvc.
fileserver-requests-storage	Requests storage for pvc.
fileserver-web-storage-class-name	Storage class name for pvc.
fluentd-targets-host	Hostname or IP address of the forward server.
gn-lb-subnet	Specify which Azure load balancer subnet to allocate from.
grafana-lb-ip	Istio ingress gateway IP.

Misc Information

Parameter	Value
grafana-url	Exposed grafana URL.
image-pull-secrets	AS number to use for the local end of the session.
istio-gw-jaeger-hostname	Jaeger host (IP or FQDN).
istio-gw-li-jaeger-hostname	Jaeger host (IP or FQDN).
istio-proxy-include-ip-ranges	Defines the subnets to be included within the Istio Service Mesh.
jaeger-host	The Jaeger host IP address.
kiali-lb-ip	Istio ingress gateway IP.
l3am-max-ppe	Maximum number of PPE pods expected in this UPF.
l3am-spread-factor	The number of L3AM Resource Blocks that will be assigned to each PPE pod. Each resource block encapsulates an N3 loopback as well as the UI Pool chunks from the UE Pools configured on the N6 network instances. The range is [1, (maxPpe - 1)] inclusive. Recommended value is (maxPpe -1).
leader-election-renew-deadline	The timeout for the leader election of istio operator.
license-host	Flexera license server host.
license-port	Flexera license server port.
metallb-enable-frr	Enable speaker FRR mode.
metallb-frr-image	FRR image name.
metallb-frr-repo	FRR image repo path.
multusDeployed	Set to true if deploying along with fed-upf, fed-agf, and fed-n3iwf. In case of fed-amf, fed-mme nad fed-amf- mme, set to true if those feds use multus for providing services. In all other cases, set to false.
n4-lb-subnet	Specify which Azure load balancer subnet to allocate from.
nfs-filepath	Fileserver remote file path.
nfs-server	Fileserver remote host address.
oam-lb-subnet	Specify which Azure load balancer subnet to allocate from.

Misc Information

Parameter	Value
pfc-p-ext-svc-name	Name of external pfc-p service.
pfc-p-ext-svc-type	Interface type for pfc-p microservice
pfc-p-lb-ip	https://kubernetes.io/docs/tasks/access-application-cluster/create-external-load-balancer/ .
pfc-p-u-external-fqdn	If specified, this is added as an annotation to the PFC-P service. Other applications can then query to find the FQDN through which external peers can reach this application.
pfc-p-u-lb-ip	Load balancer IP address.
ppe-node-zone-resiliency-enabled	Enable or disable zone/rack level resiliency for PPE failures. By default it will be disabled.
prometheus-lb-ip	Istio ingress gateway IP.
redis-limit-cpu	CPU limit.
redis-limit-mem	Memory limit.
redis-primaries	Number of primaries.
redis-replicas	Number of replicas.
redis-request-cpu	CPU requests.
redis-request-mem	Memory requests.
sshd-cipher-suite	Comma-separated ciphers to set the cipherSuiteString for sshd that terminates ssh connections (for CLI or NETCONF) on the cfgmgr pod sshd process.
storage-type	Storage type may be local, remote, or rook.
tdef-enabled - true/false	Enable/disable traffic detection enforced function.
tls-cipher-suite	Colon-separated ciphers for TLS that terminates the RESTCONF interface on the cfgmgr pod confd process.
tvS-enabled - true/false	Enable/disable transparent video shaping function.
unique-name-suffix	Federation unique name suffix.
upf-cfgmgr-lb-ip	Load balancer IP address.
upf-ingress-gw-lb-fqdn	FQDN to be added as annotation to the Ingress Gateway LoadBalancer service.
upf-ingress-gw-lb-ip	Specific IP requested for Ingress Gateway LoadBalancer service.

Misc Information

Parameter	Value
bgp-peer	
metallb-peer-name	BGP peer name.
metallb-peer-asn	AS number to expect from the remote end of the session.
metallb-peer-ip	Address to dial when establishing the session.
metallb-my-asn	AS number to use for the local end of the session.
ip-address-pool	
metallb-pool-name	IP address pool name.
metallb-pool-addresses	
metallb-ip-address-range	A single IP address range that can either be a CIDR prefix, or an explicit start-end range of IPs.
Security Parameters and Users	
Include security params and users	Enable to configure and include security parameters and users.
Enable Local Password Auth	Use of local password for user authentication.
Enable Root	Root login is disabled by default.
Users	In order to access the network function with the CLI, you must define "admin", "oper", "readonly", "liadmin", and "lioper" group users in pod-cfgmgr with either public keys and/or certificate authority keys and/or SHA- encrypted passwords, and/or configure one or more TACACS servers.
user	User name.
passwd	User password.
pubkey	User public key.
group	User group.

Multi-Node OPM Shared Parameters

Table 68. fed-opm_multi Shared Parameters

Parameter	Value
General	
docker-reg	Docker Registry path

Misc Information

Parameter	Value
upstream-docker-reg	Upstream UC Infra Docker Registry Path
image-pull-secrets	AS number to use for the local end of the session.
enable-opm-dreg	True to enable OPM Docker Registry.
enable-infra-dreg	True to enable UC Infra Docker Registry, require Cinder CSI for storage
Infra-dreg-lbip	UC Infra Docker Registry IP
Infra-dreg-port	UC Infra Docker Registry Port
Infra-dreg-secret-name	UC Infra Docker Registry Secret Name
Infra-dreg-url	UC Infra Docker Registry URL
unique-name-suffix	Federation unique name suffix.
OPM	
opm-password	OPM password.
opm-platform	OPM platform.
opm-username	OPM username.
Load Balancer	
grafana-lb-ip	Istio ingress gateway IP.
istio-lb-ip	Value for istio-lb-ip is Istio Ingress Gateway IP.
kargo-lb-ip	Load-balancer IP.
kube-prom-lb-ip	Istio ingress gateway IP.
oam-lb-subnet	Specify which Azure load balancer subnet to allocate from.
opm-lb-ip	Specific IP requested for OPM Ingress Gateway LoadBalancer service.
prometheus-lb-ip	Istio ingress gateway IP.
thanos-lb-ip	Value for thanos-lb-ip is Istio Ingress Gateway IP.
PaaS	
admin-password	Grafana password.
elasticsearch-host	Elasticsearch IP address.
fed-unique-name	Federation unique name. This parameter is hidden in OPM GUI.

Misc Information

Parameter	Value
grafana-unique-name	Federation unique name. This parameter is hidden in OPM GUI.
grafana-url	Exposed grafana URL.
istio-gw-jaeger-hostname	Jaeger host (IP or FQDN).
istio-proxy-include-ip-ranges	Defines the subnets to be included within the Istio Service Mesh.
istio-unique-name	Federation unique name. This parameter is hidden in OPM GUI.
jaeger-host	The Jaeger host IP address.
leader-election-renew-deadline	The timeout for the leader election of istio operator.
opa-unique-name	Federation unique name. This parameter is hidden in OPM GUI.
paas-helpers-unique-name	Federation unique name. This parameter is hidden in OPM GUI.
prometheus-unique-name	Federation unique name. This parameter is hidden in OPM GUI.
redis-operator-unique-name	Federation unique name. This parameter is hidden in OPM GUI.
redis-cluster-unique-name	Federation unique name. This parameter is hidden in OPM GUI.
service-reg-unique-name	Federation unique name. This parameter is hidden in OPM GUI.
storage-type	Storage type may be local, remote, or rook.
tlsctx	tlsctx configuration.
tlskey	tlskey configuration.
Mongo/Cosmos	
mongito-init-password	Database init password.
mongo-init-username	Database init username.
mongo-password	Database password.
mongo-username	Database username.
mongodb-image	Deployment image of MongoDB.
NFS	
nfs-filepath	Fileserver remote file path.

Misc Information

Parameter	Value
nfs-server	Fileserver remote host address.
Redis	
redis-limit-cpu	CPU limit.
redis-limit-mem	Memory limit.
redis-primaries	Number of primaries.
redis-replicas	Number of replicas.
redis-request-cpu	CPU requests.
redis-request-mem	Memory requests.
Metallb	
bgp-peer	
metallb-peer-asn	AS number to expect from the remote end of the session.
metallb-peer-ip	Address to dial when establishing the session.
metallb-my-asn	AS number to use for the local end of the session.
ip-address-pool	
metallb-pool-name	IP address pool name.
metallb-pool-address	
metallb-ip-address-range	A single IP address range that can either be a CIDR prefix, or an explicit start-end range of IPs.

fed-amf_mme Deployment Modes

fed-amf_mme can be deployed in the following ways:

- Stand-alone AMF
- Stand-alone MME
- Combined AMF and MME

Use the settings described in this section to deploy fed-amf_mme in your desired configuration with OPM.

Stand-alone AMF Deployment

Set the following values when using OPM to deploy fed-amf_mme as stand-alone AMF. Enter all other values as required:

Table 69. Stand-alone AMF Deployment Helm Values

Parameter	Value
Enable Amf Function	True
Enable Mme Function	False
On Prem	True
pod-amf_n2 > fqdnIpList > Fqdn	Provide FQDN
pod-amf_n2 > fqdnIpList > ipAddresses	Provide corresponding IP address
pod-cfgmgr > confd > Aes256cfb128Key	Provide confd key
pod-gtpc_agent> externalSvc > Enabled	True
pod-gtpc_agent> externalSvc > Name	n26-amf
pod-gtpc_agent> externalSvc > Load Balancer IP	Provide IP address
pod-gtpc_agent> externalSvc > Type	n26-amf
pod-lb_ppe > endpoints > ips >	Provide the IP address used for amf-n2 service endpoints
pod-lb_ppe > endpoints > sctp> ports	Provide the port used for amf-n2 service endpoints
pod-lb_ppe > sriov > Labels	Add labels used for SR-IOV

Stand-alone MME Deployment

Set the following values when using OPM to deploy fed-amf_mme as stand-alone MME. Enter all other values as required:

Table 70. Stand-alone MME Deployment Helm Values

Parameter	Value
Enable Amf Function	False
Enable Mme Function	True
On Prem	True
diameterGroups > Enabled	True

Misc Information

Parameter	Value
diameterGroups > fqdnIpList > Fqdn	Add FQDN for mme-s1, diameter-base as needed
diameterGroups > fqdnIpList > ipAddresses	Provide corresponding IP addresses
pod-cfgmgr > confd > Aes256cfb128Key	Provide confd key
pod-gtpc_agent> externalSvc > Enabled	True
pod-gtpc_agent> externalSvc > Name	s11-mme
pod-gtpc_agent> externalSvc > Load Balancer IP	Provide IP address
pod-gtpc_agent> externalSvc > Type	s11-mme
pod-gtpc_agent> externalSvc > Enabled	True
pod-gtpc_agent> externalSvc > Name	s10
pod-gtpc_agent> externalSvc > Load Balancer IP	Provide IP address
pod-gtpc_agent> externalSvc > Type	s10
pod-gtpc_agent> externalSvc > Enabled	True
pod-gtpc_agent> externalSvc > Name	n26-mme
pod-gtpc_agent> externalSvc > Load Balancer IP	Provide IP address
pod-gtpc_agent> externalSvc > Type	n26-mme
pod-mme_cc > externalGtpcSvc > Enabled	True
pod-mme_cc > externalGtpcSvc > Name	s11-mme
pod-mme_cc > externalGtpcSvc > Load Balancer IP	Provide IP address
pod-mme_cc > externalGtpcSvc > Load Balancer FQDN	s11-mme
pod-mme_cc > externalGtpcSvc > Type	s11-mme
pod-mme_cc > externalGtpcSvc > Enabled	True
pod-mme_cc > externalGtpcSvc > Name	s10
pod-mme_cc > externalGtpcSvc > Load Balancer IP	Provide IP address
pod-mme_cc > externalGtpcSvc > Load Balancer FQDN	s10
Parameter	Value
pod-mme_cc > externalGtpcSvc > Type	s10
pod-mme_cc > externalGtpcSvc > Enabled	True

Misc Information

Parameter	Value
pod-mme_cc > externalGtpcSvc > Name	n26-mme
pod-mme_cc > externalGtpcSvc > Load Balancer IP	Provide IP address
pod-mme_cc > externalGtpcSvc > Load Balancer FQDN	n26-mme
pod-mme_cc > externalGtpcSvc > Type	n26-mme
pod-diameter_base > diameterGroups > Enabled	True
pod-diameter_base > diameterGroups > External Ip Service Id	Add based on FQDN set above
pod-lb_ppe > endpoints > ips >	Provide the IP address used for mme-s1 and diameter- base service endpoints
pod-lb_ppe > endpoints > sctp> ports	Provide the port used for mme-s1 and diameter-base service endpoints
pod-lb_ppe > sriov > Labels	Add labels used for SR-IOV

Combined AMF and MME Deployment

Set the following values when using OPM to deploy fed-amf_mme as combined AMF and MME. Enter all other values as required:

Table 71. Combined AMF and MME Deployment Helm Values

Parameter	Value
Enable Amf Function	True
Enable Mme Function	True
On Prem	True
diameterGroups > Enabled	True
diameterGroups > fqdnIpList > Fqdn	Add FQDN for mme-s1, diameter-base, amf-n2 as needed
diameterGroups > fqdnIpList > ipAddresses	Provide corresponding IP addresses
pod-cfgmgr > confd > Aes256cfb128Key	Provide confd key
pod-gtpc_agent> externalSvc > Enabled	True
pod-gtpc_agent> externalSvc > Name	n26-amf
pod-gtpc_agent> externalSvc > Load Balancer IP	Provide IP address

Misc Information

Parameter	Value
Parameter	Value
pod-gtpc_agent> externalSvc > Type	n26-amf
pod-gtpc_agent> externalSvc > Enabled	True
pod-gtpc_agent> externalSvc > Name	s11-mme
pod-gtpc_agent> externalSvc > Load Balancer IP	Provide IP address
pod-gtpc_agent> externalSvc > Type	s11-mme
pod-gtpc_agent> externalSvc > Enabled	True
pod-gtpc_agent> externalSvc > Name	s10
pod-gtpc_agent> externalSvc > Load Balancer IP	Provide IP address
pod-gtpc_agent> externalSvc > Type	s10
pod-gtpc_agent> externalSvc > Enabled	True
pod-gtpc_agent> externalSvc > Name	n26-mme
pod-gtpc_agent> externalSvc > Load Balancer IP	Provide IP address
pod-gtpc_agent> externalSvc > Type	n26-mme
pod-mme_cc > externalGtpcSvc > Enabled	True
pod-mme_cc > externalGtpcSvc > Name	s11-mme
pod-mme_cc > externalGtpcSvc > Load Balancer IP	Provide IP address
pod-mme_cc > externalGtpcSvc > Load Balancer FQDN	s11-mme
pod-mme_cc > externalGtpcSvc > Type	s11-mme
pod-mme_cc > externalGtpcSvc > Enabled	True
pod-mme_cc > externalGtpcSvc > Name	s10
pod-mme_cc > externalGtpcSvc > Load Balancer IP	Provide IP address
pod-mme_cc > externalGtpcSvc > Load Balancer FQDN	s10
pod-mme_cc > externalGtpcSvc > Type	s10
pod-mme_cc > externalGtpcSvc > Enabled	True
pod-mme_cc > externalGtpcSvc > Name	n26-mme

Parameter	Value
pod-mme_cc > externalGtpcSvc > Load Balancer IP	Provide IP address
pod-mme_cc > externalGtpcSvc > Load Balancer FQDN	n26-mme
pod-mme_cc > externalGtpcSvc > Type	n26-mme
Parameter	Value
pod-diameter_base > diameterGroups > Enabled	True
pod-diameter_base > diameterGroups > External Ip Service Id	Add based on FQDN set above
pod-lb_ppe > endpoints > ips >	Provide the IP address used for amf-n2, mme-s1 and diameter-base service endpoints
pod-lb_ppe > endpoints > sctp> ports	Provide the port used for amf-n2, mme-s1 and diameter- base service endpoints
pod-lb_ppe > sriov > Labels	Add labels used for SR-IOV

UnityCloud Shutdown and Startup Procedures

This section describes the procedures for shutting down an entire UnityCloud cluster and a single UnityCloud node in a controlled fashion.

- [Whole Cluster Shutdown/Startup](#)
- [Single Platform VM Shutdown/Startup](#)
- [Shutdown a VM](#)
- [Delete a VM](#)
- [OpenStack Compute Shutdown/Startup](#)
- [Kubernetes Disaster Recovery Operations](#)
- [Troubleshoot Unplanned Node Loss](#)

Whole Cluster Shutdown/Startup

This procedure is for shutting down an entire UnityCloud cluster in a controlled fashion so it can be later brought back up safely. This is to be done before and after a cluster maintenance window or during some disaster scenarios.

Caution: Streaming data and data in memory can be lost when shutting down the cluster. External storage services may need to be stopped/started separately.

Note: For service continuity, it is required to redirect traffic from the cluster to be shut down to another cluster.
The steps in this procedure must be performed as the root user.

Graceful Cluster Shutdown

1. Backup the application-level data.
2. If you are performing maintenance on control plane nodes, create a backup of Kubernetes etcd as explained in [Backup Kubernetes etcd](#).
3. Scale all application deployments down to zero:

Note: Make note of the current replica count for each deployment. You will use these counts to scale up the deployments during cluster startup.

```
# kubectl scale deploy <deployment_name> -n <namespace> -- replicas=0
```

4. Drain all worker nodes:

```
# kubectl drain --force --delete-emptydir-data --ignore-daemonsets --  
disable-eviction=true <worker_node_name>
```

Note: The **--disable-eviction** parameter will allow pod deletion without considering PDBs.

5. Make sure that all worker nodes have the status **Ready,SchedulingDisabled**.

```
# kubectl get nodes
```

6. Shutdown worker nodes gracefully (not hard). Refer to [Shutdown a VM](#) for instructions.

7. Find out where the etcd leader is running, that is going to be the last master node to be shut down. Find out which pods are running etcd:

```
# kubectl get pods -n kube-system -o wide
```

8. Get the list of active etcd members. This will also show which master node is currently the etcd leader. The current etcd leader will have **true**.

a. Execute the command and examine output:

```
# kubectl exec -it -n kube-system <etcd_pod_name> -- etcdctl --
endpoints=https://127.0.0.1:2379 -- cacert=/etc/kubernetes/pki/etcd/ca.crt
-- cert=/etc/kubernetes/pki/etcd/healthcheck-client.crt --
key=/etc/kubernetes/pki/etcd/healthcheck-client.key member list
```

b. If above command shows all false, execute the following command for each etcd pod to determine the leader:

```
# kubectl exec -it -n kube-system <etcd_pod_name> -- etcdctl --
endpoints=https://127.0.0.1:2379 -- cacert=/etc/kubernetes/pki/etcd/ca.crt
-- cert=/etc/kubernetes/pki/etcd/healthcheck-client.crt --
key=/etc/kubernetes/pki/etcd/healthcheck-client.key endpoint status --
write-out=table
```

9. Shut down all other master nodes, leaving the current etcd leader for the last. Refer to [Shutdown a VM](#) for steps to shut down a node.

10. Shut down the etcd leader node. Refer to [Shutdown a VM](#) for steps to shut down a node.

11. Shut down HA Proxy node(s). Refer to [Shutdown a VM](#) for steps to shut down a node.

12. Shut down OPM.

If running a single-node OPM cluster, refer to [Shutdown a VM](#) for steps to shutdown the single OPM node.

If running a multi-node OPM cluster (Note that only 1 master, 3 worker node configuration is currently supported by OPM):

a. Perform steps in [Backup Kubernetes etcd](#) for OPM since backup should be stored somewhere external to the cluster.

b. Scale all application deployments down to zero.

Note: Make note of the current replica count for each deployment. You will use these counts to scale up the deployments during cluster startup.


```
# kubectl scale deploy <deployment_name> -n <namespace> -- replicas=0
```

- c. Drain all OPM worker nodes

```
# kubectl drain --delete-emptydir-data --ignore-daemonsets
<opm_worker_node_name>
```

- d. Make sure all the worker nodes have the status Ready, **SchedulingDisabled**

```
# kubectl get nodes
```

- e. Shut down OPM worker nodes gracefully (not hard). Refer to [Shutdown a VM](#) for steps to shut down a node.

- f. Shut down OPM master node. Refer to [Shutdown a VM](#) for steps to shut down a node.

Cluster Startup

1. For single node OPM, start OPM node(s) and wait until all node(s) are Ready.

```
# kubectl get nodes
```

For multi-node OPM clusters, start the OPM master node first, followed by the OPM workers. When nodes are up, uncordon each of the OPM worker nodes, until all worker nodes are in Ready state.

```
# kubectl uncordon <opm_worker_node_name>
```

```
# kubectl get nodes
```

Bring back all the OPM applications by scaling them up again, using the deployment's replica count before shutdown:

```
# kubectl scale deploy <deployment_name> -n <name-space> -- replicas=N
```

3. Start Platform master etcd leader and wait until you get status Ready.

```
# kubectl get nodes
```

4. Start the rest of the Platform master nodes, and wait for them to become Ready.

```
# kubectl get nodes
```

5. Start all the Platform workers, wait until you see them on status **Ready,SchedulingDisabled**.

```
# kubectl get nodes
```

6. Uncordon each of the Platform worker nodes, until all worker nodes are in **Ready** state:

```
# kubectl uncordon <worker_node_name>
```

7. Bring back all the applications by scaling them up again, using the deployment's replica count before shutdown:

```
# kubectl scale deploy <deployment_name> -n <name-space> -- replicas=N
```

8. Verify all UC federation pods are up and running before redirecting traffic back to cluster.

Single Platform VM Shutdown/Startup

This procedure is for gracefully shutting down (and starting up) a single UnityCloud node. This procedure should be done before and after a planned node maintenance window or during some disaster scenarios.

If rebooting or shutting down Openstack Compute nodes, or there is a plan to delete VMs, refer to [OpenStack Compute Shutdown/Startup](#).

Important Notes

- Streaming data and data in memory can be lost when shutting down a node.
- Removing a node may result in degraded workload capacity and throughput.
- Statefulset pods configured to use local storage are expected to stay in Pending when it is on a node that is shutdown or deleted. Local storage is not recommended in production.
- Statefulset pods are not automatically evicted from shutdown node. Statefulset pods can be evicted from shutdown node by following <https://v1-21.docs.kubernetes.io/docs/tasks/run-application/force-delete-stateful-set-pod/>

Caution: Evicting statefulset pods can be unsafe. Read considerations carefully.

- Statefulset pods may take 6 minutes or more to go back to Running state after a node that it was on is restarted. This is because of a hardcoded internal 6-minute timer in Kubernetes that waits for volume to be safely unmounted from node. The following Kubernetes KEP improves this behavior - [enhancements/keps/sig-storage/2268-non-graceful-shutdown at master · kubernetes/enhancements \(github.com\)](https://github.com/kubernetes/enhancements/pull/2268).
- Daemonset pods are not evicted from **NotReady** nodes. K8s daemonset pods will show as **Running** even when the node is **NotReady**. This is a known issue.
- The steps in this procedure need to be performed as the root user.

Graceful Worker VM Shutdown

Note: Before shutting down, decommissioning, or rebooting a worker node, in order to minimize any possible traffic impact, it is critical to steer traffic away from the node before shutdown by de-prefencing the node's BGP client IP on your fabric.

1. Backup any application data that might be affected by the node shutdown.
2. Create a backup of Kubernetes etcd. Refer to [Backup Kubernetes etcd](#).
3. Cordon the node:

```
# kubectl cordon <worker_node_name>
```

4. Drain worker node:

```
# kubectl drain --force --delete-emptydir-data --ignore-daemonsets --
disable-eviction=true <worker_node_name>
```

Note: The `--disable-eviction` parameter will allow pod deletion without considering PDBs.

5. Shut the node down. Refer to [Shutdown a VM](#) for steps to shut down a node.

Delete and Rejoin a Worker Node

If the worker node that is shut down needs be decommissioned permanently or is going to be offline for some time, you may delete the node from the cluster. Deleting the node will evict all pods that are still associated to the node. This node can be re-joined at a later time as a new node.

Note: For short node maintenance or planned node reboots, it is **not required** and **not recommended** to delete a node.

Note: When deleting a node, all custom node labels or annotations that have been applied on the node post-cluster installation will be lost. Review the existing labels and annotations by describing the node and reviewing what labels or annotations need to be manually applied after re-adding deleted node.

Note: When deleting a node that is running fed-rook-ceph OSDs, it is required to purge the Ceph disk so that it can be joined successfully later on a new OSD, on a new node.

```
# kubectl describe node <worker_node_name>
```

There are two ways to delete a node:

- If you are using OPM to manage the cluster, delete the worker node in the Cluster Management page.
- If you are not using OPM to manage the cluster:

```
# kubectl delete node <worker_node_name>
```

The new node is automatically added and re-joined at boot-up. to the cluster when it boots.

To add a new node:

- If you are using OPM to manage the cluster, add the node in the Cluster Management page and wait for the VM to be created and node to be re-joined.
- If you are not using OPM, manually add the new worker VM using OpenStack. For example:

```
# openstack stack create --wait cna-worker-001 -t
CNA_platform_all.multinetwork.yaml -e cna-worker-001.env
```

Note: If you have de-preferenced a node using BGP on your fabric, remove the de-prefencing after the node/cluster is healthy.

Graceful Master VM Shutdown

Note: For single-master clusters, losing the only master node results in the cluster going to a read-only state. For multi-master clusters, losing 2 of 3 (quorum) master nodes will result in the cluster going to a read-only state. No new pods can be scheduled, updated, deleted, replicated if there is no quorum. When cluster is running headless, some workload operators that depend on the API server will fail to function properly. This can cause service outages.

To shut down a master node:

1. Backup any application data that might be affected by the node shutdown.

2. Create a backup of etcd. Refer to [Backup Kubernetes etcd](#).
3. On the secondary or tertiary master, grab and store the certificate key used to encrypt the temporary kubeadm-secret. **[For single master clusters, skip this step].**

Example:

```
# grep key=\"k8_cert_key\" /var/log/configParser.log

<Property oe:key=\"k8_cert_key\"
oe:value=\"a45a0b6e3768765bea530de062b01cf89c5d705c450cd2fff507d
23893eb7bdf\"/>
```

4. Shut the node down. Refer to [Shutdown a VM](#) for steps to shut down a node.

Note: It is not required to drain a master node.

Delete and Rejoin a Master Node

If master node that is shut down needs to be decommissioned permanently or is going to be offline for some time, you may delete the node from the cluster. Deleting the node will evict all pods that are still associated to the node. This node can be re-joined later as a new node.

Note: If using OPM to manage the cluster, you can only delete the secondary and tertiary master in a 3 master-node cluster. You cannot delete the only master node in a single-master cluster. If not using OPM, you must delete the master VM from the Openstack Hypervisor aside from deleting the master node from K8s control plane.

Note: For short node maintenance or planned node reboots, it is **not required** and **not recommended** to delete a master node from Kubernetes or the VM itself.

Note: When deleting a node, all custom node labels or annotations applied on the node post- cluster installation will be lost. Review existing labels and annotations by describing the node and reviewing what labels or annotations need to be manually applied after re-adding deleted node.

Refer to [Delete a VM](#) to delete the master node VM. Perform this before proceeding to the next steps below.

```
# kubectl describe node <master_node_name>
```

To delete a node:

```
# kubectl delete node <master_node_name>
```

To re-join a master node, perform the following:

1. On another master node, re-upload control plane certificates as a new un-expired Secret to the cluster. The **<cert_key>** parameter is the certificate key you saved before shutting down the master node.

```
# kubeadm init phase upload-certs --upload-certs --certificate- key
<cert_key>
```

If the certificate key was not saved, look for the key on other master nodes. If the certificate key cannot be found, do not supply one so a new key is generated.

```
# kubeadm init phase upload-certs --upload-certs
```

2. Remove the master node from the Kubernetes etcd member list if it exists. Run the following on an etcd node:

```
# kubectl exec -it -n kube-system <etcd_pod_name> -- etcdctl --
endpoints=https://127.0.0.1:2379 -- cacert=/etc/kubernetes/pki/etcd/ca.crt
-- cert=/etc/kubernetes/pki/etcd/healthcheck-client.crt --
key=/etc/kubernetes/pki/etcd/healthcheck-client.key member list
```

where **<etcd_pod_name>** is any pod in

```
# kubectl -n kube-system get pods -- selector=component=etcd,tier=control-
plane
```

From the etcd member list, take note of the ID in the 1st column of the etcd member that was deleted and use it as **<etcd_member_id>** in command below:

```
# kubectl exec -it -n kube-system <etcd_pod_name> -- etcdctl --
endpoints=https://127.0.0.1:2379 -- cacert=/etc/kubernetes/pki/etcd/ca.crt
-- cert=/etc/kubernetes/pki/etcd/healthcheck-client.crt --
key=/etc/kubernetes/pki/etcd/healthcheck-client.key member remove
<etcd_member_id>
```

3. Power on the master node and wait for it to join the control-plane. If master VM was deleted, re-create it first.

a. Verify that added node is Ready by executing the following command:

```
# kubectl get nodes
```

b. Verify that master is showing up as Kubernetes etcd member:

```
# kubectl exec -it -n kube-system <etcd_pod_name> -- etcdctl --
endpoints=https://127.0.0.1:2379 -- cacert=/etc/kubernetes/pki/etcd/ca.crt
-- cert=/etc/kubernetes/pki/etcd/healthcheck-client.crt --
key=/etc/kubernetes/pki/etcd/healthcheck-client.key member list
```

Note: If you have de-prefenced a node using BGP on your fabric, remove the de-prefencing after the node/cluster is healthy.

Graceful OPM VM Shutdown

In single-node OPM deployments, losing an OPM node can cause data loss and service outages when the NFS server, Docker registry, or chart repository on the OPM VM are not accessible.

For single OPM node clusters, refer to [Shutdown a VM](#) for steps to shutdown the single OPM VM.

VM Startup

After the node maintenance window, perform the steps to bring the node back online.

1. Start the VM and wait for the Kubernetes node to become **Ready** (master) or **Ready,SchedulingDisabled** (worker):

```
# kubectl get nodes
```

2. If starting a worker node, uncordon it.

```
# kubectl uncordon <worker_node_name>
```

Shutdown a VM

VMs can be shutdown gracefully, using any of the following options:

1. Using worker node's SSH terminal, run the following command as root (Preferred)

```
# shutdown -h 10
```

2. Using Openstack CLI command:

```
# openstack server stop <server_name_or_uuid>
```

3. Using Openstack Horizon, click the instance's options drop-down and then select **Shut Off Instance**.

Delete a VM

If you are using OPM to manage the cluster, delete the VMs/Nodes using the Cluster Management page.

If you are not using OPM to manage the cluster, delete OpenStack instance (VMs) from OpenStack using the command below:

```
# openstack stack delete <stack id>
```

Where **<stack id>** is the Openstack stack ID of the VM to be deleted. Use **openstack stack list** to get a list of stacks.

Note: Before deleting a stack, make sure to have copies of Openstack Heat and/or environment files used to create the VM.

OpenStack Compute Shutdown/Startup

The following procedure describes how to gracefully shutdown an Openstack Compute hosting UnityCloud VMs.

Graceful Compute Node Shutdown

1. Determine the instances that are running on the compute node to be powered-off

```
# openstack server list --all-projects --host <compute_node>
```

2. Determine if any of the VMs need to be manually re-created on another compute/zone to avoid an outage or service degradation.

Note: Note down the VM types (platform master or worker, OPM or HA Proxy VMs). Make sure the instance's VM parameters and environment variables are saved for possible re-use.

3. For each VM running on the compute, refer to [Single Platform VM Shutdown/Startup](#) Single Platform Node Shutdown/Startup to gracefully shut down the VM, depending on the VM type.
4. If required, refer to [Recreate VM on Another Compute/Zone](#) for steps on how to re-create the powered-off VM on another compute to minimize service degradation.

Recreate VM on Another Compute/Zone

VMs may be re-created on another compute/zone to minimize or avoid service outages or degradation.

Before re-creating VMs on another compute/zone, the powered-off VM must be deleted. Refer to [Delete a VM](#) for steps on deleting an Openstack VM.

Note: The new compute may not be able to handle the same workloads if it has no matching hardware resources (Example: SR-IOV interfaces, NUMA, CPU pinning). This must be taken into consideration.

Platform VM (Worker)

To add a new worker node to the Kubernetes cluster:

1. If using OPM, add a new worker VM in the Cluster Management page. Change the Vm Zone if needed.

If not using OPM, add a new worker VM using the OpenStack CLI. Example:

```
openstack stack create --wait cna-worker-001 -t CNA_platform_
all.multinetwork.yaml -e cna-worker-001.env
```

2. Wait for the VM to boot and join the cluster. Verify that added node is Ready by running following command:

```
# kubectl get nodes
```

Platform VM (Master)

Follow the steps in [Delete and Rejoin a Master Node](#) but for last step, the master VM is re- created:

1. If using OPM, add the new master VM in the Cluster Management page. Change the Vm Zone if needed (**Preferred**).

Alternatively, you can log into the OPM VM and execute the following:

```
# kubectl exec -it <vnfmgr-osrm-pod> -- /bin/bash -c 'source
/root/heat/stack.rc; openstack stack create -f json <master- hostname> -t
/root/heat/CNA_platform_all.yaml -e
/root/heat/<master-hostname>.env'
```

Where <vnfmgr-osrm-pod> is the pod name from `kubectl -n vnfmgr get pods --selector=app=vnfmgr-osrm-1000` and <master-hostname> is the hostname of the master to be created.

Note: If you are using OPM, deleting or re-adding the primary master VM is not supported.

If not using the OPM, create the new master using the OpenStack CLI:

```
# openstack stack create --wait cna-master-001 -t CNA_platform_
all.multinetwork.yaml -e cna-master-001.env
```

2. Wait for the VM to boot and join the cluster. Verify that added master node is Ready by running the following command:

```
# kubectl get nodes
```

3. Verify that master is showing up as Kubernetes etcd member:

```
# kubectl exec -it -n kube-system <etcd_pod_name> -- etcdctl --
endpoints=https://127.0.0.1:2379 -- cacert=/etc/kubernetes/pki/etcd/ca.crt
-- cert=/etc/kubernetes/pki/etcd/healthcheck-client.crt --
key=/etc/kubernetes/pki/etcd/healthcheck-client.key member list
```

OPM VM

When creating a new OPM VM, make sure to re-use the same OPM IP address and domain. Otherwise, UnityCloud application federations will need to be re-deployed or re-configured with new configuration.

Sample command to create a single node OPM VM:

```
# openstack stack create --wait cna-opm-001 -t CNA_OPM_VM_ cinder.yaml -e
cna-opm-001.env
```

Compute Node Startup

After the compute node maintenance window, perform the following steps:

1. Power on the compute node and verify that the instances are all in active state:

```
# openstack server list --all-projects --host <compute_node>
```

2. Wait for the VMs to boot up. Verify if applications running on the VMs are up and healthy.

Kubernetes Disaster Recovery Operations

This procedure describes how to backup the Kubernetes etcd database to a remote store. In disaster recovery scenarios, the Kubernetes control plane nodes can be recovered by restoring the etcd snapshot.

Note: Any new data post the etcd snapshot time will be lost when the snapshot is restored.

Backup Kubernetes etcd

Prerequisite: Remote storage with available space for etcd backups.

This procedure uses the OPM NFS server to store the UnitCloud Platform Kubernetes etcd backups. While etcd snapshots can be stored on the master nodes, it is not recommended. A remote store external to cluster is preferred so even if you lose the master node, you do not lose the backup.

Note: The steps below need to be performed as the root user (not maint user).

1. Create the backup location on the NFS share.

For example, on the OPM node execute the following:

```
# mkdir -p /filestore/uc-kube-etcd-backups
```

2. On all master nodes, log in as root, create the backup directory and mount the remote store:

```
# REMOTE_STORE=<opm_ip>
```

```
# ETCD_SNAPSHOT_DIR="/root/etcd-backups" # mkdir -p ${ETCD_SNAPSHOT_DIR}
```

```
# mount -t nfs ${REMOTE_STORE}:/filestore/uc-kube-etcd-backups/
${ETCD_SNAPSHOT_DIR}
```

3. Make sure there is enough size for the backup by inspecting at the size of the database to be backed up. On a master node, the database size can be checked using the following command:

```
# ls -sh /var/lib/etcd/member/snap/db
```

4. On the first master node, execute the following commands to create the backup job template:

```
# ETCD_IMAGE=$(grep image:
```

```
/etc/kubernetes/manifests/etcd.yaml | awk '{print $2}')
```

```
# BACKUP_JOB="etcd-backup.yaml"
```

```
# SNAPDATE=$(date +%Y%m%d%H%M%S%Z)
```

```
# cat > ${BACKUP_JOB} <<EOF
```

```

apiVersion: batch/v1 kind: Job

metadata:

name: etcd-backup namespace: kube-system labels:

job: etcd-backup spec:

template: metadata:

name: etcd-backup labels:

job: etcd-backup spec:

containers:

- name: etcd-backup image: ${ETCD_IMAGE} env:

- name: ETCDCCTL_API
value: "3"

command: ["/bin/sh"] args: ["-c", "etcdctl --

endpoints=https://127.0.0.1:2379 -- cacert=/etc/kubernetes/pki/etcd/ca.crt
-- cert=/etc/kubernetes/pki/etcd/healthcheck-client.crt --
key=/etc/kubernetes/pki/etcd/healthcheck-client.key snapshot save
/backup/etcd-snapshot-${SNAPDATE}.db"]

volumeMounts:

- mountPath: /etc/kubernetes/pki/etcd name: etcd-certs

readOnly: true

- mountPath: /backup name: etcd-backup

restartPolicy: Never nodeSelector:

node-role.kubernetes.io/master: "" tolerations:

- effect: NoSchedule operator: Exists

hostNetwork: true volumes:

- name: etcd-certs hostPath:

path: /etc/kubernetes/pki/etcd type: DirectoryOrCreate

- name: etcd-backup hostPath:

path: ${ETCD_SNAPSHOT_DIR}

type: Directory

```

EOF

5. On the first master node, delete the old etcd-backup job if it exists and run a new one based on the new template.

```
# kubectl -n kube-system delete job etcd-backup --ignore-not-found=true

# kubectl create -f ${BACKUP_JOB}
```

Note: The etcd backup is created from any of the master nodes. To find the node associated with the backup, you can use the command: `kubectl -n kube-system get pods -- selector=job=etcd-backup -o wide`

6. Verify that the backup is created in the remote store.

Restore Kubernetes etcd

This procedure is for restoring Kubernetes etcd from a previous backup. This is for recovering all Kubernetes control plane nodes in disaster recovery scenarios.

Note: The Kubernetes version of the backup file and the UC cluster to recover it to, should be the same. Cross-version restoration is not supported.

Note: The etcd snapshot is restored on one master node. In multi-master setups, etcd will sync the data to all other master nodes. However, this does not mean that the nodes are automatically joined as new etcd members to the cluster. You must join one master node to the cluster, restore the database and then continue re-joining the remaining master nodes.

Note: The steps below need to be performed as the root user (not maint).

Prerequisites:

- All master nodes are up and running.
- Etcd snapshot file accessible on 1st master node

Perform the following on the first master node. Note that some steps below assign values to shell variables that are then used in succeeding steps.

1. If not already mounted, mount the remote store where the snapshot is located. Alternatively, it could be copied over to the first master node. For example, the step below mounts the OPM Filestore and displays the snapshot file. Replace variables if needed.

REMOTE_STORE: IP address of the OPM NFS server ETCD_SNAPSHOT_DIR: Local path on master node mounting the remote store

```
REMOTE_STORE=<change me>
```

```
ETCD_SNAPSHOT_DIR="/root/etcd-backups"
```

```
# mkdir -p ${ETCD_SNAPSHOT_DIR}
```

```
# mount -t nfs ${REMOTE_STORE}:/filestore/uc-kube-etcd-backups/
${ETCD_SNAPSHOT_DIR}
```

```
# ls -lh ${ETCD_SNAPSHOT_DIR}
```

Note down the filename of the snapshot to be restored.

2. Stop etcd on ALL MASTER NODES by moving the static etcd pod manifest to a directory.
Example:

```
# mkdir -p /root/etcd-manifest
```

```
# mv /etc/kubernetes/manifests/etcd.yaml /root/etcd-manifest/
```

Before proceeding to the next step, make sure that etcd container is not running anymore on all master nodes.

```
# docker ps | grep -E "etcd "
```

3. Purge existing etcd on all master nodes.

Note: Purging etcd destroys the data.

```
# rm -rf /var/lib/etcd
```

4. Login back to the first master node if not already on it.
5. Restore the etcd snapshot on the first master node. Replace variables if needed:

ETCD_IMAGE: The etcd image repo path

SNAPSHOT: The filename of the snapshot to be restored noted in Step 1.

MASTER1_NODE_IP: The IP of the node. For multi-network clusters, this is the private/internal cluster network IP address of the first master node.

NODE_NAME: The master node's hostname. Command below uses \$(hostname)

Note: *NODE_NAME and MASTER1_NODE_IP should be the same values after --initial- cluster argument in the static etcd pod manifest file that was moved.*

Check /root/etcd-manifest/etcd.yaml

```
# ETCD_IMAGE=$(grep image: /root/etcd-manifest/etcd.yaml | awk '{print $2}')
```

```
SNAPSHOT=<change me> MASTER1_NODE_IP=<change me> NODE_NAME=$(hostname)
docker run\
```

```
--rm\
```

```
-v ${ETCD_SNAPSHOT_DIR}:/etcd_snapshot\
```

```
-v /var/lib:/var/lib\
```

```
--entrypoint "" ${ETCD_IMAGE} /bin/sh -c "\
```

```
ETCDCTL_API=3 etcdctl snapshot restore /etcd_snapshot/${SNAPSHOT}\
```

```
--data-dir /var/lib/etcd\
```

```
--name ${NODE_NAME}\
```

```
--initial-cluster ${NODE_NAME}=https://${MASTER1_NODE_IP}:2380\
```

```
--initial-advertise-peer-urls https://${MASTER1_NODE_IP}:2380"
```

6. Start etcd on the first master:

```
# mv /root/etcd-manifest/etcd.yaml /etc/kubernetes/manifests/
```

Verify that the 1st master joined the etcd cluster. If etcd is still not up, an error is returned. # sudo docker run\

```
--rm\
```

```
-v /etc/kubernetes/pki/etcd:/etc/kubernetes/pki/etcd\
```

```
--entrypoint "" ${ETCD_IMAGE} /bin/sh -c "\ ETCDCTL_API=3 etcdctl\
```

```
--endpoints=https://${MASTER1_NODE_IP}:2379\
```

Misc Information

```
--cacert=/etc/kubernetes/pki/etcd/ca.crt\  
--cert=/etc/kubernetes/pki/etcd/healthcheck-client.crt\  
--key=/etc/kubernetes/pki/etcd/healthcheck-client.key member list"
```

Example of successful output showing started etcd member:

```
1ee91ce3e139d620, started, master-1-node, https://10.0.0.1:2380,  
https://10.0.0.1:2379, false
```

7. For recovering multi-master clusters, you need to add one master node at a time. Otherwise, proceed to [Verify etcd Backup Restoration](#).

Login to the next master node to be joined if proceeding.

8. On the next master node, replace the variables in command below and join the new etcd member.

Note: Make sure that you did not miss stopping etcd and purging etcd data on this node (Step 2 and 3).

ETCD_IMAGE: The etcd image repo path

MASTER1_NODE_IP: The IP of the first master node already joined to the cluster. For multi-network clusters, this is the private/internal cluster network IP address of the first master node.

NODE_NAME: Hostname of the master node to be joined. Command below uses \$(hostname)

NODE_IP: The IP of the master node to be joined. For multi-network clusters, this is the private/internal cluster network IP address of the master node.

```
# ETCD_IMAGE=$(grep image: /root/etcd-manifest/etcd.yaml | awk '{print  
$2}')
```

```
MASTER1_NODE_IP=<change me>
```

```
NODE_IP=<change me>
```

```
NODE_NAME=$(hostname)
```

```
# sudo docker run\  
--rm\  
-v /etc/kubernetes/pki/etcd:/etc/kubernetes/pki/etcd\  
--entrypoint "" ${ETCD_IMAGE} /bin/bash -c "\ ETCDCTL_API=3 etcdctl\  

```



```
--endpoints=https://${MASTER1_NODE_IP}:2379\
--cacert=/etc/kubernetes/pki/etcd/ca.crt\
--cert=/etc/kubernetes/pki/etcd/healthcheck-client.crt\
--key=/etc/kubernetes/pki/etcd/healthcheck-client.key\ member add
${NODE_NAME} --peer-urls=https://${NODE_IP}:2380"
```

9. Start etcd on master node to be joined:

```
# mv /root/etcd-manifest/etcd.yaml /etc/kubernetes/manifests/
```

10. Verify that the master joined the etcd cluster:

```
# sudo docker run\
--rm\
-v /etc/kubernetes/pki/etcd:/etc/kubernetes/pki/etcd\
--entrypoint "" ${ETCD_IMAGE} /bin/bash -c "\ ETCDCTL_API=3 etcdctl\
--endpoints=https://${MASTER1_NODE_IP}:2379\
--cacert=/etc/kubernetes/pki/etcd/ca.crt\
--cert=/etc/kubernetes/pki/etcd/healthcheck-client.crt\
--key=/etc/kubernetes/pki/etcd/healthcheck-client.key member list"
```

11. Repeat steps 9, 10, and 11 on the remaining master nodes.

Verify etcd Backup Restoration

Use the following procedure confirms successful etcd backup restoration.

1. On the first master, list etcd members:

ETCD_IMAGE: The etcd image repo path.

MASTER1_NODE_IP: The IP of the first master node already joined to the cluster. For multi-network clusters, this is the private/internal cluster network IP address of the first master node.

```
# ETCD_IMAGE=$(grep image: /etc/kubernetes/manifests/etcd.yaml | awk
'{print $2}')
```

MASTER1_NODE_IP=<change me>

```
# sudo docker run\
```

```
--rm\
```

```
-v /etc/kubernetes/pki/etcd:/etc/kubernetes/pki/etcd\
--entrypoint "" ${ETCD_IMAGE} /bin/sh -c "\ ETCDCTL_API=3 etcdctl\
--endpoints=https://${MASTER1_NODE_IP}:2379\
--cacert=/etc/kubernetes/pki/etcd/ca.crt\
--cert=/etc/kubernetes/pki/etcd/healthcheck-client.crt\
--key=/etc/kubernetes/pki/etcd/healthcheck-client.key member list"
```

All members should be present and showing started state:

```
1ee91ce3e139d620, started, master-1-node, https://10.0.0.1:2380,
https://10.0.0.1:2379, false
```

```
96737f78c1829e9c, started, master-2-node, https://10.0.0.2:2380,
https://10.0.0.2:2379, false
```

```
9a5603a153f5832c, started, master-3-node, https://10.0.0.3:2380,
https://10.0.0.3:2379, false
```

2. Verify if all Kubernetes master nodes are showing Ready:

```
# kubectl get nodes
```

3. Verify all kube-system pods are Running:

```
# kubectl -n kube-system get pods
```

Troubleshoot Unplanned Node Loss

When there is an unplanned node loss, the primary course of action is to bring the node back as soon as possible. A lost K8s node will show up with NotReady state.

To verify node status, execute the following command:

```
kubectl get nodes
```

When a node is lost, some UnityCloud applications may not be able to automatically migrate to another available node. The following troubleshooting steps describe the UnityCloud federations that may require special handling or considerations during unplanned node loss scenarios.

Note: When a UnityCloud worker node is deemed un-recoverable, or if a node may be fixed but will require being offline for an extended period of time, the K8s worker node may be deleted from K8s control plane and the VM from OpenStack. Refer to [Delete and Rejoin a Worker Node](#) for more details.

Fed-kafka-strimzi

When using Kafka with the storage type set to "local," data is not persistent because it relies on ephemeral disks. To ensure data persistence, you should use an external storage back-end that is topology-aware, such as Cinder, FedRook, or Ceph. In this case, you don't need to select the storage type as "local."

The MirrorMaker component under the pod-kafka_strimzi is optional and disabled by default. It only needs to be enabled when you require data migration from an existing Kafka cluster.

Fed-service-reg (Consul)

If the node is quickly recovered, the Consul pod(s) will automatically restart on the same node.

Caution: The following operation is un-safe. This has the potential to violate properties inherent to K8s Statefulsets and cause data loss. Bringing the node back online should be the top priority.

1. If the node goes down and fails the Consul pod, forcefully delete the service-reg-server pod:

```
# kubectl delete pod <fed_service_reg_pod> --grace-period=0 -- force --
namespace fed-service-reg
```

2. Verify the new pod is created on another node and pod state is running:

```
# kubectl -n fed-service-reg get pods
```

Fed-elastic

If the node is quickly recovered, the Elasticsearch pod(s) will automatically restart on the same node.

Caution: The following operation is un-safe. This has the potential to violate properties inherent to K8s Statefulsets and cause data loss. Bringing the node back online should be the top priority.

1. If the node goes down and fails the Elasticsearch pod, forcefully delete the service-reg-server pod:

```
# kubectl delete pod <fed_elastic_pod> --grace-period=0 -force --namespace
fed-elastic
```

2. Verify the new pod is created on another node and pod state is running:

```
# kubectl -n fed-elastic get pods
```

Fed-rook-ceph

In clusters using UnityCloud Fed-rook-ceph storage, Ceph MON pods will take 15-20 minutes to failover when a node is lost or drained and is not recovered in time.

K8s Daemonsets

When a K8s node is lost, kube-system daemonset pods still shows that they are **Running**. The following kube-system pods will appear running even if the node they were on is **NotReady**.

- kube-proxy
- calico-node
- node-local-dns
- kube-multus-ds (will report as not ready)

This is a known issue in K8s. Bringing the node back online or deleting it will resolve the mis-reporting of the daemonset pods status.

Hashicorp Vault Configuration with cert-manager

This topic provides step-by-step instructions to integrate Hashicorp vault with cert-manager to utilize the HC vault as an issuer for certificates in Kubernetes. Cert-manager automates the management and issuance of TLS certificates from various sources, including HC vault. The information in this topic allows users to integrate HC vault with cert-manager and use HC vault as an issuer for certificates in Kubernetes.

Prerequisites

Hashicorp vault configuration is performed by Vault administrators. The configuration shown here only serves as a guidance and is subject to change based on customer requirements.

Initialize and Unseal Hashicorp vault

Use the following commands to initialize and unseal Hashicorp vault:

```
$kc exec -it -n <vault-namespace> vault-0 - sh
$vault operator init -key-shares=1 -key-threshold=1
$vault operator unseal <unseal key>
$vault login <root token>
```

Enable JWT Auth

Use the following command to enable JWT Auth. This is required for Kubernetes serviceaccount token validation.

```
$vault auth enable jwt
```

Configure JWT Auth

JWT auth config's `jwt_validation_pubkeys` is updated with `sa.pub` or with certificate created with JWK for AKS clusters. See the following example:

```
$vault write auth/jwt/config \
jwt_validation_pubkeys="-----BEGIN PUBLIC KEY-----
<Full Content of pub key>
-----END PUBLIC KEY-----"
```

Note: sa.pub can be found at /etc/kubernetes/pki/sa.pub on K8s master node, for AKS clusters, use below command to fetch the JWK and convert the output to a public key using <https://8gwifi.org/jwkconvertfunctions.jsp>

```
$ kubectl get --raw "$(kubectl get --raw /.well-known/openid-configuration | jq -r '.jwks_uri' | sed -r 's/.*\.[^/]+(.*)/\1/')" | jq '.keys | .[-1]'
```

Enable and Configure PKI Engines

Use the following commands to enable and configure the PKI engines for OAM, LI and SBI on respective paths.

```
$vault secrets enable --path=pki/oam/private/int pki
$vault secrets enable --path=pki/sbi/private/int pki
$vault secrets enable --path=pki/li/private/int pki

$vault secrets tune -max-lease-ttl=87600h pki/oam/private/int
$vault secrets tune -max-lease-ttl=87600h pki/sbi/private/int
$vault secrets tune -max-lease-ttl=87600h pki/li/private/int

$vault write pki/oam/private/int/root/generate/internal common_name=oamvault.com ttl=87600h
$vault write pki/sbi/private/int/root/generate/internal common_name=sbivault.com ttl=87600h
$vault write pki/li/private/int/root/generate/internal common_name=livault.com ttl=87600h

$vault write pki/oam/private/int/config/urls issuing_certificates="=<vault-service-url>/v1/pki/ca"
crl_distribution_points="=<vault-service-url>/v1/pki/crl"
$vault write pki/sbi/private/int/config/urls issuing_certificates="=<vault-service-url>/v1/pki/ca"
crl_distribution_points="=<vault-service-url>/v1/pki/crl"
$vault write pki/li/private/int/config/urls issuing_certificates="=<vault-service-url>/v1/pki/ca"
crl_distribution_points="=<vault-service-url>/v1/pki/crl"
```

Create JWT roles

Use the following commands to create the JWT roles for each clusterissuer:

```
$vault write auth/jwt/role/5g.oam.private.jwt role_type="jwt"
bound_audiences="vault://affirmed-clusterissuer-oam" user_claim="sub"
bound_subject="system:serviceaccount:fed-paas-helpers:vault-issuer"
policies="pki-policy-oam" ttl="1h"

$vault write auth/jwt/role/5g.sbi.private.jwt role_type="jwt"
bound_audiences="vault://affirmed-clusterissuer-sbi" user_claim="sub"
```

```
bound_subject="system:serviceaccount:fed-paas-helpers:vault-issuer"
policies="pki-policy-sbi" ttl="1h"

$vault write auth/jwt/role/5g.li.private.jwt role_type="jwt"
bound_audiences="vault://affirmed-clusterissuer-li" user_claim="sub"
bound_subject="system:serviceaccount:fed-paas-helpers:vault-issuer"
policies="pki-policy-li" ttl="1h"
```

Create PKI roles

Use the following commands to create the required PKI roles:

```
$vault write pki/oam/private/int/roles/5g.oam.private
allow_any_name=true allow_subdomains=true max_ttl=72h

$vault write pki/sbi/private/int/roles/5g.sbi.private
allow_any_name=true allow_subdomains=true max_ttl=72h

$vault write pki/li/private/int/roles/5g.li.private allow_any_name=true
allow_subdomains=true max_ttl=72h
```

Create PKI policies

Use the following commands to create the required PKI policies:

```
$vault policy write pki-policy-oam - <<EOF
path "pki/oam/private/int*" { capabilities =
["read", "list"] }
path "pki/oam/private/int/roles/5g.oam.private" { capabilities =
["create", "update"] }
path "pki/oam/private/int/sign/5g.oam.private" { capabilities =
["create", "update"] }
path "pki/oam/private/int/issue/5g.oam.private" { capabilities =
["create", "update", "read", "list"] }
EOF

$vault policy write pki-policy-sbi - <<EOF
path "pki/sbi/private/int*" { capabilities =
["read", "list"] }
path "pki/sbi/private/int/roles/5g.sbi.private" { capabilities =
["create", "update"] }
path "pki/sbi/private/int/sign/5g.sbi.private" { capabilities =
["create", "update"] }
path "pki/sbi/private/int/issue/5g.sbi.private" { capabilities =
["create", "update", "read", "list"] }
EOF

$vault policy write pki-policy-li - <<EOF
path "pki/li/private/int*" { capabilities =
["read", "list"] }
path "pki/li/private/int/roles/5g.li.private" { capabilities =
["create", "update"] }
path "pki/li/private/int/sign/5g.li.private" { capabilities =
["create", "update"] }
```

```
path "pki/li/private/int/issue/5g.li.private" { capabilities =
["create", "update", "read", "list"] }
EOF
```

Configuration - fed-paas_helpers deployment

Install the fed-paas_helpers chart with the following configuration. Modify the configuration according to the values used while configuring Hashicorp vault for PKI paths, Role Name, Issuer Name and JWT auth.

```
global: ...
vault:
enabled: true
hashicorpVault:
enabled: true
#Replace with vault server URL
serverUrl: http://vault-svc.vault:8200
azureKeyVault:
enabled: false
...
pod-cert_manager:
enabled: true
issuerTypes:
hashicorpVault:
enabled: true
serviceAccount: vault-issuer
namespace: ""
issuers:
#Replace with clusterissuer name that is configured in HC vault
- issuerName: affirmed-clusterissuer-sbi
  #Replace with pki path that is configured in HC vault
  pkiPath: "pki/sbi/private/int/sign/5g.sbi.private"
  #Replace with JWT role that is configured in HC vault
  jwtRole: "5g.sbi.private.jwt"
  #Replace with jwt auth path that is configured in HC vault
  jwtMountPath: "/v1/auth/jwt"
- issuerName: affirmed-clusterissuer-li
  pkiPath: "pki/li/private/int/sign/5g.li.private"
```



```

    jwtRole: "5g.li.private.jwt"
    jwtMountPath: "/v1/auth/jwt"
-   issuerName: affirmed-clusterissuer-oam
    pkiPath: "pki/oam/private/int/sign/5g.oam.private"
    jwtRole: "5g.oam.private.jwt"
    jwtMountPath: "/v1/auth/jwt"
...

```

Verify Configuration

Follow these steps to verify the configuration:

1. Verify the configuration by checking the status of clusterissuers.

```

$kc get clusterissuers
NAME                                READY   AGE
affirmed-clusterissuer-li          True    7d
affirmed-clusterissuer-oam         True    7d
affirmed-clusterissuer-sbi         True    7d

```

2. Verify that certificate creation results in the creation of a Kubernetes secret recourse with certificates issued from vault

```

# Create a sample certificate resource
$kubectl create -f - <<EOF
apiVersion: cert-manager.io/v1
kind: Certificate
metadata:
  name: test-certificate
spec:
  commonName: test.com
  dnsNames: - test.com
  duration: 10h0m0s
  issuerRef:
    kind: ClusterIssuer
    name: affirmed-clusterissuer-oam
  renewBefore: 1h0m0s
  secretName: test-secret
  subject:
    organizations:
    - affirmednetworks
  usages:
    - server auth
    - client auth
EOF

#Check if the certificate is created and is in Ready state
$kc get certificate
NAME READY SECRET AGE
test-certificate True test-secret 12s

```

Hashicorp Vault Config with cert-manager

```
# Check if the secret is created
$kc get secret
NAME TYPE DATA AGE
test-secret kubernetes.io/tls 3 14s

#cleanup of test resources
$kc delete certificate test-certificate
certificate.cert-manager.io "test-certificate" deleted

$kc delete secret test-secret
secret "test-secret" deleted
```

Hashicorp Vault Configuration for Secrets Management

This topic provides step-by-step instructions to integrate Hashicorp vault with UC Clusters for secrets retrievals by PAAS and NF federations. The secrets are stored in Hashicorp Vault and based on configuration in UC Cluster, specific secrets are retrieved from vault and stored in the form of Kubernetes secrets using CSI(Container Storage Interface) secret store volumes.

Prerequisites:

Hashicorp vault configuration is performed by Vault administrators. The configuration shown here only serves as guidance and is subject to change based on customer requirements.

Vault Initialization

Please refer to this [section](#).

Enable JWT Auth

Please refer to this [section](#).

Config JWT Auth

Please refer to this [section](#).

Enable and Configure Secrets Engine

1. Run the following command on the NF Cluster to get ClusterIssuer URL:

```
kubectl get --raw /.well-known/openid-configuration | jq -r '.issuer'
```

Execute subsequent commands to configure Vault.

2. Use the following commands to enable and configure the secrets engine on Vault. This command enabled secrets engine at a specific path “secrets” in vault of type “kv” engine

```
vault secrets enable -version=2 -path="secrets" kv
```

3. Configure a vault policy to access secrets engine path. This is generic policy and can access any data within secrets/ path with read permission. It can be made as specific as possible

```

vault policy write reader-policy - <<EOF
path "secrets/*" {
  capabilities = ["read"]
}
EOF

```

4. Use the following commands to create the JWT roles for each federation which is trying to retrieve secrets from vault.

```

vault write auth/jwt/role/<fed-namespace>-reader \
role_type="jwt" \
bound_audiences="<clusterIssuer URL>" \ #ClusterIssuer URL from step(1)
user_claim="sub" \
bound_subject="system:serviceaccount:<fed-namespace>:vault-sa" \
policies="reader-policy" \ # Policy name from step(2)
ttl="1h"

```

For example, if pods in fed-smf are configured to retrieve secrets from vault, the following jwt-policy must be created in vault:

If cluster issuer URL is "https://kubernetes.default.svc.cluster.local" from step (1). Issue following command using vault binary:

```

vault write auth/jwt/role/fed-smf-reader \
role_type="jwt" \
bound_audiences="https://kubernetes.default.svc.cluster.local" \
user_claim="sub" \
bound_subject="system:serviceaccount:fed-smf:vault-sa" \
policies="reader-policy" \
ttl="1h"

```

A similar role must be created for each federation trying to retrieve secrets from vault, by creating unique name for each role and specify appropriate <fed-namespace>. This role will be referred in PAAS/NF federation during deployment.

Creation of Secret in Vault

Secrets can be created in vault using following format inside vault:

```

vault kv put secrets/<secretName> <key>=<value>

```

To keep it simple and atomic, it is recommended that vault secret have only one sensitive data. Also, attribute(<key>) within secret is always set to "value" which holds sensitive data, so it can be referenced in a common way in SecretProviderClass.

For example (test values for reference only):

```
vault kv put secrets/grafanaUser value="test"
vault kv put secrets/grafanaPwd value="$secretPa$$ord "
```

Enable CSI Driver and Provider in fed-kube-addons

Enable deployment of CSI driver and provider in fed-kube-addons when any of PAAS and/or NF federation requires retrieval of secrets from vault.

Set the following values during fed-kube-addons deployment:

```
global:
...
# Vault Parameters
vault:
  enabled: true
  hashicorpVault:
    enabled: true
```

Verify Configuration

When fed-kube-addons is deployed along with changes mentioned above, CSI driver and provider pods will be running in kube-system NS and can be verified from below command.

```
kubect1 get ds -n kube-system | egrep -i "^Name|csi"
```

Configuration Manager support for Hashicorp Vault

When vault is enabled for NF federations: fed-smf, fed-upf(where pod-cfgmgr is running)..etc, using the following helm values:

```
global:
  vault:
    enabled: true
    hashicorpVault:
      enabled: true
...
```

Few sensitive values used by configuration manager must be stored in Vault and references of the path should be provided in helm values for security.

Helm variable: aes256cfb128Key

The value of “aes256cfb128Key” set in values.yaml at below path:

```
pod-cfgmgr:
  enabled: true
  confd:
    aes256cfb128Key: ""
```

The secret must be created in vault using command[sample command]:

```
vault kv put secrets/confdAes256cfb128Key value=<value>
```

Then, the vault path of the secret must be provided in the following path:

```
global:
  ...
  secretNames:
    cfgmgr:
      confdKey: secrets/data/confdAes256cfb128Key
```

If secret path in vault is secrets/confdAes256cfb128Key, then specify path in values.yaml as secrets/data/confdAes256cfb128Key [add data/ after secrets/]

Helm variable: encryptConfigKey

If encryptConfigKey.enabled is set to true and vault is enabled, the key of encrypt config should be created in vault and vault path of secret has to be specified in values.yaml

```
vault kv put secrets/encryptkey value=<value>
```

Then, the path of the secret must be provided in following path:

```
global:
  ...
  secretNames:
    cfgmgr:
      confdKey: secrets/data/confdAes256cfb128Key
      encryptConfigKeySecretName: secrets/data/encryptkey
  ...
```

Similarly, other **optional** parameters/secrets used by configuration manager can be stored in vault and path of these secrets can be specified in:

The following parameters for configuration manager which are set today in helm values.yaml :

```
pod-cfgmgr:
  security:
    root_passwd:
    root_pubkey:
    local_admin_users:
      - user: ""
        passwd: ""
```

```

    pubkey: ""
  local_oper_users:
    - user: ""
      passwd: ""
      pubkey: ""
  local_readonly_users:
    - user: ""
      passwd: ""
      pubkey: ""
  local_liadmin_users:
    - user: ""
      passwd: ""
      pubkey: ""
  local_lioper_users:
    - user: ""
      passwd: ""
      pubkey: ""
  local_maint_users:
    - user: ""
      passwd: ""
      pubkey: ""
  root_domain_ca_key: ""
  admin_domain_ca_key: ""
  liadmin_domain_ca_key: ""
  maint_domain_ca_key: ""

```

These security parameters can be stored in vault and the vault path can be given for (optional) for each argument in

```

global:
  secretNames:
    cfgmgr:
    ..

```

Table 72. Attribute Information

Attributes	Equivalent Vault Path for attribute
adminDomainCAkeySecretName	secrets/data/adminDomainCAkey
liadminDomainCAkeySecretName	secrets/data/liadminDomainCAkey
maintDomainCAkeySecretName	secrets/data/liadminDomainCAkey
limaintDomainCAkeySecretName	secrets/data/limaintDomainCAkey
rootDomainCAkeySecretName	secrets/data/rootDomainCAkey
rootPasswdSecretName	secrets/data/rootPasswd
rootPubkeySecretName	secrets/data/rootPubkey
m2mPubkeySecretName	secrets/data/m2mPubkey

Hashicorp Vault Config for Secrets Management

Attributes	Equivalent Vault Path for attribute
adminUsers:	List of adminUsers.
- usernameSecretName	secrets/data/admin-user0
passwdSecretName	secrets/data/admin-pwd0
pubkeySecretName	secrets/data/admin-pub0
	Each user can have passwd or pubkey. username is mandatory
liadminUsers:	List of liadminUser.
- usernameSecretName	secrets/data/liadmin-user0
passwdSecretName	secrets/data/liadmin-pwd0
pubkeySecretName	secrets/data/liadmin-pub0
	Each user can have passwd or pubkey. username is mandatory
operUsers:	List of operUsers.
- usernameSecretName	secrets/data/oper-user0
passwdSecretName	secrets/data/oper-pwd0
pubkeySecretName	secrets/data/oper-pub0
	Each user can have passwd or pubkey. username is mandatory
lioperUsers:	List of lioperUsers.
- usernameSecretName	secrets/data/lioper-user0
passwdSecretName	secrets/data/lioper-pwd0
pubkeySecretName	secrets/data/lioper-pub0
	Each user can have passwd or pubkey. username is mandatory
readonlyUsers:	List of readonlyUsers.
- usernameSecretName	secrets/data/ro-user0
passwdSecretName	secrets/data/ro-pwd0
pubkeySecretName	secrets/data/ro-pub0
	Each user can have passwd or pubkey. username is mandatory
userProvided:	Vault path of userProvided certificates used for RESTCONF.
tls crtSecretName	Mandatory, if pod-cfgmgr.userProvidedVault.enabled is true
tlskeySecretName	
cacertSecretName	

For example, If lioperUsers has to be configured, create secret in vault with required username and password

Hashicorp Vault Config for Secrets Management

```
vault kv put secrets/lioper-user0 value=testUser
```

and configure

```
global:
```

```
  secretNames:
```

```
    cfgmgr:
```

```
      liOperUsers:
```

```
        - UsernameSecretName: secrets/data/lioper-user0
```

Halo-E integration with UC GUIs

This section is for integration of the following UC GUIs with Halo-E.

- Grafana (deployed in fed-grafana)
- Grafana (deployed in fed-kube-prom)
- Prometheus (deployed in fed-kube-prom)
- Alert Manager (deployed in fed-kube-prom)
- Alert Manager (deployed in fed-prometheus)
- Thanos (deployed in fed-prometheus)
- Jaeger
- Kibana

Pre-requisites

- A Hashicorp Vault cluster should be set up.
- Fed-paas-helpers should be configured to pull secrets from the Hashicorp Vault. This can be done by setting the following in the values.yaml of fed-paas-helpers.

```
→ global.vault.enabled = true
→ global.vault.hashicorpVault.enabled = true
→ global.vault.hashicorpVault.serverUrl = <Hashicorp Vault URL>
→ global.vault.hashicorpVault.reader =
  <Hashicorp Vault role to access secrets>
→ global.vault.hashicorpVault.authMountPath =
  <Hashicorp Vault auth mount path>
→ global.vault.hashicorpVault.vaultNamespace = <Hashicorp Vault
  namespace>
```

Integration steps

Feature Enablement

By default, this feature is disabled in the applicable feds such as fed-paas-helpers, fed-kube-prom, fed-grafana, fed-prometheus, fed-jaeger, fed-kibana, fed-otel-collector, fed-elastic and fed-elastalert.

The feature is enabled by setting the flag *global.authType* = “haloe” in the values.yaml of the respective feds.

Application registration with HaloE

- Only ONE application needs to be registered with HaloE which will represent all the above listed UC GUIs.
- A ClientID representing the application and a ClientSecret would be required to complete the registration.
- The Redirect URIs of all the GUIs that need to be integrated with Halo-E should be configured in Halo-E.

The Redirect URIs should be in the following format

<https://<FQDN>:<port>/oauth2/callback>

Sample redirect URI:

<https://grafana.fed-kube-prom.att.com:3000/oauth2/callback>

Note: Since Halo-E only accepts FQDNs for the Redirect URIs, the configuration can be done even prior to the deployment of the GUIs. Post deployment, the DNS entries should be updated to resolve the FQDNs to the appropriate external IPs of the respective services.

Provision of required values in Hashicorp vault

- The ClientID and ClientSecret values used in the application’s registration with HaloE should be configured as **separate** key-value pairs in Hashicorp Vault.

Note: If a self-signed CA is used to sign the HaloE certificates, then the CA certificate should also be provisioned as a key-value pair in Hashicorp Vault. This step is NOT required if Halo-E certificates are signed by well-known CAs.

Installation of fed-paas-helpers

- In addition to the values.yaml parameters mentioned in the Pre-requisites section above, the following should be configured.
 - The 'key' names for the ClientID and ClientSecret provisioned in Hashicorp Vault should be provided in the fed-paas-helpers values.yaml.
 - The OIDC endpoints of Halo-E deployment should be provisioned in the fed-paas-helpers values.yaml
 - Additionally, the 'key' name of the CA certificate, if provisioned in Hashicorp Vault, should also be provided in the fed-paas-helpers values.yaml.

Note: Whenever any of the above secrets in Hashicorp Vault are rotated, then the oauth2-proxy pod running in fed-paas-helpers needs to be restarted to pick-up the new value(s). The restart of this pod should not impact the call traffic.

```
global:
  secretNames:
    oidc:
      oidcClientIDSecretName: ""
      oidcClientSecretName: ""
      caCertSecretName: ""
```

```
pod-oauth2_proxy:
  oauth2proxy:
    oidcIssuerUrl: ""
    oidcAuthorizeUrl: ""
    oidcUserInfoUrl: ""
    oidcTokenUrl: ""
```

If Vault is enabled, then below secret config should be added for fluentd.

```
global:
  secretNames:
    fluentd:
      mountCSI: true
      usernameSecretName: ""
      passwordSecretName: ""
```

Regarding continuity of the single-sign-on session [OPTIONAL]

It is possible to configure parameters in `fed-paas-helper's values.yaml` that control the continuity of the single-sign-on session once authentication is successful. These parameters are already set with 'default' values, so this step is optional.

- The time after which the 'cookie', that `oauth2-proxy` creates, will expire. The recommendation is to set this to Halo-E's "Refresh-Token lifespan". However, this is set to '0' by default which means that the cookie will be a 'session-cookie' which will expire when the browser is closed, which is recommended from a 'security' perspective.

```
pod-oauth2_proxy:
  cookieExpire: <value> # Default is "0"
```

- If this parameter is set to a non-zero value, the cookie remains active for the specified time, which means the session will be active even after closing the browser. Format for setting this is in hms, e.g., 2h3m4s or 3h or 50m or 5s

- OAuth2-proxy tries to refresh an Access-Token after the duration set in the parameter below. The recommendation is to set this to Halo-E's "Access-Token lifespan - 1m". By default, it is set to 59m. Format for setting this is in hms, e.g., 2h3m4s or 3h or 50m or 5s

```
pod-oauth2_proxy:
  cookieRefresh: <value> # Default is "59m"
```

Certificate Management in Istio Ingress and Egress Gateways

Istio Ingress and Egress Gateway (both LI and non-LI) provides mTLS encryption through the TLS certificates provisioned to these gateways in the form of K8s secrets. TLS certificates in the form of secrets are provisioned by cert-manager which interacts with Hashicorp Vault for fetching the certificates.

TLS Certificates Provisioning through Cert-Manager/Hashicorp Vault

Istio Ingress and Egress Gateways helm chart provides the interface for configuring the certificates provisioned through cert-manager/HashiCorp Vault mode. The following is example configuration:

```
global:
# PaaS Service Mesh Selector
  mesh:
    istio:
      mtlsEnabled: false          <===== Change
this to true
# This controls use of TLS in GWs
  ingressGwTLS:
    enabled: false              <===== Change
this to true
# Set the tls mode
# SIMPLE --> to enable only TLS
# MUTUAL --> to enable mTLS (default)
    mode: MUTUAL

# The below section covers the TLS certificates provisioning for
# Istio Ingress Gateway
  certManager:
    enabled: true
    certDuration: 10h
    renewBefore: 1h
    organization: affirmednetworks
    issuerName: affirmed-clusterissuer-sbi
```

Certificate Management in Istio Ingress and Egress Gateways

```
    issuerType: "vault"                                     <===== This must
be set to vault for the cert-manager to fetch the
certificates from HashiCorp Vault

    egressGwTLS:

        enabled: false                                     <===== Change
this to true

# Set the tls and mTLS mode
# SIMPLE --> to enable only TLS
# MUTUAL --> to enable mTLS (default)

    mode: MUTUAL

# The below section covers the TLS certificates provisioning for
# Istio Egress Gateway

    certManager:

        enabled: true

        certDuration: 10h

        renewBefore: 1h

        organization: affirmednetworks

        issuerName: affirmed-clusterissuer-sbi

        issuerType: "vault"                               <===== This must
be set to vault for the cert-manager to fetch the certificates from
HashiCorp Vault
```

When Istio Ingress and Egress Gateways are configured during the deployment like the preceding, users can verify if the certificates are provisioned correctly using the following command:

```
rmangala@dev-rmangala:~$ kc get Certificates -n fed-smf
```

NAME	READY	SECRET	AGE
smf-com-egress-certificate controller 25d	True	egress-cert-manager-cert-	
smf-com-ingress-certificate cert-controller 25d	True	smf-com-ingress-cert-manager-	
smf-com-li-egress-certificate controller 25d	True	egress-li-cert-manager-cert-	
smfli-com-li-ingress-certificate manager-cert-controller 25d	True	smfli-com-li-ingress-cert-	

In the preceding output, Certificates provisioning is considered to be successful only when READY status is set to true. With the READY status set to true, egress-cert-manager-cert-controller, smf-com-ingress-cert-manager-cert-controller, egress-li-cert-manager-cert-controller and smfli-com-li-ingress-cert-manager-cert-controller are the secrets created with the TLS certificates present inside it for Istio Egress Gateway, Istio Ingress Gateway, Istio LI Egress Gateway and Istio LI Ingress Gateway respectively.

The following is an example secret that is created by cert-manager:

```

apiVersion: v1
data:
  ca.crt: <...>
  tls.crt: <...>
  tls.key: <...>
kind: Secret
metadata:
  annotations:
    cert-manager.io/alt-names: smf.com
    cert-manager.io/certificate-name: smf-com-egress-certificate
    cert-manager.io/common-name: smf.com
    cert-manager.io/ip-sans: ""
    cert-manager.io/issuer-group: ""
    cert-manager.io/issuer-kind: ClusterIssuer
    cert-manager.io/issuer-name: affirmed-clusterissuer-sbi
    cert-manager.io/uri-sans: ""
  creationTimestamp: "2024-01-16T15:17:11Z"
  labels:
    controller.cert-manager.io/fao: "true"
  name: egress-cert-manager-cert-controller
  namespace: fed-smf
  resourceVersion: "39692174"
  uid: f507a170-ee77-46c6-8906-2c2e1b816337
type: kubernetes.io/tls

```


Ingress and Egress Gateways mTLS Encryption Observability

Istio Ingress and Egress Gateways have the responsibility of interfacing with the external hosts over HTTP/TCP communication. It is these gateways that secure communication using the mTLS authentication and encryption agnostic to the applications that communicate with the external hosts. This is applicable to both LI as well as non-LI gateways.

Both the Istio gateways provide observability about the mTLS authentication and encryption in the event of communication failures using the Prometheus metrics. Observability through logs is disabled by default as it is CPU intensive to enable them in a production environment. Observability through Prometheus metrics is enabled by default.

Istio Ingress Gateway KPIs

The following are the Prometheus metrics exposed by Istio Li and non-LI Ingress Gateways:

- `envoy_listener_ssl_connection_error` - Total TLS connection errors not including failed certificate verifications.
- `envoy_listener_ssl_handshake` - Total successful TLS connection handshakes.
- `envoy_listener_ssl_session_reused` - Total successful TLS session resumptions.
- `envoy_listener_ssl_no_certificate` - Total successful TLS connections with no client certificate.
- `envoy_listener_ssl_fail_verify_no_cert` - Total TLS connections that failed because of missing client certificate.
- `envoy_listener_ssl_fail_verify_error` - Total TLS connections that failed CA verification.
- `envoy_listener_ssl_fail_verify_san` - Total TLS connections that failed SAN verification.
- `envoy_listener_ssl_fail_verify_cert_hash` - Total TLS connections that failed certificate pinning verification.
- `envoy_listener_ssl_ocsp_staple_failed` - Total TLS connections that failed compliance with the OCSP policy.
- `envoy_listener_ssl_ocsp_staple_omitted` - Total TLS connections that succeeded without stapling an OCSP response.
- `envoy_listener_ssl_ocsp_staple_responses` - Total TLS connections where a valid OCSP response was available (irrespective of whether the client requested stapling).
- `envoy_listener_ssl_ocsp_staple_requests` - Total TLS connections where the client requested an OCSP staple.

- `envoy_listener_ssl_ciphers_<cipher>` - Total successful TLS connections that used cipher `<cipher>`. Example cipher: `TLS_AES_128_GCM_SHA256`.
- `envoy_listener_ssl_curves_<curve>` - Total successful TLS connections that used ECDHE curve `<curve>`. Example curve: `X25519`.
- `envoy_listener_ssl_sigalgs_<sigalg>` - Total successful TLS connections that used signature algorithm `<sigalg>`. Example sigalg: `rsa_pss_rsae_sha256`.
- `envoy_listener_ssl_versions_<version>` - Total successful TLS connections that used protocol version `<version>`. Example version: `TLSv1_3`.

Following is an example of retrieving the SSL metrics from the Istio Ingress Gateways:

```
$kc exec -it istio-ingressgateway-554b4bdd7b-gbt7m -c istio-proxy bash -n
fed-smf -- curl http://127.0.0.1:15090/stats/prometheus | grep ssl

# TYPE
envoy_listener_server_ssl_socket_factory_downstream_context_secrets_not_re
ady counter
envoy_listener_server_ssl_socket_factory_downstream_context_secrets_not_re
ady{listener_address="0.0.0.0_8443"} 0
# TYPE envoy_listener_server_ssl_socket_factory_ssl_context_update_by_sds
counter
envoy_listener_server_ssl_socket_factory_ssl_context_update_by_sds{listene
r_address="0.0.0.0_8443"} 22
# TYPE
envoy_listener_server_ssl_socket_factory_upstream_context_secrets_not_read
y counter
envoy_listener_server_ssl_socket_factory_upstream_context_secrets_not_read
y{listener_address="0.0.0.0_8443"} 0
# TYPE envoy_listener_ssl_ciphers_TLS_AES_128_GCM_SHA256 counter
envoy_listener_ssl_ciphers_TLS_AES_128_GCM_SHA256{listener_address="0.0.0.
0_8443"} 5
# TYPE envoy_listener_ssl_connection_error counter
envoy_listener_ssl_connection_error{listener_address="0.0.0.0_8443"} 2
# TYPE envoy_listener_ssl_curves_X25519 counter
envoy_listener_ssl_curves_X25519{listener_address="0.0.0.0_8443"} 5
# TYPE envoy_listener_ssl_fail_verify_cert_hash counter
envoy_listener_ssl_fail_verify_cert_hash{listener_address="0.0.0.0_8443"}
0
# TYPE envoy_listener_ssl_fail_verify_error counter
envoy_listener_ssl_fail_verify_error{listener_address="0.0.0.0_8443"} 0
# TYPE envoy_listener_ssl_fail_verify_no_cert counter
envoy_listener_ssl_fail_verify_no_cert{listener_address="0.0.0.0_8443"} 0
# TYPE envoy_listener_ssl_fail_verify_san counter
envoy_listener_ssl_fail_verify_san{listener_address="0.0.0.0_8443"} 0
# TYPE envoy_listener_ssl_handshake counter
envoy_listener_ssl_handshake{listener_address="0.0.0.0_8443"} 5
# TYPE envoy_listener_ssl_no_certificate counter
envoy_listener_ssl_no_certificate{listener_address="0.0.0.0_8443"} 0
# TYPE envoy_listener_ssl_ocsp_staple_failed counter
envoy_listener_ssl_ocsp_staple_failed{listener_address="0.0.0.0_8443"} 0
# TYPE envoy_listener_ssl_ocsp_staple_omitted counter
```

```
envoy_listener_ssl_ocsp_staple_omitted{listener_address="0.0.0.0_8443"} 0
# TYPE envoy_listener_ssl_ocsp_staple_requests counter
envoy_listener_ssl_ocsp_staple_requests{listener_address="0.0.0.0_8443"} 0
# TYPE envoy_listener_ssl_ocsp_staple_responses counter
envoy_listener_ssl_ocsp_staple_responses{listener_address="0.0.0.0_8443"}
0
# TYPE envoy_listener_ssl_session_reused counter
envoy_listener_ssl_session_reused{listener_address="0.0.0.0_8443"} 2
# TYPE envoy_listener_ssl_sigalgs_rsa_pss_rsae_sha256 counter
envoy_listener_ssl_sigalgs_rsa_pss_rsae_sha256{listener_address="0.0.0.0_8443"} 5
# TYPE envoy_listener_ssl_versions_TLSv1_3 counter
envoy_listener_ssl_versions_TLSv1_3{listener_address="0.0.0.0_8443"} 5
```

Istio Egress Gateway KPIs

The following are the Prometheus metrics exposed by Istio LI and non-LI Egress Gateways:

- `envoy_cluster_com_ssl_connection_error` - Total TLS connection errors not including failed certificate verifications.
- `envoy_cluster_com_ssl_handshake` - Total successful TLS connection handshakes.
- `envoy_cluster_com_ssl_session_reused` - Total successful TLS session resumptions.
- `envoy_cluster_com_ssl_no_certificate` - Total successful TLS connections with no client certificate.
- `envoy_cluster_com_ssl_fail_verify_no_cert` - Total TLS connections that failed because of missing client certificate.
- `envoy_cluster_com_ssl_fail_verify_error` - Total TLS connections that failed CA verification.
- `envoy_cluster_com_ssl_fail_verify_san` - Total TLS connections that failed SAN verification.
- `envoy_cluster_com_ssl_fail_verify_cert_hash` - Total TLS connections that failed certificate pinning verification.
- `envoy_cluster_com_ssl_ocsp_staple_failed` - Total TLS connections that failed compliance with the OCSP policy.
- `envoy_cluster_com_ssl_ocsp_staple_omitted` - Total TLS connections that succeeded without stapling an OCSP response.
- `envoy_cluster_com_ssl_ocsp_staple_responses` - Total TLS connections where a valid OCSP response was available (irrespective of whether the client requested stapling).
- `envoy_cluster_com_ssl_ocsp_staple_requests` - Total TLS connections where the client requested an OCSP staple.

- `envoy_cluster_com_ssl_ciphers_<cipher>` - Total successful TLS connections that used cipher `<cipher>`. Example cipher: `TLS_AES_128_GCM_SHA256`.
- `envoy_cluster_com_ssl_curves_<curve>` - Total successful TLS connections that used ECDHE curve `<curve>`. Example curve: `X25519`.
- `envoy_cluster_com_ssl_sigalgs_<sigalg>` - Total successful TLS connections that used signature algorithm `<sigalg>`. Example sigalg: `rsa_pss_rsae_sha256`.
- `envoy_cluster_com_ssl_versions_<version>` - Total successful TLS connections that used protocol version `<version>`. Example version: `TLSv1_3`.

The preceding Prometheus metrics can be observed in Prometheus GUI or can be retrieved from the individual Istio Gateway pod instance as well.

The following is an example of fetching the SSL metrics from Istio Egress Gateway:

```
$kc exec -it istio-egressgateway-57db8ff9-lmk4w -c istio-proxy bash -n
fed-smf -- curl http://127.0.0.1:15090/stats/prometheus | grep ssl

# TYPE envoy_cluster_com_ssl_ciphers_TLS_AES_128_GCM_SHA256 counter
envoy_cluster_com_ssl_ciphers_TLS_AES_128_GCM_SHA256{cluster_name="outbound|443||3sep1j5"} 3
# TYPE envoy_cluster_com_ssl_connection_error counter
envoy_cluster_com_ssl_connection_error{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_connection_error{cluster_name="outbound|443||3sep1j5"} 0
# TYPE envoy_cluster_com_ssl_curves_X25519 counter
envoy_cluster_com_ssl_curves_X25519{cluster_name="outbound|443||3sep1j5"} 3
# TYPE envoy_cluster_com_ssl_fail_verify_cert_hash counter
envoy_cluster_com_ssl_fail_verify_cert_hash{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_fail_verify_cert_hash{cluster_name="outbound|443||3sep1j5"} 0
# TYPE envoy_cluster_com_ssl_fail_verify_error counter
envoy_cluster_com_ssl_fail_verify_error{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_fail_verify_error{cluster_name="outbound|443||3sep1j5"} 0
# TYPE envoy_cluster_com_ssl_fail_verify_no_cert counter
envoy_cluster_com_ssl_fail_verify_no_cert{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_fail_verify_no_cert{cluster_name="outbound|443||3sep1j5"} 0
# TYPE envoy_cluster_com_ssl_fail_verify_san counter
envoy_cluster_com_ssl_fail_verify_san{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_fail_verify_san{cluster_name="outbound|443||3sep1j5"} 0
# TYPE envoy_cluster_com_ssl_handshake counter
```

```

envoy_cluster_com_ssl_handshake{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_handshake{cluster_name="outbound|443||3sep1j5"} 3
# TYPE envoy_cluster_com_ssl_no_certificate counter
envoy_cluster_com_ssl_no_certificate{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_no_certificate{cluster_name="outbound|443||3sep1j5"} 0
# TYPE envoy_cluster_com_ssl_ocsp_staple_failed counter
envoy_cluster_com_ssl_ocsp_staple_failed{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_ocsp_staple_failed{cluster_name="outbound|443||3sep1j5"} 0
# TYPE envoy_cluster_com_ssl_ocsp_staple_omitted counter
envoy_cluster_com_ssl_ocsp_staple_omitted{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_ocsp_staple_omitted{cluster_name="outbound|443||3sep1j5"} 0
# TYPE envoy_cluster_com_ssl_ocsp_staple_requests counter
envoy_cluster_com_ssl_ocsp_staple_requests{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_ocsp_staple_requests{cluster_name="outbound|443||3sep1j5"} 0
# TYPE envoy_cluster_com_ssl_ocsp_staple_responses counter
envoy_cluster_com_ssl_ocsp_staple_responses{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_ocsp_staple_responses{cluster_name="outbound|443||3sep1j5"} 0
# TYPE envoy_cluster_com_ssl_session_reused counter
envoy_cluster_com_ssl_session_reused{cluster_name="outbound|443||35322e3137312e36342e3230313a343433"} 0
envoy_cluster_com_ssl_session_reused{cluster_name="outbound|443||3sep1j5"} 1
# TYPE envoy_cluster_com_ssl_sigalgs_rsa_pss_rsae_sha256 counter
envoy_cluster_com_ssl_sigalgs_rsa_pss_rsae_sha256{cluster_name="outbound|443||3sep1j5"} 3
# TYPE envoy_cluster_com_ssl_versions_TLSv1_3 counter
envoy_cluster_com_ssl_versions_TLSv1_3{cluster_name="outbound|443||3sep1j5"} 3

```

In this Prometheus metrics output, the `cluster_name` label indicates external host with which the TLS communication is enabled. To know more about the `cluster_name` and how to map it to the external host defined in the K8s cluster, follow these instructions:

1. Copy the string which is defined after the destination port defined in the `cluster_name` label value:

```
cluster_name="outbound|443||3sep1j5"
```

2. Run the following command:

```
$kc get ServiceEntry -n fed-smf | grep 3sep1j510-91-164-11-external-host-se ["3sep1j5.com"]    STATIC    7d14h
```

3. Copy the ServiceEntry name and run the following command to know more the external host:

Ingress and Egress Gateways mTLS Encryption Observability

```
$kc get ServiceEntry -n fed-smf **10-91-164-11-external-host-se** -oyaml
apiVersion: networking.istio.io/v1beta1
kind: ServiceEntry
metadata:
  creationTimestamp: "2024-02-05T17:16:15Z"
  generation: 1
  name: 10-91-164-11-external-host-se
  namespace: fed-smf
  ownerReferences:
    - apiVersion: istioconfig.affirmednetworks.com/v1alpha1
      blockOwnerDeletion: true
      controller: true
      kind: ExternalHost
      name: istio-config-10-91-164-11
      uid: 49e974be-cd3c-4fac-bc7a-2779e2d58d78
  resourceVersion: "29457415"
  uid: 99b40c70-0a91-4538-9bf0-6733b386aaa6
spec:
  endpoints:
    - address: 10.91.164.11
  exportTo:
    - .
  hosts:
    - 3sep1j5.com
  ports:
    - name: https-443
      number: 443
      protocol: HTTPS
      resolution: STATIC
```

TLS Communication Failure Alerts

Istio Ingress and Egress Gateways define two metrics-based alerts indicating mTLS communication failure for the following reasons:

1. SSL Connection Error

- 2. Failed certificate pinning verification
- 3. Failed CA verification
- 4. Failed because of missing client certificate
- 5. Failed SAN verification
- 6. TLS connections with no client certificate
- 7. Failed compliance with the OCSP policy

Ingress TLS Communication Failure Alert

Figure 105. Ingress TLS Communication Recording Rule

alert: <code>ingressTLSCommunicationFailureAlert</code> expr: <code>increase(ingress_tls_communication_failures_total[1m]) > 0</code> for: 2m labels: cluster: 10.91.164.8 prometheus: fed-prometheus/prometheus severity: critical shard: 0 annotations: description: This alert indicates that the Istio Ingress Gateway is facing TLS communication failures while establishing TLS sessions from its clients. The alert is based on failure metrics documented here ["https://www.envoyproxy.io/docs/envoy/latest/configuration/listeners/stats#tls-statistics"]. Refer to job label to identify which gateway is reporting the failures in its communication. podname: ({{ \$labels.pod }}) potentialImpact: TLS communication failures at Istio Ingress Gateway could result in peers (Eg. 3GPP NF) unable to communicate could result in call drops causing service impact. repairAction: Check the certificates configuration at the Istio Ingress Gateway or the TLS peers for correction. summary: TLS communication failures in Istio Ingress Gateways encountered.	OK	19.169s ago	0.348ms
---	----	-------------	---------

Egress TLS Communication Failure Alert Definition

Figure 106. Egress TLS Communication Failure Alert Definition

Rule	State	Error	Last Evaluation	Evaluation Time
alert: <code>egressTLSCommunicationFailureAlert</code> expr: <code>increase(egress_tls_communication_failures_total[1m]) > 0</code> for: 2m labels: cluster: 10.91.164.8 prometheus: fed-prometheus/prometheus severity: critical shard: 0 annotations: description: This alert indicates that there are TLS communication failures at the Istio Egress Gateways(including UI gateway). This could indicate certificates misconfiguration in the gateways or at the TLS peers. This alert is based on failure metrics documented here ["https://www.envoyproxy.io/docs/envoy/latest/configuration/upstream/cluster_manager/cluster_stats#tls-statistics"]. Refer to job label to identify which gateway is reporting the failures in its communication. podname: ({{ \$labels.pod }}) potentialImpact: The Egress Gateways would not be able to communicate with the peers (Eg. 3GPP NF) which could result in service impact. repairAction: Check the certificates configuration at the Istio Egress Gateway or the TLS peers for correction. summary: TLS communication failures in Istio Egress Gateways encountered.	OK		19.166s ago	0.382ms

The following is an example of the Egress TLS Communication Failure Alert raised.

Figure 107. Egress TLS Communication Failure Alert



MNOPM HashiCorp Vault Integration

The integration of HashiCorp Vault with MNOPM (Multi node OPM) federation is a strategic initiative aimed at enhancing the security and management of sensitive information within the Unity Cloud solution from UC 5.2 release. This integration focuses on migrating sensitive data such as usernames, passwords and connection strings from local storage to the HashiCorp Vault, ensuring secure and centralized management of secrets.

Key Features

Vault Configuration: Helm specific value JSON allowing users to enable or disable the HashiCorp Vault, set the server URL, define roles, and specify authentication methods.

Sensitive Data Management: All sensitive data used in the fed-opm_multi (multi-node) deployment, including credentials will be stored in the HashiCorp Vault. This eliminates plaintext sensitive data from Helm values and replaces them with vault path references.

Automatic Secret Rotation: OPM will manage the rotation of secrets configured in the Vault server automatically, reducing the need for manual intervention.

Exclusions: Single Node OPM is excluded from this integration due to its lightweight nature and the impracticality of using Kube add-ons and a running K8 cluster

Pre-requisites

To ensure a smooth integration of HashiCorp Vault with OPM, the following pre-requisites must be met:

Vault Server Management: Each customer will manage their own HashiCorp Vault server.

Centralized Vault Server: A centralized (external) Vault server will be used, with multiple Kubernetes clusters interacting with it.

Authentication Method: JWT authentication method with static keys is preferred.

Secret Setup: OPM secret information must be set up in the Vault before deployment.

Vault Availability and Configuration: The availability and configuration of the Vault server will be managed by the customer's HashiCorp Vault team.

Secrets to be migrated to HashiCorp Vault

The following table lists the OPM secrets to be migrated to HashiCorp.

Table 73. OPM Secrets

Name	Namespace	Data	Description
mongo-auth	vnfmgr	username, password, connectionString	Mongo auth credentials
mongo-init-auth	vnfmgr	username, password, connectionString	Mongo Init auth credentials
opm-auth	vnfmgr	username, password	OPM auth credentials

MNOPM Deployment Order

- The deployment of CSI Driver and CSI Provider occurs via `fed-kube_addons`. As `fed-opm-multi` relies on `fed-kube-addons`, the CSI components will be prepared prior to the installation of `Fed-opm-multi`.
- Since the CSI volume utilizing `SecretProviderClass (SPC)` is mounted onto the `vault-helper` pod, it is necessary to establish a dependency on the `pod-vault_helper` as well.
- In summary, both **fed-kube-addons** and **fed-paas-helpers** need to be deployed before the deployment of `fed-opm-multi` to enable the use of HashiCorp Vault.

GUI interface

The following configuration snippet allows users to enable or disable the vault and set the required parameters. By default, HashiCorp Vault is disabled; users can choose to enable it and configure the necessary parameters.

Figure 108. Enable and Configure HashiCorp Vault parameters



 fed-opm_multi - 0.0.0-466-main-int values.yaml

vault

Enabled * ☒ ⓘ

hashicorpVault

Enabled * ☒ ⓘ

Role ⓘ

Vault Namespace ⓘ

Server Url ⓘ

Auth Mount Path ⓘ

Below is a configuration snippet for users to set secret paths in the vault. The listed values are defaults; users can configure different paths in the vault server and GUI when deploying MNOPM with HashiCorp vault enabled.

Figure 109. Configure the secret vault paths for the OPM secrets

fed-opm_multi - 0.0.0-466-main-int values.yaml

secretNames

fedopmmulti

Opm Username Secret Name	secrets/data/opmUserne	Opm Password Secret Name	secrets/data/opmPassw
Mongo Username Secret Name	secrets/data/mongoUse	Mongo Password Secret Name	secrets/data/mongoPas
Mongo Init Username Secret Name	secrets/data/mongoInitU	Mongo Init Password Secret Name	secrets/data/mongoInitP
Mongo Connection String Secret Name	secrets/data/mongoCon		

Changes required in fed-opm-multi helm chart

The following snippet configures the vault parameters in the fed-opm-multi helm chart. The user inputs from the GUI will be placed in this section during deployment.

Figure 110. HashiCorp Vault Integration – Required changes in chart

```
# Vault specific values
vault:
  # Global switch to enable / disable vault configuration
  enabled: false
  # Any one type of vault should be enable at a time
  hashicorpVault:
    # Switch to enable / disable Hashicorp vault
    enabled: false
    # Hashicorp Vault server URL
    serverUrl: http://fed-hc-vault-test-svc-internal.fed-hc-vault-test:8200
    # Hashicorp Vault role to access secrets
    role: "reader"
    # Hashicorp Vault Auth mount path
    authMountPath: "jwt"
    # Hashicorp Vault namespace
    vaultNamespace: ""
```

Figure 111. HashiCorp Vault Parameter definition

Variables	Description
<code>enable-global-vault</code>	To enable the global vault option for the charts. Default is false.
<code>enable-hashicorp-vault</code>	To enable the HashiCorp Vault to read the secrets from the vault, by default false
<code>hashicorp-vault-server-url</code>	HashiCorp Vault server URL, basically vault server endpoint. For example, <code>http://fed-hc-vault-test-svc-internal.fed-hc-vault-test:8200</code>
<code>role</code>	HashiCorp Vault role to access secrets, default is reader role (Optional)
<code>authMountPath</code>	HashiCorp Vault Auth mount path, default is jwt (Optional)
<code>vaultNamespace</code>	HashiCorp Vault namespace, default is empty (Optional)

Below is the snippet for configuring the secret vault path

Figure 112. Snippet – secret vault path

```
secretNames:
  fedopmmulti:
    opmUsernameSecretName: secrets/data/opmUsernameSecretName
    opmPasswordSecretName: secrets/data/opmPasswordSecretName
    mongoUsernameSecretName: secrets/data/mongoUsernameSecretName
    mongoPasswordSecretName: secrets/data/mongoPasswordSecretName
    mongoInitUsernameSecretName: secrets/data/mongoInitUsernameSecretName
    mongoInitPasswordSecretName: secrets/data/mongoInitPasswordSecretName
    mongoConnectionStringSecretName: secrets/data/mongoConnectionStringSecretName
```

Security Provider for MNOPM

SPC integrates with HashiCorp Vault by defining which secrets to fetch. According to the **SecretNames** definition, one SPC at the federation level gathers secrets from the HC vault for both the federation and pods. The SPC class name must match the parameter name in the **secretNames** section in values.yaml. A new SPC named **fedopmmulti-spc.yaml** will be deployed with fed-opm-multi. The SPC file is located at `/fed-opm_multi/interface/templates/`.

In service auto secrets rotation upon rotation in Vault

The CSI driver polls the HashiCorp Vault server every 2 minutes to update K8 secrets with the latest information. The **OpmSecretWatcher** daemon runs alongside the existing TLS certificate monitor in the k8rm pod to detect rotated secrets in the Vault. It updates user secrets in the K8 registry if there are changes in opm-auth, and updates credentials in Mongo-DB and mongo-init. This dynamic update may cause minor service disruptions but requires no manual intervention.

Changes in the User Management

With the admin user migrating to the vault, it cannot be managed by the MNOPM GUI when HashiCorp Vault is enabled. If a user attempts to update the admin from the GUI, a message will state that updates must be made through HashiCorp Vault.

Other users can be created or updated in the GUI when the vault is enabled.

Verify the HashiCorp Vault Integration

Below are the steps to verify whether the secrets are being read from the HashiCorp Vault after the successful deployment / upgrade of MNOPM federation. Execute the following command on the master node using the command line interface (CLI).

```
kc get secrets -n vnfmgr
```

The TYPE of the MNOPM secrets (opm-auth, mongo-auth, and mongo-init-auth) should be "Opaque".

Figure 113. Verify the secrets are populated from HashiCorp Vault

NAME	TYPE	DATA	AGE
kubeconfig-def765-0123456789abcdef	Opaque	1	12d
m2m-private-keys	Opaque	0	12d
mongo-auth	Opaque	3	12d
mongo-bootstrap-auth-data	Opaque	1	12d
mongo-init-auth	Opaque	3	12d
opm-auth	Opaque	2	12d
opm-unitycloud-com-ingress-cert-manager	kubernetes.io/tls	3	12d
sh.helm.release.v1.fed-opm-multi.v1	helm.sh/release.v1	1	12d
users	Opaque	1	12d